Skill India
कौशल भारत - कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

SCGJ
SKILL COUNCIL FOR GREEN JOBS
ISO 9001 : 2015 CERTIFIED

Participant Handbook

Sector
Green Jobs

Sub Sector
Renewable Energy

Occupation
Entrepreneur

Reference Id
SGJ/Q0901, Version 3.0; NSQF Level 4

Please Scan/ Click QR Code to access e book

Solar Photovoltaic Entrepreneur

This book is sponsored by

Skill Council for Green Jobs

3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri, New Delhi – 110021 Email: info@sscgj.in

Website: https://sscgj.in/
Phone: 011.41792866

Disclaimer

The information contained herein has been obtained from sources reliable to Skill Council for Green Jobs (SCGJ). SCGJ disclaims all warranties to the accuracy, completeness or adequacy of such information. SCGJ shall have no liability for errors, omissions or inadequacies in the information contained herein or for interpretations thereof. Every effort has been made to trace the owners of the copyright material included in the book. The publishers would be grateful for any omissions brought to their notice for acknowledgements in future editions of the book. No entity in SCGJ shall be responsible for any loss whatsoever, sustained by any person who relies on this material. All pictures shown are for illustration purpose only. The coded boxes in the book called Quick Response Code (QR Code) will help to access the e. resources linked to the content. These QR codes are generated from links and youtube video resources available on internet for knowledge enhancement on the topic and are not created by SCGJ. Embedding of the link or QR code in the content should not be assumed endorsement of any kind.

SCGJ is not responsible for the views expressed or content or reliability of linked videos. SCGJ can not guarantee that these links/QR codes will work all the time as we have no control over availability of linked pages.

> Skilling is building a better India.
> If we have to move India towards
> development then Skill Development
> should be our mission.

Shri Narendra Modi
Prime Minister of India

Certificate

COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

SKILL COUNCIL FOR GREEN JOBS

for

SKILLING CONTENT: PARTICIPANT HANDBOOK

Complying to National Occupational Standards of
Job Role/ Qualification Pack: '**Solar Photovoltaic Entrepreneur**' QP No. '**SGJ/ Q 0901 NSQF Level 4**'

Date of Issuance: 08/05/2025
Valid up to*: 07/05/2028

Valid up to the next review date of the Qualification Pack or the 'Valid up to' date mentioned above (whichever is earlier)

Authorised Signatory
(Skill Council for Green Jobs)

Acknowledgement

Institute of Solar Power Technologies & Vocational Training Development

Kontent Edge Private Limited

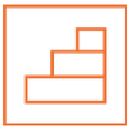
About this book

Government of India is aiming towards a capacity installation of over 300 GW of solar PV out (includes both rooftop and ground mounted projects) of cumulative 450 GW of renewable energy targets by 2030. In 2015, a target of 175 GW of renewable energy capacity was planned to be achieved by 2022 out of which 100 GW was from Solar Energy (40 GW by Solar rooftop and 60 GW by Ground mounted projects). As per an estimate of a recent study carried out by CEEW-NRDC and SCGJ, over 3 Million new jobs can be created in domestic renewable energy sector (mainly solar and wind energy) by 2030. Considering the huge technically trained workforce requirement to meet this ambitious goal, Skill Council for Green Jobs has designed and implemented a range of skilling interventions with a focus on Project construction, system Installation and Operations & Maintenance related job roles for both solar rooftop and ground mounted solar projects. Solar PV Entrepreneur is a key job role who ventures into Solar market to lead an enterprise. He/She has the understanding of solar business models, market, technical knowledge of rooftop solar PV plants, along with components procurement and financing. Solar Photovoltaic Entrepreneur can prepare the feasibility study report and perform basic energy generation forecasting using simulation software and is also responsible for the managing the complete Solar PV project lifecycle. He/she is required by the solar industry to drive capacity addition in both solar rooftop and ground mounted projects. Multiple trainings have also been undertaken under PMKVY scheme, State funding's and other market mode to ensure that skilled workforce are readily made available to meet the requirements of the fast growing Indian solar industry. This Participant Handbook is designed to enable training for the specific Qualification Pack (QP). The contents of this book are in simple language, without going into too much theoretical details and calculations. It is envisaged that this training manual will provide the participants with the knowledge and skills required for installing and maintaining a Solar Photovoltaic System with a focus on solar rooftop, complying with all applicable codes, standards, and safety requirements; and enable them to actively participate in the growing solar rooftop market.

Units and symbols used in the book have been listed below

Symbols Used

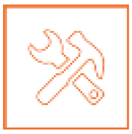
Key Learning Outcomes



Steps

Role Play

Tips



Practical

Unit Objectives

Activity

Exercise

Table of Contents

1. Introduction to Solar PV Sector in India

SGJ/N0111

Key Learning Outcomes

At the end of this module, participants will be able to:

1. Discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India.
2. Discuss various market research reports and industrial magazines present in the market.
3. Explain different type of ground mounted and Rooftop Solar PV Power Plants and their working principles.
4. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
5. Explain type of off- grid Solar PV Power devices and their working principles.
6. Explain various components of a solar PV system and their operating principles.
7. Explain solar energy and power sector landscape in the country.
8. Discuss benefits of solar energy over conventional sources of energy.
9. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country.
10. Analyse the basics of electrical concepts like voltage, current, power, energy, etc
11. Explain typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components.
12. Demonstrate how to interpret sample Electricity bill of a customer, notices and/or cautions.
13. Explain how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans.
14. Discuss emerging solar entrepreneurship landscape including success stories of various start- ups.

UNIT 1.1: Overview of Solar PV Technology

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India.
2. Explain different type of ground mounted and Rooftop Solar PV Power Plants and their working principles.
3. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
4. Explain type of off- grid Solar PV Power devices and their working principles.
5. Explain various components of a solar PV system and their operating principles.
6. Explain solar energy and power sector landscape in the country.
7. Discuss benefits of solar energy over conventional sources of energy.

1.1.1 Introduction

The Sun is the natural source of energy on Earth. The energy received from the Sun is in the form of heat energy and light energy. Voltage generated using Sunlight is called Solar Photovoltaics / PV & using Heat energy directly for different purposes is called Solar Thermal. These energies are received in the form of electromagnetic radiations. The incoming solar radiations are captured using solar panels, that convert solar energy to electrical energy. The electrical energy generated by solar panels is stored in a battery in an Off-Grid SPV System or pumped into the grid directly in a Grid Tie/Grid Interactive SPV System, and is then used to power various electronic devices in the house as shown in the following figure:

Fig. 1.1.1: Solar Energy

The solar energy is also a type of renewable energy or green energy.

Solar Energy is being utilized by both Direct and Indirect conversion as shown in the following figure:

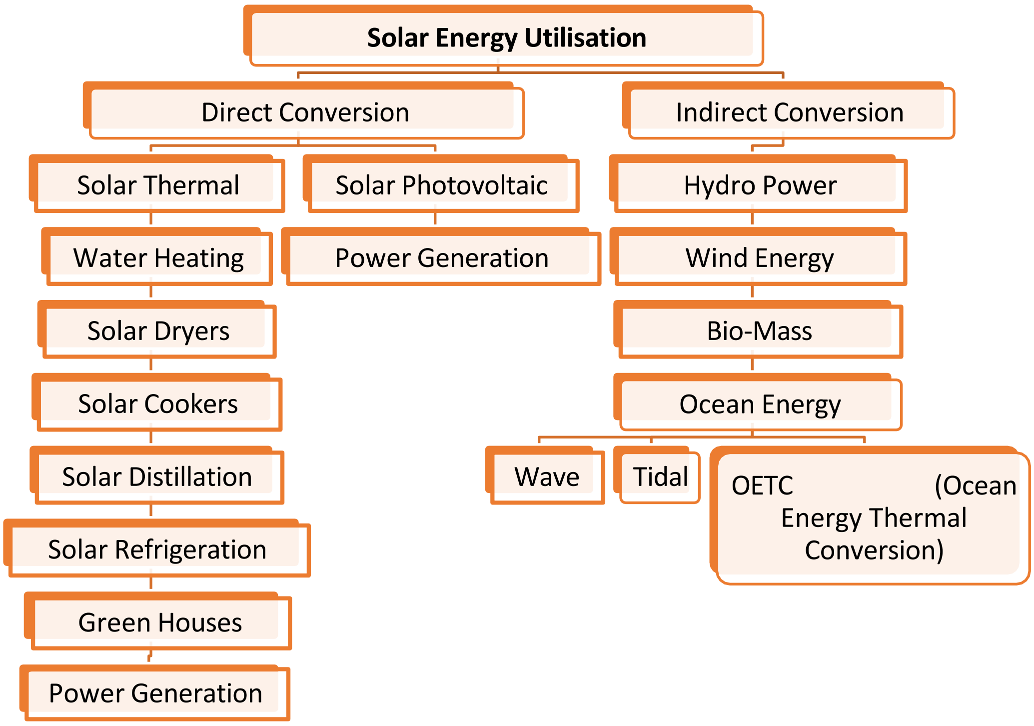


Fig. 1.1.2: Conversion of Solar Radiation into Various Forms

The following figure represents the different types of solar PV applications:

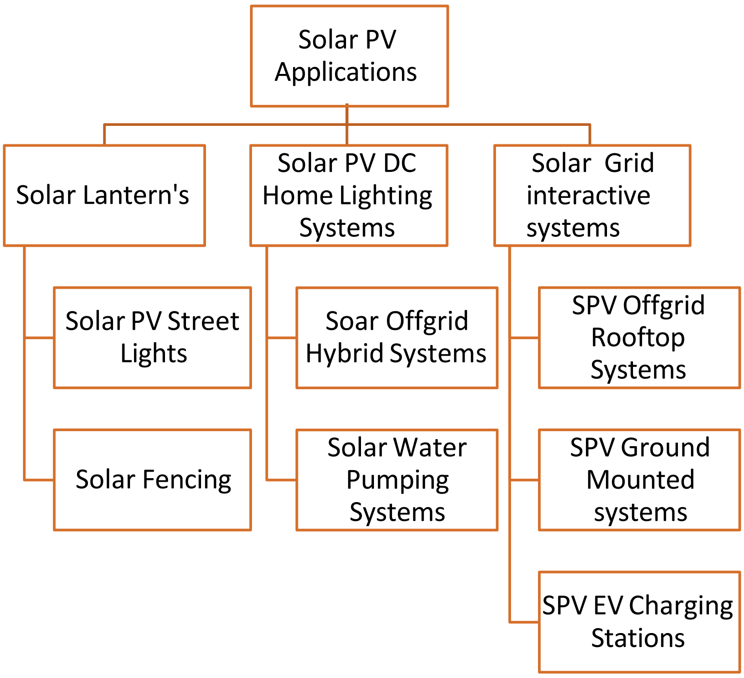


Fig. 1.1.3 Different Types of Solar PV Applications

1.1.2 Solar Photovoltaic (PV) System

A Solar Photovoltaic (PV) System is a setup designed to supply the available solar energy from the Sun in the form of electricity. The different components of Solar PV System are:

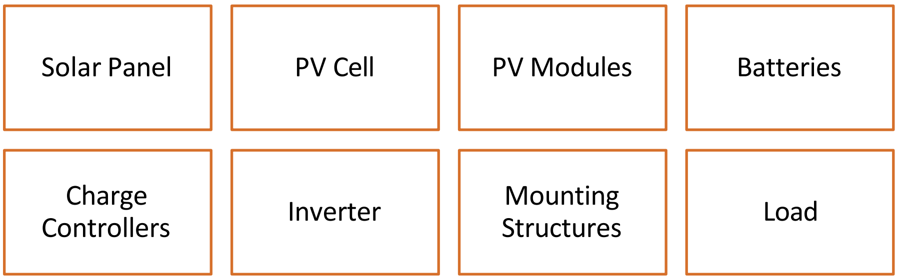


Fig. 1.1.4: Parts of Solar PV System

Solar Module

Photovoltaic cells convert the Sunlight into electrical energy by exposing the Pn junction to photons in the sun’s rays. Each cell has a 0.5volts generating capacity and in order to get the appropriate system voltage these cells are connected in series which is called string. Multiple strings are connected into parallel to generate the rated current and voltage. These multiple strings which are connected in parallel are put into an enclosure and framed to protect it from damage which is called module. Today modules are available from 1wp to 570wp sizes.

A group of modules inter connected are called as Array. A basket of Photovoltaic technologies have emerged in recent times that include crystalline and amorphous technologies.

Based on the process of growth of silicon crystal these crystals have been defined as Mono-Crystalline & Poly-Crystalline. As the crystal forms with one single layer it is hence called as Mono which means one. Likewise As the crystal forms with multiple layers it is called as Poly which means more than one. Amorphous / Thin Film modules are classified based on the compounds used in making the modules. These thin film modules are based on CdTe(Cadmium Telluride) and CIGS(Copper Indium Gallium Diselenide) are available commercially. The following table represents the type, efficiency and area requirement for crystalline and thin film solar panels.

Type	Efficiency	Area(Sqft/Kwp)
Crystalline		
Monocrystalline	16-22%	80-100
Polycrystalline	16-19%	100
Thin film		
CdTe	9-16%	130
CIGS	9-14%	130

Table 1.1.1: Types, Efficiency and Area Requirement of Crystalline and Thin Film Solar Panels

Crystalline cells have better efficiencies than the thin films. In recent years the price difference between these technologies have almost become negligible. In SPVRT systems crystalline modules are the used as they are efficient and also occupy less space giving scope for higher capacities. It is also important to note that module efficiency will not only impact the area requirement.

Requirement for a particular capacity (or wattage) and will not change the energy generation from same capacity at same location. This means the modules of thin film and crystalline technologies, which have different efficiency, installed at same location would have same specific energy generation (kWh / kWp), Of course, there may be difference due to other factors like temperature coefficient and useful radiation spectrum and so on. The following image shows the parts of a solar panel:

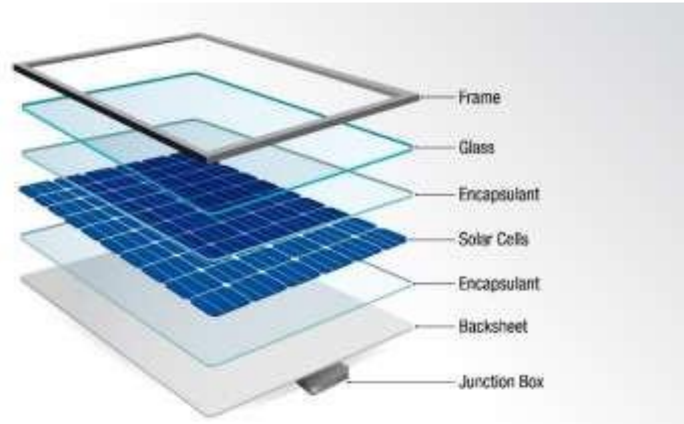


Fig. 1.1.3: Parts of a Solar Panel

Solar PV modules are the most important part of the system. They are also called the power generators of the system. In addition to PV modules, a PV system also includes various other parts.

Batteries

For PV systems that operate at night or during the absence of sunlight, storage of energy is important. Batteries are used to store electricity.

Charge Controller

The PV module output depends on the intensity of sunlight and the temperature of the cell. Charge controllers or regulators are the components which control the DC output and deliver that to the grid, batteries, and/or loads and ensures smooth operation of the PV system. A PV charge controller prevents overcharge by reducing the flow of energy to your battery once it reaches a certain voltage. Once the voltage drops when the sun intensity is lower or there is an increase in electrical usage, the controller will allow for the maximum charge possible.

Inverter

For applications that are run by AC power, the DC/AC inverters are required to be installed in PV systems.

Mounting Structures

They are required to fix the PV modules and to ensure that the modules are directed towards the sun.

Load

The household appliances and equipment that require to be powered by the PV solar system are called load.

The following image shows different components of a Solar PV System:

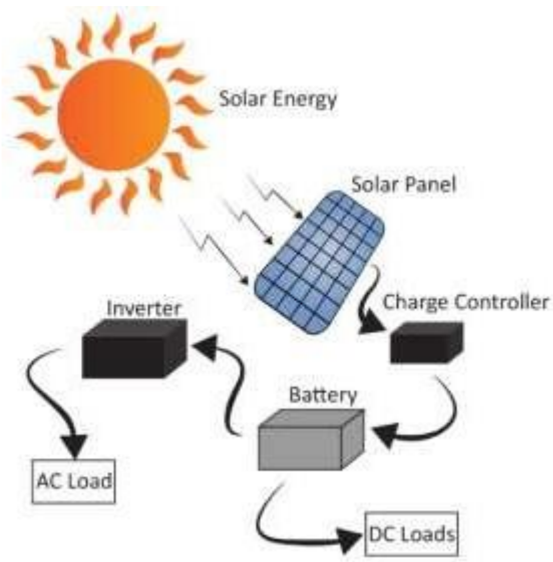


Fig. 1.1.4: Components of a Solar PV System

Types of Solar PV System

Based on the method of generating energy, a solar energy or solar PV system can be of different types. Based on the system configuration, PV systems can be classified as:

Stand-alone	Grid-connected	Hybrid

Fig. 1.1.5: Types of PV Systems

The basic principle behind PV systems remains same but different types of systems provide energy solution for different requirements.

Standalone Solar PV System

A simple *stand-alone PV system* is an automatic solar system that produces electrical power to charge banks of batteries during the day for use at night when the suns energy is unavailable. A stand-alone small scale PV system employs rechargeable batteries to store the electrical energy supplied by a PV panels or array.

Stand-alone PV systems are ideal for remote rural areas and applications where other power sources are either impractical or are unavailable to provide power for lighting, appliances and other uses.Since batteries are used, it is important to use electronic devices such as a charge controller to protect them. Also, for conversion of DC electricity from PV modules and battery into AC electricity, an inverter must be used.

The examples of standalone solar PV systems are:
- Solar lantern
- Solar PV home lighting system
- Solar PV water pumping system

The following figure shows a standalone solar PV system:

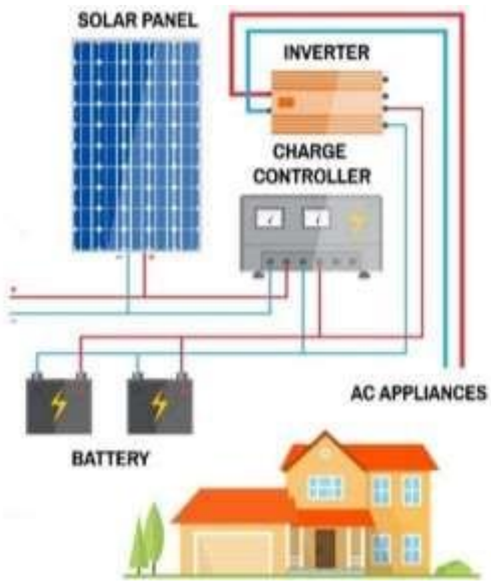


Fig. 1.1.6: Stand-alone Solar PV System

Grid-connected Solar PV System

In this type of PV system, the system is connected to a nearby, available electricity grid and the generated electricity is fed into the grid after converting the DC electricity into AC electricity. No batteries are required in this type of PV system. The figure represents a typical arrangement of grid-connected solar PV system:

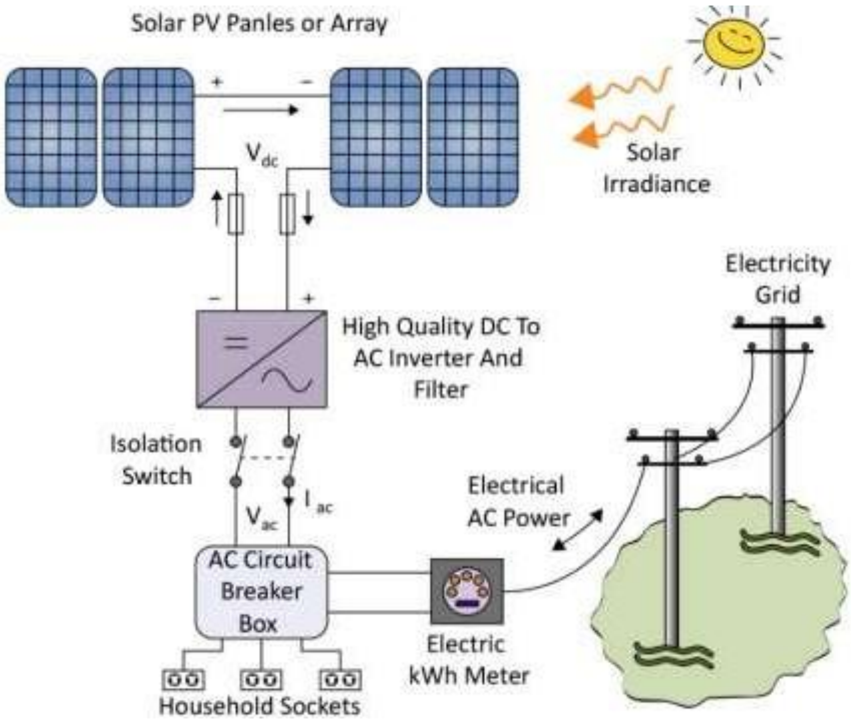


Fig. 1.1.7: Grid-connected solar PV system

Electricity grid voltage and frequency are well defined and, therefore, the PV electricity can be fed to electricity grid only after proper power conditioning, that is converting PV generated electricity to appropriate voltage and frequency level.

In grid-connected solar PV system, the inverter performs the following function:

- DC to AC conversion
- Grid synchronization, which is related to adjusting the generated PV energy to appropriate voltage and frequency level.

This type of PV system is used in India for large scale solar PV power plants.

Hybrid Solar PV System

In this type of PV system, an auxiliary source of energy such as a diesel generator is used in addition to solar PV modules and/or grid. This needs to be done when solar PV modules are not designed to supply the full energy required by the load (may be due to cost reason). In such cases, auxiliary source of solar PV systems is used, hence, they are called hybrid solar PV system.

The following figure shows a hybrid solar PV system:

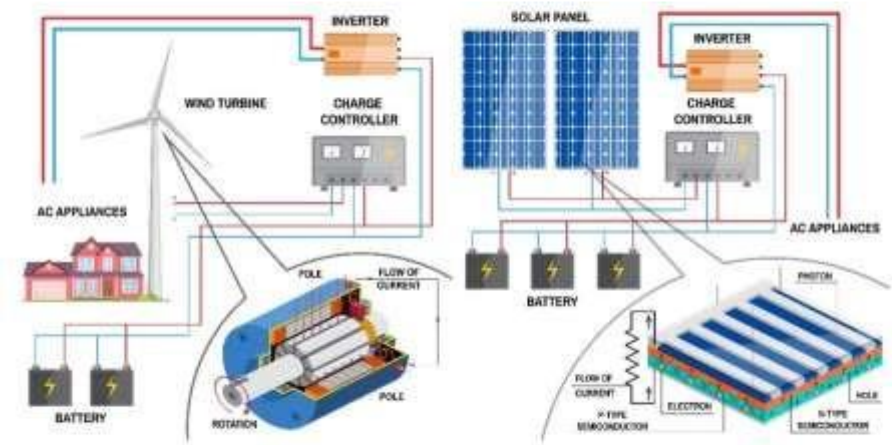


Fig. 1.1.8: Hybrid Solar PV System

1.1.3 Solar PV Technology

Photovoltaic Effect – The Working Principle of PV Technology

The solar panel works on the principle of photovoltaic effect, the emission of electrons from a material when it is exposed to light. The following figure represents the photovoltaic effect:

Fig. 1.1.9: Working of Solar Panel (Photovoltaic Effect)

The photovoltaic effect is explained below:

The photovoltaic effect is a process that generates voltage or electric current in a photovoltaic cell when it is exposed to sunlight. It is this effect that makes solar panels useful, as it is how the cells within the panel convert sunlight to electrical energy. The photovoltaic effect was first discovered in 1839 by Edmond Becquerel.

The photovoltaic effect occurs in solar cells. These solar cells are composed of two different types of semiconductors - a p-type and an n-type - that are joined together to create a p-n junction. By joining these two types of semiconductors, an electric field is formed in the region of the junction as electrons move to the positive p-side and holes move to the negative n-side. This field causes negatively charged particles to move in one direction and positively charged particles in the other direction. Light is composed of photons, which are simply small bundles of electromagnetic radiation or energy. These photons can be absorbed by a photovoltaic cell - the type of cell that composes solar panels. When light of a suitable wavelength is incident on these cells, energy from the photon is transferred to an atom of the semiconducting material in the p-n junction. Specifically, the energy is transferred to the electrons in the material. This causes the electrons to jump to a higher energy state known as the conduction band. This leaves behind a "hole" in the valence band that the electron jumped up from. This movement of the electron as a result of added energy creates two charge carriers, an electron-hole pair.

When unexcited, electrons hold the semiconducting material together by forming bonds with surrounding atoms, and thus they cannot move. However in their excited state in the conduction band, these electrons are free to move through the material. Because of the electric field that exists as a result of the p-n junction, electrons and holes move in the opposite direction as expected. Instead of being attracted to the p-side, the freed electron tends to move to the n-side. This motion of the electron creates an electric current in the cell. Once the electron moves, there's a "hole" that is left. This hole can also move, but in the opposite direction to the p-side. This produces electric current which is used to operate appliances, satellites and various other objects.

Types of Solar Panel

Based on the solar cell types, Solar panels are categorised as:

Fig. 1.1.10: Types of Solar Panels

Polycrystalline Solar Panel

Each PV cell is made of multiple silicon crystal fragments that are melded together during manufacturing and are also known as multi-crystalline cells that comprises of several small crystals called crystallite. The raw silicon is melted and kept in square mould. Then, it is cooled and sliced to make square wafers. Polycrystalline solar panels are of rectangular shape without having any rounded edges.

These panels are less efficient and less expensive than mono crystalline panels but like the latter, require to be mounted in a rigid frame.

Monocrystalline Solar Panel

Each solar PV cell is made of a single silicon crystal, which contain pure silicon and are in cylindrical shape. To lower the cost and optimize the performance of a monocrystalline cell, four sides of the cylinder are cut out to make silicon wafers. It provides the monocrystalline solar panels their appearance. These cells can be easily recognized from their uniform look and even, coloured external surface because of high-purity silicon contents.

These cells are highly efficient and must be mounted in a rigid frame to ensure protection.

Thin-film Solar Panel

Thin-film panels contain amorphous cells. Amorphous cells are formed by placing a thin film of non-crystalline (amorphous) silicon onto a wide range of surfaces. Due to their amorphous nature, these are flexible.

Amorphous silicon cells usually have low efficiency but are eco-friendly as they do not utilize harmful heavy metals such as cadmium or lead.

The power output of amorphous cells reduces over time, especially during the first few months, after that they are generally stable. The specified output of the amorphous cells should be one that is generated after the stabilization stage.

Power Rating of Solar Modules

Output of module is tested under controlled conditions for uniform comparison of available technologies and different manufacturers and models. These conditions are call standard test conditions (STC) and the modules are rated based on this testing STC parameters are as follows:

- Cell temperature 25 deg C
- Irradiance of 1000W/m^2
- Air mass of 1.5

This implies that for measuring performance of a module, cell temperature is considered which is different from ambient temperature. Cell temperature is always higher than ambient temperature and can reach about 25deg C higher (Tcell = Tair +25degC).

The real field conditions are different from standard test conditions and hence most of manufactures test their equipment's' performance closer to real-world conditions known as normal operating cell temperature (NOCT) and provide rating on the following conditions:

- Ambient air temperature 200C
- Irradiance of 800W/m^2

- Wind speed 1m/s
- Electrically open circuit

Performance Degradation Over Life Cycle

The performance of a PV module decreases over time. The degradation rate is higher in the first year upon initial exposure to light and then stabilizes with time. Following factors affects the degree of degradation:

- Quality of materials used in manufacture
- Manufacturing process
- Quality of assembly
- Packaging of the cells into the module
- Maintenance levels employed at the site

Generally, the degradation of a good quality module is considered to be maximum 20% during the module life of 25 years at the rate of 0.7% to 1% per year.

It is important to identify the PV Module performance and qualification standards adopted by MNRE:

- STC performance IEC60904-I
- IEC61215(Revised)/IS14286 Crystalline silicon modules
- IEC61853 Energy rating
- IEC61646 Thin Film modules
- IEC62108 Concentrated Photovoltaic receivers and module design qualification type approval
- IEC61730-I & II Module safety construction & Qualification

Note:
To know more about types of Solar PV Systems, you may refer to the following link:
https://www.youtube.com/watch?v=FurOeegcA3Q

Title of Video: Types of Solar PV System

By: Electrical Engineering
Duration: 9 m 17 sec

Note:
To know more about types of Solar Panel, you may refer to the following link:
https://www.youtube.com/watch?v=qIpRaQkzqAw

Title of Video: Types of Solar Panels

By: MECH Tech.
Duration: 7 m 31 sec

Mounting Structures

Solar panel mounting structures or systems are made of aluminium, galvanized iron (GI) and mild steel (MS) material. The following image shows the basic solar panel mounting systems:

Ground Mounting

Roof Mounting

Shade Structure Mount

Building Integrated PV (BIPV)

Fig. 1.1.11: Basic solar panel mounting systems

The following table lists some common types of PV array mounting systems:

Structure	Description	Image
Ground-mounted PV arrays	• Generally used for larger PV systems or where the rooftop installations are not practical. • The racks, poles and other foundations can be used for supporting the arrays. • They are usually more vulnerable to damage than rooftop arrays, but the constraints for their orientation and location is less as compared to the rooftop installations.	

Integrated mount	• They have panels attached directly to rafters and replace roofing materials. • They require modules are integrated into the exterior of the building or roofing Sometimes these systems are referred to building integrated PV.	
Stand-off mount/Flush mount	• They use standoffs attached to the roof for supporting rails on which the PV modules are mounted. • They allow sufficient gap to provide air flow as solar panels usually need air flow to give best performance. • They are recommended and popular mounts for residential purposes.	

Structure	Description	Image
Standard panel fixture	• They allow panels to be fixed in the customer place as it is. • There is no requirement of adjustments in these fixtures.	
Non- standard panel fixtures	• They require pictorial representation of customer place where the panel is to be installed. • They require the panel fixture be designed separately as per the customer roof model.	

Ballasted mounting systems	• They are used in flat roof commercial projects. • They are dependent on the weight of the array, racking system and material such as concrete paver to fix the array to the roof or ground. • They do not require penetration to roof or ground.	

Structure	**Description**	**Image**
Roof top	• They are very popular for installing solar PV arrays. • They offer less physical protection and limited access to the PV array for safety. • They generally provide better sun exposure. • They do not occupy space on the ground. Hence the ground may be needed for other purposes. • They must be secured structurally and any attachment and penetration must be weather sealed properly. • They should not be blocked from sun by any shades.	
Direct mount	• They allow panels to be attached directly to the roof. • They are cheap and easy to install.	

Structure	Description	Image
Pole mount	• They are generally used with manufactured racks that are mounted on the top or attached to the side of a steel pole. • They are popular for off-grid residential photovoltaic systems, as the weight of PV array is balanced over the pole, seasonal adjustments can be easily done. • They are very common in small applications having one or two modules where the entire system mounted on a single pole. • They offer better cooling for the panels as compared to roof mounting.	
Rack mounting structure	• They are used for non-tracking system at ground level. • They are also used in large commercials or utility scale arrays.	

Structure	Description	Image
Sun-tracker systems	• They follow the sun daily. • They are generally mounted on poles and allow the system to receive greater amount of solar energy. • They allow tracking to enhance the summer gain by 30% or more, but in winter the gain is 15% or less. • They allow tracking in two axes to achieve maximum performance. • They can also be of single-axis to provide simplicity and reliability.	
Shade structure	• They work as patio covers to provide shade. • For example, carports can have solar panels and cover as many parking spots as the project requires. These carports could have the provision of electric vehicle charging stations as an added benefit to the cars parked there.	

Table 1.1.1: PV Mounting Systems

Types of Ground-Mounted Solar PV Power Plants and their working principles:

The different types of ground-mounted solar PV Power Plants are:

Fig 1.1.12: Types of Ground-Mounted Solar PV Power Plants

Fixed-tilt Solar PV Power Plants

Fixed-tilt solar PV power plants are made with a stationary angle of inclination, which means that the solar panels' angle with respect to the ground doesn't change. The simplest and most economical method for producing solar energy is this kind of arrangement. The solar panels in this kind of solar plant are installed at a set angle and orientation, usually facing south. During installation, the angle of inclination is established and set. The panels in this form of plant do not follow the path of the sun, making it less efficient but the easiest and most affordable.

Working Principle:

Fixed-tilt solar PV power plants' operating theory is based on the science of how solar panels transform light into electricity. A flow of electrons is produced when sunlight strikes the solar panels; these electrons can be collected and transformed into usable electricity. The angle at which the solar panels are facing the sun's rays determines how effective this process will be.

Based on the latitude of the installation location, fixed-tilt solar PV power plants are created to optimise the angle of the solar panels. The panels are installed at an angle, usually between 25 and 35 degrees, which is ideal for gathering the most sunshine all year long.

The solar panels remain stationary after they are positioned at the ideal angle. The solar panels' angle with respect to the ground remains constant during the day as the sun moves across the sky. This means that the panels are not able to capture the maximum amount of sunlight possible at all times, as they would with a tracking system.

Hence, the working principle of fixed-tilt solar PV power plants is relatively simple. Based on the latitude of the installation site, the angle of the solar panels is adjusted for maximum energy absorption, and the panels are fixed in place. While this approach is less efficient than tracking systems, it is also less expensive to install and maintain, making it a popular choice for many solar energy projects.

Single-axis Solar PV Power Plants

In Single-axis solar PV Power plant, the solar panels are mounted on a single-axis tracking system that follows the sun's path from east to west. This tracking system allows for maximum sunlight absorption, making it more efficient than fixed-tilt plants.
These are designed to track the movement of the sun across the sky, maximizing the amount of sunlight that hits the solar panels throughout the day. The tracking mechanism typically consists of a motorized system that adjusts the angle of the solar panels to maintain their orientation perpendicular to the sun's rays.

Working Principle:

The working principle of single-axis solar PV power plants is relatively straightforward. When the sun rises in the morning, the solar panels gradually rotate to retain their orientation as the day goes on and the sun moves across the sky, ensuring that they are always perpendicular to the sun's rays. This makes it feasible for solar panels to collect the most energy, regardless of the sun's position in the sky.

To maintain the solar panels aligned with the sun, the tracking system normally comprises of a number of sensors and motors. The motors adjust the angle of the solar panels based on the sensors' detection of the sun's position in the sky. The panels are then positioned to look straight at the sunset after this procedure has continued throughout the day.

Overall, the working principle of single-axis solar PV power plants is designed to maximize the efficiency of solar energy collection by tracking the movement of the sun across the sky. This technology is an important tool for increasing the viability of solar energy as a renewable source of power.

Dual-Axis Solar PV Power Plants

In dual-axis solar PV power plant, the solar panels are mounted on a dual-axis tracking system that follows the sun's path in both east-west and north-south directions. This tracking system allows for even greater sunlight absorption, making it the most efficient type of ground-mounted solar plant.
In order to maximise the quantity of sunshine that reaches the solar panels throughout the day, dual-axis solar PV power plants are made to track the sun's movement in two dimensions. In comparison with single-axis systems, which only track the sun's movement in one dimension (often from east to west) Dual-axis systems additionally account for the sun's changing angle throughout the year.

Working Principle:

Dual-axis solar PV power plants have a more complicated operating principle than single-axis systems, but the fundamental concept is the same. A motorised mechanism that alters the angle of the solar panels in both the horizontal and vertical planes works in conjunction with a series of sensors to identify where the sun is in the sky.

The motorised mechanism changes the angle of the solar panels in response to the sensors' detection of the sun's location in both the east-west and north-south directions. This enables the panels to stay perpendicular to the sun's rays regardless of where it is in the sky.

Dual-axis solar PV power plants track the sun's position throughout the day and also make adjustments according to the sun's varying angular position throughout the year. This is achieved by altering the solar panels' angle according to the installation site's latitude as well as the season.

Thus, by tracking the sun's movement in two dimensions, dual-axis systems can maximize the amount of energy captured by solar panels, making them an important tool for increasing the viability of solar energy as a renewable source of power.

Types of Ground-Mounted Solar PV Power Plants and their working principles

There are two main types of rooftop solar PV power plants; on-grid and off-grid.

On-grid rooftop solar PV power plants

These are solar power systems installed on the rooftop of a building that are connected to the electric grid.

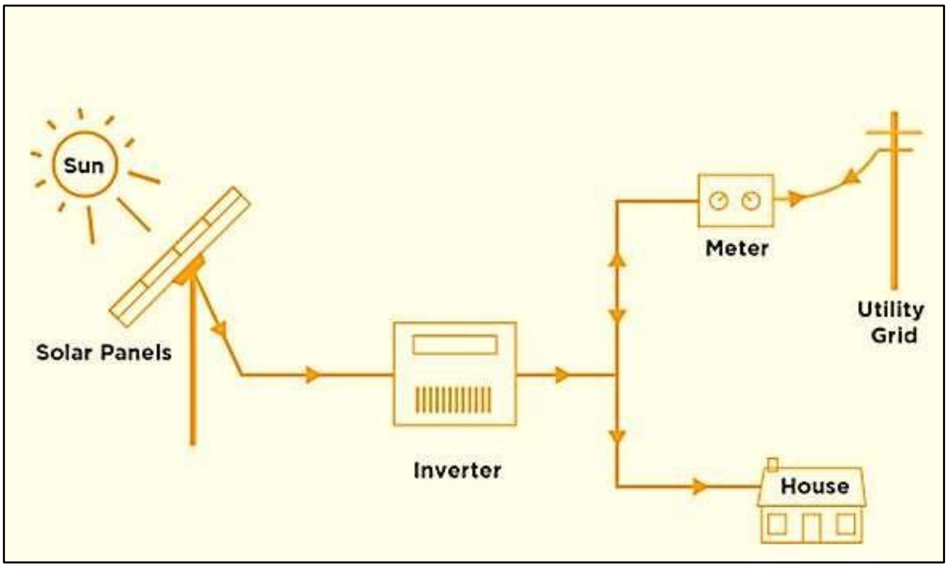


Fig. On-grid rooftop solar PV Power Plant

Working Principle:

The working principle of an on-grid rooftop solar PV power plant involves several components and processes that work together to generate electricity from sunlight and feed it into the electric grid. Here's how it works:

- **Solar Panels**: On-grid rooftop solar PV power plants use photovoltaic (PV) panels to convert sunlight into electricity. These panels are made up of multiple solar cells, which are connected in series and/or parallel to produce the desired voltage and current output. When sunlight hits the solar cells, it excites electrons in the cells, creating a flow of DC (direct current) electricity.
- **Inverter**: The DC electricity generated by the solar panels is sent to an inverter, which converts it into AC (alternating current) electricity that is compatible with the electric grid. The inverter also regulates the voltage and frequency of the AC electricity to match the standards of the grid.
- **Electrical Meter**: The AC electricity produced by the inverter is sent to the building's electrical panel, which distributes the power to the building's electrical loads (appliances, lights, etc.). If the solar panels produce more electricity than the building's electrical loads require, the excess electricity is sent back to the electric grid.
- **Electric Grid**: The electric grid acts as a storage system for the excess electricity generated by the solar panels. When the solar panels are not producing enough electricity to meet the building's electrical needs, the electric grid supplies the additional power required. This process is known as net metering, and it allows the building owner to earn credits on their electricity bill for the excess electricity that they generate.

Off-grid rooftop solar PV power plants

The working principle Off-grid rooftop solar PV power plants are designed to operate independently of the electric grid, providing power to a building without any connection to the utility grid.

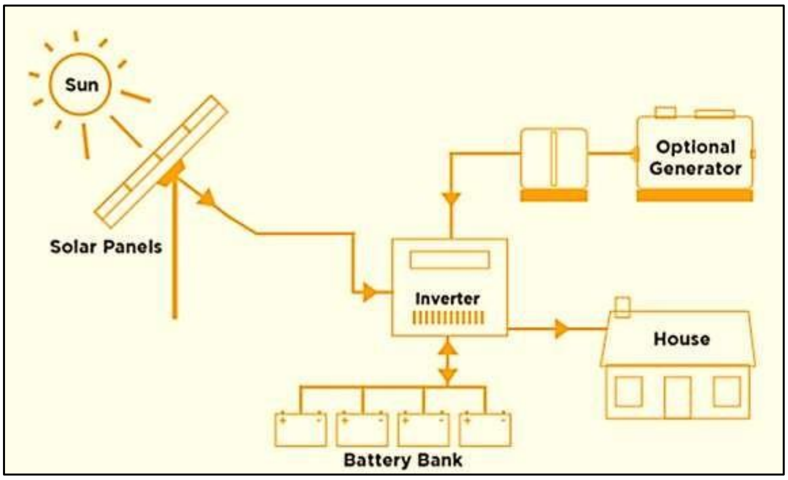


Fig. Off-grid rooftop solar PV Power Plant

Working Principle:

The working principle of Off-grid rooftop solar PV power plant is as follows

- **Solar Panels**: Off-grid rooftop solar PV power plants use photovoltaic (PV) panels to convert sunlight into electricity, just like on-grid systems. These panels are mounted on the rooftop of the building and are connected to a charge controller, which manages the charging of the batteries.

- **Inverter**: The inverter converts the DC electricity from the batteries into AC electricity that is compatible with the building's electrical system. The inverter also regulates the voltage and frequency of the AC electricity to match the standards of the building's electrical loads.
- **Charge Controller**: The charge controller regulates the flow of electricity from the solar panels to the batteries, preventing overcharging and extending the life of the batteries. It also converts the DC electricity from the solar panels into a suitable voltage and current for the batteries.
- **Batteries**: The electricity generated by the solar panels is stored in batteries, which are used to power the building's electrical loads when there is no sunlight. The batteries are connected to an inverter, which converts the DC electricity from the batteries into AC electricity that can be used by the building's electrical devices.
- **Electrical Panel**: The AC electricity produced by the inverter is sent to the building's electrical panel, which distributes the power to the building's electrical loads (appliances, lights, etc.). If the solar panels are not producing enough electricity to meet the building's electrical needs, the battery backup system provides additional power.
- **Optional Generator**: It is often used as a backup power source in an off-grid solar PV power plant. This generator can be used to supplement the power generated by the solar panels and batteries during periods of low sunlight, or when the electrical demand exceeds the capacity of the solar PV system.

1.1.4 Overview of Solar PV technology, Rooftop and Ground Mounted Solar, Grid Interactive and Off- Grid Solar Sector in India

In recent years, solar photovoltaic (PV) technology has gained popularity as a technique of producing renewable energy, and India is one of the nation leading this trend.

India's rooftop solar market has expanded significantly in recent years as a result of government incentives and policy. Numerous rooftop solar projects have been developed in both the commercial and residential sectors as a result of the government's goal of reaching 500 GW of Renewable Energy by 2030. Rising public knowledge of the advantages of renewable energy sources and declining solar panel costs, in addition to government initiatives, have aided in the expansion of rooftop solar in India.
In the ground-mounted solar sector, India has also made significant progress, with large-scale solar parks being developed across the country. These solar parks, which typically cover several square kilometers, are connected to the grid and supply electricity to both urban and rural areas. The development of solar parks has been facilitated by government policies and initiatives, such as the Solar Park Scheme and the National Solar Mission.

Numerous financial tools, including subsidies, tax breaks, and low-interest loans, have helped India's adoption of solar PV technology. The government has also started a number of programmes to encourage the use of solar PV, such as the International Solar Alliance (ISA), which aims to encourage the use of solar energy in developing nations, and the Solar Energy Corporation of India (SECI), which makes it easier to develop solar projects through mechanisms like auctions.

Solar energy can be used in India through both off-grid and grid-interactive technologies. Grid-interactive solar systems are linked to the electric grid and send any extra power they produce back into it. On the other hand, off-grid solar systems are not linked to the power grid and are intended to supply electricity to specific homes or communities.

Due to government initiatives and laws, India's grid-interactive solar market has experienced rapid expansion in recent years. The ambitious goal of installing 20 GW of grid-connected solar power by 2022 was established by the Jawaharlal Nehru National Solar Mission (JNNSM), which was announced in 2010. Since then, the goal has been raised to 100 GW. The JNNSM aided in the creation of solar parks and other massive projects by offering financial incentives and subsidies for the installation of grid-connected solar systems.

In recent years, India's off-grid solar industry has also experienced tremendous expansion, especially in rural areas where grid electricity is unavailable or unreliable. Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY), the government's rural electrification scheme, has offered financial support for the installation of off-grid solar systems in distant locations, assisting in the provision of power to millions of families.

A number of private sector efforts, such as microfinance schemes and pay-as-you-go business models that enable people to buy solar systems in instalments, have also benefited the off-grid solar sector. Off-grid solar pump installation for agricultural usage has also received help from the government's subsidy programme, the Pradhan Mantri Kisan Urja Suraksha evam Utthan Mahabhiyan (PM KUSUM).

Despite these developments, challenges remain in the solar PV sector in India, including issues related to grid integration, land acquisition, and the availability of finance. However, the overall trend is positive, and with continued government support and technological advancements, the solar PV sector in India is expected to grow significantly in the coming years.

UNIT 1.2: Broader Business, Finance, Technology and Regulatory Landscape of Solar Energy in the Country

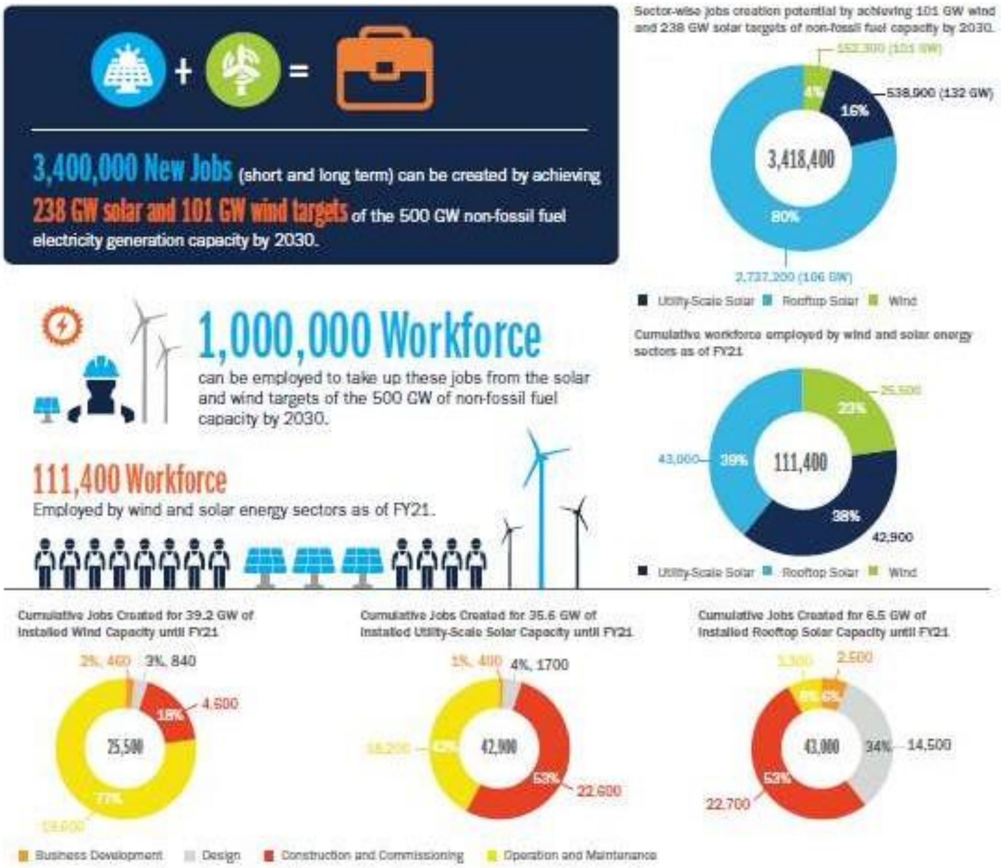
Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss various market research reports and industrial magazines present in the market.
2. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country.
3. Discuss benefits of solar energy over conventional sources of energy.

1.2.1 Market Research

The demand of skilled manpower in the Renewable Energy sector (mainly Solar Photovoltaics Industry) in India and worldwide is a subject under study which has been undertaken by various organizations. As per the report on India's Expanding Clean Energy Workforce, by the Council on Energy, Environment and Water, Natural Resources Defense Council, and Skill Council for Green Jobs; as of August 2021, the cumulative 81 GW renewable capacity addition in the country has employed an estimated 111,400 workers across various RE project phases like business development, design, construction and commissioning, and operation and maintenance (see illustration). 77 percent of these workers were in the solar sector (85,900) corresponding to 42 GW of installed capacity and remaining in wind sector (25,500) with 39 GW capacity. The rooftop solar segment, despite contributing 15 percent of the total solar capacity (6.6 GW), employed more workers (43,000) than utility-scale solar (42,900) with 35.6 GW. The distributed nature of rooftop solar makes these projects more labour intensive than utility-scale projects. It is important to note that majority of the jobs created in solar domain is for the pre-construction/ construction, installation & commissioning and Operations & Management (O&M) stages, which are suitable for a certified Solar PV Installer (Suryamitra). Growth in the renewable energy sector (particularly solar) has already created thousands of jobs in India and as the country aspire to become a global solar leader with installed capacity of over 300 GW by 2030, huge opportunities for job creation particularly in installation and O&M related stages shall be available.



Source: Akanksha Tyagi, Charu Lata, Jessica Korsh, Ankit Nagarwal, Deepak Rai, Sameer Kwatra, Neeraj Kuldeep, and Praveen Saxena. 2022. India's Expanding Clean Energy Workforce. Council on Energy, Environment and Water, Natural Resources Defense Council, and Skill Council for Green Jobs.

Market Research Studies and Industrial Magazines

There are various market research reports and industrial magazines available in India that provide insights into the solar PV industry. These market research studies and industrial magazines offer insightful information about the Indian solar photovoltaic sector, including the most recent trends in the market, technological advancements, and regulatory changes. Investors, decision-makers, and business experts who want to comprehend the dynamics of the Indian solar market might benefit from them. Some of them are as follows:

- **Bridge to India**: Bridge to India is a leading consulting and knowledge services provider in the Indian renewable energy market. They publish reports and newsletters that cover various aspects of the Indian solar industry, including policy, regulation, and market trends.
- **Mercom India**: Mercom India is a research and consulting firm that specializes in the renewable energy sector. They provide market intelligence reports, industry news, and analysis on the Indian solar market.

- **Solar Quarter**: It is an Indian magazine that covers the latest developments in the solar industry, including technology, policy, and market trends. They also organize conferences and events related to solar energy.
- **PV Magazine India**: PV Magazine India is a leading online magazine that covers news, analysis, and market trends related to the solar PV industry in India. They also provide insights into the latest technological developments and policy changes in the Indian solar market.
- **Indian Solar Market**: It is a market research and consulting firm that provides in-depth analysis and insights into the Indian solar industry. They offer market research reports, customized research services, and consulting services to clients in the solar industry.

India's shift to a greener and more sustainable energy future features solar energy as a major motivator. The Indian government has recently put into effect a number of policies and programmes designed to encourage the use of solar energy throughout the nation. This has improved the corporate, financial, technological, and governmental environments for solar energy in India.

1.2.2 Business Landscape

India's solar energy industry has grown significantly over the past few years as a result of favourable government policies, falling solar module prices, and rising investor interest. Large companies like Adani, Tata Power, and ReNew Power, as well as a number of smaller ones, dominate the Indian solar business. The market for solar energy in India has attracted considerable international investment from nations including China, Japan, and the US. Here are some of the key features of the business landscape of solar energy in India:

Dominance of Large Players: The Indian solar market is dominated by large players such as Adani, Tata Power, ReNew Power, Greenko, and Azure Power. These companies have a significant share of the installed solar capacity in India and are expanding their operations to tap into the growing demand for solar energy in the country.

Increasing Investment: The Indian solar market has attracted significant foreign investment, with countries such as Japan, the US, and China investing in the sector. According to the Ministry of New and Renewable Energy (MNRE), foreign direct investment (FDI) inflows into the renewable energy sector in India reached USD 9.78 billion between April 2000 and December 2020.

Growing Role of Small and Medium Enterprises: Along with large players, the solar energy sector in India is also seeing an increasing number of small and medium enterprises (SMEs) entering the market. These companies are primarily involved in the manufacturing of solar modules, inverters, and other equipment used in solar power plants.

Government Support: The Indian government has been actively promoting the use of solar energy through various policy initiatives, such as the Jawaharlal Nehru National Solar Mission and the National Solar Energy Federation of India. The government has also provided incentives such as tax breaks, subsidies, and low-interest loans to encourage the growth of the solar industry in the country.

Increasing Demand: The demand for solar energy in India is expected to continue to grow in the coming years, driven by factors such as rising energy demand, increasing awareness about climate change, and decreasing costs of solar technology. According to the International Energy Agency, solar energy is expected to account for around 38% of India's total installed power capacity by 2040.

1.2.3 Finance Landscape

To encourage the use of solar energy in the nation, the Indian government has put in place a number of finance schemes. Subsidies, tax breaks, and concessional loans are a few examples of these. The government has also established a number of finance organisations, like the Solar Energy Corporation of India (SECI) and the Indian Renewable Energy Development Agency (IREDA), to offer financial support to the industry. The expansion of the Indian solar industry has also been significantly aided by private sector financing such as project financing, private equity, and venture capital.

In 2019, MNRE launched the Pradhan Mantri Kisan Urja Suraksha Utthan Mahabhiyan (PM-KUSUM) scheme. The objective is to provide assistance in installing off-grid solar pumps in rural areas. The idea behind this initiative is to increase farmers' income by selling extra electricity generated and reducing the over-dependence on expensive diesel. The central government plans to invest INR 34,422 Crores in this scheme.

The aim of KUSUM scheme is to solarize and de-dieselization of agricultural section. Under this scheme, farmers, co-operative groups and panchayats can apply for the installation of solar pumps. The cost of the solar pumps is divided into following categories:

Government to provide a 60% subsidy directly to farmers.

30% will be provided through soft loans to farmers.

10% actual cost to be incurred by farmers.

Fig. 1.2.1: Categories of Cost under KUSUM Scheme

Following are three main components of the KUSUM scheme:

- **Component-A**: For Setting up of 10,000 MW of Decentralized Grid Connected Renewable Energy Power Plants on barren land. Renewable energy-based power plants (REPP) of capacity 500 kW to 2 MW will be setup by individual farmers/ group of farmers/ cooperatives/ panchayats/ Farmer Producer Organisations (FPO)/Water User associations (WUA) on barren/fallow land. These power plants can also be installed on cultivable land on stilts where crops can also be grown below the solar panels. The renewable energy power project will be installed within five km radius of the sub-stations in order to avoid high cost of sub-transmission lines and to reduce transmission losses. The power generated will be purchased by local DISCOM at pre-fixed tariff

- **Component-B**: For Installation of 17.50 Lakh stand-alone solar agriculture pumps. Individual farmers will be supported to install standalone solar agriculture pumps of capacity up to 7.5 HP for replacement of existing diesel agriculture pumps / irrigation systems in off-grid areas, where grid supply is not available. Pumps of capacity higher than 7.5 HP can also be installed, however, the financial support will be limited to 7.5 HP capacity

- **Component-C**: For Solarisation of 10 Lakh Grid Connected Agriculture Pumps. Individual farmers having grid connected agriculture pump will be supported to solarise pumps. The farmer will be able to use the generated solar power to meet the irrigation needs and the excess solar power will be sold to DISCOMs at pre-fixed tariff

The scheme also motivates to use the barren land in rural areas and cultivable fields for producing renewable energy. With both state and central government's financial assistance, the farmers' financial burden is kept to the minimum by providing subsidy and financing the solar water pumping system.

Under KUSUM scheme, total 60% subsidy is provided on solar water pump. The remaining 30% will be financed.

Fig. 1.2.2: Subsidies and financing of solar water pumping system

1.2.3 Technology Landscape

The Indian solar industry has witnessed rapid technological advancements in recent years, driven by innovation and research and development. Advances in solar PV technology, such as the development of higher efficiency solar cells, have made solar energy more affordable and efficient. The adoption of energy storage technologies, such as batteries and pumped hydro storage, is also becoming increasingly important in the Indian solar industry, as it can help address the issue of intermittency and variability of solar power. Here are some of the key features of the technology landscape of solar energy in India:

- Solar Photovoltaic (PV) Technology: Solar PV technology is the most widely used technology for generating solar energy in India. It involves converting sunlight into electricity using solar panels made of silicon cells. The efficiency of solar PV technology has been improving steadily, and new materials such as perovskite and organic solar cells are being developed to further increase efficiency.

- Concentrated Solar Power (CSP) Technology: CSP technology involves using mirrors or lenses to concentrate sunlight onto a small area, which generates heat that can be used to produce electricity. While CSP technology has not been widely adopted in India due to high costs, there have been recent developments in the sector, including the construction of a 750 MW CSP project in Rajasthan.

- **Energy Storage Technology**: Energy storage technology is becoming increasingly important in the solar energy sector, as it allows solar power to be used even when the sun is not shining. In India, lithium-ion batteries are the most widely used energy storage technology, but other technologies such as flow batteries and thermal energy storage are also being explored.
- **Smart Grid Technology**: Smart grid technology involves using advanced sensors, communication systems, and data analytics to manage the distribution of electricity from solar power plants. The use of smart grid technology can help to optimize energy usage and improve the reliability and efficiency of solar power plants.
- **Internet of Things (IoT) Technology**: IoT technology involves using sensors and connected devices to monitor and control various aspects of solar power plants, such as energy production, storage, and distribution. The use of IoT technology can help to improve the performance and efficiency of solar power plants.

1.2.4 Regulatory Landscape

The Indian government has implemented several policies and initiatives aimed at promoting the adoption of solar energy in the country. These include the National Solar Mission, which aims to install 100 GW of solar capacity by 2022, and various state-level policies and regulations aimed at promoting solar energy adoption. The government has also implemented several measures to encourage domestic manufacturing of solar equipment, such as the imposition of safeguard duties on imported solar modules.

The regulatory landscape of solar energy in India is complex, and involves a range of government policies, regulations, and incentives that are aimed at promoting the development and deployment of solar energy in the country. Here are some of the key features of the regulatory landscape of solar energy in India:

- **National Solar Mission**: The National Solar Mission is a government initiative aimed at promoting the development and deployment of solar energy in India. The mission has set a target of achieving 100 GW of solar power capacity by 2022.
- **Renewable Purchase Obligation (RPO)**: The RPO is a policy that requires electricity distribution companies to purchase a certain percentage of their power from renewable energy sources, including solar energy.
- **Net Metering Policy**: The net metering policy allows households and businesses that generate solar energy to sell excess energy back to the grid and receive credits on their electricity bills.
- **Solar Parks**: The government has established several solar parks throughout the country, which provide infrastructure and support for the development of solar power projects.
- **Incentives and subsidies**: The government provides various incentives and subsidies for the development of solar energy projects, including capital subsidies, interest rate subsidies, and tax incentives.
- **Regulatory bodies**: Several regulatory bodies are involved in overseeing the development and deployment of solar energy in India, including the Ministry of New and Renewable Energy (MNRE), the Central Electricity Regulatory Commission (CERC), and state electricity regulatory commissions.

Note:
To know more about Business Landscape, you may refer to the following link:
https://www.youtube.com/watch?v=wAxxyI-3_uc

Title of Video:	Business Matters | Where does India stand on solar energy targets?
By:	The Hindu
Duration:	5 m 10 sec

Note:
To know more about financing Landscape, you may refer to the following link:
https://www.youtube.com/watch?v=n0MF0iZEOes

Title of Video:	Introduction to the Renewable Energy Financial Model
By:	Financial modeling
Duration:	5 m 10 sec

1.2.5 Benefits of Solar Energy Over Conventional Sources of Energy

There are several benefits of solar energy over conventional sources of energy such as:

- **Renewable**: Solar energy is a renewable source of energy, unlike fossil fuels, which are finite and will eventually run out. This means that solar energy can provide a sustainable source of power for the future.
- **Clean**: Solar energy is a clean source of energy, with no emissions of pollutants or greenhouse gases. This makes it an environmentally friendly alternative to conventional sources of energy, which contribute to air and water pollution and climate change.
- **Cost-effective**: The cost of solar energy has been declining rapidly over the past few years, making it increasingly cost-effective. In many parts of the world, solar energy is already cheaper than fossil fuels.
- **Reduces dependence on fossil fuels:** Solar energy can reduce dependence on fossil fuels, which are often imported and subject to price fluctuations. This can improve energy security and reduce the risk of supply disruptions.
- **Scalable**: Solar energy can be deployed on a small scale, such as on rooftops, or on a large scale, such as in utility-scale solar power plants. This makes it a scalable solution to meet a wide range of energy needs.
- **Job creation**: The solar energy sector is a source of job creation, providing employment opportunities in manufacturing, installation, and maintenance of solar panels and related equipment.

Note:
To know more about benefits of solar energy , you may refer to the following link:
https://www.youtube.com/watch?v=u-cq7TwK3Xk

Title of Video:	Solar Energy vs. Fossil Fuels: A Comparison
By:	Simplified Solar
Duration:	3 m 07 sec

UNIT 1.3: Overview of Grid Interactive and Off-Grid Solar Sector in India

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss and provide an overview of grid interactive and off- grid Solar Sector in India.
2. Explain types of off-grid solar PV power devices and their working principles.

1.3.1 Overview of Grid Interactive and Off- Grid Solar Sector in India

To achieve 175 GW of renewable energy capacity by 2022, India has set high goals, of which 100 GW are anticipated to come from solar power. To entice investments from domestic and foreign businesses, the government is also encouraging the building of large-scale solar parks.

India is a part of the International Solar Alliance (ISA), an alliance of more than 120 nations that strives to encourage the use of solar power and lower its price. By 2030, the ISA hopes to have deployed 1,000 GW of solar power worldwide.

India's grid-interactive and off-grid solar industries are both expanding quickly and are anticipated to play a key role in supplying the nation's energy requirements in a sustainable and eco-friendly way.

Grid-Interactive Solar Sector

The grid-interactive solar sector refers to solar projects that are connected to the main grid and generate electricity that can be fed into the grid. These solar power plants are designed to supply electricity to the grid and are installed in large-scale utility projects or commercial/industrial rooftops.

In India, the grid-interactive solar sector has been rapidly growing, driven by favourable government policies, declining solar module prices, and increasing investor interest. Grid-interactive solar projects have been promoted by the Ministry of New and Renewable Energy (MNRE) through a number of programmes and incentives, including the Rooftop Solar Programme, National Solar Mission, and Solar Parks Scheme. As of February 2021, India's grid-connected solar power capacity was 40.09 GW, and the government has set a goal to reach 100 GW of solar capacity by 2022.

Off-grid Solar Sector:

Off-grid solar sector in India refers to the generation and utilization of solar power in remote or rural areas, which are not connected to the main power grid. Since more than 30% of Indians still don't have access to electricity, the off-grid solar market there is especially crucial. India's off-grid solar market can be broadly divided into two types:

- **Standalone Solar Systems**: These solar power systems were created to supply all the energy requirements of a single home or small company; they are not linked to the main power grid. These systems can be as simple as those that power lighting and mobile devices or as complex as those that can run fans, televisions, and refrigerators.
- **Mini-Grids**: These are distributed solar systems made to supply a collection of homes or small enterprises. Mini-grids, which can supply dependable and reasonably priced electricity to off-grid communities, typically consist of a solar power plant, battery storage, and distribution infrastructure.

India's off-grid solar industry has expanded significantly in recent years, propelled by government programmes like the Saubhagya scheme and the Deen Dayal Upadhyaya Grammen Jyoti Yojana, which seek to give all families in the nation access to energy. Increasing investor interest, falling solar module prices, and the availability of cutting-edge financing options like pay-as-you-go systems have all benefited the industry.

Despite these encouraging improvements, India's off-grid solar industry still confronts a number of obstacles, such as restricted access to funding, a shortage of experienced labour, and insufficient legislative and regulatory assistance. However, the off-grid solar industry in India has the potential to significantly contribute to the nation's sustainable development goals while also satisfying its energy demands, given the government's sustained backing and the growing interest of private sector companies.

Note:
To know more about types of Mounting Systems, you may refer to the following link:
https://www.youtube.com/watch?v=zr9AlfA85ME

Title of Video: Understanding Grid Interactive Systems

By: NAZ Solar Electric

Duration: 4 m 18 sec

1.3.2 Types of off-grid solar PV power devices and their working principles

Off-grid solar PV power devices are designed to operate independently of the electric grid, providing electricity in remote locations where grid connection is not feasible or economical. There are several types of off-grid solar PV power devices, including:

- **Solar Home Lighting Systems (SHLS)**: These systems are designed to provide lighting and basic electricity for households in remote areas. SHLS typically include a solar panel, a battery, a charge controller, and one or more LED lights.

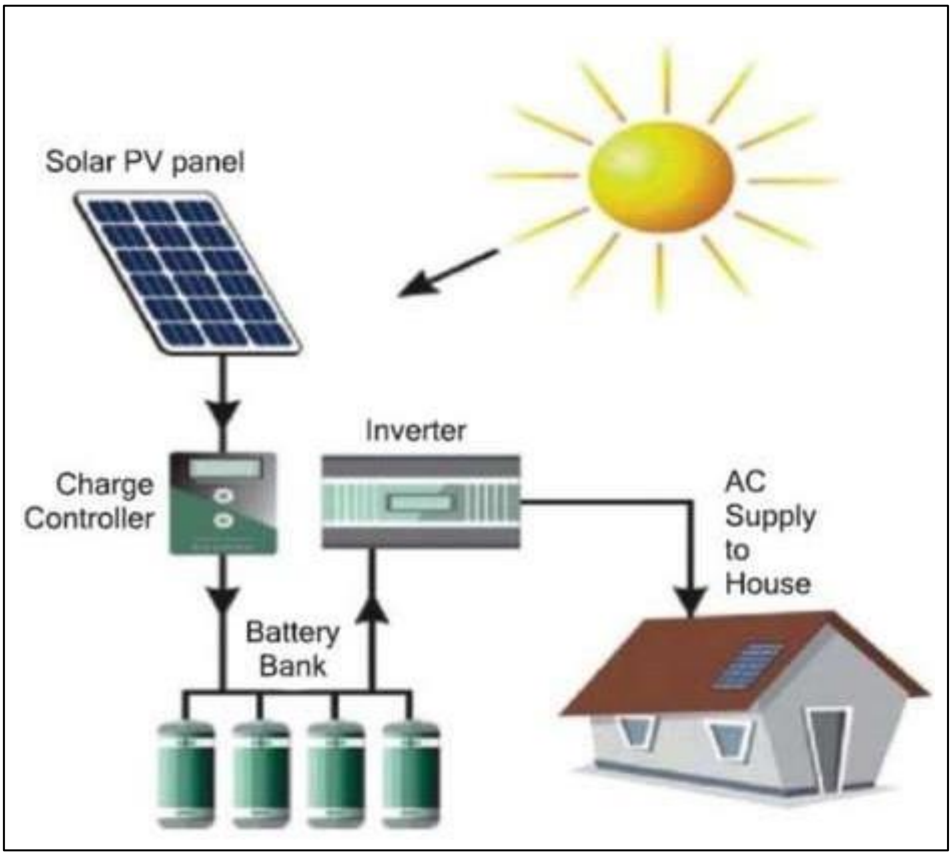


Image Source: https://pavanempowersolutions.com/wp-content/uploads/2019/11/Solar-Panel-3.jpg

Here's how the system works:

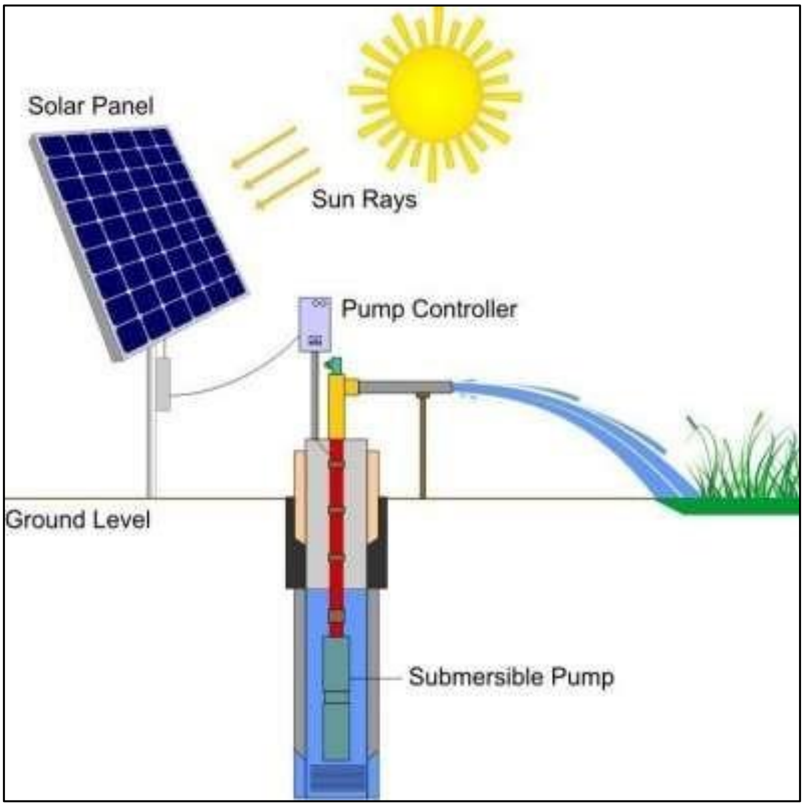
- **Solar Panels**: Solar panels are installed on the roof or any other location where they can receive sunlight. These panels convert sunlight into DC electricity.
- **Charge Controller**: A charge controller regulates the flow of electricity from the solar panels to the battery. It prevents overcharging and ensures that the battery is charged efficiently.
- **Battery**: The battery stores the DC electricity generated by the solar panels. The battery capacity is selected based on the power requirements of the lights and other devices that will be powered by the system.
- **LED Lights**: LED lights are energy-efficient and designed to use DC electricity. These lights are connected to the battery and provide light for the home.

During the day, solar panels absorb sunlight and convert it into electricity. The charge controller regulates the flow of electricity and charges the battery. When it gets dark, the LED lights are powered by the battery, providing light for the home. The cycle repeats the next day when the solar panels again begin to absorb sunlight.

- **Solar Lanterns**: A solar lantern is a portable lighting device that uses solar energy to power the light. It typically consists of a solar panel, a rechargeable battery, an LED bulb, and a charge controller.
  Here's a brief explanation of how solar lanterns work:

- **Solar panel**: Solar lanterns have a small solar panel on the top or side that collects sunlight during the day. The solar panel is made up of photovoltaic cells that convert sunlight into electrical energy.
- **Battery**: The electrical energy generated by the solar panel is stored in a rechargeable battery that is located inside the lantern. The battery is usually a lithium-ion or nickel-metal hydride type that can store the energy efficiently.
- **LED lights**: The solar lantern has one or more LED (Light Emitting Diode) bulbs that provide the light. LED bulbs are energy-efficient and last longer than traditional incandescent bulbs. They can be designed to provide different levels of brightness and color temperatures.
- **Switch**: Solar lanterns usually have an on/off switch that can be used to control the light. Some lanterns also have a low/high brightness switch that can be used to adjust the brightness level.
- **Charging time**: The solar panel on the lantern needs to be exposed to sunlight for a certain amount of time to charge the battery fully. The charging time varies depending on the size of the solar panel, the capacity of the battery, and the amount of sunlight available.

- **Solar Water Pumps**: These are used for pumping water in remote areas without electricity. The solar water pump system consists of a solar panel, a pump, a motor, and a controller.



The working of an off-grid solar water pump is as follows:

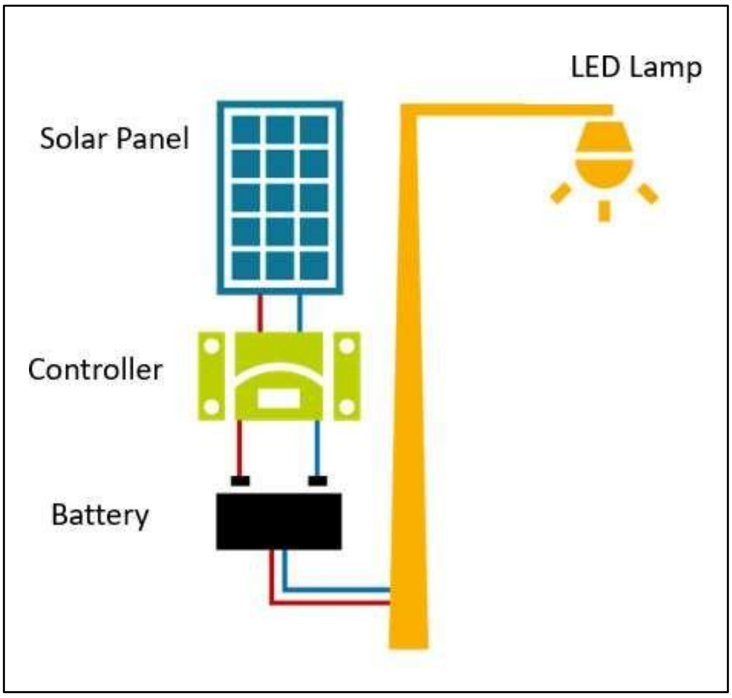
- **Solar Panels**: These are used to convert solar energy into electrical energy. These solar panels are installed in an open area where they can receive maximum sunlight.

- **Controller**: It is used to regulate the amount of energy that is being supplied to the pump. It ensures that the motor is running at the optimum speed and that the solar panels are not being overcharged.
- **Motor**: It is responsible for converting the electrical energy into mechanical energy. The motor is usually a brushless DC motor, which is more efficient and reliable than a traditional AC motor.
- **Pump**: The pump is used to lift the water from the ground to the surface. The pump can be of various types such as a centrifugal pump or a submersible pump.

When sunlight falls on the solar panels, it produces direct current (DC) electricity. This DC electricity is sent to the controller, which regulates the amount of energy that is being supplied to the motor. The controller ensures that the motor is running at the optimum speed, which ensures maximum efficiency. The motor then drives the pump, which lifts the water from the ground to the surface.

The amount of water that can be pumped by an off-grid solar water pump depends on the capacity of the solar panels, the controller, the motor, and the pump. The efficiency of these components also plays a crucial role in determining the amount of water that can be pumped. Off-grid solar water pumps have a lower operating cost than conventional pumps as they do not require fuel or electricity from the grid.

- **Solar Street Lights**: These are used for lighting public areas like streets, parks, and playgrounds. Solar street lights consist of a solar panel, battery, charge controller, LED lamp, and a pole.



The working principle of off-grid solar street lights can be explained as follows:

- **Solar panels**: The solar panels are installed on top of the street light and they convert the sun's energy into electricity. The solar panel absorbs sunlight during the day and charges the battery.
- **Battery**: It is a storage device that stores the electricity generated by the solar panels. It is usually a lead-acid or lithium-ion battery. The battery provides power to the light at night.
- **Controller**: It manages the flow of electricity between the solar panel, battery, and light. It regulates the charging and discharging of the battery and prevents overcharging or over-discharging.
- **LED light**: The LED light is the source of light in the solar street light. It is energy-efficient and provides bright light.

During the day, the solar panels absorb sunlight and charge the battery. At night, the controller detects the absence of sunlight and turns on the LED light. The light remains on until the battery is depleted or until the controller turns it off in the morning when the solar panels start charging the battery again.

UNIT 1.4: Basics Of Electrical Concepts

Unit Objectives

At the end of this unit, participant will be able to:

1. Analyse the basics of electrical concepts like voltage, current, power, energy, etc
2. Explain typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components.
3. Demonstrate how to interpret sample Electricity bill of a customer, notices and/or cautions.

1.4.1 Current

How do we express electric current? Electric current is expressed by the amount of charge flowing through a particular area in unit time. In other words, it is the rate of flow of electric charges.

How does an electrical appliance use current? A switch makes a conducting link between the cell and the appliance (say, an LED lamp). A continuous and closed path of an electric current is called an electric circuit. Now, if the circuit is broken anywhere (or the switch of the LED lamp is turned off), the current stops flowing and the LED does not glow.

The electric current is expressed by a unit called ampere (A). An instrument called ammeter measures electric current

in a circuit. It is always connected in series in a circuit through which the current is to be measured.

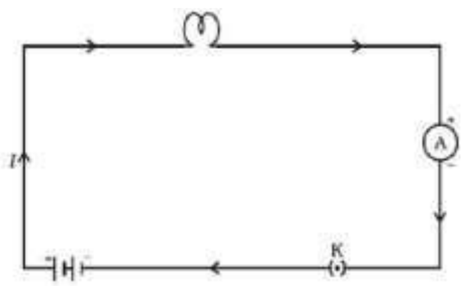


Fig 1.4.1 Schematic diagram of an electric circuit comprising – cell, LED bulb, ammeter and plug key

Note that the electric current flows in the circuit from the positive terminal of the cell to the negative terminal of the cell through the bulb and ammeter.

1.4.2 Direct Current (DC) and Alternating Current (AC)

AC is characterized by 'frequency'. In India, 50 Hertz (cycles per second) is the Alternating Current (AC) frequency. In some countries like USA and Canada, 60 Hertz (cycles per second) is the Alternating Current (AC) frequency.

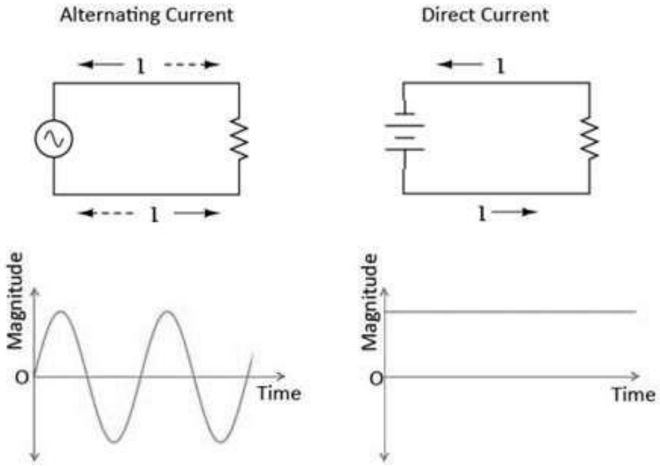


Fig 1.4.2 Alternating Current and Direct Current - Circuit and Waveforms

Read the Electricity Bill

Knowing how to read your electricity bill is critical for several reasons. Firstly, it helps you to understand your energy consumption and how much energy you are using. By identifying areas where you can reduce your energy consumption, you can save money on your electricity bill. Secondly, reviewing your electricity bill allows you to verify the accuracy of the charges. This helps you identify any billing errors and resolve them with the electricity provider. Thirdly, reading your electricity bill helps you to budget your energy costs and plan for future expenses, which is particularly useful for businesses that need to manage their energy costs to ensure profitability. Fourthly, by understanding the details of your electricity bill, you can compare energy plans offered by different providers and choose the one that best suits your needs and budget. Lastly, by understanding your energy consumption patterns, you can identify opportunities for energy efficiency measures such as upgrading to more efficient appliances or installing solar panels. Overall, being able to read your electricity bill is essential for managing your energy costs, making informed decisions about energy usage, and identifying opportunities for energy efficiency. To read the electricity bill, you need to perform the following tasks:

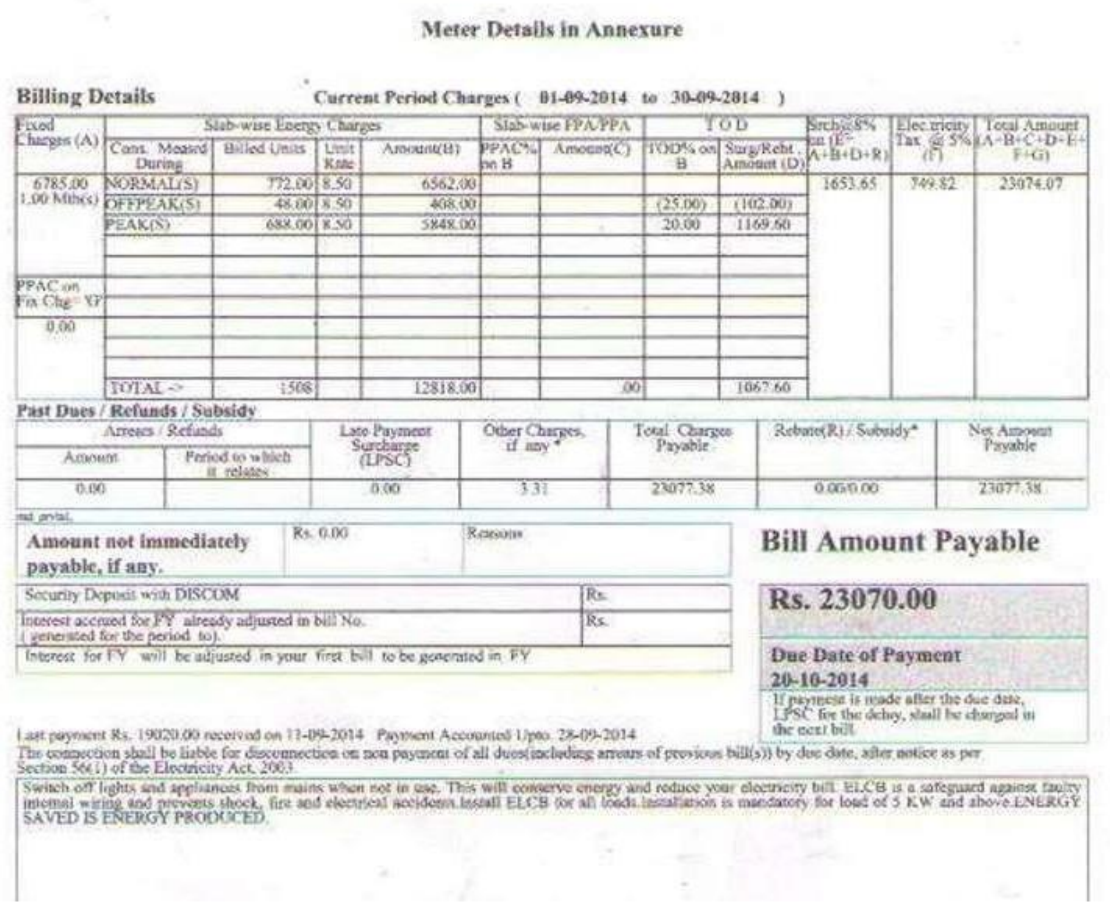
- Try to capture the bills of at least the 3 most recent months
- Record the units consumed in each month
- Record the unit electricity charges mentioned in the bill

Also, note the additional surcharges that have been applied by the distribution company. This will be useful when later explaining the payback to the consumer

Extract and record information as shown in the sample table

Month	Units Consumed	Unit Rate Applied	Surcharges
May			
June			
July			

Table 1.4.1: Information recorded from electricity bill



Meter Details in Annexure

Billing Details Current Period Charges (01-09-2014 to 30-09-2014)

Fixed Charges (A)	Slab-wise Energy Charges				Slab-wise FPA/PPA		TOD		Srch@8% on (E= A+B+D+R)	Elec tricity Tax @ 5% (F)	Total Amount (A+B+C+D+E+F+G)
	Cons. Measrd During	Billed Units	Unit Rate	Amount(B)	PPAC% on B	Amount(C)	TOD% on B	Surg/Rebt. Amount (D)			
6785.00 1.00 Mth(s)	NORMAL(S)	772.00	8.50	6562.00					1653.65	749.82	23074.07
	OFFPEAK(S)	48.00	8.50	408.00			(25.00)	(102.00)			
	PEAK(S)	688.00	8.50	5848.00			20.00	1169.60			
PPAC on Fix Chg= YP 0.00											
	TOTAL ->	1508		12818.00		.00		1067.60			

Past Dues / Refunds / Subsidy

Arrears / Refunds		Late Payment Surcharge (LPSC)	Other Charges, if any *	Total Charges Payable	Rebate(R) / Subsidy*	Net Amount Payable
Amount	Period to which it relates					
0.00		0.00	3.31	23077.38	0.00/0.00	23077.38

nd prvtal.

Amount not immediately payable, if any.	Rs. 0.00	Reasons

Security Deposit with DISCOM	Rs.
Interest accrued for FY already adjusted in bill No. (generated for the period to).	Rs.
Interest for FY will be adjusted in your first bill to be generated in FY	

Bill Amount Payable

Rs. 23070.00

Due Date of Payment
20-10-2014

If payment is made after the due date, LPSC for the delay, shall be charged in the next bill

Last payment Rs. 19020.00 received on 11-09-2014 Payment Accounted Upto. 28-09-2014

The connection shall be liable for disconnection on non payment of all dues(including arrears of previous bill(s)) by due date, after notice as per Section 56(1) of the Electricity Act, 2003.

Switch off lights and appliances from mains when not in use. This will conserve energy and reduce your electricity bill. ELCB is a safeguard against faulty internal wiring and prevents shock, fire and electrical accidents.Install ELCB for all loads.Installation is mandatory for load of 5 KW and above.ENERGY SAVED IS ENERGY PRODUCED.

Fig. 1.4.2 Sample electricity bill demonstrating consumption patterns of the customer during normal, peak and off-peak hours

UNIT 1.5: Financial Institutions and Banks Involved in Financing Solar Power Projects

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans.
2. Discuss emerging solar entrepreneurship landscape including success stories of various start- ups.

1.5.1 Financial Institutions and Banks Involved in Lending for Solar Power Projects

In India, there are numerous banks and financial institutions that provide loans for solar energy projects. These financial organisations offer loans for a range of solar power project types, including rooftop, utility-scale, and solar park projects. They provide various loan products with various interest rates, payback schedules, and other terms and conditions, such as term loans, working capital loans, and project finance loans.

These financial institutions additionally offer other financial services and products for solar power projects, including equipment finance, lease financing, refinancing, and bridge financing. Additionally, they offer project management support, technical assistance, and advisory services to help borrowers plan and carry out their solar power projects.

Some of the prominent financial institutions and banks involved in lending solar power projects are:

- Indian Renewable Energy Development Agency (IREDA): IREDA is a dedicated financial institution that provides loans for renewable energy projects in India, including solar power projects. The interest rates for IREDA loans range from 7-9% depending on the project's size and duration.
  State Bank of India: SBI offers loans for solar power projects under its Green Channel Counter Scheme. The bank provides loans up to 75% of the project cost, with a repayment period of up to 15 years. The interest rate for these loans is usually around 8-10%.
- ICICI Bank: ICICI Bank offers loans for solar power projects under its Corporate and Commercial Banking division. The bank provides loans up to 70% of the project cost, with a repayment period of up to 12 years. The interest rate for these loans is usually around 8-10%.
- Axis Bank: Axis Bank provides loans for solar power projects under its Renewable Energy Financing Scheme. The bank offers loans up to 70% of the project cost, with a repayment period of up to 12 years. The interest rate for these loans is usually around 8-10%.

- Yes Bank: Yes Bank offers loans for solar power projects under its Renewable Energy Financing Scheme. The bank provides loans up to 70% of the project cost, with a repayment period of up to 12 years. The interest rate for these loans is usually around 8-10%.

1.5.2 Terms and Conditions Associated with Loans

The terms and conditions associated with these loans may vary depending on the financial institution or bank. However, some common terms and conditions include:

- **Eligibility criteria**: Most banks and financial institutions have eligibility criteria for solar power project loans, which may include:
  - **Project Size**: Most financial institutions and banks have a minimum project size requirement for lending loans for solar power projects. The minimum project size may range from 100 kWp to 1 MWp, depending on the lender.
  - **Creditworthiness**: Financial institutions and banks consider the creditworthiness of the borrower, which includes the borrower's financial history, credit score, and repayment capacity. Borrowers with a good credit history and a high credit score are more likely to be eligible for loans.
  - **Project Location**: The location of the solar power project is also an important eligibility criterion. Financial institutions and banks may prefer projects located in areas with high solar irradiation and easy grid connectivity.
  - **Project Technology**: The technology used in the solar power project is also an important factor. Financial institutions and banks prefer projects that use proven and reliable technologies that have a track record of successful implementation.
  - **Project Viability**: Financial institutions and banks conduct a detailed feasibility study of the solar power project to assess its viability, technical feasibility, and financial viability. Projects that are found to be technically and financially viable are more likely to be eligible for loans.
  - **Collateral**: Most financial institutions and banks require collateral for solar power project loans, which may include land, buildings, or any other assets owned by the borrower.
  - **Other Requirements**: Financial institutions and banks may have other eligibility criteria, such as the borrower's industry experience, business plan, and legal compliance.
- **Interest Rates:** The interest rates on solar power project loans may vary depending on the lender, loan amount, and tenure. Depending on the borrower's creditworthiness, the feasibility of the project, and other variables, the interest rates may be fixed or floating and range from 8% to 12% annually.
- **Loan Tenure:** Depending on the lender and the type of loan, the loan term for solar power projects may range from 5 years to 20 years. For large-scale solar power projects, certain lenders might offer longer loan terms.

- **Loan Amount:** The loan amount for solar power projects may depend on the project cost, project size, and the borrower's creditworthiness. Some lenders may offer up to 75% to 85% of the project cost as a loan.
- **Collateral:** Loans for solar power projects typically demand collateral, which may include land, buildings, or any other assets owned by the borrower. The collateral value may be equivalent to the loan amount or a percentage of the loan amount.
- **Processing Fees:** Lenders may charge processing fees for solar power project loans, which may range from 0.5% to 1% of the loan amount. Some lenders may waive processing fees for certain borrowers or loan types.
- **Prepayment Charges:** Some lenders may charge prepayment charges if the borrower chooses to repay the loan before the end of the loan tenure. The prepayment charges may range from 1% to 5% of the outstanding loan amount.
- **Disbursement and Repayment:** Depending on how the solar power project is coming along, lenders may disburse the loan amount in instalments. The lender might want regular updates and progress reports from borrowers. Depending on the loan type and tenure, repayment may take the form of equivalent monthly instalments (EMIs) or lump sum payments.
- **Other Conditions:** Lenders may have other conditions for solar power project loans, such as insurance requirements, technical specifications, and compliance with regulatory requirements.

Before submitting an application for one of these loans, it is crucial to read and comprehend all of the terms and conditions that are attached to it. To choose the best alternative for their solar power project, borrowers can also analyse the terms and conditions of various banks and financial organisations.

Emerging solar entrepreneurship landscape including success stories of various start- ups.

The solar energy sector has witnessed significant growth in recent years, fostering the emergence of numerous innovative startups globally. Here are some success stories of startups contributing to the evolving landscape of solar entrepreneurship:

Sunrun: Founded in 2007, Sunrun is one of the largest residential solar companies in the United States. It offers solar installation services and innovative financing options like solar leases and power purchase agreements (PPAs) to make solar more accessible to homeowners.
Tesla (formerly SolarCity): Acquired by Tesla in 2016, SolarCity was a pioneer in providing solar energy systems for residential and commercial properties. Tesla continues to develop solar roof tiles and offers solar solutions integrated with their energy storage products, like the Powerwall.
Oolu: Operating in West Africa, Oolu focuses on providing solar energy solutions to off-grid communities. It offers pay-as-you-go solar kits to households, allowing them to access clean and affordable energy without the need for a grid connection.
M-KOPA: Based in Kenya, M-KOPA provides solar home systems on a rent-to-own basis. The company leverages mobile payment technology to enable low-income households to affordably access solar power, replacing kerosene for lighting and charging mobile devices.

Exercise

1. Explain the photovoltaic effect.

____

____

____

____

2. Explain the different types of solar mounting structures.

____

____

____

____

3. Explain the different types of solar panel.

____

____

____

____

2. Set up new venture for Solar Photovoltaic System

SGJ/N0111

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Describe the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms.
2. Discuss various evolving opportunities across solar solutions, services and market ecosystem.
3. Discuss how to identify the key ingredients of a business plan.
4. Discuss how to distinguish between fixed and working capital requirements.
5. Describe the components of a loan application.
6. Discuss how to demonstrate good etiquettes and manners while communicating with the client.
7. Discuss how to demonstrate the importance of time management.
8. Discuss how to demonstrate leadership skills and effective resource management techniques.
9. Discuss the use of MS Word and Excel for preparing a business proposal
10. Discuss to prepare a workable presentation for marketing and business development.
11. Discuss how to choose the right buyer in a given situation of market parameters.
12. Identify the challenges and risks for new entrepreneurs and the possible mitigation measures.
13. Explain various taxation and related compliances to set up and operate a solar business
14. Discuss the key aspects to successfully run a solar business.
15. Explain how evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape.
16. Explain how to demonstrate responsible and disciplined behaviour and promote a safe and interactive environment at work while respecting gender and disability concerns.

UNIT 2.1: Setting Up a New Venture

Unit Objectives

At the end of this unit, participant will be able to:

- Describe the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms.
- Discuss various evolving opportunities across solar solutions, services and market ecosystem.
- Discuss how to identify the key ingredients of a business plan.
- Discuss how to distinguish between fixed and working capital requirements.
- Describe the components of a loan application.

2.1.1 Entrepreneurship

Entrepreneurship is the act of one person or a group of people starting a business from an idea with the goal of making money, whether in the manufacturing or service industries.

Choosing the Industry or Area of operation is the first step in successfully putting an idea into practise. Which industry would be best for me? Is it the agri-processing sector, the auto ancillary industry, the production of solar PV modules, or the module mounting structure? Whichever produces the highest profit is the fabrication unit or the fabrication unit that will serve the solar energy sector, whether it be MMS, Solar Fencing poles, Solar water heater structures, etc. If it is a service sector, it may be EPC, Installation, O&M of Solar PV Power Packs, Software Development, Consulting, Product Design (R&D), etc. Making the optimal selection is aided by a market analysis. Entrepreneur is one who wishes to start and succeed in his/her enterprise, and plays different roles at different stages. Some essential qualities for entrepreneurs are:

Desire to Win	Persistent	Moderate risk taker	Ability to find and explore opportunity
Analytical ability	Using feedback	Facing uncertainty	Flexibility
Planner	Interpersonal skills	Motivator	Stress taker
	Positive self-concept	Orienttation for future	

Fig: 2.1.1: Some essential qualities for entrepreneurs

- **Desire to Win**: A strong desire to win. Most individuals dream of success, but they hardly ever take any action to make it a reality. Entrepreneurs, on the other hand, have a tremendous ambition to continually achieve new goals and won't stop until they succeed.
- **Persistent**: A never-say-die attitude. Once an entrepreneur commits to a goal and a plan of action, they never change their mind. They persevere and work hard until the entire project is completed effectively, despite obstacles.
- **Moderate Risk Taker**: Entrepreneurs prefer a middle-of-the-road strategy while handling tricky situations. They don't take big risks and aren't gamblers. They like a modest risk over a wild bet that has a decent probability of winning and is high enough to be thrilling.
- **Ability to Find and Explore Opportunities**: Alert to opportunities and seize them to their advantage. Entrepreneurs are creative and can turn crises into opportunities. However, they are grounded in reality enough to see how the chance fits into achieving their objectives.
- **Analytical Ability**: They tackle issues with objectivity. Entrepreneurs will make business decisions based on the facts, regardless of their personal preferences. Instead of friends and family, they turn to experts for help. They do not act impulsively or emotionally when making decisions.
- **Using Feedback**: As they work towards their objectives, it is critical for them to be aware of their progress. Entrepreneurs accept fast feedback on their performance and favour timely and precise information, whether or not it is favourable. Unfavourable news motivates them to make changes so they can achieve their objectives.
- **Facing Uncertainty:** Even under challenging yet intriguing circumstances, entrepreneurs are not deterred. Despite the odds, they believe that they can succeed given their wealth of skills. They continue to march unabated, taking advantage of the wonderful possibilities that come their way despite the lack of rules. They swiftly adapt to the new surroundings and come to terms with it.
- **Flexibility**: Successful entrepreneurs are flexible and are not afraid to alter their plans if the circumstances warrant it after assessing the pros and downsides.
- **Planner**: Entrepreneurs anticipate needs and make plans for the future. The majority of successful people have goals that they plan to achieve over a period of time.
- **Interpersonal Skills**: Entrepreneurs can deal with people at all levels. An entrepreneur encounters a wide variety of people. To assist him achieve his goals, he must convince people to work for and with him. He enjoys working with people and is capable of handling them.
- **Motivator**: A successful entrepreneur can influence others and motivate them to think and act in his way.
- **Stress Taker**: They have the stamina to spend long hours while tackling multiple issues. The entrepreneur is a vital character in his business and must handle multiple problems at once while making the best judgements possible, even if it causes him physical and emotional hardship. If one can work hard for a long time while maintaining composure, this is achievable.
- **Positive Self-Concept**: They have self-awareness. An achiever channels his dreams into worthwhile, reachable goals and establishes criteria for perfection (POSITIVE SELF-CONCEPT). He is capable of doing this because he is aware of his skills and flaws and takes a positive attitude. He rarely has a bad attitude.

- **Orientation for Future**: They have the ability to look into the future. They won't allow the past to bother them and think only of the present and future. "Bygones are bygones, what of now?" This is their usual response.

An individual might not possess all of these traits, but the majority do. Making a list of one's personality qualities is the first step for someone who wants to start their own business. He/she will be able to strengthen his weaknesses with the aid of his self-awareness and analysis.

2.1.2 Process of Setting Up a New Venture

Following are the basic steps involved in setting up a new venture:

Conduct market research:

Before you start your business, you need to research the market to identify the potential demand for your solar PV products or services. You should also look at the competition and assess ways to differentiate yourself.

Develop a business plan:

A business plan is a crucial step in starting any business, including a solar PV business. Your plan should include a description of your business, your target market, your marketing and sales strategy, your financial projections, and your goals.

Obtain necessary certifications and licenses:

Depending on where you plan to operate your solar PV business, you may need to obtain certifications and licenses from local or national authorities.

Secure financing:

Starting a solar PV business can be capital intensive, so you may need to secure financing to get started. This could include personal savings, loans, or investors.

Choose a location:

Once you have secured financing, you need to choose a location for your business. Consider factors such as the cost of real estate, accessibility, and proximity to potential customers.

Purchase equipment:

You will need to purchase solar panels, inverters, mounting systems, and other equipment necessary for installing solar PV systems.

Hire and train employees:

To operate your business, you will need to hire and train employees to help with sales, installation, and maintenance.

Develop marketing strategies:

Develop marketing strategies to reach your target market, such as social media, search engine optimization, and advertising.

Install and maintain systems:

Once you have customers, you will need to install and maintain their solar PV systems, ensuring they are working effectively and efficiently.

Fig:2.1.2 Basic Steps Involved in Setting Up a New Venture

Types of Organisations and Required Registration

As an entrepreneur you should be aware of the several types of organizations and the respective registration requirements. Following are some of the most common types of organizations and their registration requirements:

- **Sole Proprietorship Concern:** The sole proprietorship is nothing but a company run by a single person. Usually, in sole proprietorship companies, the owner is responsible for all profits or losses. It's an individual company and pretty easy to set up the business. Mostly work from home or businesses run from home by a single person prefer this type of registration.

Documents required for registration of sole proprietorship concern are:

- Utility bills of the company
- KYC details
- Company Establishment License
- Income tax data

- **One Person Company (OPC):** One Person Company Registration just made an entry into the Indian market. This type of registration is used by the majority of small enterprises and startups conducted by a single individual. The owners are protected from liability under this registration; therefore, no partnerships are even necessary. As one person manages every area of the organisation, it's fairly simple to maintain, oversee, and run. In a nutshell, it's a hybrid of a private limited company and a sole proprietorship. The minimum capital need to be eligible for this registration is around Rs. 1,000,000. Additionally, the person needs to be an Indian citizen. Note that the if you're into a finance business, then you aren't qualified to get an OPC registration.

  Documents required for registration of OPC (By the company owner)
  - PAN or Passport
  - Aadhar Card
  - ID proof
  - Bank statements, electricity, gas, or phone bill
  - Photocopy
  - Signature of the specimen

  Documents required for the registration office

  - Bank statement or bills
  - Rental agreements
  - NOC of a property owner
  - Copy of property deed (in case of own properties)

- **Partnership Concern:** The number of partners makes a big difference between a sole proprietorship and a partnership. Two or more persons can form a partnership business, and the agreement outlines each member's responsibilities in detail. In the meantime, the partners likewise split the profits in accordance with the contract. The partners, on the other hand, are likewise accountable for bearing losses comparable to those of profits. If these enterprises have a registered Partnership Deed, they can operate legally even without a licence. The Indian Partnership Act of 1932 governs the partnership companies.

  Documents required for the registration of Partnership concern
  - Partnership Deed copy including the name of the organization or entity, company's operations, primary business location, additional company locations, date of partnership, name of the partners and address, time period from which the company has established.
  - Partners' PAN card and Aadhar card/Voter ID card
  - Ownership/rent/lease of the business location

- **Limited Liability Partnership Company (LLP):** Assets in an LLP are completely distinct from personal assets and offer the highest level of limited liability protection. The partners in the company rely on the amount of share capital to cover the liabilities. One of the key prerequisites for forming an LLP is maintaining a minimum capital of 1,00,000 with at least one partner being a citizen of this nation.

Documents required for registration of LLP

- PAN card or identity proof such as Aadhar Card, Driver's License, or Voter ID
- Bank statements, phone bills, mobile bills, and gas bills
- Passport size photocopy
- Foreigners or NRIs require a passport
- DSC and registered office address proof documents

- **Private Limited Company:** In order to protect their investments, the shareholders typically share in the liability. The total number of shares held by each shareholder constitutes the net capital of these companies. The shares of a private limited business cannot be traded or transferred openly, in contrast to those of many other corporations. As per the Companies Act of 2013, an enterprise is said to be a private limited company if it meets the following criteria:
  - It should consist of at least two directors and utmost fifteen directors.
  - One director must be an Indian resident.
  - The minimum and maximum shareholders of the company should be two and 200, respectively.
  - It should possess a reliable capital fee of a minimum of ₹1,00,000.
  - It should have a registered office address within India.

Documents required for the registration of Private Limited Company

- PAN card of all company directors
- Electricity and water bills of the company
- Passport-size photo IDs of all directors
- Director's Aadhar card/Voter Id card
- Property documents
- Rental agreements
- Authorization letter from the landowners

It is essential to register the business under the Micro, Small, and Medium Enterprise (MSME) category as soon as the entrepreneur registers his organisation as a concern or a corporation, depending on the type of the business and the amount of investment. Activities are categorised into two sectors; the service sector, and the manufacturing sector.

2.1.3 Evolving opportunities across solar solutions, services and market ecosystem

The solar industry is constantly evolving, and there are many emerging opportunities across solar solutions, services, and the market ecosystem. Here are some of the significant evolving opportunities:

- **Energy Storage**: As solar energy generation becomes more widespread, there is a growing need for energy storage solutions. Battery storage solutions enable solar systems to store excess energy generated during the day and use it during periods of low or no sunlight.
- **Smart Grids**: The integration of solar energy into smart grids is another significant opportunity. Smart grids can optimize the distribution of solar energy and reduce the need for traditional power generation, reducing greenhouse gas emissions.
- **Solar Asset Management**: The increasing number of solar installations has created a demand for asset management services. Solar asset management involves the management of solar projects to optimize energy production, reduce costs, and increase profitability.
- **Solar O&M Services**: Solar operations and maintenance (O&M) services are also an emerging opportunity. These services involve the maintenance and repair of solar installations to ensure they operate efficiently and maximize energy production.
- **Community Solar**: Community solar projects allow individuals and businesses to purchase a portion of a solar installation's output. This provides access to solar energy for those who may not be able to install solar panels on their own property.
- **Solar Financing**: Solar financing has become more accessible, with many innovative financing models available. These models include power purchase agreements (PPAs), leasing, and community solar financing.
- **Solar Manufacturing**: With the increasing demand for solar panels and other solar equipment, there is an opportunity for manufacturing solar components locally. This can create jobs and support local economies.
- **Solar-powered Transportation**: Solar power can be used to power transportation, including electric vehicles and solar-powered boats, trains, and planes. These solutions can reduce greenhouse gas emissions and increase energy independence.

2.1.4 Key Ingredients of a Business Plan

A business plan is a critical document that outlines the goals, strategies, and operational details of a business. The key ingredients of a business plan will vary depending on the type and scope of the business, but generally include the following:

- **Executive Summary**: The executive summary is the first section of a business plan and provides an overview of the entire document. It should be concise, yet compelling, and capture the reader's attention.
- **Business Description**: The business description should provide an overview of the business, including its mission, values, and goals. It should also describe the products or services offered, the target market, and the competitive landscape.
- **Market Analysis**: The market analysis should provide an in-depth understanding of the target market, including its size, growth potential, and trends. It should also include a competitive analysis that identifies the strengths and weaknesses of key competitors.
- **Marketing and Sales Strategy**: The marketing and sales strategy should outline the methods used to promote the business and sell its products or services. This section should include details on pricing, distribution channels, and advertising.
- **Operations and Management**: The operations and management section should provide details on how the business will be organized and managed. It should include information on staffing, facilities, equipment, and technology.

- **Financial Projections**: The financial projections section should provide a detailed forecast of the business's revenue, expenses, and profits over a set period. It should include cash flow projections, balance sheets, and income statements.
- **Funding Requirements**: The funding requirements section should outline the amount of money required to start and operate the business. It should also detail how the funds will be used and identify potential sources of funding.

The following figure represents the generic structure of a solar rooftop business model:

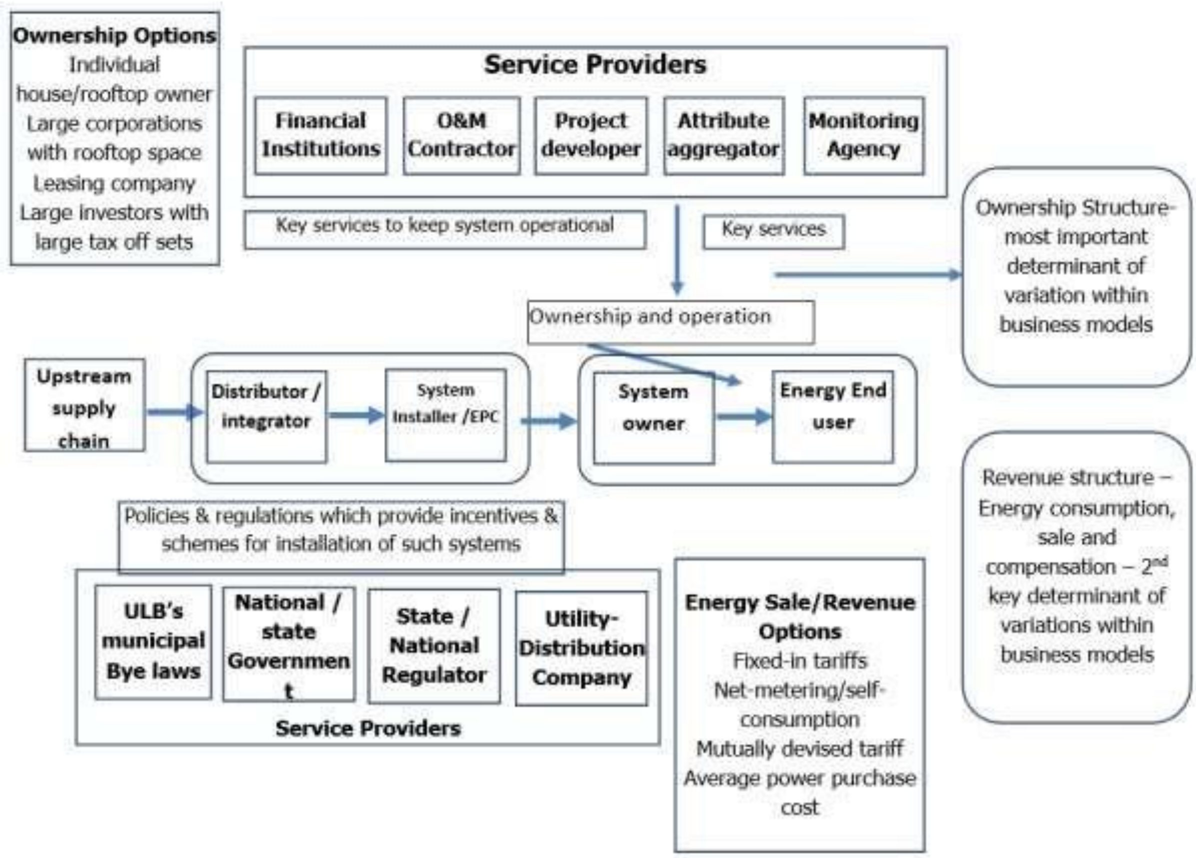


Fig2.1.3: Generic structure of a solar rooftop business model

Fixed and Working Capital Requirements

Fixed and working capital are two types of capital that businesses need to operate, but they serve different purposes and are used for different things. Distinguishing between fixed and working capital is essential for effective financial planning and management

Fixed Capital

A business's long-term capital investments to buy assets that will be employed for a considerable amount of time, usually more than a year, are referred to as fixed capital. These assets are not meant for resale; rather, they are used in the creation of commodities or services. A few examples of fixed capital investments are the acquisition of land, structures, tools, equipment, and vehicles.

Working Capital

The money required to finance a business's ongoing operations is referred to as working capital, on the other hand. To pay for costs like payroll, rent, utilities, inventory, and accounts payable, working capital is required. Working capital is a short-term capital requirement that is normally determined by subtracting a company's current obligations from its current assets.

Distinguish Between Fixed Capital and Working Capital

The distinction between fixed and working capital is important because the two types of capital have various effects on a company's cash flow management and financial health. Fixed capital investments have a high initial cost but produce income over a long period of time. Working capital, on the other hand, is used to meet ongoing costs and demand variations, and it needs to be carefully managed to maintain adequate cash flow to cover costs.

To distinguish between fixed and working capital requirements, businesses should consider the nature of their operations, their current assets and liabilities, and their long-term goals.

Businesses with a high level of fixed capital investment may need to make a larger initial expenditure but have lower ongoing costs, whereas companies with a high need for working capital may need to make larger cash infusions more frequently to maintain cash flow.

2.1.5 Components of Loan Application

When applying for a loan for a solar PV project, there are several components that the lender will typically require to evaluate the project's feasibility and assess the risk. The exact components of the loan application may vary depending on the lender and the specific project, but the following are some common components:

Fig.2.1.4: Some Common Components of Loan Application for a Solar PV Project

- **Executive Summary**: This is a brief overview of the project, including the project's goals, location, size, and technology used. It should also include the project's expected financial returns and key stakeholders.
- **Project Description**: A detailed description of the project, including its technical specifications, capacity, and expected output. It should also include information on the site, such as site analysis, environmental impact, and permitting requirements.
- **Financial Projections**: Detailed financial projections for the project, including capital costs, operating costs, revenue projections, and cash flow projections. This section should also include a sensitivity analysis to demonstrate the project's resilience to changes in key assumptions.
- **Project Schedule**: A timeline for the project, including the construction phase, commissioning, and operation. This section should include a detailed breakdown of the key milestones and deliverables.
- **Energy Offtake Agreement**: If the project has an energy off-take agreement, this should be included in the application. The agreement should include details on the term of the agreement, the price of electricity, and the counterparty's creditworthiness.
- **Team and Management**: The loan application should include information on the project team and management, including their relevant experience and qualifications. This section should also include information on the project's contractors, suppliers, and other key stakeholders.
- **Collateral and Guarantees**: The lender may require collateral or guarantees to secure the loan, such as a mortgage on the project's assets or a guarantee from a third-party. This section should include details on the collateral or guarantees being offered.

Thus, a loan application should provide a detailed and comprehensive overview of the solar PV project, including technical, financial, and operational details. The lender will use this information to evaluate the project's feasibility and assess the risk of providing financing.

UNIT 2.2: Workplace Etiquette

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to demonstrate good etiquettes and manners while communicating with the client.

2.2.1 Forwarding Accurate Information

One must understand that their behaviour serves as an example for other people at the workplace. An individual's work ethic and etiquette will set standards for how others are expected to behave.

One must ensure that accurate information is passed and recorded in the books. To properly and accurately fill record books and/or make reports and ensure that correct and timely information is passed, one must:

Be truthful in data representation.	Acknowledge errors and do not distort any fact or incident.	Always be honest and truthful.
Always be on time.	Always respect your co-workers and managers. Do not intimidate and/or bully other workers.	Always use professional language in reports and in professional meetings.
Avoid the use of cell phones on duty unless it is an emergency.	Accept personal responsibility for your actions and do not criticise others untruthfully.	Do not use loud, offensive or abusive language.

Fig. 2.2.1: Recording and Forwarding Accurate Information

Every organization has an obligation to keep certain information confidential and the kind of information which needs to be shared with the public.

The promotional materials or PR materials can be shared with public include:

Fig. 2.2.2: Public Information

2.2.2 Assisting Others

Since one mostly works in teams, it is very important to develop good habits to work effectively with co-workers. Some characteristics that are required to work effectively with co-workers and while assisting them are:

- Be honest and straight-forward
- Do share the load equally and assist others if required
- Treat others with respect and dignity
- Be an active listener and speak with discretion when communicating with your colleagues
- Acknowledge the contribution of your team members

Every new employee should be trained properly before the commencement of work. Conduct periodic reviews on co-workers work and assist wherever required.

2.2.3 Interact with Others without Any Bias

Employees should be treated equitably and with respect, participation without favouritism is the key. Every employee should feel valued, they should believe their individuality is respected by others and at the same time should feel connected with the team

- Every employee should feel safe to speak up and be open without feeling judged or embarrassed
- When employees are confident with themselves, they feel empowered and give their best in the job which helps them grow along.
- General points to be kept in mind:
- Do use a common language understood by all while speaking to a group
- Do speak to everyone with respect in the workspace, interact with all co-workers without discrimination
- Do not bully or make anyone feel inferior or left out
- Do have open discussions and involve everyone
- If possible, conduct informal get-togethers and celebrate festivals. It will help in engaging every employee, improving the organization's culture and making the employees feel appreciated.

2.2.4 Communicating Effectively

One should keep the following points in mind when having a conversation with a colleague or client:

- Use positive language while reporting a complaint to senior.
- Remain assertive when under pressure.
- Ask questions calmly. Do not intimidate or interrogate.
- Aim to achieve a win/win outcome for both the parties.
- Honour the commitments and promises made to resolve the issues.

Often, one needs to communicate with people at different levels, such as labourers, supervisors, team members and so on. Effective communication has an important role to play in fulfilling responsibilities. These forms of communication are:

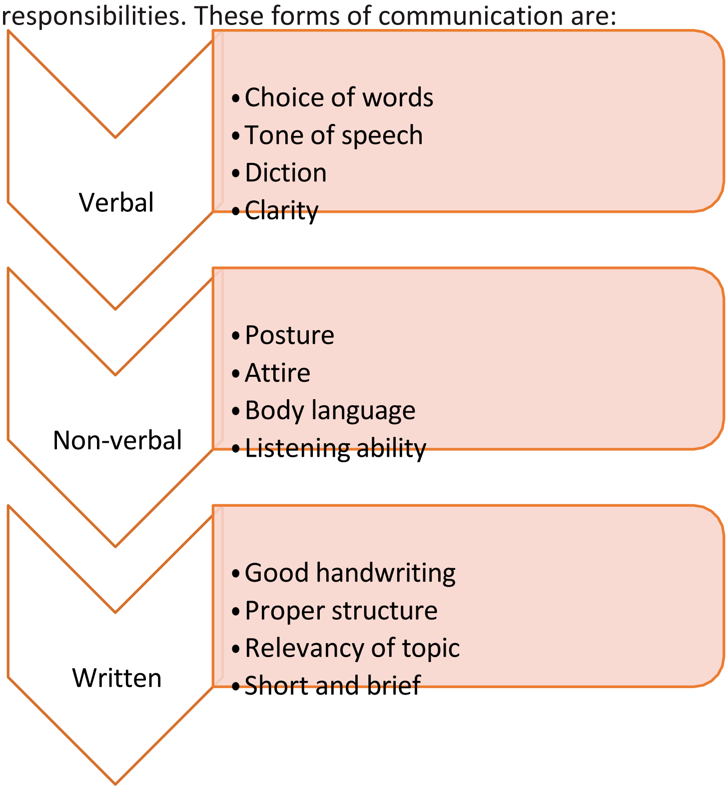


Fig. 2.2.4: Forms of communication

Some language and listening barriers are:

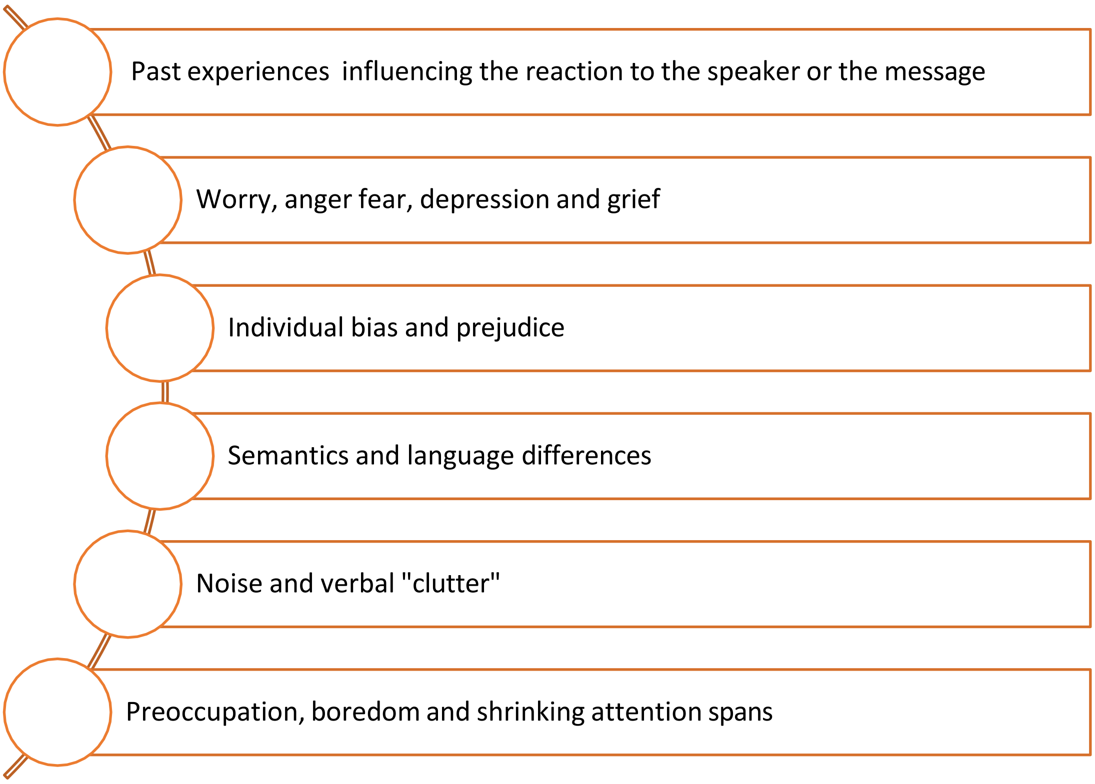


Fig. 2.2.5: Language and listening barriers

Effective communication depends on active and effective listening. The following figure shows some ways to develop an active listening skill:

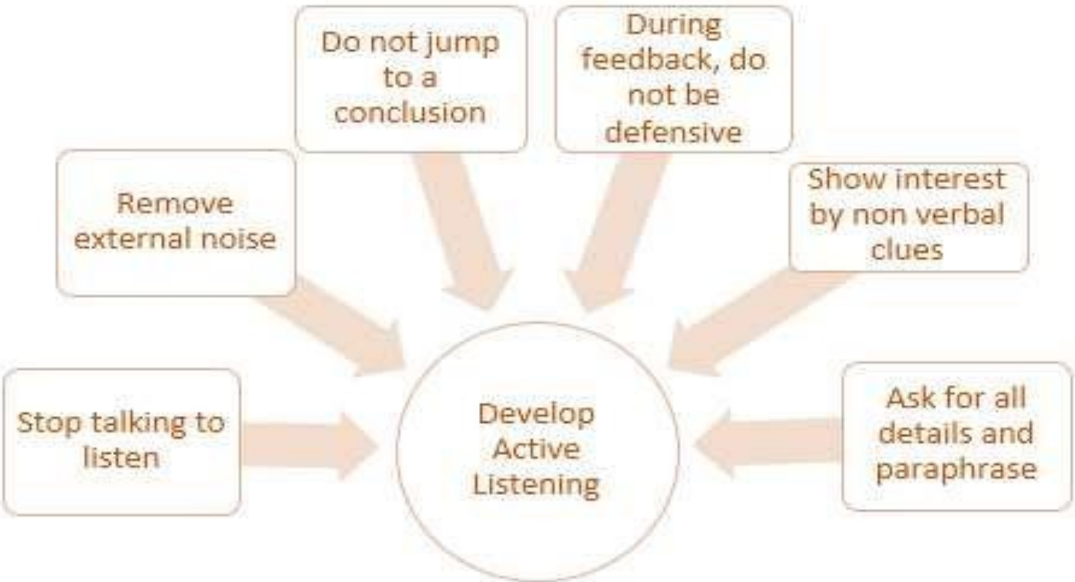


Fig. 2.2.6: Active listening skills

2.2.5 Interacting Effectively with Clients & Colleagues

While interacting with clients and colleagues, one should always maintain a decorum and follow proper manners. Etiquette and manners can be easily developed by following certain practices. These practices are:

- Be attentive
- Keep an eye contact while facing the speaker
- Pay-attention to non-verbal cues – gestures, facial expressions, tone and posture
- Do not interrupt or try to impose "solution"
- Wait for the speaker to pause if he or she allows any question to be clarified

Fig. 2.2.7: Etiquette and Best practices

The conflicts that arise due to perception barriers can be solved by the following methods:

Perception barriers

- Inform someone senior
- Ignore the conflict
- Try to deal with it
- Compromise with it
- Clarify viewpoint with the concerned person
- Try to negotiate

Fig. 2.2.8: Conflicts

UNIT 2.3: Time Management, Leadership Skills and Resource Management

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss how to demonstrate the importance of time management.
- Discuss how to demonstrate leadership skills and effective resource management techniques.

2.3.1 Time Management

Time management is the process organizing your time, and deciding how to allocate your time between different activities. Good time management is the difference between working smart (getting more work done in less time) and working hard (working for more time to get more done).

Effective time management leads to an efficient work output, even when you are faced with tight deadlines and high-pressure situations. On the other hand, not managing your time effectively results in inefficient output and increases stress and anxiety.

Benefits of Time Management

Time management can lead to huge benefits like:

- Greater productivity
- Higher efficiency
- Better professional reputation
- Reduced stress
- Higher chances for career advancement
- Greater opportunities to achieve goals

Not managing time effectively can result in undesirable consequences like:

- Missing deadlines
- Inefficient work output
- Substandard work quality
- Poor professional reputation
- Stalled career
- Increase in stress and anxiety

Some traits of effective time managers are:

- They begin projects early
- They set daily objectives

- They modify plans if required, to achieve better results
- They are flexible and open-minded
- They inform people in advance if their help will be required
- They know how to say no
- They break tasks into steps with specific deadlines
- They continually review long term goals
- They think of alternate solutions if and when required
- They ask for help when required
- They create backup plans

2.3.2 Effective Time Management Techniques

You can manage your time better by putting into practice certain time management techniques. Some helpful tips are:

- **Plan out your day as well as plan for interruptions.** Give yourself at least 30 minutes to figure out your time plan. In your plan, schedule some time for interruptions.
- **Put up a "Do Not Disturb" sign** when you absolutely have to complete a certain amount of work.
- **Close your mind to all distractions.** Train yourself to ignore ringing phones, don't reply to chat messages and disconnect from social media sites.
- **Delegate your work.** This will not only help your work get done faster, but will also show you the unique skills and abilities of those around you.
- **Stop procrastinating.** Remind yourself that procrastination typically arises due to the fear of failure or the belief that you cannot do things as perfectly as you wish to do them.
- **Prioritize.** List each task to be completed in order of its urgency or importance level. Then focus on completing each task, one by one.
- **Maintain a log of your work activities.** Analyse the log to help you understand how efficient you are, and how much time is wasted every day.
- **Create time management goals** to reduce time wastage.

Tips

- Always complete the most important tasks first.
- Get at least 7 – 8 hours of sleep every day.
- Start your day early.
- Don't waste too much time on small, unimportant details.
- Set a time limit for every task that you will undertake.
- Give yourself some time to unwind between tasks.

2.3.3 Leadership Qualities That All Entrepreneurs Need

Leadership means setting an example for others to follow. Setting a good example means not asking someone to do something that you wouldn't willingly want to do yourself. Leadership is about figuring out what to do in order to win as a team, and as a company.

Leaders believe in doing the right things. They also believe in helping others to do the right things. An effective leader is someone who:

- Creates an inspiring vision of the future.
- Motivates and inspires his team to pursue that vision.

Building a successful enterprise is only possible if the entrepreneur in charge possesses excellent leadership qualities. Some critical leadership skills that every entrepreneur must have are:

1. **Pragmatism:** This means having the ability to highlight all obstacles and challenges, in order to resolve issues and reduce risks.
2. **Humility:** This means admitting to mistakes often and early, and being quick to take responsibility for your actions. Mistakes should be viewed as challenges to overcome, not opportunities to point blame.
3. **Flexibility:** It is critical for a good leader to be very flexible and quickly adapt to change. It is equally critical to know when to adapt and when not to.
4. **Authenticity:** This means showing both, your strengths and your weaknesses. It means being human and showing others that you are human.
5. **Reinvention:** This means refreshing or changing your leadership style when necessary. To do this, it's important to learn where your leadership gaps lie and find out what resources are required to close them.
6. **Awareness:** This means taking the time to recognize how others view you. It means understanding how your presence affects those around you.

Benefits of Effective Leadership

Effective leadership results in numerous benefits. Great leadership leads to the leader successfully:

- Gaining the loyalty and commitment of the team members
- Motivating the team to work towards achieving the company's goals and objectives
- Building morale and instilling confidence in the team members
- Fostering mutual understanding and team-spirit among team members
- Convincing team members about the need to change when a situation requires adaptability

2.3.4 Resource Management

Effective resource management involves making the most efficient use of available resources to achieve the desired outcomes. To demonstrate effective resource management techniques, Following are some steps you can follow:

- Define your resources: Start by identifying the resources you have at your disposal, including financial, human, technological, and physical resources.
- Prioritize your resources: Determine which resources are most critical to achieving your goals and prioritize their allocation. You can follow the following prioritization framework to prioritize your task:

Fig2.3.1: Prioritization Framework

- Set clear goals: Establish clear, specific, measurable, achievable, relevant, and time-bound (SMART) goals for the use of your resources.

S	M	A	R	T
Specific	Measurable	Attainable	Relevant	Time-Bound
Add in as many details as possible. What will you do? Why and by when?	Make sure your goal is trackable. How will you measure your goal?	Take time to reflect. Can you realistically accomplish this goal within a certain timeline?	Think about what is important to you. Does this goal align with your values and larger objectives and goals?	Keep yourself accountable. By when do you want to accomplish this goal? How long will it take?

- Monitor and evaluate resource usage: Regularly track your resource usage and evaluate how well you're using them to achieve your goals.
- Implement cost-saving measures: Identify the ways to reduce costs while maintaining quality, such as implementing energy-saving measures, consolidating services, or negotiating better rates with suppliers.
- Invest in technology and automation: Consider using technology and automation to streamline processes, reduce costs, and improve efficiency.
- Train and empower employees: Provide training and support to employees to help them make the most effective use of available resources.
- Communicate effectively: Establish clear communication channels with stakeholders to ensure that everyone is aware of resource allocation decisions and understands the rationale behind them.
- Continuously improve: Continuously seek feedback and make improvements to your resource management approach to optimize your results.

UNIT 2.4: Productivity Tools: MS Word, MS Excel, PowerPoint and Presentations

Unit Objectives

At the end of this unit, participant will be able to:

- Discuss the use of MS Word and Excel for preparing a business proposal
- Discuss to prepare a workable presentation for marketing and business development.

2.4.1 MS Word

MS Word is a word processor used to make documents, letters, reports, etc. It has features like spell check, grammar check, text and font formatting, advanced page layout, and more.

Steps to create MS Word document are as follows:

- **Launch the MS Word Program:**
  - Click on the Windows Start Button to open Microsoft Word. Select "Word" from listed programs as shown in the following screen:

Fig. 2.4.1: Launching MS Word

- **Create a document:**
  - Once you open Word, select New on left hand side as shown in the image below:

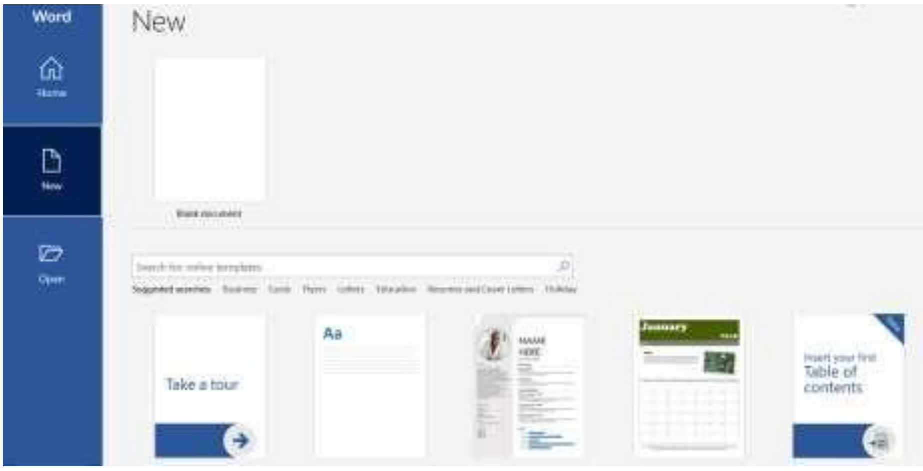


Fig. 2.4.2: MS Word

  - You can either select a **template** or a plain document to work by selecting "**Blank document**" to create your Word document. Once you select Blank document option, a new blank document will open up as shown in the image below:

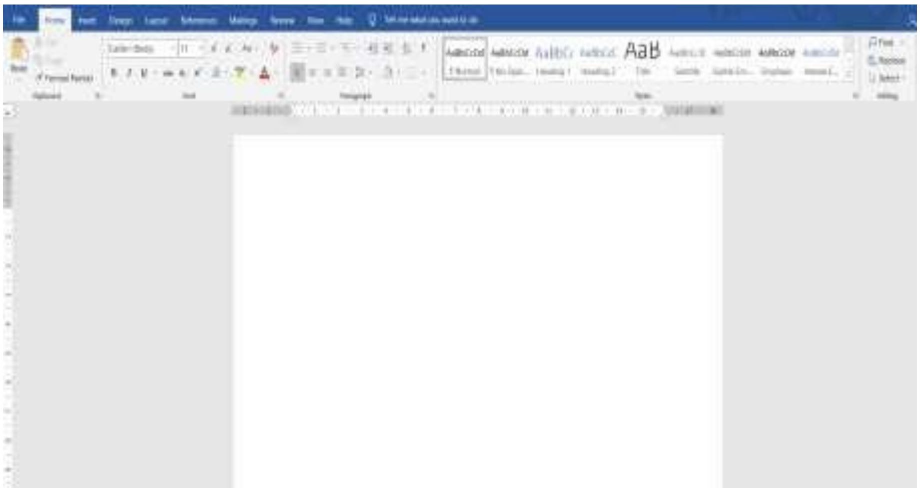


Fig. 2.4.3: MS Word - Blank Document

- **Add and format text, add pictures, graphs, shapes, etc.:**
  - You can add text by simply typing on the blank space and format the text using different options given in the toolbar in **Home** tab.

Fig. 2.4.4: MS Word - Home Tab

- You can add pictures, graphs, shapes, etc. to your document. This can be done by clicking on **Insert** tab and choose the required feature.

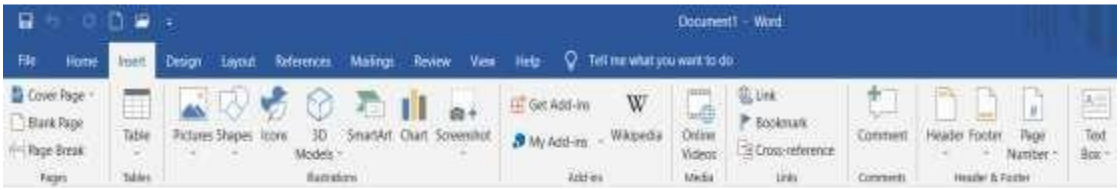


Fig. 2.4.5: MS Word - Insert Tab

- **Save Your Document:**
  - You can save your document using **Save** option on **File** tab.
  - Once you click save, you need to browse a folder where you want to save your document.
  - Type the name of your document in the **File** name box and then choose **Save**.

2.4.2 MS Excel

MS Excel is a spreadsheet application used to maintain numerical, tabular and statistical data. It contains a number of rows and columns. The rectangular box on each intersection of a column and a row is called a "cell". Each cell has one piece of information.

Steps to create MS Excel sheet are as follows:

- **Launch the MS Excel Program:**
  - Click on the Windows Start Button to open Microsoft Excel. Select "Excel" from listed programs as shown in the following image:

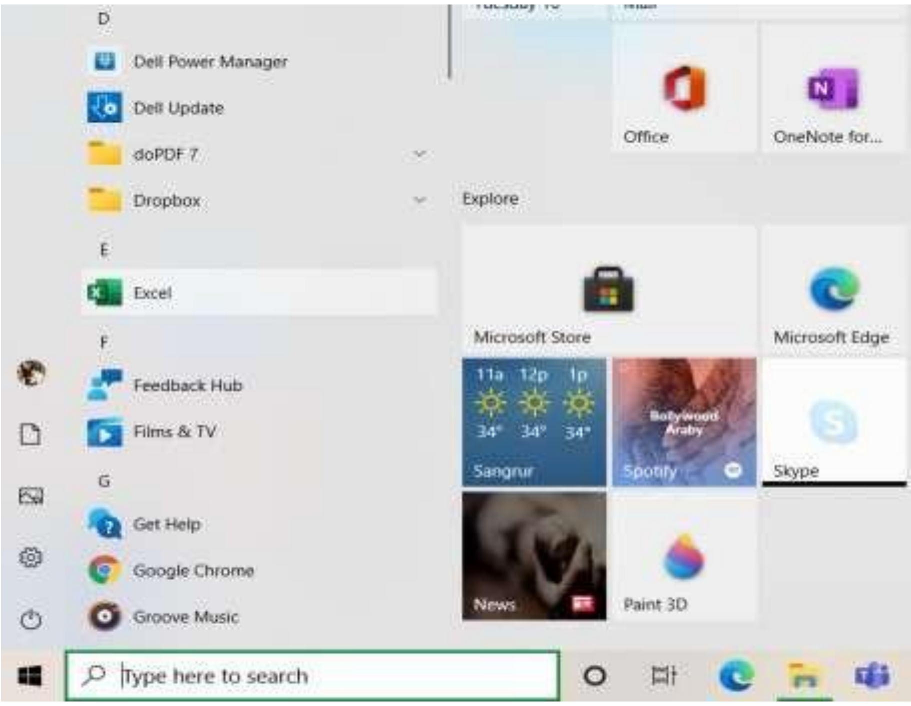


Fig. 2.4.6: Windows Start Button

- **Create a workbook:**
  - Once you open Excel, select New on left hand side as shown in the image below:

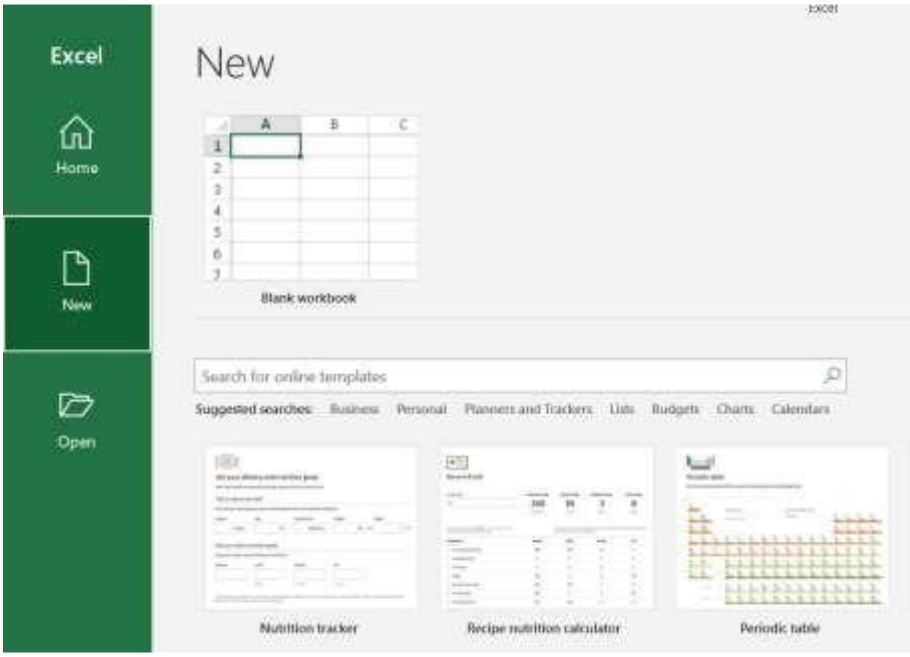


Fig. 2.4.7: MS Excel

- **Create a workbook:**
  - You can either select a **template** or a blank spreadsheet to work by selecting “**Blank workbook**” to create your Excel sheet. Once you select Blank workbook option, a new blank spreadsheet will open up as shown in the image below:

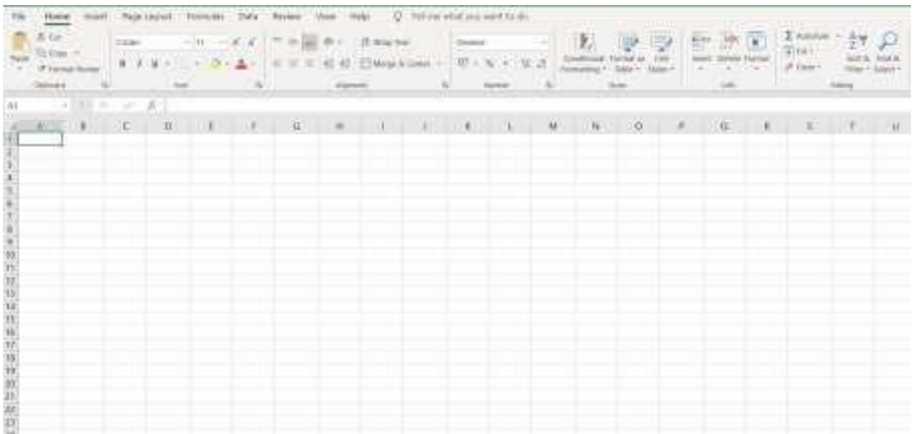


Fig. 2.4.8: MS Excel - Blank Workbook

- **Working with spreadsheet:**
  - Columns are represented with alphabets and rows are represented by numbers. Each cell is identified by unique address which consists of a column header followed by a row header. Data is added to these cells.

- For example: In case of first cell, the column header is A, and row header is 1, so the cell address is A1.

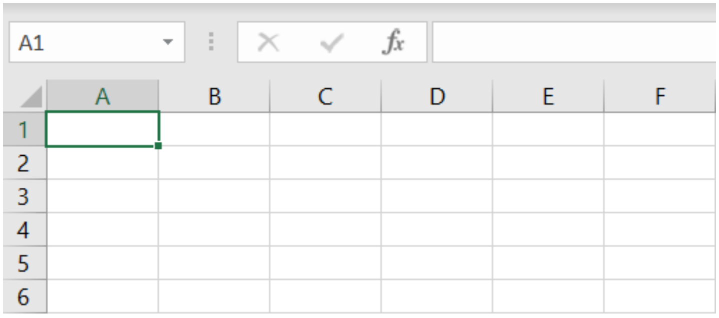


Fig. 2.4.9: MS Excel - Cell

- **Format excel spreadsheet:**
  - You can enter the required data in columns and rows. This data can be formatted using different options available in "**Home**" tab such as font, size, bold, italics, and many more.

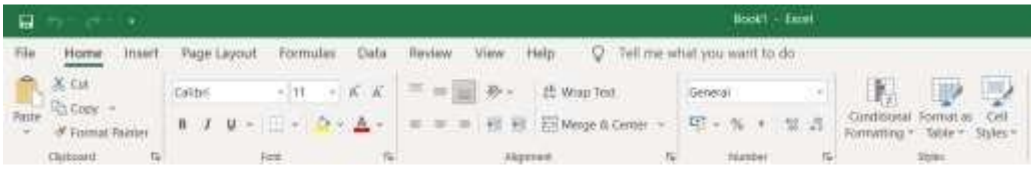


Fig. 2.4.10: MS Excel - Home Tab

  - You can also add pictures, pivot charts, etc. in excel sheet using different options available in "**Insert**" tab.

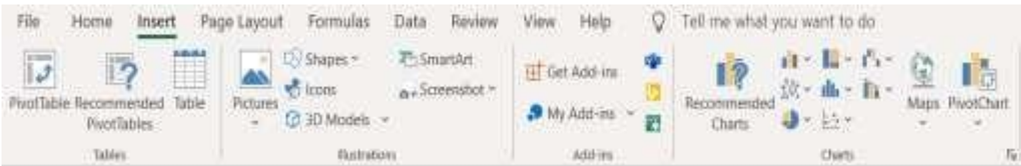


Fig. 2.4.11: MS Excel – Insert Tab

Activity

Use MS Word and MS Excel and its applications to prepare business proposal that should contain the following details:

- Area available
- Partner's Details
- Sanction of Loan (if any) by financial institution
- Machinery
- Recruitment of technical team
- Provision of facilities like water, electricity etc.
- License details
- Compliance letter

2.4.3 MS PowerPoint

Microsoft PowerPoint is a software program to create presentations for professional or personal use. These are software programs designed to increase productivity of computer users while working. These tools make work easier and more efficient to complete. This is sometimes done directly and sometimes indirectly.

Steps to create PowerPoint presentation are as follows:

- **Launch the PowerPoint Program:**
  - When you open PowerPoint, you can either select a blank presentation or a template as shown below:

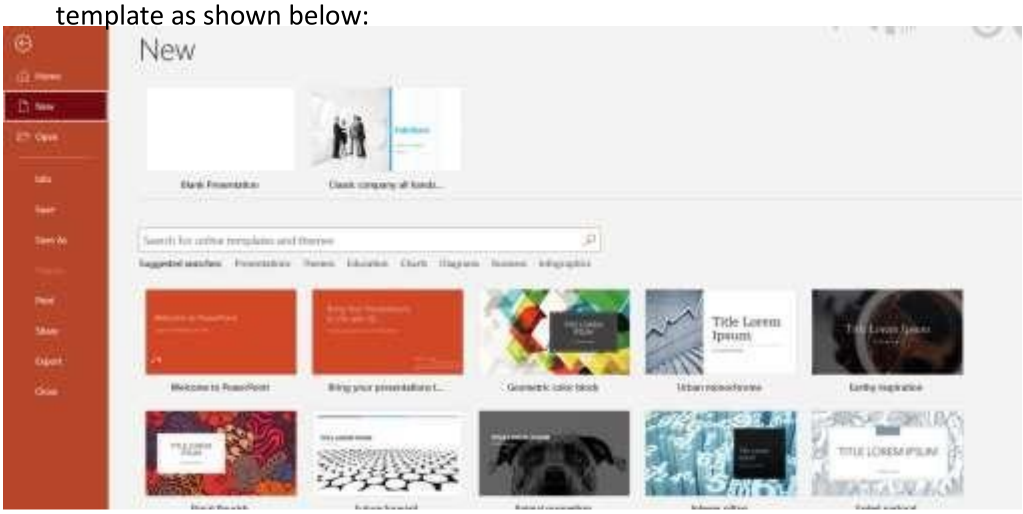


Fig. 2.4.12: MS PowerPoint

- **Choose Design:**
  - Once you select a blank presentation, click on ‘Design’ tab to select a design for your presentation.

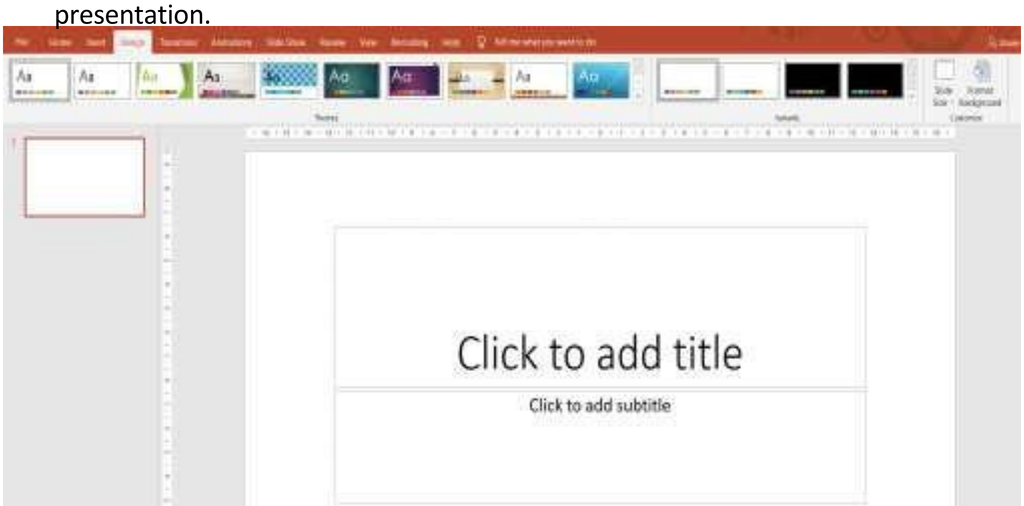


Fig. 2.2.13: MS PowerPoint - Design Tab

- **Save Presentation:**
  - You can save your presentation using **Save** option on **File** tab.
  - Once you click save, you need to browse a folder where you want to save your presentation.
  - Type the name of your presentation in the **File name** box and then choose **Save**.
  - Keep saving your work while making the presentation to avoid any loss of work.
- **Create Title Page:**
  - Now you can create title page by adding text in the given text boxes. You can also change the text font, colour, size using toolbar.
- **Add more slides:**
  - You can add more slides to your presentation by clicking "New Slide" option. There will be different choices and you can add the slide as per your requirement.
- **Add pictures, graphs, shapes, etc.:**
  - You can add pictures, graphs, shapes, etc. to your presentation. This can be done by clicking on Insert tab and choose the required feature.
  - You can add image to clicking on Pictures option and choosing the required image using the given options.

Activity

Prepare a presentation for marketing and business development of rooftop Solar PV project.

UNIT 2.5: Challenges and Risks for New Entrepreneurs

Unit Objectives

At the end of this unit, the participant will be able to:

- Identify the challenges and risks for new entrepreneurs and the possible mitigation measures.
- Discuss how to choose the right buyer in a given situation of market parameters.

2.5.1 Challenges and Risk for New Entrepreneurs

Starting a new business is an exciting endeavour, but it comes with several challenges and risks. Following are some of the common challenges and risks for new entrepreneurs, along with possible mitigation measures:

Challenges and Risk	Description	Mitigation Measures
Lack of experience and skills	Starting a new business requires a broad range of skills, including marketing, finance, and management. New entrepreneurs may lack experience or skills in these areas.	Conduct market research to gain insights into the industry and seek advice from experts in the field. Consider attending training or workshops to improve your skills.
Financial constraints	Starting a new business requires significant financial investment. Entrepreneurs may face financial constraints in the initial stages of the business.	Create a detailed business plan and secure funding from investors or lenders. Consider starting with a small-scale business or starting part-time while keeping another source of income.
Competition	Entering a market with established competitors can be challenging, particularly for small businesses.	Conduct thorough market research to identify gaps in the market that can be exploited. Focus on building a unique brand that stands out from

Challenges and Risk	Description	Mitigation Measures
		competitors and provides a unique value proposition.
Legal and regulatory challenges	Entrepreneurs must comply with legal and regulatory requirements, including permits, licenses, and taxes.	Consult with a lawyer or accountant to ensure compliance with legal and regulatory requirements. Seek advice from the relevant regulatory bodies or trade associations.
Business model failure	A poorly designed business model may lead to failure or stagnation of the business.	Develop a comprehensive business plan that outlines the business model and includes detailed financial projections. Continuously monitor and adjust the business model based on market trends and customer feedback.
Failure to generate customers	Attracting and retaining customers is critical for any business.	Conduct market research to understand customer needs and preferences. Develop a strong brand that resonates with customers and provides a unique value proposition. Use social media and other marketing channels to reach potential customers.

Table 2.5.1: Common Challenges and Risks for New Entrepreneurs, along With Possible Mitigation Measures

2.5.2 Choose the Right Buyer

Choosing the right buyer in India can be a challenging task, especially when the market parameters are uncertain. Following are some steps to consider when selecting the right buyer in a given situation of market parameters:

- **Identify your target market**: Determine the market segment that is most likely to purchase your product or service. Conduct market research to gain insights into the needs, preferences, and behaviour of your target market.
- **Evaluate buyer's reputation and financial stability**: Assess the potential buyer's reputation and financial stability. Check for past business transactions, credit ratings, and reviews from other sellers.
- **Analyse buyer's purchasing power**: Evaluate the buyer's purchasing power, including their budget, payment terms, and ability to pay on time. Look for buyers who have a stable financial position and a record of prompt payment.
- **Consider buyer's location and distribution network**: Assess the buyer's location and distribution network to determine if they have the ability to reach your target market. Evaluate the buyer's shipping capabilities, delivery times, and customer service.
- **Negotiate terms and conditions**: Negotiate the terms and conditions of the transaction with the buyer, including the price, payment terms, delivery terms, and warranties. Ensure that the terms are fair and reasonable for both parties.
- **Monitor market trends**: Continuously monitor market trends to stay updated with changes in demand, competition, and pricing. Adjust your strategy accordingly to remain competitive in the market.

UNIT 2.6: Taxes and Compliances

Unit Objectives

At the end of this unit, the participant will be able to:

- Explain various taxation and related compliances to set up and operate a solar business

2.6.1 Tax and Related Compliances

Taxation and related compliances are essential for setting up and operating a solar business. Compliance with tax laws and regulations can help a business in legal compliance, tax benefits, financial planning, accounting and record-keeping, and reputation building.

In India, there are various taxation and related compliances that a solar business must comply with. Following are some of the essential ones:

- **Goods and Services Tax (GST)**: A solar business must register under GST if its annual turnover is above a certain threshold. GST is a value-added tax that is levied on the sale of goods and services. The tax rate for solar panels and related equipment is 5%.
- **Income Tax**: A solar business is required to file income tax returns every year. The tax rate for a company in India is 25% for companies with a turnover up to INR 400 crores and 30% for companies with a turnover above INR 400 crores.
- **Customs Duty**: If a solar business imports equipment from other countries, it must pay customs duty. The customs duty rate varies depending on the type of equipment and the country of origin.
- **State Level Taxes**: In addition to GST, a solar business may be required to pay state-level taxes such as Value Added Tax (VAT) and Central Sales Tax (CST). The tax rate varies from state to state.
- **Renewable Purchase Obligations (RPO)**: Electricity distribution companies in India are required to purchase a certain percentage of their electricity from renewable sources. A solar business can sell its electricity to these distribution companies and receive Renewable Energy Certificates (RECs) for the electricity produced.
- **Compliance with Environmental and Safety Regulations**: A solar business must comply with environmental and safety regulations such as the Water (Prevention and Control of Pollution) Act, 1974 and the Air (Prevention and Control of Pollution) Act, 1981.
- **Labour Laws**: A solar business must comply with labour laws such as the Employees' Provident Funds and Miscellaneous Provisions Act, 1952 and the Employees' State Insurance Act, 1948. These laws govern the welfare of employees and require employers to make contributions to various funds.

UNIT 2.7: Aspects of Successful Solar Business

Unit Objectives

At the end of this unit, the participant will be able to:

- Discuss the key aspects to successfully run a solar business.
- Explain how evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape.

2.7.1 Key Aspects to Successfully Run a Solar Business

Running a solar business in India can be a lucrative opportunity given the country's focus on renewable energy. Here are some key aspects to consider for successfully running a solar business in India:

- **Legal and regulatory compliance**: The solar industry in India is regulated by several government bodies. To successfully run a solar business, it is essential to comply with legal and regulatory requirements, including obtaining necessary permits and licenses.
- **Market research and analysis**: Conducting market research and analysis can help identify gaps in the market and understand consumer demand. This information can be used to develop a unique value proposition and target the right customers.
- **Technological expertise**: Solar businesses require specialized technological expertise, including knowledge of solar photovoltaic technology and installation, operation, and maintenance of solar systems. Hiring experienced personnel or partnering with experts can help ensure the quality of the product or service.
- **Financial management**: Managing finances is essential to ensure the sustainability of the business. Develop a comprehensive business plan, secure funding from investors or lenders, and maintain accurate financial records.
- **Marketing and sales**: Effective marketing and sales strategies can help create brand awareness and generate leads. Develop a strong brand, build relationships with customers and suppliers, and use digital marketing tools to reach potential customers.
- **Partnership and collaboration**: Collaboration with other companies and organizations can help build networks, share resources, and expand the reach of the business. Consider partnering with other renewable energy companies, research institutions, and government bodies.
- **Quality assurance**: Quality assurance is crucial to ensure the reliability and safety of solar systems. Ensure that the products or services meet industry standards and regulations, and conduct regular quality checks and maintenance.

Impact of Evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio in improving overall solar landscape

The evolving start-ups in the solar consultancy, O&M (operation and maintenance), installation, and EPC (engineering, procurement, and construction) space are playing a significant role in improving the overall solar landscape in India. These start-ups are bringing innovation and new ideas to the industry, which is leading to better services, products, and cost-effective solutions.

One way in which start-ups are contributing to the solar landscape in India is by diversifying their solar portfolio. This means that they are not only focusing on solar energy generation but also on other areas such as storage, electric vehicle charging infrastructure, and energy efficiency. By diversifying their portfolio, these start-ups are creating a holistic approach to the energy sector, which is necessary for the sustainable growth of the industry.

Furthermore, start-ups in the solar consultancy, O&M, installation, and EPC space are bringing in new technologies and processes to the industry. They are using advanced analytics, machine learning, and artificial intelligence to improve the efficiency of solar plants and reduce downtime. These new technologies are also helping in predictive maintenance, which is reducing the need for reactive maintenance and, in turn, lowering the overall costs.

Another way start-ups are improving the solar landscape in India is by providing customized solutions for clients. Start-ups are better able to understand the unique requirements of individual clients and offer tailored solutions to meet their needs. This personalized approach is helping to increase customer satisfaction, which is essential for the growth of the industry.

Evolution of Solar PV based Decentralized Business models

- 1st generation-end User-owned System Supply
- 2nd Generation 3rd Party-owned & operated
- 3rd Generation -Fully Integrated with USBM

Key characteristics

1st generation

- Pioneering business model
- Focus on direct ownership of systems by building/rooftop owners
- End user is either the user or the utility through feed in tariff
- Institutions limited to financing, designing, deploying, & maintaining systems

2nd Generation

- Business model addresses through third parties market barriers which were not there with individual owners-scale, technical expertise, optimization
- Third parties install systems on residential & commercial rooftops, earn a lease payment/tariff from owner for use of energy generated

3rd Generation

- Hybridized business models emerging with grater variation in ownership, control and operation
- Movement by utilities to capture more value through involvement like utility's own FiT, bidding processes
- Power fed into the grid where it becomes a part of the utilities supply

Utility involvement and incentive strategy

1st generation

- Utility role largely limited and passive -providing net-metering / feed-in-tariff payments and interconnection
- Model driven by feed-in tariffs or net metering combined with subsidies/tax rebates, capital subsidy
- E.g Germany, Japan, United states, WBREDA pilots in kolkata

2nd Generation

- Utility role still limited, in some ways the same as that of the earlier generation
- Incentives range from variations in tax rebates to lease payments, net metering, feed in tariffs, ToU metering, capital subsidy etc.,
- Focus on reducing host's energy bill while leveraging scale to earn better returns
- E.g. United States.

3rd Generation

- Utility driving these models to actively become a part of the value chain and earn returns
- Focus on utilities either developing projects as third parties & feed into the grid or earning returns through solar REC's e.g Solar shares, customer sited PV systems, services Agreements.

Owner focused development

- Rooftop owner owned
- Active involvement of up-stream stakeholders

Entry of market intermediaries

- Ownership shifts to external stakeholders
- Increased focus on scale-up and efficiency

Entry of IPP's and utilities further scale up

- Ownership shifts to utilities/communities /IPP's
- Ensuring returns from solar for utilities

UNIT 2.8: Respecting Gender and Disability Concerns

Unit Objectives

At the end of this unit, the participant will be able to:

- Explain how to demonstrate responsible and disciplined behaviour and promote a safe and interactive environment at work while respecting gender and disability concerns.

2.8.1 Respecting Diversity and Gender

One can be working in national and international workspaces with employees who come from a wide variety of backgrounds. Diversity typically includes differences in race, gender, sexual orientation, gender identity or expression, political and religious affiliation, socioeconomic background, cultural background, geographic location, physical disabilities and abilities, relationship status, veteran status and age.

When one is closely involved with seniors and subordinates, they can play an important role in delivering workspace inclusion by being vocal about the benefits of working with people from diverse backgrounds and making others believe that workspace inclusion can drive stronger better performance and results.

The following diagram shows the key elements of a diverse and inclusive workspace that must be considered to mitigate risks that can arise out of working with co-workers from diverse backgrounds.

Fig. 2.8.1: Four Key elements of Diversity and Inclusion

General points to be kept in mind:

- Do use a common language understood by all while speaking to a group.
- Do speak to everyone with respect in the workspace, interact with all co-workers without discrimination.
- Do not bully or make anyone feel inferior or left out.
- Do have open discussions and involve everyone.
- If possible, conduct informal get-togethers, celebrate festivals. It will help in engaging every employee, improving the organization culture and making the employees feel appreciated.

2.8.2 Working in a Diverse Environment

Every employee should feel valued, they should believe their individuality is respected by others and at the same time should feel connected with the team. Every employee should feel safe to speak up and be open without feeling judged or embarrassed. When employees are confident with themselves, they feel empowered and give their best in the job which helps them grow along.

Maintaining a good environment and culture is beneficial for both the organisation and the employees. Some ways to improve the environment and culture of the organisation are:

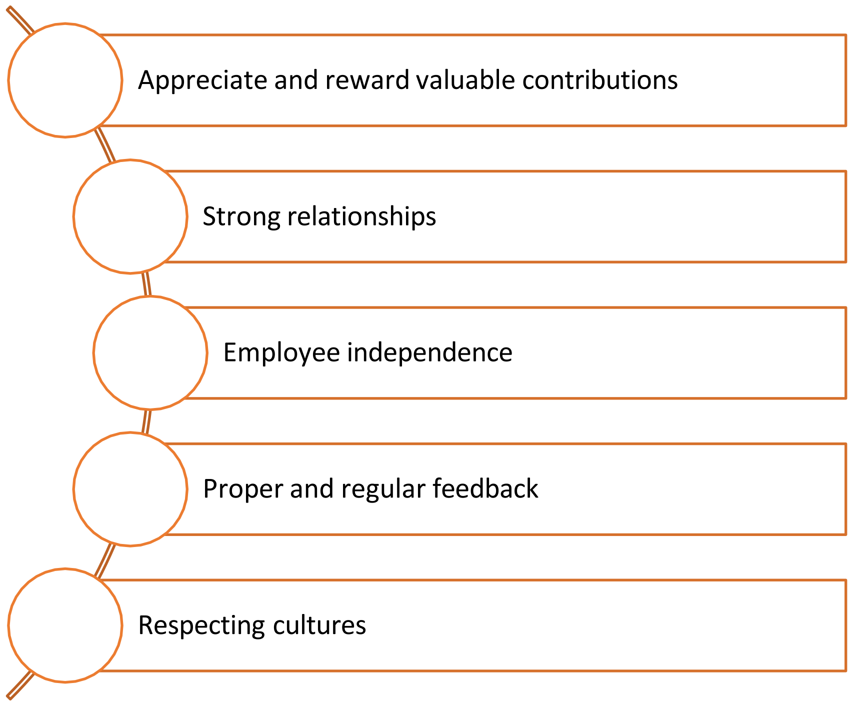


Fig. 2.8.2: Ways to Improve the Environment and Culture of Organisation

2.8.3 Gender Sensitivity and PwD (Person with Disability) Sensitivity

World Health Organisation defines Gender as:

***Gender refers to the roles, behaviours, activities, attributes and opportunities that any society considers appropriate for girls and boys, and women and men*.**

Gender sensitivity refers to providing equal rights, benefits, and opportunities to both women and men. It refers to sensitive and complicated issues like unequal pay, sexual harassment, gender bias differences in access to resources, preferring men on higher roles, racism etc. Gender sensitivity is one of the major issues been observed in our society. It is not about women fighting against men. In fact, it provides education about gender sensitive issues faced by both the sexes and provides provisions for its solution. It is about complimenting each other and building a strong society.

As mentioned by UNESCO:

'Gender sensitivity is not about pitting women against men. On the contrary, education that is gender sensitive benefits members of both sexes. It helps them determine which assumptions in matters of gender are valid and which are stereotyped generalizations. Gender awareness requires not only intellectual effort but empathy and open-mindedness. It opens up the widest possible range of life options both for women and men.'

In addition to the gender sensitivity, there is one major issue that needs to be addressed and that is PwD (Person with Disability) Sensitivity.

All individuals' women, men, transgender despite of their physical abilities/disabilities (Pwd) forms an integral part of a system and should be treated and benefited equally. It is the responsibility of the society that regardless of the gender status and physical ability status, all individuals should be benefits equally.

A society should be capable enough to identify and address these gender discrimination/inequality issues and PwD issues.

As per UNESCO, all individual which includes women, men, transgender and PwD have following rights:

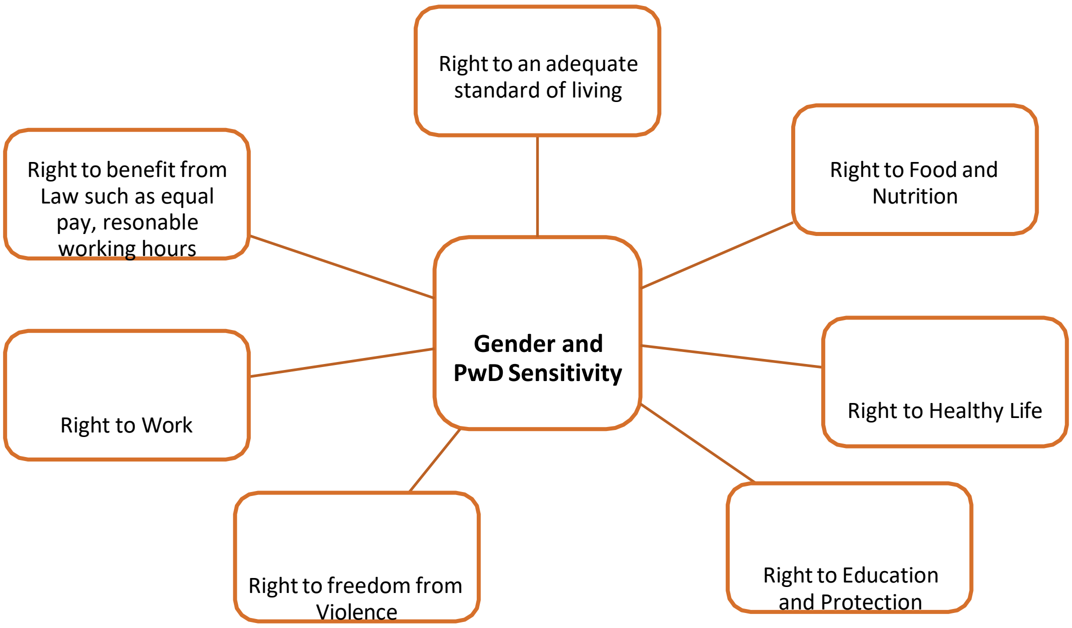


Fig. 2.8.3: Gender and PwD Sensitivity

The modification in the thought process and providing awareness at the childhood level (school levels) can help addressing such issues. Keeping this in mind Government of India has taken the step towards it by improvising the NCERT and CBSE school curriculum by replacing gender bias statements in the textbooks.

Not only updating but the National Council of Educational Research and Training National Council of Educational Research and Training (NCERT) have also developed content which has been incorporated in the textbooks across the subjects to promote gender sensitization in the school curriculum.

As responsible organisation, many have taken significant steps to address the gender sensitivity issues. One of its kinds was taken by The Tata Tea (India) in collaboration with NGOs in 2017. The name of the awareness program was “Jaago Re 2.0”. The aim of the program was to deal with gender sensitization and women empowerment.

To overcome gender disparity, various schemes are launched by Government of India and are implemented by Ministry of Women and Child Development are as follows:

Beti Bacho Beti Padhao (BBBP)	• to improve the Child Sex Ratio and enabling education for the girl children
Swadhar Greh Scheme	• to provide relief and rehabilitation to destitute women and women in distress
Ujjawala	• a Comprehensive Scheme for prevention of trafficking and rescue, rehabilitation and re-integration of victims of trafficking and commercial sexual exploitation
Rashtriya Mahila Kosh (RMK)	• to provide loan to poor women through Intermediary Microfinancing Organisations (IMOs), Non-Governmental Organisations (NGOs) to promote their socio-economic development
Working Women Hostels (WWH)	• For ensuringsafe accomodation for women working away from their home town or place of residence
One Stop Centre	• to provide integrated support and assistance to women affected by violence

Fig. 2.8.3: Schemes by Government of India

Government has also launched many schemes to address PwD sensitivity issues. You may visit the following link to explore these schemes:

http://www.swavlambancard.gov.in/schemes/search

Code of Conduct Principles

Code of Conduct principles helps organisations to form a framework and enhance gender sensitization and PwD sensitivity at workplace. Code of Conduct to be followed by are:

1. **Discrimination**:
   - On the basis of marital status, both women and men employees should protect against discrimination. It has been observed in many known organisation that married/newly married women/single parent are not preferred for hiring. But with Government CSR initiate such kind of discriminations has been addressed.
   - Pregnancy tests should not be used as a condition of hiring or of continued employment.
   - All women employees should be protected against threats of losing their jobs or not considering for promotions due to their pregnancy or their marital status.
   - Both women and men should be given equal opportunities for professional development and trainings at workplace.
   - Both women and men with family responsibilities, specifically in case of the only earning member of the family should be protected against discrimination with respect to dismissal
   - Transgender if qualified for the required position should also be given preferences.
   - The employee should not be discriminated on the bases of his/her physical disabilities.

Non-Discrimination Compliance

Under federal and state laws, it is in contradiction of the law for proprietors to differentiate staffs and job applicants and/or harassment to occur with their organizations. It is also against the law to treat people unethically or bother them because of the age, ***disability***, ***homosexuality***, ***marital*** *or* ***domestic status, race***, ***sex*** *or* ***transgender status*** of any relative, friend or colleague of a job applicant or employee Employers, managers and supervisors must treat all their job applicants and employees on the basis of their individual merit and not because of irrelevant personal characteristics. They must also do their best to make sure that their employees are not harassing any other job applicant or employee.

2. **Wages**:
    - All employees should be paid according to the comparable wages, hours, and benefits as per the Labour Act.
    - Each employee has right to benefit from insurance policies that covers health/or any injury at workplace.
    - As per the Maternity Act, all employees are entitled to maternity protection and benefits. Such as 6 months maternity leave for women and 10-15 days for parental leave for men.

3. **Forced Labour:**
    - All the migrant workers (both women and men) have right to entitlements same as local employees. Any security deposit fee from migrant workers is prohibited.
    - The employer cannot keep the original certification or any original documents such as Aadhar Card, Passport etc. of the employees.
    - Any commissions and other fees in connection with employment of migrant workers must be covered by the employer. The employer must not require the employee to submit his or her identification documents. Deposits are not allowed.
    - If a task includes uplifting of heavy loads, the physical conditions should be taken into consideration for both women and men.
    - The employee with physical disability should not be forced to work beyond his/her physical capability

4. **Working Hours**:
    - An employer cannot force an employee for overtime. Any kind of such imposed threating is an illegal offence. This is applicable to both women and men.
    - Both women and men despite of their physical disability working on same position should be paid equally.

5. **Harassment and Abuse**
    - It is mandatory for every organisation to form a sexual harassment committee at workplace where the employee can report for such incidence.
    - The harassment and abuse includes verbal, physical, psychological or sexual abuse. It should be addressed seriously then and there.

- Any kind of lewd remarks also considered as an abuse.
- All employees either permanent or on contract, either women or men should be protected against harassment and abuses at work place.
- Harassment and Abuses are not acceptable even in official transportation and dormitories.
- Any kind of comment on the physical disability of the individual is also considered as harassment and abuse.

6. Health and Safety

- Separate toilets should be arranged for women and men at workplace. Ensuring that these toilets are at reasonable distance and are safe, clean and hygienic to use.
- All employees should have access to basic healthcare services as per laws.
- Pregnant women should not be placed at hazardous places. For example, where chemicals are used.
- All employees should be trained for healthy and safety measured necessary for workplace.
- A person with disability (PwD) is entitled to all special facilities at workplace. An employer is responsible to provide the same.

Exercise

1. List the documents required to run business transactions.

2. Prepare list of schemes provided by Government of India.

3. What all needs to be ensured before starting a business?

4. Suggest Ways to Improve the Environment and Culture of Organisation.

5. List the documents required for business license.

Exercise

1. Which of the following is promotional or PR material that can be shared with the public?
   a. Brochures and Flyers
   b. E-newsletters
   c. Policy Documents
   d. All of the Above
2. Working in a diverse environment ensures which of the following?
   a. Strong Relationships
   b. Respecting Individuality
   c. Reward valuable contribution
   d. All of the above
3. Which of the following is not the key element of a diverse and inclusive workplace?
   a. Belongingness
   b. Safety and Openness
   c. Fairness
   d. Groupism

Skill India
कौशल भारत-कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

SCGJ | SKILL COUNCIL FOR GREEN JOBS

3. Business Model in Solar Photovoltaic

UNIT 3.1: Indian Solar Industry Overview
UNIT 3.2: Solar Business Models
UNIT 3.3: Financial Performance Indicators
UNIT 3.4: Policy and Regulatory Framework

SGJ/N4125

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain the Indian Solar PV market and key stakeholders.
2. Differentiate between CAPEX, OPEX/RESCO, Lease, BOOT and Hybrid business models.
3. Select suitable business models for different customer segments.
4. Understand PPAs, tariffs, ownership structures and financing mechanisms.
5. Calculate payback period and IRR for solar projects.
6. Understand the role of MNRE, State Nodal Agencies and DISCOM regulations.

UNIT 3.1: Indian Solar Industry Overview

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain about the Solar industry and governments.

Solar Industry Overview

India is one of the world's fastest-growing solar markets. Government initiatives, declining solar module prices, and increasing electricity tariffs have accelerated solar adoption.

Key Stakeholders

Government (MNRE, State Agencies)
↓
Developers
↓
EPC Contractors
↓
Consumers
↑
DISCOMs & Regulators
↑
Banks & Financiers

Fig. 3.1.1 Key Satkeholders

Stakeholder Roles

Stakeholder	Role
MNRE	Policy, subsidies, schemes
Developers	Invest and develop projects
EPC Companies	Design, procure and construct systems
DISCOMs	Grid connection and net metering
Banks/NBFCs	Financing and loans
Consumers	Use solar energy

Market Drivers

- Rising electricity tariffs
- Government subsidies
- Net-metering policies
- Sustainability goals
- Corporate ESG commitments
- Energy security

UNIT 3.2: Solar Business Models

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain different types of business models
2. Explain the comparison of business Models

Solar Business Models

A solar business model defines how a company creates, delivers, and captures value in the solar energy sector. It typically involves activities such as solar project development, system design, procurement of equipment, installation, operation and maintenance, and after-sales support. Solar companies may generate revenue through direct sales of solar power systems, Engineering, Procurement and Construction (EPC) contracts, rooftop leasing, Power Purchase Agreements (PPAs), or Operations and Maintenance (O&M) services. The choice of business model depends on the target customer segment—residential, commercial, industrial, or utility-scale—and factors such as investment capacity, regulatory framework, and market demand. A well-structured solar business model ensures sustainable growth while helping customers reduce energy costs and transition to clean, renewable energy.

Comparison of Solar Business Models

Model	Ownership	Investment	O&M Responsibility
CAPEX	Customer	Customer	Customer
OPEX/RESCO	Developer	Developer	Developer
Lease	Developer	Developer	Shared
BOOT	Developer initially	Developer	Developer
Hybrid	Shared	Shared	Shared

A. CAPEX Model

In the CAPEX model, the customer purchases the solar system outright.

Features

- Customer owns the system.
- Customer bears initial investment.
- Energy generated is free after payback.
- Suitable for long-term savings.

Advantages

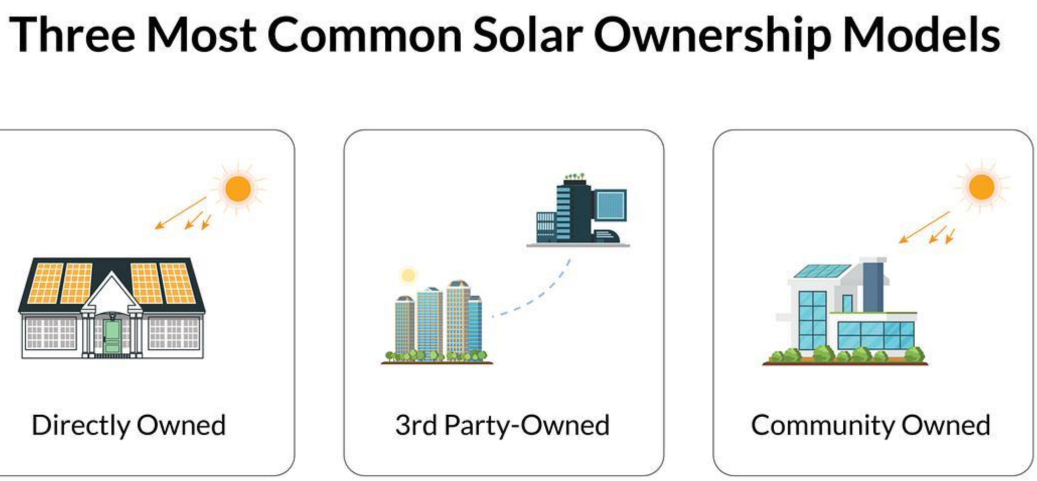
✓ Higher lifetime savings

✓ Eligible for subsidies (where applicable)

✓ Asset ownership

Typical Users

- Residential customers
- Schools
- Government buildings
- Industries with available capital



B. OPEX / RESCO Model

RESCO = Renewable Energy Service Company

The developer installs and owns the solar plant while the consumer purchases electricity through a Power Purchase Agreement (PPA).

Key Components

- No upfront investment by customer
- Developer owns asset
- Customer pays per kWh generated
- Long-term PPA (10–25 years)

Example PPA Structure

Item	Value
Contract Duration	15 Years
Tariff	₹4.50/kWh
Escalation	2% per year
Billing Frequency	Monthly

Stakeholder Roles

- Developer → Investment & O&M
- Customer → Purchase power
- DISCOM → Grid approvals

C. Lease Model

- Customer leases solar system.
- Fixed monthly rental.
- Ownership remains with lessor.
- Lower upfront cost.

D. BOOT Model

(Build-Own-Operate-Transfer)

1. Developer builds project.
2. Operates for agreed period.
3. Ownership transferred to customer after contract completion.

E. Hybrid Model

Combination of CAPEX and OPEX.

Example:

- Customer invests 40%
- Developer invests 60%

Used in large commercial projects.

3. Community Solar and Shared Solar

Community solar allows multiple consumers to benefit from a common solar project.

Benefits

- No rooftop required
- Lower investment
- Suitable for apartments and housing societies
- Shared energy benefits

Community Solar Plant
↓
Multiple Consumers
↓
Energy Credit Allocation

4. Financial Performance Indicators

A. Payback Period

Time required to recover initial investment.

$$Payback\ Period = \frac{Initial\ Investment}{Annual\ Savings}$$

Example:

- Investment = ₹5,00,000
- Annual Savings = ₹1,00,000

Payback = 5 Years

B. Internal Rate of Return (IRR)

IRR indicates project profitability.

Higher IRR = Better investment.

Typical rooftop solar IRR:

- Residential: 12–18%
- Commercial: 15–25%

C. Net Savings

Net Savings =
Electricity Cost Saved
– Maintenance Cost

Practical Activity

Use Excel:

- Enter project cost.
- Enter annual savings.
- Use IRR() function.
- Calculate simple payback period.

5. Policy and Regulatory Framework

Role of MNRE

MNRE Official Website

- National solar policies
- PM Surya Ghar Scheme
- Standards and guidelines
- Capacity targets

State Nodal Agencies

Implement state-level solar programs.

Examples:

- Delhi Solar Energy Policy
- Rajasthan Renewable Energy Corporation

DISCOM Regulations

- Net Metering
- Grid Connectivity
- Billing and Energy Accounting
- Safety Compliance

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

4. Site Feasibility Study for Rooftop Solar power plant

SCGJ | SKILL COUNCIL FOR GREEN JOBS

UNIT 4.1: Site Assessment and System Sizing
UNIT 4.2: Solar Resource Assessment, Energy Yield and Feasibility Report

SGJ/N0111SG

Key Learning Outcomes

At the end of this module, the participant will be able to:

- Identify suitable locations for rooftop solar installations.
- Assess site conditions and structural requirements.
- Perform shading analysis and prepare site maps.
- Estimate plant capacity and battery backup requirements.
- Obtain solar resource data and estimate energy generation.
- Identify project risks and prepare a feasibility study report

UNIT 4.1: Site Assessment and System Sizing

Unit Objectives

At the end of this unit, participant will be able to:

3. Explain about Site Assessment and System Sizing

Identifying the Optimum Rooftop Location

A Site Feasibility Study is the first and most critical step in designing a rooftop solar PV system. It helps determine whether a site is suitable for solar installation, identifies technical constraints, estimates system size and energy generation, and assesses project risks before investment.

An ideal rooftop should have:

✓ Maximum exposure to sunlight

✓ Minimum shading throughout the year

✓ Adequate roof area

✓ Good structural strength

✓ Easy accessibility for maintenance

Factors Affecting Site Selection

Parameter	Requirement
Roof Orientation	South-facing preferred (India)
Roof Tilt	Near local latitude
Shadow-Free Area	Maximum possible
Structural Strength	Suitable for additional load
Accessibility	Safe access for maintenance

1.2 Site-Level Pre-Requisites

Before installation, verify:

Technical Requirements

- Available roof area
- Roof condition and age
- Electrical panel location
- Earthing availability
- Grid connectivity

Structural Requirements

- Roof load-bearing capacity
- Water-proofing condition
- Wind resistance considerations

Safety Requirements

- Working at height provisions
- Safe access pathways
- Fire safety clearance

1.3 Selection of Mounting Structure

The mounting type depends on roof type and customer requirements.

Roof Type	Mounting Type
RCC Flat Roof	Fixed Tilt Structure
Metal Sheet Roof	Rail-Based Mounting
Industrial Shed	Elevated Structure
Car Parking Area	Solar Carport
Ground Area	Ground Mounted Structure

Mounting Selection Considerations

- Roof type
- Wind load
- Available area
- Maintenance access
- Future expansion requirement

1.4 Shading Assessment

Shading significantly affects energy generation.

Common Shading Obstacles

- Water tanks
- Adjacent buildings
- Trees
- Telecom towers
- Parapet walls
- HVAC equipment

Shadow Inspection Methods

1. Visual Site Survey
2. Sun Path Analysis
3. Shadow Mapping
4. Solar Pathfinder
5. Drone Survey

Basic Site Mapping

A site map should indicate:

North Direction

↑

Building Layout

Solar Installation Area

Water Tanks

Trees

Electrical Room

Access Path

Activity

Prepare a simple rooftop sketch showing:

- Roof dimensions
- North direction
- Obstacles
- Proposed module placement

1.5 Load Assessment and Load Profile

The solar plant size should be based on energy consumption.

Information Required

- Monthly electricity bills
- Connected load
- Operating hours
- Critical loads

Sample Load Profile

Load	**Quantity**	**Rating (W)**	**Hours/Day**	**Energy (Wh)**
Lights	20	20	8	3,200
Fans	15	75	10	11,250
Computers	10	120	8	9,600
AC Units	5	1500	6	45,000
Total				69,050

Estimating Solar Plant Capacity

Plant Capacity (kW)

=

Daily Energy Consumption (kWh)

÷

Peak Sun Hours

Example:

Daily consumption = 40 kWh

Peak Sun Hours = 5

Required Plant Size = 8 kW

UNIT 4.2: Solar Resource Assessment, Energy Yield and Feasibility Report

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain resource assessment, Energy Yield and Feasibility Report

4.2.1 Battery Backup Assessment

Battery sizing depends on:

- Grid reliability
- Critical load requirement
- Backup duration expected
- Budget

Battery Sizing Formula

$$Battery\ Capacity(Ah) = \frac{Load(W) \times Backup\ Hours}{Battery\ Voltage \times DOD}$$

Example

Critical Load = 2 kW

Backup = 4 Hours

Battery Voltage = 48 V

DOD = 80%

Battery Capacity ≈ 208 Ah

2.2 Solar Resource Assessment

Solar resource data is essential for accurate energy prediction.

Important Parameters

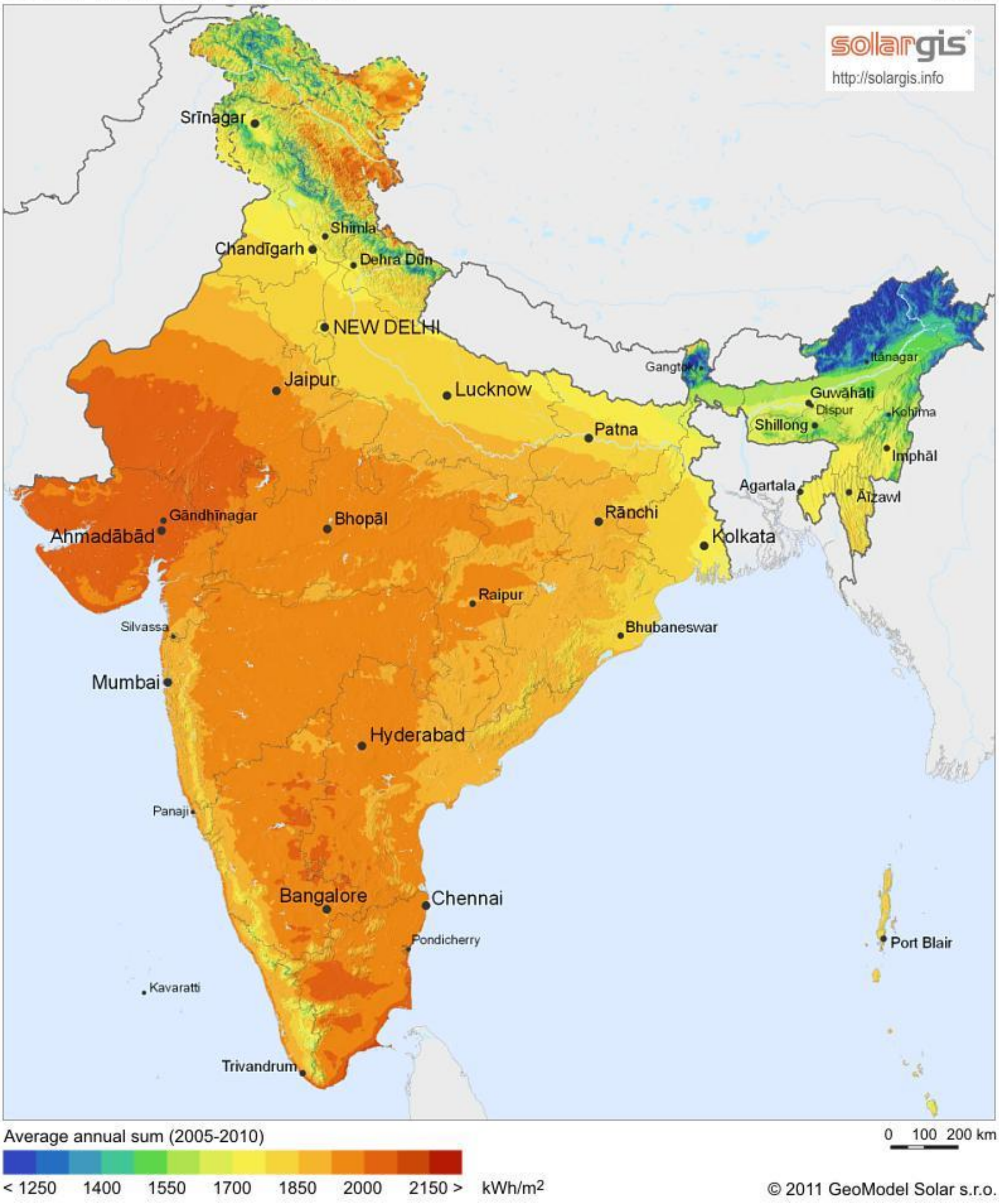
Parameter	**Description**
GHI	Global Horizontal Irradiance
DNI	Direct Normal Irradiance
DHI	Diffuse Horizontal Irradiance
Temperature	Affects module performance
Wind Speed	Influences cooling and structure design

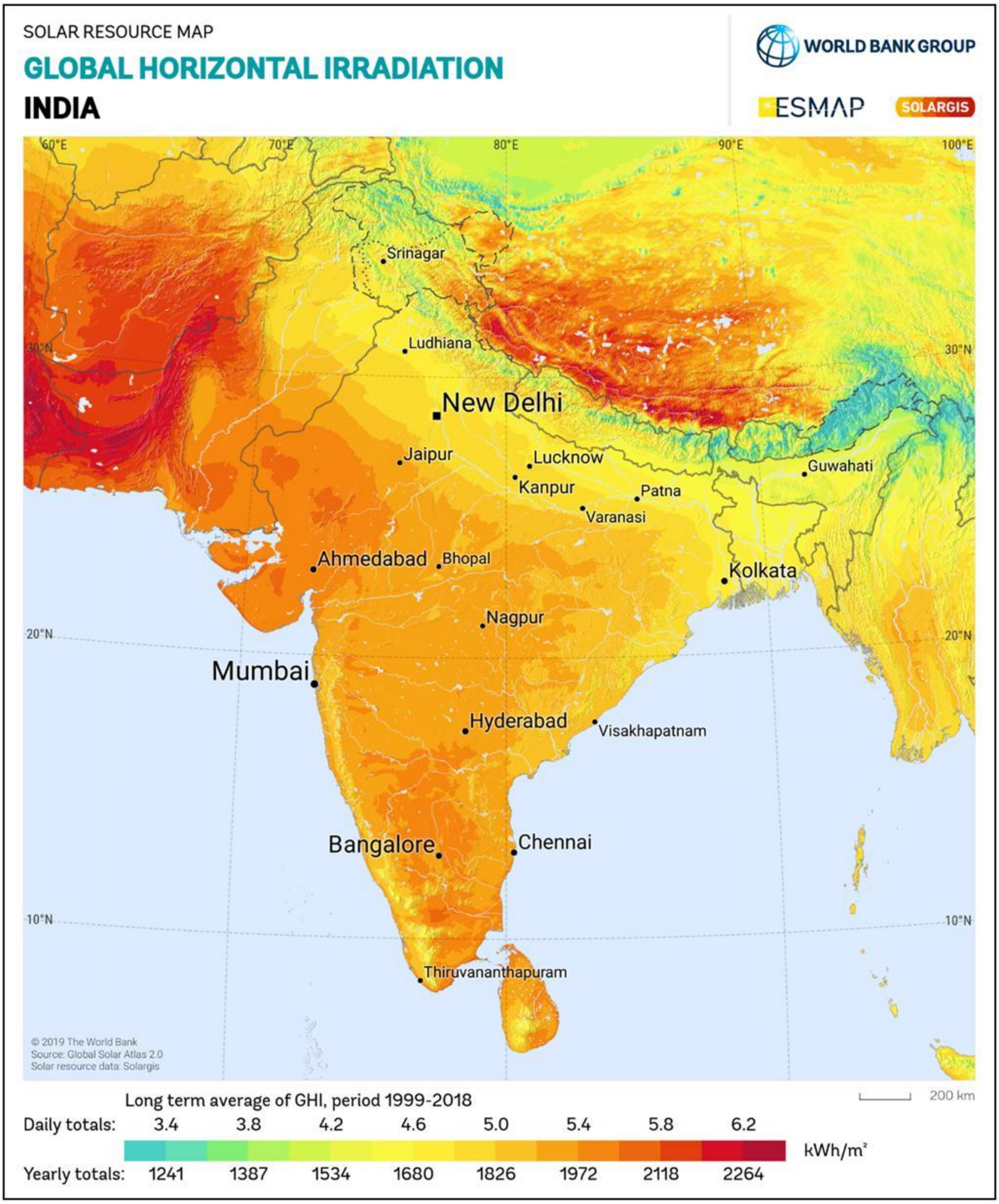
Sources of Solar Resource Data

- PVsyst Meteonorm Database
- NASA SSE Database
- Global Solar Atlas
- NIWE Data
- Site Weather Stations

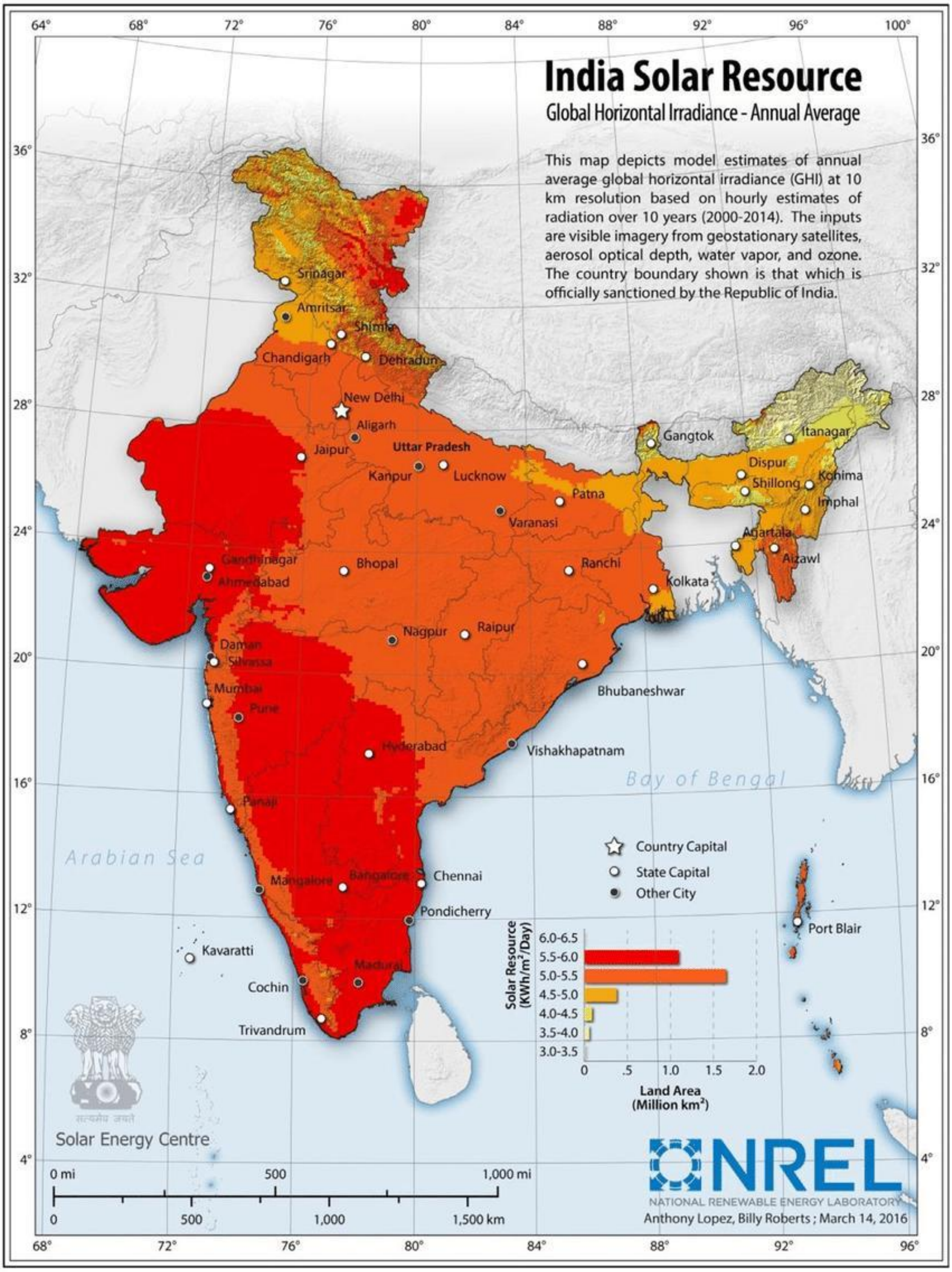
Global horizontal irradiation

India





SOLAR RESOURCE MAP
GLOBAL HORIZONTAL IRRADIATION
INDIA
WORLD BANK GROUP
ESMAP
SOLARGIS
60°E
70°E
80°E
90°E
100°E
30°N
20°N
10°N
Srinagar
Ludhiana
New Delhi
Jaipur
Lucknow
Kanpur
Patna
Varanasi
Guwahati
Ahmedabad
Bhopal
Kolkata
Nagpur
Mumbai
Hyderabad
Visakhapatnam
Bangalore
Chennai
Thiruvananthapuram
© 2019 The World Bank
Source: Global Solar Atlas 2.0
Solar resource data: Solargis
200 km
Long term average of GHI, period 1999-2018
Daily totals: 3.4 3.8 4.2 4.6 5.0 5.4 5.8 6.2
kWh/m²
Yearly totals: 1241 1387 1534 1680 1826 1972 2118 2264


India Solar Resource
Global Horizontal Irradiance - Annual Average
This map depicts model estimates of annual average global horizontal irradiance (GHI) at 10 km resolution based on hourly estimates of radiation over 10 years (2000-2014). The inputs are visible imagery from geostationary satellites, aerosol optical depth, water vapor, and ozone. The country boundary shown is that which is officially sanctioned by the Republic of India.
Country Capital
State Capital
Other City
Solar Resource (KWh/m²/Day)
6.0-6.5
5.5-6.0
5.0-5.5
4.5-5.0
4.0-4.5
3.5-4.0
3.0-3.5
Land Area (Million km²)
0
.5
1.0
1.5
2.0
Srinagar
Amritsar
Shimla
Chandigarh
Dehradun
New Delhi
Aligarh
Jaipur
Uttar Pradesh
Kanpur
Lucknow
Patna
Varanasi
Gangtok
Itanagar
Dispur
Shillong
Kohima
Imphal
Agartala
Aizawl
Kolkata
Ranchi
Bhopal
Gandhinagar
Ahmedabad
Nagpur
Raipur
Daman
Silvassa
Mumbai
Pune
Bhubaneshwar
Hyderabad
Vishakhapatnam
Bay of Bengal
Panaji
Arabian Sea
Mangalore
Bangalore
Chennai
Pondicherry
Port Blair
Kavaratti
Madurai
Cochin
Trivandrum
Solar Energy Centre
0 mi
500
1,000 mi
0
500
1,000
1,500 km
NREL
NATIONAL RENEWABLE ENERGY LABORATORY
Anthony Lopez, Billy Roberts ; March 14, 2016

2.3 Energy Generation Estimation

Energy generation can be estimated using software such as:

- PVsyst
- PV*SOL
- Helioscope
- Solargis

Inputs Required

- Plant capacity
- Module specifications
- Inverter specifications
- Tilt angle
- Azimuth angle
- Solar resource data
- System losses

Typical Losses Considered

Loss Type	Typical Value
Soiling	2–5%
Temperature Loss	3–8%
DC Cable Loss	1–2%
AC Cable Loss	1–2%
Inverter Loss	2–3%
Shading Loss	Site Specific

Energy Yield Formula

$$Annual\ Energy = Plant\ Capacity \times Specific\ Yield$$

Example:

10 kW Plant × 1500 kWh/kW/year

Annual Generation = 15,000 kWh

2.4 Risk Assessment

Technical Risks

- Roof failure
- Excessive shading
- Equipment failure
- Water leakage

Environmental Risks

- High wind speeds
- Dust accumulation
- Corrosion near coastal areas

Commercial Risks

- Change in electricity tariff
- Regulatory changes
- Delayed approvals

Risk Mitigation Measures

✓ Detailed site survey
✓ Structural analysis
✓ Quality equipment
✓ Proper O&M plan
✓ Insurance coverage

2.5 Site Feasibility Study Report

A feasibility report summarizes all findings and recommendations.

Contents of Feasibility Report

1. Site Details
2. Client Information
3. Roof Assessment
4. Load Analysis
5. Solar Resource Data
6. Shading Analysis
7. Proposed System Size
8. Battery Requirement
9. Energy Yield Estimation
10. Financial Analysis
11. Risk Assessment
12. Recommendations

Typical Feasibility Study Workflow

Site Visit
↓
Data Collection
↓
Load Assessment
↓
Shading Analysis
↓
System Design
↓
Energy Simulation
↓
Risk Assessment
↓
Feasibility Report

5. Site Feasibility Study for Ground Mount Solar power plant

UNIT 5.1: Site Feasibility Study for Ground Mount Solar power plant
UNIT 5.2: Soil Testing for Ground Mount Solar power plant.
UNIT 5.3: Topography, Contour Mapping and Its Importance in Solar Plant Design
UNIT 5.4: Electrical Feasibility Report for Ground Mount Solar Power Plant
UNIT 5.5: Estimate Energy Generation from the Solar PV Power Plant

SGJ/N4125

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess irradiation and other geological data
2. Explain how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.
3. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation
4. Explain how to conduct the soil testing and contour mapping.
5. Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant with
6. elevations and topography contour mapping etc.
7. Discuss how to assess physical site accessibility
8. Discuss to identify the policy, regulations and procedures for ground mounted solar sector.
9. Discuss how to assess transmission line & power evacuation

UNIT 5.1: Site Feasibility

Unit Objectives

At the end of this unit, participant will be able to:

4. Explain how to assess irradiation and other geological data
5. Explain how to assess the daily, monthly and annual solar resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.
6. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation

Assess Irradiation and Other Geological Data

Solar Irradiance is the amount of light energy from Sun radiations that hit a square meter of Earth per second. It can be of three types; Reflected, Direct, and Diffuse, as shown in the following figure

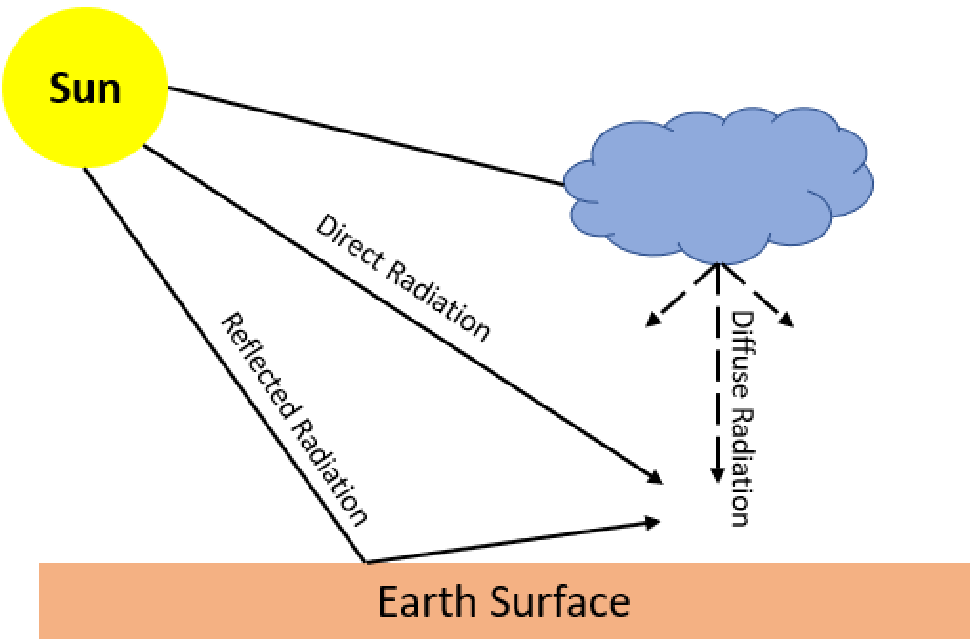


Fig. 14.1.1: Radiations

The different components of Solar Irradiance are as follows:

- **DNI:** It stands for Diffused Normal Irradiance. It represents the amount of sunlight coming perpendicular to a surface. DNI refers to sunrays that come in a straight line from the direction of the sun at its current position in the sky. The Solar PV panels maximize DNI by means of tilting or rotating with the angle of sun.
- **DHI:** It stands for Diffused Horizontal Irradiance. It represents solar radiation that does

not arrive on a direct path from the sun, but is scattered by clouds or other particles in the atmosphere and comes equally from all directions.

- **GHI:** It stands for Global Horizontal Irradiance. It represents the total amount of shortwave radiation received from above by a surface which is horizontal to the surface.

  As per mathematical equation:

  - **Global Horizontal Irradiance (GHI) = Direct Normal Irradiance (DNI)* cos (solar zenith angle) + Diffused Horizontal Irradiance (DHI)**

Geological data plays an important role in the design and construction of a ground-mounted solar power plant. Following are some examples of the geological data that should be considered:

- **Soil type and stability:** When deciding on the solar panel foundation design, the site's stability and soil type and its composition are crucial factors to take into account. To determine the composition and bearing capability of the soil, soil samples can be collected and examined.
- **Topography:** The topography of the site affects the solar panel layout and the grading and excavation required for construction. Accurate elevation data can be collected using surveying tools such as GPS and LiDAR, or through the topographic maps and data for the respective region maintained by the concerned authorities.
- **Drainage:** Proper drainage is essential to prevent erosion and ensure the stability of the site. Hydrological data such as rainfall patterns and water table levels can be analyzed to determine the best drainage strategies.
- **Seismic activity:** The movement of the Earth's crust resulting from the release of energy from tectonic plates under the surface is referred to as seismic activity. Events like earthquakes, volcanic eruptions, and tsunamis may result from this. Therefore, the probability for seismic activity should be evaluated to ensure that the solar power plant is designed to withstand earthquakes and other seismic/geological events.
- **Environmental impact:** The site should be evaluated for any potential environmental impacts such as protected habitats or sensitive ecosystems.

Thus, considering, evaluating and addressing these geological factors, the solar power plant can be designed and constructed to minimize environmental impact and maximize efficiency and stability.

Albedo

Albedo is the measure of the reflective properties of a surface or we can say the measure of the reflectivity of a surface. It refers to the fraction of incoming solar radiation that is reflected back into space by a particular surface. It can either be expressed in a ratio or as a percentage.

- The higher the value of albedo, the more energy is reflected to the source i.e. the surface with high albedo such as snow, ice, sand etc. will absorb less solar radiation.
- Lower the value of albedo, less energy will be reflected i.e. the surface with lower albedo value such as soil, rocks, water etc. will absorb more incoming solar radiations.
- High albedo surface reflects approx. 80% of the incoming solar radiation, whereas Low albedo reflects only 10% of the incoming solar radiation.

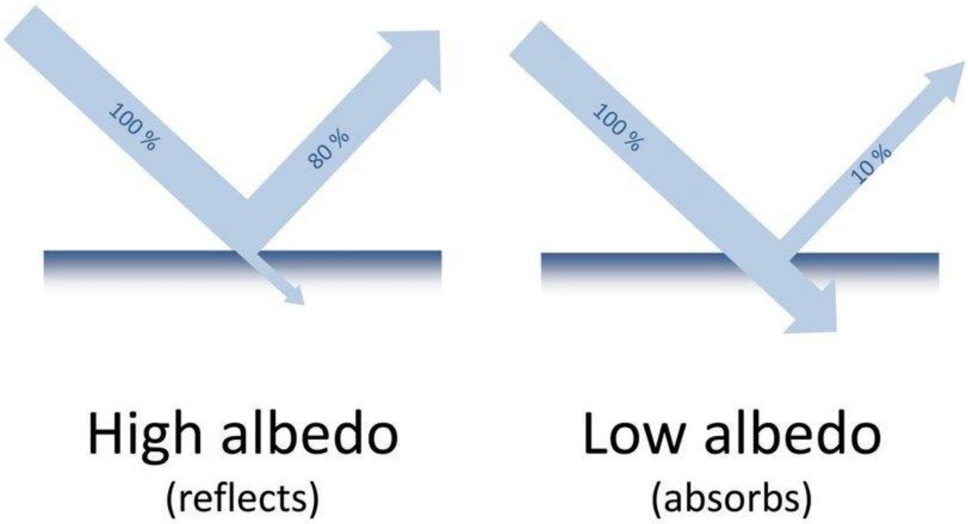


Fig14.1.2: Albedo Principle
Source https://www.researchgate.net/profile/Athena-Van-Der-Perre/publication/322665165/figure/fig24/AS:586042390355984@1516734802257/Example-of-the-albedo-principle-C-KMKG-MRAH.png

The following figure represents the generic steps to measure the albedo:

Point the lightmeter directly to the incoming light source (sun or lamp). Avoid measurements if a shadow covers the lightmeter.

Record the incoming illuminance on your data sheet.

Point the lightmeter directly at the surface that you want to measure.

Record the reflected illuminance on your data sheet.

Calculate the albedo for the surface by determining the ratio of the outgoing illuminance to the incoming illuminance.

Fig 14.1.3. Generic Steps to Measure the Albedo

The formula to calculate the albedo for a surface by determining the ratio of the outgoing illuminance to the incoming illuminance is:

- Albedo = Outgoing illuminance / Incoming illuminance

The incoming illuminance represents the amount of solar radiation that is incident on the surface, while the outgoing illuminance represents the amount of radiation that is reflected back into the atmosphere. By dividing the outgoing illuminance by the incoming illuminance, the albedo of the surface can be calculated as a fraction or percentage.
For example, if the outgoing illuminance is 50 watts per square meter and the incoming illuminance is 100 watts per square meter, the albedo of the surface would be:
Albedo = 50 / 100 = 0.5 or 50%

Collecting Data for Local Weather Conditions

Collecting data on local weather conditions is an important part of assessing the impact of climatic conditions on solar energy generation. It will help in maximising energy production and lowering the possibility of downtime brought on by adverse weather conditions. Following are some of the methods that can be used to collect data on local weather conditions:

- **Weather Stations**: Around the country/world, there are a large number of weather stations that keep track of several weather factors like temperature, humidity, wind speed, and precipitation. Many of these weather stations offer online access to real-time data. The local weather patterns and any extreme weather occurrences that can affect the production of solar energy can be understood using this data.
- **Satellite Imagery**: The weather patterns across a greater area can be seen more clearly with satellite images. Weather patterns that can have an impact on the production of solar energy, such as storms, hurricanes, and cloud cover, can be tracked using satellite imagery.
- **Ground-based Sensors**: Ground-based sensors can be installed on the solar power plant site to collect data on weather conditions such as temperature, humidity, wind speed, and solar radiation. These sensors can be connected to a data acquisition system that records and stores the data for analysis.
- **Historical Data**: Governmental organisations and other sources can provide historical information on the local weather. In order to forecast any extreme weather occurrences that can have an impact on the production of solar energy, this data can be used to determine trends and patterns in the local weather conditions.
- **Install Weather Stations**: You can set up weather stations at the location of your solar PV power plant to gather information on the temperature, humidity, wind direction and speed, and precipitation. Equipment for weather stations is available from a variety of suppliers, and it can be installed locally. The information can then be gathered and kept in a database for evaluation.
- **Use Online Weather Sources**: There are numerous online weather resources that offer real-time data on regional weather conditions. You may find information on temperature, humidity, wind speed, and precipitation by visiting websites like Weather Underground or AccuWeather. A few of these websites furthermore offer historical information that is beneficial for trend research.
- **Mobile Sensors**: Real-time data collection on weather conditions is possible with mobile sensors. To gather information on characteristics such as temperature, humidity, wind speed, and others, these sensors can be mounted on vehicles or drones.

After the data has been gathered, it can be analysed to determine how weather conditions affect the production of solar energy. For instance, the effectiveness of solar panels can be impacted by temperature variations and humidity levels, while the infrastructure of solar power plants might be negatively impacted by flooding and snow. The amount of solar radiation that reaches the solar panels can also be influenced by wind speed and pollution levels, which can have an impact on energy production. The performance and output of the solar power plant can be optimised by considering the local meteorological conditions and their effect on solar energy generation.

Note:

To know more about Albedo, you may refer to the following link:
https://www.youtube.com/watch?v=Vhvwp_XnOf8

Title of Video:	What is Albedo & Why it is Important
By:	SimplyInfo
Duration:	4 m 42 sec

UNIT 4.2: Soil Testing

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to conduct the soil testing and contour mapping.
2. Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant.

Soil testing is an important part of site preparation for solar PV power plants. It helps to assess the suitability of the soil for the construction and installation of the solar PV system. The following are some of the tests that can be conducted to evaluate the soil at the site:

- **Soil Composition**: The composition of soil can be tested to determine the percentage of sand, silt, and clay. This information is important for determining the soil's ability to support the weight of the solar PV system and for planning the foundation design.
- **Soil Bearing Capacity**: The bearing capacity of the soil refers to the amount of weight that it can support without excessive settlement or failure. This test is critical for determining the foundation design of the solar PV system.
- **Soil Resistivity**: Soil resistivity is the measure of the soil's ability to resist the flow of electrical current. This test is important for grounding systems that protect the solar PV system from lightning strikes.
- **Soil pH**: Soil pH is the measure of the acidity or alkalinity of the soil. This information is important for plant growth and for assessing the potential for soil erosion. This can be tested using the pH testing kit.
- **Soil Contamination**: Soil contamination tests are performed to identify the presence of any harmful substances in the soil such as heavy metals or hazardous chemicals. These tests are important for ensuring the safety of workers and the environment.

Soil testing can be performed by a qualified geotechnical engineer or soil testing laboratory. The results of the soil tests can then be used to determine the suitability of the site for solar PV installation, identify any potential challenges, and develop a plan to mitigate them.

Some Common Methods for Soil Testing

The foundation and stability of a solar power plant depend on the quality of the soil, making soil testing an essential part of the installation process. The most commonly used soil testing techniques for solar power plant installation include the following:

- **Standard Penetration Test (SPT):** This method is widely used in India for soil testing. It involves driving a split-spoon sampler into the soil using a hammer and counting the number of blows required to advance the sampler a certain distance. The data obtained can be used to determine the soil's strength and density.

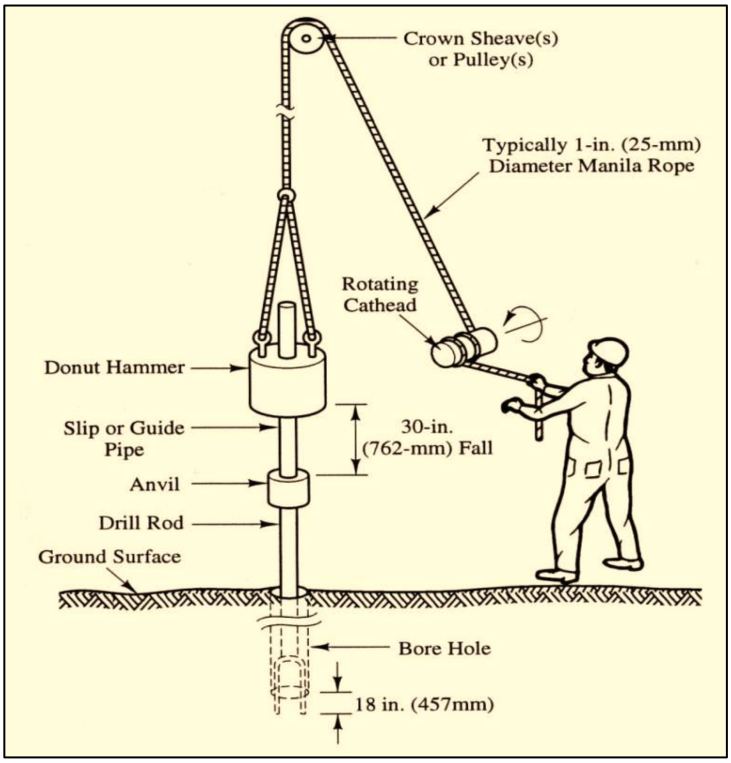


Fig14.2.1. Standard Penetration Test

Image Source: https://www.researchgate.net/profile/Debojit-Sarker/publication/280572148/figure/fig3/AS:328389186605059@1455305491498/Standard-Penetration-Test-Arrangements.png

- **Cone Penetration Test (CPT):** This method involves inserting a cone-shaped device into the soil and measuring the resistance to penetration. The CPT delivers continuous data as the probe is pushed into the soil, allowing for a thorough evaluation of the soil layers and their characteristics. The test's results can be used to create soil profiles and estimate factors like the soil type, shear strength, and bearing capacity soil's strength, stiffness, and permeability.

Note:
To know more about Standard Penetration Test, you may refer to the following link:
https://www.youtube.com/watch?v=DjWDOqQjsyQ

Title of Video:	How to conduct SPT
By:	Sangeetha Sundar
Duration:	4 m 49 sec

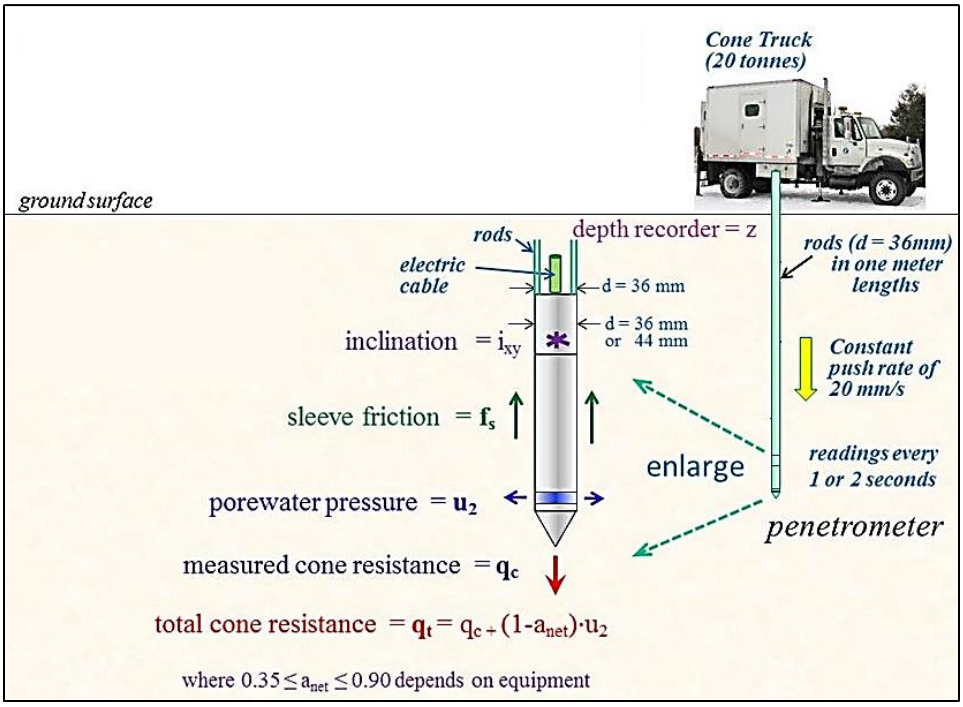


Fig14.2.2.: Cone Penetration Test

- **Plate Load Test**: This method involves placing a plate on the ground and measuring the settlement of the plate under a known load. The data obtained can be used to determine the soil's bearing capacity.

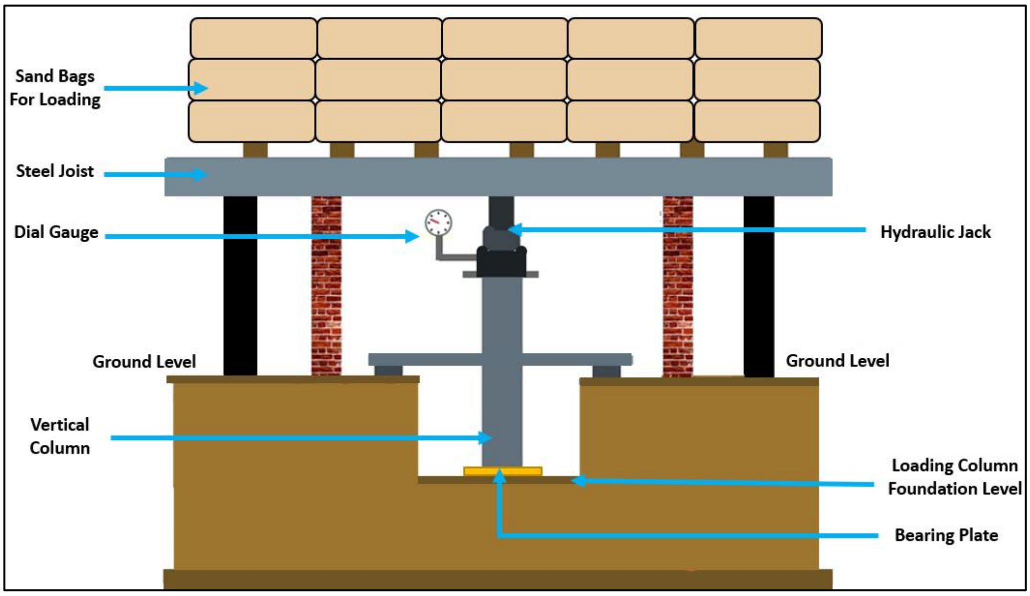


Fig14.2.3. Plate Load Test

- **Pull-Out Test**: The pull-out test is a type of soil testing method used to determine the bond strength between the soil and the reinforcement or anchor. The test involves inserting a steel rod or anchor into the ground and then pulling it out with a hydraulic jack while measuring the force required to pull it out.

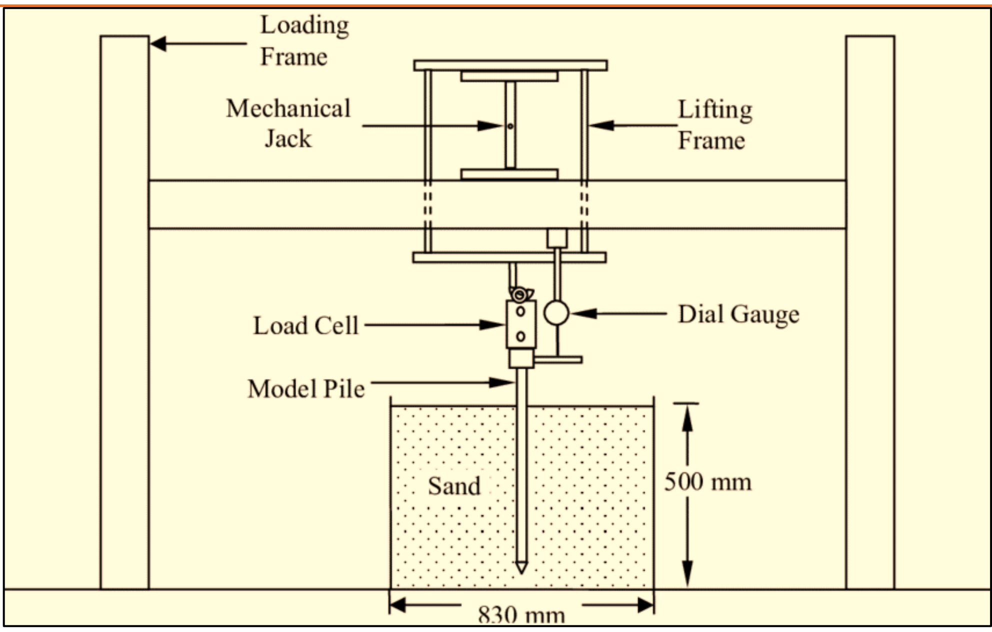


Fig14.2.4. Pull-Out Test

UNIT 5.3: Topography, Contour Mapping and Its Importance in Solar Plant Design

The term "topography" describes the topographical aspects and traits of a certain piece of land, such as its elevation, relief, and slope. The shape and features of the Earth's surface are studied, including both natural and man-made elements such mountains, valleys, rivers, lakes, forests, and structures.

Importance of Topography in Solar Design

For a number of reasons, topography is crucial to the design of a solar power plant such as:

- We are aware that when solar panels are oriented at the sun at the ideal angle, they perform at their best. Solar panels' orientation and tilt can be impacted by topography, which can reduce their output and efficiency. For example, it might be required to modify the tilt and orientation of the panels if the terrain is steep in order to ensure that they receive the most sunlight possible.
- Second, topography may have an effect on how the solar power plant is designed. To level the land and create a solid foundation for the solar panels, it could be required to construct retaining walls or other structures, for instance, if the terrain slopes.
- Thirdly, geography may have an effect on how the solar power plant is built and installed. For instance, installing the foundations for the solar panels may be more challenging if the site is rocky, which could raise the project's cost.
- Finally, topography can affect the solar power plant's overall operation and safety. To avoid damage, it could be essential to establish drainage systems or construct the solar panels on elevated platforms, for instance, if the area is prone to flooding.

Contour Mapping and its Importance in Solar Design

A two-dimensional map can be used to depict the topography or surface features of a specific area of land using the contour mapping technique. The procedure entails drawing contour lines to connect areas of the land surface with equivalent elevations, which enables a clear visualisation of the topography.

Contour mapping is a crucial component of solar plant design since it offers critical details on the terrain and slope of the area where the solar plant is planned to be built. It offers useful data that can assist designers in choosing the best location for the plant, optimising solar panel orientation, designing efficient drainage and erosion control measures, preparing the site for construction, and producing precise cost estimates. The following list of factors illustrates the significance of contour mapping in solar plant design:

- **Site Selection**: Contour mapping can help in locating the solar plant's ideal location. The contour lines can be used by solar plant designers to assess a site's suitability as a location for the facility and to identify any potential difficulties that might arise during construction.
- **Maximum Solar Energy Yield and Optimal Solar Panel Orientation**: Contour maps assist in locating the regions on the site where the solar panels can be positioned for maximum solar energy yield and optimum orientation. The designers can choose the best tilt and orientation for the solar panels to produce the most solar energy by using the contour lines, which display the differences in elevation across the site.
- **Drainage and erosion control**: Contour mapping helps in locating regions with a steep incline and the site's natural water flow. Designing efficient drainage systems and erosion control techniques is essential for reducing soil erosion and sedimentation.

- **Site Preparation**: Contour mapping is useful in locating the solar panel installation locations that need to be graded and levelled. Designers can identify the places that need earthworks to level the ground and build a solid foundation for the solar panels by examining the contour lines.
- **Accurate Cost Estimation**: Contour mapping facilitates the development of accurate estimates of costs for the solar plant's construction. Designers can precisely predict the amount of earthwork needed, the price of retaining walls, and other building expenditures thanks to the contour map's extensive information on the topography of the site.

Exercise

1. Explain different components of Solar Irradiance.

____

____

____

____

2. Explain Methods for Soil Testing.

____

____

____

____

3. Explain the Importance of Topography in Solar Design.

____

____

____

____

Module 14 - Site Feasibility Study for Ground Mount Solar power plant	Unit 14.1	What is Albedo & Why it is Important	https://www.youtube.com/watch?v=Vhvwp_XnOf8	

Module 14 - Site Feasibility Study for Ground Mount Solar power plant	Unit 14.2	How to conduct SPT	https://www.youtube.com/watch?v=DjWDOqQjsyQ	

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to assess site electrical feasibility
2. Explain how to assess cable root
3. Discuss how to identify inverter location
4. Discuss how to identify the transformer and other HT equipment location
5. Explain how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site
6. Explain how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV

UNIT 5.4: Electrical Feasibility Report for Ground Mount Solar Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to assess site electrical feasibility
- Explain how to assess cable root
- Discuss how to identify inverter location
- Discuss how to identify the transformer and other HT equipment location

5.4.1 Electrical Feasibility

An electrical feasibility study for a ground mount solar power plant refers to A study refers to an assessment carried out to ascertain the technical and financial viability of installing and running a solar power plant at a certain location from an electrical standpoint. To establish whether the solar project can be implemented, the research assesses the electrical infrastructure, grid connectivity, and other electrical characteristics.

Electrical Feasibility Report for Ground Mount Solar Power Plant

An Electrical Feasibility Report for a Ground Mount Solar Power Plant is a comprehensive document that assesses the electrical aspects and feasibility of implementing a solar power plant at a specific location. It provides a detailed analysis of the electrical infrastructure, grid connectivity, system design, and other electrical considerations related to the project.

Importance of Electrical Feasibility Report

An electrical feasibility report holds significant importance for an entrepreneur considering the development of a ground mount solar power plant. Here are the key reasons why an electrical feasibility report is crucial for an entrepreneur:

- **Technical Viability:** The report provides an assessment of the technical feasibility of implementing a solar power plant at the proposed site. It evaluates the electrical infrastructure, grid connectivity, solar resource availability, and other technical factors. This information enables the entrepreneur to determine if the site is suitable for the solar project and if it can effectively integrate with the existing electrical grid.
- **Cost Estimation and Financial Planning**: The electrical feasibility report includes cost estimates for grid interconnection, infrastructure upgrades, electrical equipment, and other relevant expenses. This information helps the entrepreneur develop an accurate financial plan for the solar power plant. It allows them to assess the project's financial viability, estimate the return on investment, and make informed decisions regarding financing options and budget allocation.
- **Grid Integration and Interconnection**: The report assesses the grid availability and capacity for power evacuation. It identifies the nearest substation, evaluates transmission line capacity, and considers grid constraints. This information is crucial for determining the feasibility of connecting the solar power plant to the grid and understanding the requirements and costs associated with grid interconnection.
- **Risk Assessment and Mitigation**: The electrical feasibility report highlights potential risks and challenges related to the electrical aspects of the solar power plant. It identifies technical constraints, grid stability concerns, voltage regulation issues, and other risks that may affect the project's success. By understanding these risks in advance, the entrepreneur

can develop appropriate mitigation strategies and contingency plans to ensure smooth project implementation.

- **Compliance and Regulatory Considerations**: The report outlines the regulatory requirements, codes, and standards that govern the installation and operation of the solar power plant. It ensures that the entrepreneur is aware of and complies with the legal and regulatory obligations. This helps avoid legal complications, delays, and potential penalties in the future.
- **Decision-Making and Project Planning**: The electrical feasibility report provides the entrepreneur with comprehensive information and analysis necessary for decision-making and project planning. It enables them to evaluate the project's viability, set realistic goals and timelines, allocate resources effectively, and make informed decisions regarding equipment selection, grid connectivity, and system design.
- **Stakeholder Engagement and Financing:** The report serves as a valuable tool for engaging stakeholders, including investors, lenders, and potential partners. It provides them with a clear understanding of the electrical feasibility of the project and its potential for success. A robust electrical feasibility report enhances the entrepreneur's credibility, increases investor confidence, and facilitates the process of securing financing and partnerships.

Parameters Included in an Electrical Feasibility Report for Ground Mount Solar Power Plant

Following are some of the parameters considered while preparing an electrical feasibility report:

- **Solar Resource Assessment**: Evaluate the solar resource availability at the site through historical data, solar radiation measurements, and shading analysis. This helps determine the potential energy generation capacity of the solar power plant.
- **Site Electrical Feasibility Assessment**: It includes the following
  - **Site Survey**: Conduct a physical survey of the site to assess its electrical infrastructure, including the presence of existing power lines, substations, and distribution networks.
  - **Load Assessment**: Analyse the local electricity demand and load profile to determine the suitability of the solar power plant in meeting the energy requirements.
  - **Grid Connectivity Analysis**: Evaluate the proximity and accessibility of the electrical grid infrastructure to the site. Determine the technical requirements for grid interconnection and assess the capacity of the grid to accommodate the additional power generation.
- **Cable Route Assessment**:
  - Conduct a thorough survey of the site to identify the optimal cable routing from the solar power plant to the grid connection point or substation.
  - Consider factors such as distance, terrain, environmental considerations, and any potential obstacles that may affect the cable installation.

- **Inverter Location Identification**:
  - Evaluate the layout and design of the solar power plant to identify suitable locations for inverter installation.
  - Consider factors such as proximity to the solar arrays, access for maintenance, and electrical safety requirements.
- **Transformer and HT Equipment Location Identification**:

    - Determine the optimal location for transformers and other high-tension (HT) equipment based on the site layout, electrical requirements, and grid connection specifications.
    - Consider factors such as electrical losses, voltage regulation, accessibility, and safety considerations.
- **Grid Availability Assessment**:
    - Identify the nearest substation and assess its capacity to accommodate the additional power generated by the solar power plant.
    - Evaluate the distance between the project site and the substation, considering transmission line capacity and potential grid constraints.
    - Verify the availability of grid connection points and the feasibility of connecting the solar power plant to the grid.
- **Evacuation Voltage Level Assessment**:
    - Determine the appropriate evacuation voltage level based on the project size, grid requirements, and local regulations.
    - Consider factors such as the project's capacity, transmission line capacity, and the compatibility of the evacuation voltage level with the electrical grid infrastructure.

Thus, These parameters and assessments are crucial components of an electrical feasibility report for a ground mount solar power plant. They help determine the technical and economic viability of the project and provide valuable insights for decision-making and project planning.

5.1.2 Assessing Site Electrical Feasibility

Assessing the site electrical feasibility for a ground mount solar power plant involves a thorough evaluation of the electrical infrastructure and conditions at the proposed site. Here are the steps to conduct a site electrical feasibility assessment:

- **Site Survey with respect to existing electrical infrastructure**: Conduct a physical survey of the site to gather information about the existing electrical infrastructure, such as power lines, substations, and distribution networks. Identify the location of nearby electrical infrastructure that can potentially facilitate grid connectivity.
- **Load Assessment**: Analyse the electricity consumption patterns and load profile of the proposed site or nearby load centers. Determine the energy demand and peak load requirements to understand the electrical needs of the area.
- **Solar Resource Assessment**: Evaluate the solar resource potential at the site by analysing historical solar radiation data, conducting on-site solar measurements, and performing shading analysis. This assessment helps estimate the solar energy generation capacity of the site.
- **Grid Connectivity Analysis**: Evaluate the proximity and accessibility of the electrical grid infrastructure to the proposed site. Determine the distance to the nearest substation and assess the capacity of the grid to handle the additional power generated by the solar power plant.
- **Grid Interconnection Requirements**: Identify the technical requirements for grid interconnection, such as voltage levels, power factor correction, and grid protection standards. Understand the regulations, guidelines, and procedures set by the local utility or grid operator for connecting renewable energy systems to the grid.
- **System Design Considerations**: Analyze the site layout and consider factors such as orientation, available land area, and potential shading from nearby structures or vegetation.

Optimize the solar power plant's design, including the arrangement of solar panels, stringing configurations, and electrical wiring, to maximize energy generation.

- **Environmental Considerations**: Assess any environmental factors that may impact the electrical feasibility of the project, such as temperature variations, wind conditions, or exposure to corrosive environments. Consider the impact of such factors on equipment performance and lifespan.
- **Safety and Compliance**: Ensure that the proposed site meets safety standards and regulatory requirements. Evaluate factors such as clearance distances, grounding systems, and compliance with electrical codes and regulations.
- **Cost Analysis for Electrical Infrastructure**: Estimate the costs associated with electrical infrastructure upgrades, grid interconnection, and equipment installation. Consider the financial feasibility of the project by assessing the potential return on investment and payback period.

5.1.3 Assessing Cable Root

Assessing the cable route for a ground mount solar power plant involves evaluating the optimal path for routing electrical cables from the solar arrays to the desired connection point, such as a substation or grid interconnection point. Here are the steps to assess the cable route:

- **Site Survey to Identify Potential Obstacles or Constraints That May Affect the Cable Route**: Conduct a detailed survey of the site to understand the topography, terrain, and any potential obstacles or constraints that may affect the cable route. Identify existing infrastructure, such as roads, buildings, or other structures, that could impact cable installation.
- **Distance and Layout**: Determine the distance between the solar arrays and the connection point. Consider factors such as the layout of the solar power plant, the desired connection point's location, and any geographical features that may influence the cable route.
- **Cable Sizing and Voltage Drop**: Calculate the appropriate cable sizing based on the current capacity and voltage requirements of the solar power plant. Consider the length of the cable route and the permissible voltage drop to ensure efficient power transmission and minimize losses.
- **Terrain and Environmental Factors**: Evaluate the terrain conditions along the cable route, including slopes, elevation changes, and any potential hazards. Consider the impact of environmental factors such as extreme weather conditions, temperature variations, or exposure to corrosive elements.
- **Cable Protection and Safety**: Determine the necessary measures to protect the cables from damage, including physical protection (e.g., conduit, ducts) and appropriate burial depths. Consider safety requirements, such as maintaining proper clearances from other utilities or potential hazards.
- **Accessibility and Maintenance of Cable Route:** Assess the accessibility of the cable route for future maintenance and repairs. Consider factors such as ease of access, proximity to roads or paths, and the availability of service vehicles or equipment for maintenance purposes.
- **Regulatory Compliance as per Government Electrical Codes**: Ensure compliance with local regulations and electrical codes governing cable installation. Consider any permits or approvals required for trenching, excavation, or underground cable installation.
- **Cost Analysis for cable installation**: Estimate the costs associated with cable installation, including materials, labour, trenching, and any necessary equipment. Consider factors such

as the cable length, terrain complexity, and any additional requirements for cable protection or routing.

- **Documentation and Design of cable route**: Develop detailed drawings or diagrams illustrating the cable route, including trenching plans, cable sizes, and connection points. Document the assessment findings and include them in the overall project documentation for reference and future maintenance purposes.

By assessing the cable route for a ground mount solar power plant, stakeholders can ensure efficient and reliable electrical connectivity from the solar arrays to the grid connection point. This assessment helps optimize cable sizing, mitigate risks, ensure compliance with regulations, and minimize installation and maintenance costs.

Note:
To know more about cable route for a ground mount solar power plant, you may refer to the following link:
https://www.youtube.com/watch?v=kazwzLyKD2A

Title of Video:	Installing PV Wire And Conduit For Our Solar System
By:	Country Living Experience
Duration:	11 m 57 sec

5.1.4 Identifying Inverter Location

Identifying the optimal inverter location for a ground-mount solar power plant is crucial for ensuring efficient energy conversion and effective system operation. The following are the parameters that should be considered when identifying the inverter location:

- **System Design and Layout**: Understand the layout and design of the ground mount solar power plant, including the arrangement of solar panels, the orientation of the arrays, and the overall configuration. Consider factors such as available space, shading analysis, and the desired electrical connections.
- **Proximity to Solar Arrays**: Locate the inverters in close proximity to the solar arrays to minimize cable lengths and associated losses. This helps optimize the system's overall efficiency and reduces the cost of cabling.
- **Environmental Considerations**: Assess environmental factors that may impact the performance and lifespan of the inverters. Avoid locations prone to excessive heat, humidity, or other adverse conditions. Ensure proper ventilation and cooling for the inverters to maintain optimal operating temperatures.
- **Accessibility and Maintenance**: Choose an inverter location that allows for easy accessibility and maintenance. Consider factors such as proximity to service paths or roads, clearance space for equipment, and the availability of adequate working space for maintenance and troubleshooting.

- **Electrical Infrastructure**: Determine the requirements for electrical connections between the solar arrays and the inverters. Consider the location of the point of interconnection, whether it's a distribution panel, main switchboard, or other electrical equipment. Ensure that the chosen inverter location aligns with the electrical infrastructure and connection requirements.
- **Inverter Capacity and Expansion**: Consider the capacity and scalability of the inverters. Evaluate the future expansion potential of the solar power plant and select an inverter location that allows for easy integration of additional inverters if needed.
- **Electrical Safety**: Ensure compliance with electrical safety standards and regulations when selecting the inverter location. Consider factors such as proper grounding, clearances, and electrical safety equipment requirements.
- **Communication Requirements and Monitoring**: Assess the communication requirements for the inverters, such as data monitoring, control signals, and remote access capabilities. Choose a location that allows for reliable communication between the inverters and the monitoring system.
- **Aesthetics and Land Use**: Consider the visual impact and land use implications of the inverter location. Select the location that minimizes the visual presence of the inverters and integrates well with the overall landscape.
- **Cost Analysis as Per the Inverter Location**: Evaluate the costs associated with the inverter installation, including cabling, equipment, and any necessary infrastructure upgrades. Consider the overall balance between cost-effectiveness and system performance.

5.4.5 Identifying Transformer and Other HT equipment location

Identifying the optimal location for transformers and other high-tension (HT) equipment in a ground mount solar power plant is crucial for efficient power distribution and grid connectivity. Following are the parameters considered when identifying the location for transformers and HT equipment:

- Electrical System Design
- Load Distribution and Voltage Regulation
- Proximity to Inverters and Solar Arrays
- Grid Connection Point
- Electrical Safety and Clearances
- Space Availability and Accessibility
- Environmental Considerations
- Noise and Aesthetic Considerations
- Cost Analysis

Fig.17.1.1 Parameters Considered When Identifying the Location for Transformers and HT Equipment

- **Electrical System Design**: Understand the electrical system design of the solar power plant, including the capacity of the system, voltage levels, and the expected power output. Consider factors such as the number of inverters, the layout of the solar arrays, and the desired electrical connections.
- **Load Distribution and Voltage Regulation**: Determine the distribution of loads within the solar power plant and analyze the voltage drop along the electrical network. Identify areas with higher loads or longer cable lengths that may require voltage regulation through transformers.
- **Proximity to Inverters and Solar Arrays**: Locate the transformers and HT equipment in close proximity to the inverters and solar arrays to minimize cable lengths and associated losses. This helps optimize system efficiency and reduces the cost of cabling.
- **Grid Connection Point**: Identify the location of the grid connection point, such as a substation or distribution panel, and assess the distance between the grid connection point and the transformers/HT equipment. Consider the capacity and requirements of the grid connection point for seamless integration.
- **Electrical Safety and Clearances**: Ensure compliance with electrical safety standards and regulations when selecting the location for transformers and HT equipment. Consider factors such as clearances, grounding requirements, and safety equipment necessary for the operation and maintenance of the equipment.
- **Space Availability and Accessibility**: Evaluate the available space at the solar power plant site for installing the transformers and HT equipment. Ensure that there is sufficient space for equipment installation, ventilation, and maintenance. Consider accessibility for equipment installation and future maintenance requirements.
- **Environmental Considerations**: Assess environmental factors that may impact the performance and lifespan of the transformers and HT equipment. Consider factors such as temperature variations, exposure to corrosive elements, and protection against adverse weather conditions.
- **Noise and Aesthetic Considerations**: Take into account the noise levels generated by transformers and HT equipment and their impact on the surrounding environment. Consider measures to mitigate noise, if necessary. Additionally, ensure that the equipment location aligns with the aesthetic requirements and overall landscape of the solar power plant.
- **Cost Analysis**: Evaluate the costs associated with the installation of transformers and HT equipment, including equipment procurement, infrastructure upgrades, and cabling. Consider the overall balance between cost-effectiveness and system performance.

By carefully considering these factors, stakeholders can identify the most suitable location for transformers and other HT equipment in a ground mount solar power plant. This ensures efficient power distribution, voltage regulation, and seamless integration with the electrical grid, ultimately optimizing the performance and reliability of the solar power plant.

Note:
To know more about transformer in solar PV Plant, you may refer to the following link:
https://www.youtube.com/watch?v=oeJDOznZyIc
Title of Video: Transformer information in solar plant
By: ROSHAN SANGARE, Duration: 9 m 37 sec

UNIT 5.2: Estimate Energy Generation from the Solar PV Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to assess grid availability for power evacuation including nearest substation and transmission line capacity as well as distance from project site
- Explain how to assess the evacuation voltage level i.e. 230 V/440 V/11 KV /33 KV66 kV

5.2.1 Estimate Energy Generation

Estimating the energy generation from a solar PV power plant is of utmost importance for several reasons. The first benefit is that it helps with project planning and sizing, enabling designers to create a system that satisfies targets and demand for electricity. For investment optimisation and ensuring that the plant produces enough energy, accurate estimations are essential. Furthermore, estimations of energy generation are essential for conducting financial analyses and assessing the project's financial viability. They influence decisions about investments and return on investment by assisting in the assessment of revenue potential, project costs, and profitability. Such estimations also make grid integration and power purchase agreements easier to negotiate, enabling grid operators to gauge capacity.

Estimating energy generation from a solar PV power plant typically involves the use of specialized tools and software designed for solar energy analysis. Following are the tasks involved in estimating energy generation:

- **Gather project information**: Collect all relevant information about the solar PV power plant project, including the location, PV system specifications (such as panel type, capacity, efficiency), orientation and tilt angle of the panels, and expected operational parameters (such as system losses, shading, soiling, and temperature coefficients).
- **Select solar energy analysis software**: There are several software options available for solar energy analysis, such as PVsyst, SAM (System Advisor Model), PVSol, and Helioscope. Choose a software tool that suits your requirements and is capable of providing accurate energy generation estimates.
- **Input project details**: Launch the selected software and input the project details into the software interface. This may include the geographical location, panel specifications, system configuration, and any other relevant data required by the software.
- **Define meteorological data**: Solar energy analysis software relies on accurate meteorological data to estimate energy generation. Input the appropriate meteorological data for the project location, including solar irradiance, temperature, wind speed, and other relevant climate information. This data is often available from weather stations or can be obtained from trusted sources or databases.
- **Configure system parameters**: Set the system parameters within the software, such as system losses (due to wiring, inverter efficiency, mismatch losses, etc.), shading factors (if applicable), and other efficiency parameters based on the project's specific design and expected operating conditions.

- **Run simulations**: Use the software to simulate the solar energy generation based on the input parameters and system configuration. The software will perform calculations based on algorithms, weather data, and system characteristics to estimate the energy generation potential of the solar PV power plant.
- **Review results**: Once the simulation is complete, review the results provided by the software. The software will typically provide detailed information on energy generation, including monthly, daily, and hourly energy profiles, system performance ratios, and other relevant metrics.
- **Analyze and validate the results**: Analyze the generated energy profiles and metrics to assess the expected performance of the solar PV power plant. Compare the results against industry standards, benchmark data, or similar projects to validate the accuracy of the estimates.
- **Refine the analysis if needed**: If the estimated energy generation does not meet your expectations or desired targets, review the input parameters, system configuration, or other factors that may affect energy generation. Make adjustments to the analysis in the software and re-run simulations to refine the estimates.
- **Document the results**: Document the energy generation estimates and any assumptions or caveats associated with the analysis. These results will be valuable for project planning, financial modeling, and communication with stakeholders.

Calculating Energy Generation from Solar PV Plant

The following figure represents the various steps involved in estimating/calculating the energy generation from the Solar PV Plant:

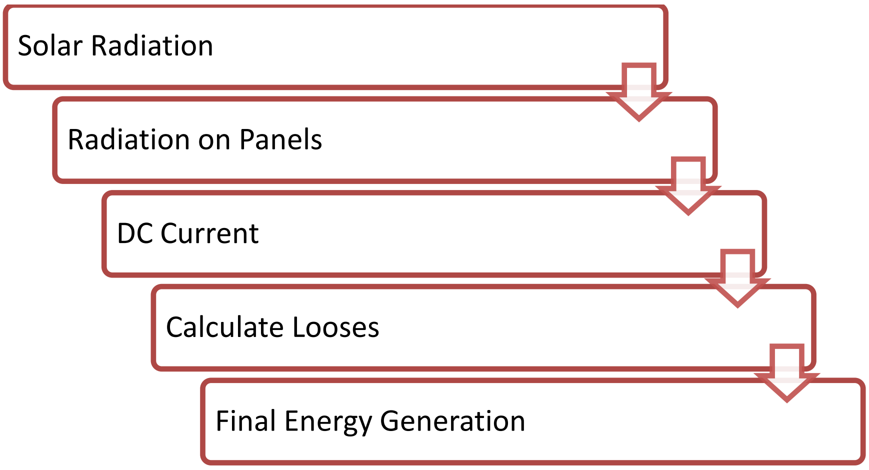


Fig.17.2.1: Steps Involved in Estimating/Calculating the Energy Generation from The Solar PV Plant

To estimate energy generation in a solar power plant, you can calculate the following parameters:

- **Solar Radiation:** To calculate solar radiation, you can use various models and formulas. One commonly used formula is the Hottel-Whillier-Bliss (HWB) model, which estimates the total incident solar radiation on a horizontal surface. The formula for the HWB model is:

  Solar radiation ($kWh/m^2/day$) = Isc × (1 + a × Tilt) × (1 - b × |Lat - Lat0|)

  Where:
  - Isc = Extraterrestrial radiation on a horizontal surface (typically available from solar radiation databases)
  - Tilt = Tilt angle of the solar panel (in degrees)
  - a, b = Empirical coefficients for the latitude adjustment factor

- Lat = Latitude of the project location
- Lat0 = Latitude of the reference location

- **Radiation on Panels:** To calculate the radiation on panels, you need to consider the panel's orientation and tilt angle. The formula for calculating the radiation on panels based on the HWB model is:

  Radiation on panels (kWh/m²/day) = Solar radiation (kWh/m²/day) × cos(Incidence Angle)
  Where:
  - Incidence Angle = Angle between the solar radiation vector and the normal vector of the panel surface
- **DC Current:** To calculate the DC current generated by the solar panels, you need to consider the panel's power rating and efficiency. The formula for DC current is:

  DC Current (A) = Panel Power Output (W) / Panel Voltage (V)
- **Losses:** Losses in a solar power plant can occur due to various factors, such as wiring losses, inverter losses, module mismatch losses, soiling, shading, and temperature losses. The total losses can be calculated by summing up the losses from each individual component or factor. The formula for calculating losses will vary depending on the specific component or factor being considered. For example, to calculate wiring losses, you can use the formula:
  Wiring Losses (%) = (I^2 × R) / (P × 100)
  Where:
  - I = Current (A)
  - R = Resistance (ohms)
  - P = Power (W)
- **Final Energy Generation:** The final energy generation is the product of the effective radiation on the panels, the DC current, and the time period for which the panels are exposed to sunlight. The formula for calculating final energy generation is:

Final Energy Generation (kWh) = Effective Radiation on Panels (kWh/m²/day) × DC Current (A) × Time period (hours)
These formulas provide a general understanding of the calculations involved in estimating energy generation in a solar power plant. However, it's important to note that specific factors, system configurations, and additional losses need to be considered for accurate estimations. Specialized software tools like PVsyst, SAM, or PVSol can simplify these calculations and provide more comprehensive results based on the specific parameters of your solar power plant.

5.2.2 Assess Grid Availability

Assessing grid availability for power evacuation and determining the nearest substation, transmission line capacity, and distance from the project site for a ground-mounted solar power plant involves several steps. Here's a guide on how to approach this assessment:

- **Identify the project location**: Determine the exact coordinates or address of the proposed solar power plant site. This information will be necessary for conducting the assessment accurately.
- **Obtain grid maps and data**: Contact the local utility company or relevant grid operator to request grid maps and data for the area surrounding the project site. This information may include the existing substation locations, transmission lines, and their capacities.

- **Review grid maps and substation locations**: Study the grid maps and identify the nearest existing substations to your project site. Determine their locations in relation to the site and note their capacities. This will help you understand the available grid infrastructure and the potential for power evacuation.
- **Evaluate transmission line capacity**: Assess the transmission lines connecting the identified substations to the main grid. Determine the capacity of these lines to ensure they can handle the additional power generated by your solar power plant. If the existing transmission line capacity is insufficient, you may need to consider grid upgrades or alternative options for power evacuation.
- **Consider voltage levels**: Examine the voltage levels of the transmission lines and substations. Ensure that they are compatible with the solar power plant's requirements. Voltage transformation may be necessary if there is a mismatch between the project's output and the grid's voltage.
- **Assess distance and connectivity**: Measure the distance between the project site and the nearest substation. Consider the accessibility of the site to determine the feasibility of connecting to the grid. Evaluate the existing infrastructure, such as roads and pathways, and assess any potential challenges that may affect the cost and effort required for grid connectivity.
- **Engage with the grid operator**: Contact the local utility company or grid operator to discuss your project and obtain more specific information. Share your plans, electricity generation estimates, and power evacuation requirements. The grid operator can provide guidance on the technical feasibility, connection process, and any necessary permits or agreements required for interconnection.
- **Conduct a feasibility study**: Based on the gathered information and discussions with the grid operator, conduct a feasibility study to assess the overall viability of connecting your solar power plant to the grid. Consider factors such as grid capacity, distance, costs, and potential grid expansion plans in the area.
- **Engage with experts if needed**: If you lack expertise in grid assessment or require more detailed analysis, consider engaging with renewable energy consultants, electrical engineers, or grid integration specialists. They can provide in-depth assessments, simulations, and guidance to ensure optimal power evacuation and grid integration for your solar power plant.

Remember that the process may vary based on the specific regulations, grid structure, and regional considerations. It's essential to work closely with the local grid operator and relevant authorities to ensure compliance with local regulations and guidelines.

5.2.3 Assess the Evacuation Voltage Level

Evacuation voltage, also known as grid voltage or grid connection voltage, refers to the electrical voltage level at which a solar PV power plant is connected to the electrical grid for power transmission and distribution. It represents the voltage at which the generated electricity from the solar power plant is fed into the grid and supplied to consumers.
The evacuation voltage level depends on various factors, such as:

- Local grid infrastructure
- Grid connection standards
- Regulatory requirements.

Common evacuation voltage levels for solar PV power plants can range from low voltage (e.g., 230V or 440V) to medium voltage (e.g., 11 kV, 33 kV, or 66 kV) or even higher, depending on the scale and capacity of the solar power plant and the capabilities of the local electrical grid.

Selecting the appropriate evacuation voltage level is crucial to ensure efficient and reliable power evacuation from the solar PV power plant. It should be compatible with the grid infrastructure, transmission lines, and substations available in the vicinity of the project site. Additionally, it needs to comply with the local electrical regulations and standards set by the grid operator and relevant authorities.

Determining the evacuation voltage level involves conducting feasibility studies, coordinating with the local grid operator, and performing power system analyses to assess grid capacity and stability. It is typically done in consultation with electrical engineers and experts who have expertise in power system design, grid integration, and compliance with electrical regulations.

Thus, the evacuation voltage is the voltage level at which the solar PV power plant connects to the electrical grid, allowing for the transfer of electricity from the plant to the grid for distribution to consumers.

Assessing Evacuation Voltage

When assessing the evacuation voltage level for a solar PV power plant, several parameters need to be considered. These parameters help determine the appropriate voltage level for grid connection. Following are some of the key parameters used in the assessment:

- **Grid Connection Standards and Regulations**: Review the local grid connection standards and regulations set by the relevant authorities and grid operator. These standards specify the allowable voltage levels, equipment requirements, and technical guidelines for connecting the solar PV power plant to the grid.
- **Grid Infrastructure**: Evaluate the capacity and capabilities of the local electrical grid infrastructure. Consider factors such as the nearest substation's voltage level, the availability of transmission lines, and the grid's stability and capacity to accommodate the additional power generation from the solar PV power plant.
- **Project Size and Capacity**: Determine the size and capacity of the solar PV power plant, including the number of panels, total power output, and expected energy generation. The project's size will impact the required evacuation voltage level to effectively transmit the generated electricity to the grid.
- **Load Demand**: Assess the electricity demand or load requirements of the project and the surrounding area. Consider the power consumption of the solar power plant itself, as well as any additional electrical loads that will be supplied by the grid. This evaluation helps ensure that the chosen evacuation voltage can meet the demand and provide reliable power supply.
- **Voltage Stability and Power Quality**: Evaluate the voltage stability and power quality requirements of the grid and the solar PV power plant. Voltage stability analysis helps ensure that the evacuation voltage level remains within acceptable limits during normal and abnormal operating conditions.
- **Safety Considerations**: Consider safety factors related to the evacuation voltage, including electrical safety regulations, protection mechanisms, and isolation requirements. The chosen evacuation voltage should comply with safety standards and minimize risks associated with electrical shock, equipment damage, and operational hazards.
- **Economic Feasibility**: Assess the economic feasibility of different evacuation voltage options. Consider the cost of grid connection equipment, such as transformers and inverters, as well as any potential grid upgrade requirements. Balancing the costs and benefits of different voltage levels will help determine the most economically viable solution.

Exercise

1. List out the parameters included in an Electrical Feasibility Report for Ground Mount Solar Power Plant.

2. Identify the optimal inverter location for a ground-mount solar power plant.

3. Explain the parameters to be considered While Identifying the Location for Transformers.

Module 17 - Electrical Feasibility Study for Ground Mount Solar Power Plant	Unit 17.1	Installing PV Wire And Conduit For Our Solar System	75	https://www.youtube.com/watch?v=kazwzLyKD2A	
Module 17 - Electrical Feasibility Study for Ground Mount Solar Power Plant	Unit 17.1	Transformer information in solar plant		https://www.youtube.com/watch?v=oeJDOznZyIc	

6. Installation of Solar Rooftop PV Plant

Unit 6.1 Civil, Mechanical and Electrical Installation

Unit 6.2 Installation Supervision, Testing and Commissioning

SGJ/N4125

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Oversee the installation of rooftop solar PV systems,
2. Understand the installation of mounting structures and electrical equipment
3. Ensure the availability and proper use of installation tools,
4. Supervise the testing and commissioning of the solar power plant
5. Able to Test and Commission Differentiate between CAPEX, OPEX/RESCO, Lease, BOOT and Hybrid business models.

UNIT 6.1: Civil, Mechanical and Electrical Installation

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain about Civil, Mechanical and Electrical Installation

Civil, Mechanical and Electrical Installation

The installation of a rooftop solar PV plant involves the proper execution of civil, mechanical, and electrical activities to ensure the safety, reliability, and performance of the system. A Solar Photovoltaic Entrepreneur should understand the installation process, supervise installation activities, ensure the availability of tools and materials, and verify that all equipment is installed according to design specifications and safety standards. This module provides an understanding of the key activities involved in the installation, testing, and commissioning of a rooftop solar PV system

The installation process begins with site preparation and civil works. Before installing the solar modules and associated equipment, it is important to ensure that the roof is structurally suitable and capable of supporting the additional load imposed by the solar PV system. The civil foundation work generally includes the construction of footings, pedestals, or ballast foundations that support the mounting structures. During the casting of civil foundations, proper dimensions, alignment, and curing practices should be followed to achieve the required strength and stability. The foundation must be capable of withstanding wind loads, environmental conditions, and the weight of the solar installation throughout its operational life.

The Solar Photovoltaic Entrepreneur should ensure that all required installation tools and equipment are available at the site before the commencement of work. The availability of appropriate tools improves productivity, installation quality, and worker safety. Commonly used tools include measuring tapes, spirit levels, torque wrenches, drilling machines, impact drivers, crimping tools, cable cutters, multimeters, insulation testers, clamp meters, earth resistance testers, safety harnesses, ladders, and personal protective equipment. Proper maintenance and calibration of these tools are essential to ensure accurate installation and testing.

Once the foundation work is completed, the mounting structures are installed according to the approved design and layout. The mounting structure acts as the support framework for the solar modules and must be installed with proper orientation, tilt angle, spacing, and alignment. During the installation process, all structural members should be securely fastened and checked for mechanical stability. The installer must verify that the mounting structure complies with design specifications and can safely withstand operational and environmental loads. Proper installation of mounting structures helps maximize solar energy generation while ensuring long-term structural integrity.

Following the installation of mounting structures, foundations for major electrical equipment such as inverters, AC Distribution Boards (ACDBs), DC Distribution Boards (DCDBs), batteries, and other accessories are prepared. These foundations should provide adequate support, protection from water ingress, and easy accessibility for operation and maintenance. Equipment placement should consider ventilation requirements, cable routing convenience, safety clearances, and future maintenance activities.

The installation of energy meters is an important activity in rooftop solar projects, particularly for grid-connected systems. Energy meters are used to measure electricity generation, consumption, import, and export. During installation, the meter should be mounted securely and connected according to the utility and regulatory requirements. Proper sealing, wiring, and verification of meter functionality are essential to ensure accurate measurement and billing.

The AC isolator is another important component that provides a safe means of disconnecting the inverter from the electrical system during maintenance or emergencies. The isolator should be installed at an easily accessible location and should be capable of safely handling the operating current and voltage of the system. Proper labeling and identification of the isolator help ensure safe operation and maintenance.

The installation of earthing and lightning protection systems is critical for the safety and protection of the solar PV plant. Earthing provides a low-resistance path for fault currents and helps protect equipment and personnel from electrical hazards. Separate earthing arrangements are generally provided for solar modules, mounting structures, inverters, and lightning arrestors. The lightning arrestor protects the plant from lightning strikes by safely directing surge currents to the ground. Proper installation and periodic testing of the earthing and lightning protection systems help improve system reliability and safety.

Component	Purpose	Typical Installation Location
Solar PV Modules	Convert sunlight into electricity	Rooftop mounting structure
Mounting Structure	Supports solar modules	Roof surface
Inverter	Converts DC power into AC power	Electrical room/wall-mounted location
DC Distribution Board (DCDB)	DC protection and isolation	Near inverter
AC Distribution Board (ACDB)	AC protection and distribution	Near inverter or main LT panel
Energy Meter	Measures generation/import/export energy	Metering location
AC Isolator	Safe disconnection of AC supply	Near inverter
Earthing System	Electrical safety and fault protection	Ground/earthing pit
Lightning Arrestor	Protection against lightning surges	Highest point of installation

Table 6.1.1: Major Components Installed in a Rooftop Solar PV Plant

Tool/Equipment	Application
Measuring Tape	Measuring roof dimensions and spacing
Spirit Level	Checking alignment and level of structures
Torque Wrench	Tightening bolts to specified torque
Drilling Machine	Drilling holes for mounting installation
Crimping Tool	Cable lug and connector crimping
Cable Cutter	Cutting DC and AC cables
Multimeter	Voltage and continuity measurement
Clamp Meter	Current measurement
Insulation Resistance Tester	Insulation testing of cables and equipment
Safety Harness	Protection during work at height

Table 6.1.2: Common Tools Used During Installation

UNIT 6.2: Installation Supervision, Testing and Commissioning

Unit Objectives

At the end of this unit, participant will be able to:

7. Explain installation, supervision, Testing and Commissioning

Installation Supervision, Testing and Commissioning

A Solar Photovoltaic Entrepreneur plays an important role in supervising the installation activities and ensuring that all work is performed according to design specifications, safety standards, and quality requirements. During installation, regular inspections should be conducted to verify alignment, cable routing, equipment placement, torque settings, and workmanship quality. Any deviations from the approved design should be identified and corrected before commissioning. Effective supervision helps reduce installation errors, improve system performance, and ensure long-term reliability.

The electrical installation phase involves the interconnection of solar modules, DC cables, combiner boxes, inverters, ACDBs, protection devices, and grid connection points. Proper cable sizing, cable dressing, termination practices, and protection arrangements should be followed to minimize losses and ensure safe operation. Cable routes should be planned to avoid mechanical damage, excessive bending, and exposure to harsh environmental conditions. All electrical connections must be tightened according to manufacturer specifications and checked for continuity and insulation integrity.

Before energizing the solar PV plant, comprehensive testing activities are carried out on both the DC and AC sides of the system. Testing on the DC side includes visual inspection of modules and wiring, polarity verification, string voltage measurement, continuity testing, insulation resistance testing, and verification of module string configurations. These tests help identify installation defects and ensure that the DC system is operating according to design requirements.

Testing on the AC side includes verification of inverter operation, phase sequence checks, continuity testing, insulation resistance testing, earthing verification, protection device functionality, and synchronization with the utility grid where applicable. The operation of AC isolators, circuit breakers, energy meters, and monitoring systems should also be checked to confirm proper functionality. Any abnormalities observed during testing should be rectified before commissioning.

Commissioning is the final stage of the installation process and confirms that the solar PV plant is ready for operation. During commissioning, the inverter is energized, system parameters are verified, and performance data is recorded. Generation readings, inverter settings, protection functions, communication systems, and monitoring platforms are checked to ensure proper operation. The commissioning process also includes documenting test results, preparing commissioning reports, and obtaining necessary approvals from the client and utility authorities.

After successful commissioning, a final inspection is conducted to verify the overall quality of installation and compliance with applicable standards. The client is provided with system documentation, operating manuals, warranty information, safety instructions, and maintenance guidelines. A properly installed and commissioned rooftop solar PV plant delivers reliable energy generation, improved safety, and long-term performance, thereby ensuring customer satisfaction and project success.

Test Activity	Purpose
Visual Inspection	Check module condition and wiring quality
Polarity Test	Verify correct positive and negative connections
String Voltage Measurement	Confirm proper string configuration
Continuity Test	Verify electrical continuity of conductors
Insulation Resistance Test	Detect insulation faults
Connector Inspection	Check MC4 and cable connections
Earthing Verification	Ensure effective grounding
String Current Measurement	Verify module string performance

Table 6.2.1: DC Side Testing Activities

Inspection/Test	Objective
Inverter Functional Check	Verify inverter operation
AC Voltage Measurement	Confirm output voltage levels
Phase Sequence Check	Ensure correct phase sequence
AC Cable Continuity Test	Verify electrical continuity
Insulation Resistance Test	Check insulation health
Protection Device Testing	Verify MCB/MCCB and protection functions
Energy Meter Verification	Confirm accurate energy recording
Grid Synchronization Check	Verify proper synchronization with utility grid
Monitoring System Verification	Confirm data communication and monitoring
Final Commissioning Report	Document successful plant commissioning

Table 6.2.2: AC Side Testing and Commissioning Checklist

7. Installation of Solar Ground Mounted PV Plant

Unit 7.1 Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant
Unit 7.2 Manage Ground Mounted Solar Civil Construction work
Unit 7.3 Interpret drawings of DC and AC side of electrical Component of Ground Mount Solar Power Plant
Unit 7.4 Manage Electrical installation in Ground Mount Solar Plant

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain how to prepare a project development plan.
2. Discuss how to assess soil characteristics in accordance to mounting structure design requirement.
3. Discuss the type of foundation suitable for module mounting structures, inverters, transformers etc. based on the soil test report.
4. Discuss how to prepare structural design report.
5. Discuss how to prepare the layout for cable trenching, Electrical control panel, Transformer, VCB etc.
6. Explain how to prepare Pile foundation, column post and other civil foundation.
7. Explain the design of the pile foundation as per the soil assessment report.
8. Explain how to assess design of the civil/ structural allied works of the solar PV power plant including compound wall/entry gate internal plant roads walkways between different rows of modules water distribution network water drainage system, etc.
9. Explain the importance of soil test report for designing the Mounting structure.
10. Discuss civil/ structural drawings for the ground mount solar power plant.
11. Discuss the reinforcement material for civil construction of various foundation.

UNIT 7.1: Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to prepare a project development plan.

7.1.1 Project Development Plan

A project development plan is a crucial instrument for assuring a project's success. It offers a schedule that details the major checkpoints, available materials, budget, and project-related hazards, as well as the duties and obligations of the project team. The project development plan makes sure that all stakeholders are aligned and working towards a single aim by establishing the project's scope and objectives. It assists in reducing risks and eliminating potential impediments while also ensuring that the project is done effectively, efficiently, and within budget.

A project development plan is an essential instrument for assuring a project's success. It provides a roadmap that details the major checkpoints, available materials, budget, and project-related hazards, as well as the duties and obligations of the project team. The project development plan makes sure that all stakeholders are aligned and working towards a single aim by establishing the project's scope and objectives. It assists in reducing risks and eliminating potential impediments while also ensuring that the project is done effectively, efficiently, and within budget.

In order to accomplish the project successfully, efficiently, and within budget, it is necessary to prepare a project development plan for a ground-mount solar power plant. The following processes can be used to create a project development plan for a ground-mounted solar power plant:

Creating a Project Development Plan

- Define the Project Scope
- Conduct a Site Assessment
- Develop a Project Timeline
- Identify Project Resources
- Define Roles and Responsibilities
- Assess Project Risks
- Develop a Budget
- Create a Work Plan
- Establish Communication Plan
- Define Quality Standards
- Obtain Necessary Permits and Approvals
- Procure Equipment and Materials
- Construct the Solar Power Plant
- Commission the Solar Power Plant
- Monitor and Maintain the Solar Power Plant

Fig 15.1.1. Steps to Create Project Development Plan

- **Define the project**: The first step is to define the scope and objectives of the project. This includes identifying the purpose of the solar power plant, such as generating clean energy for a specific community or business, and determining the desired outcomes, such as reducing energy costs or carbon emissions.
- **Develop a project timeline**: A project timeline should be created to outline the key milestones and deadlines for the project. For a ground mount solar power plant, this may include securing permits, procuring materials and equipment, completing construction, and commissioning the plant.
- **Identify project resources**: Resources needed for the project should be identified, including personnel, materials, equipment, and budget. This includes identifying a project manager, engineers, electricians, construction workers, and any necessary equipment and materials.
- **Define roles and responsibilities**: The roles and responsibilities of the project team should be clearly defined. This includes the project manager, stakeholders, external partners, and subcontractors. The project manager should be responsible for overseeing the project and ensuring that all team members are fulfilling their duties.
- **Assess project risks**: Risks associated with the project should be assessed and contingency plans should be developed to address potential obstacles. For a ground mount solar power plant, potential risks may include changes in regulations or permit requirements, weather-related delays, or equipment failures.
- **Develop a budget**: A budget should be created that outlines the costs associated with personnel, materials, equipment, and any external partners. The budget should be regularly reviewed and updated to ensure that the project remains within budget.
- **Create a work plan**: A detailed work plan should be created that outlines the steps required to complete the project, including timelines and milestones. This should include a detailed construction plan, testing and commissioning plan, and a plan for ongoing maintenance and monitoring.
- **Establish communication plan**: A communication plan should be established to ensure that all stakeholders are kept informed of project progress and any changes. This should include regular project meetings, status reports, and a plan for communicating with external partners and regulatory agencies.
- **Define quality standards**: Quality standards should be defined for the project, including testing and evaluation requirements and quality control procedures should be established to ensure that the project meets these standards. For a ground mount solar power plant, this may include inspections and testing of the solar panels, mounting hardware, and electrical equipment.
- **Monitor and evaluate project progress**: Project progress should be monitored and evaluated regularly to ensure that the project is on track and meeting its objectives. This includes regular site inspections, testing and commissioning of equipment, and ongoing maintenance and monitoring of the plant. Any necessary adjustments should be made to the project plan to ensure that the project is completed successfully.

Note:
To know more about Planning and Building Solar Farm, you may refer to the following link:
https://www.youtube.com/watch?v=C2xYV57HVX8

Title of Video:	Planning and Building a Solar PV Farm
By:	responsAbility Investments AG
Duration:	2 m 27 sec

7.1.2 Assess Soil Characteristics

Soil characteristics refer to the physical and chemical properties of soil that influence its behaviour and suitability for different applications. Some examples of soil characteristics include:

- **Composition**: The composition of soil refers to the types and amounts of mineral particles, organic matter, and other materials present in the soil. The following figure represents the soil composition:

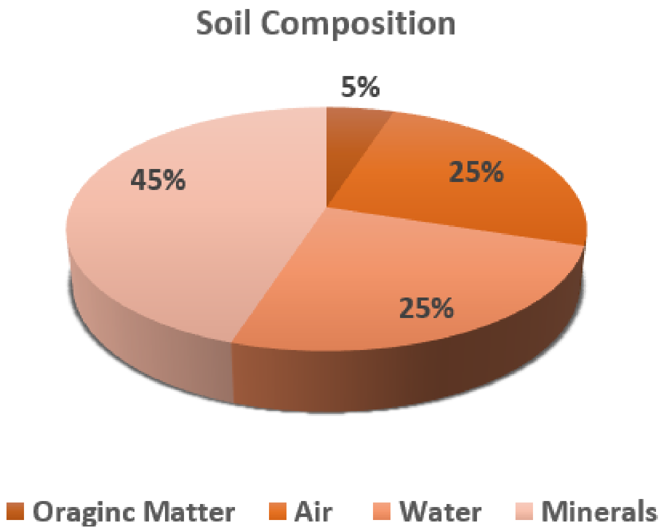


Fig 15.2.1. Soil Composition by Volume

- **Texture**: Soil texture refers to the size distribution of mineral particles in the soil, such as sand, silt, and clay. The texture is important because it influences:

Amount of water the soil can hold

Rate of water movement through the soil

How fertile the soil is

Fig 15.2.2.: Soil texture influences

- **Structure**: Soil structure refers to the arrangement of soil particles into aggregates or clusters. The following figure represents some of the different types of soil structures:

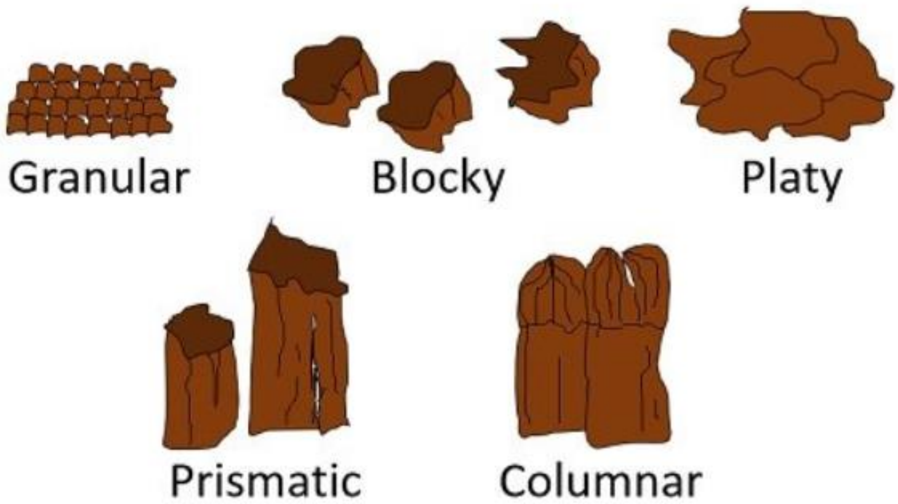


Fig 15.2.3. Types of Soil Structure

- **Granular**: In this structure, individual soil particles are not bonded together and exist independently. This type of structure is common in sandy soils where the individual particles are relatively large and do not stick together.
- **Blocky structure:** In this structure, soil particles are arranged into large, angular blocks or clods. This type of structure is common in well-aggregated soils with good drainage, and it can allow for easy water and air movement through the soil.
- **Platy structure:** In this structure, soil particles are arranged in thin, flat layers or plates. This type of structure is common in compacted soils and can limit the movement of water and air through the soil.
- **Prismatic structure:** In this structure, soil particles are arranged into vertical columns or prisms. This type of structure is common in soils with a high clay content and good drainage.
- **Columnar structure:** It is also known as peds or columns, is a type of soil structure where the soil particles are arranged vertically in columns or pillars. These columns are separated by vertical cracks or fissures, which give the soil a distinctive appearance. It is formed due to the drying and wetting of the soil. It is usually found in arid or semi-arid regions where there is limited water and high evaporation rates.

- **Soil porosity**: It refers to the amount and distribution of pores or spaces between soil particles, which can affect water and air movement through the soil.

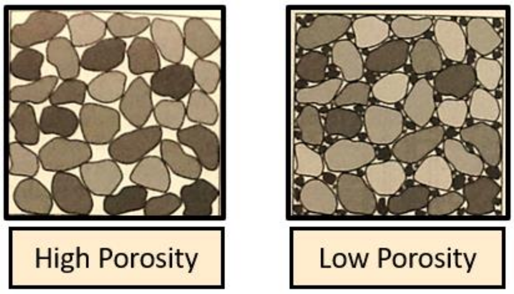


Fig 15.2.4. Soil Porosity

- **Permeability**: Soil permeability refers to the rate at which water can move through the soil. The following figure represents the duration of water drainage across different soil types.

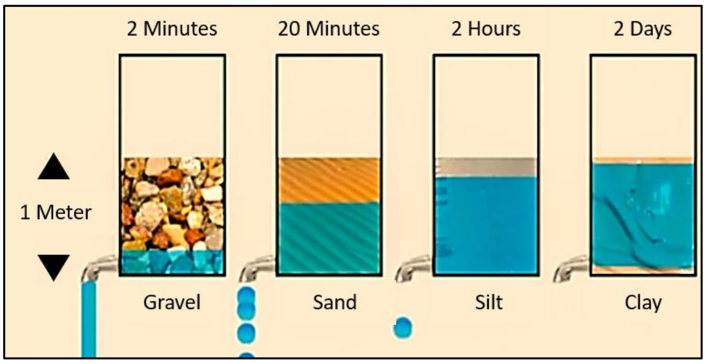


Fig 15.2.5. Soil Permeability

- **Bearing capacity**: Soil bearing capacity refers to the ability of soil to support a load without excessive deformation or failure. The following table represents bearing capacity of different types of soil:

Types of Soil	Bearing Capacity (Kg/m²)
t, Wet Clay or Muddy Clay	5000
t Clay	10000
e Loose and Dry Sand	10000
ck Cotton Soil	15000
ist Clay and Sand Clay Mixture	15000
se Gravel	25000
dium Clay	25000
dium, Compact and Dry Sand	25000
mpact Clay	45000
mpact Sand	45000
mpact Gravel	45000
t Rocks	45000
ninated Rock Such as Sand Stone and Lime Stone	165000
rd Rock such as Granite, Diorite, Trap	330000

Fig 15.2.6. Soil Bearing Capacity

- **Shear strength**: Soil shear strength refers to the ability of soil to resist sliding or failure along a plane.

Assessing soil characteristics is an important step in determining the appropriate mounting structure for a ground-mount solar power plant. Following are the steps to assess soil characteristics in accordance with mounting structure design requirements:

- **Determine the soil characteristics**: To ascertain the properties of the soil, such as its composition, bearing capacity, and water content, a soil test should be performed. A trained geotechnical engineer should conduct the soil test; the expert will collect soil samples at various depths and analyse the samples in a lab.
- **Determine the design requirements**: Based on the site conditions and the specifications, the design requirements for the mounting structure should be established. This comprises the dimensions and weight of the solar panels, the mounting structure's height and spacing, and the wind and snow loads to which it will be subjected.
- **Analyse soil test results**: The findings of the soil test should be examined to identify the soil's bearing capacity, shear strength, and compressibility. The mounting structure's foundation will be designed using this information.
- **Select the appropriate foundation type**: Based on the soil test results and the design requirements, the appropriate foundation type should be selected.
- **Determine foundation dimensions**: The design specifications and the findings of the soil test should be used to calculate the foundation's dimensions. This comprises the foundation's width and depth as well as any necessary reinforcing.
- **Verify foundation design**: The foundation design should be verified by a qualified structural engineer to ensure that it meets the design requirements and will provide adequate support for the mounting structure.

7.1.3 Type of Foundations

Following are the different types of foundations for ground mounting structures at solar plants:

- **Concrete foundations:** The weight of the solar panel system is supported by a form of foundation called a concrete foundation that uses reinforced concrete. Typically, it is made up of a number of piers or footings that are built to withstand the weight of the solar panels and disperse it to the earth below. Due to its strength, longevity, and low maintenance needs, concrete foundations are a common choice for ground-mounted solar power systems.

Fig 15.2.7. Concrete Foundation

- **Steel Foundations**: Another common option for ground-mounted solar power plants is steel foundations, commonly referred to as driven piles. They are pushed deeply into the ground to provide stable support for the solar panel system, making them particularly well-suited for locations with soft or unstable soil conditions.

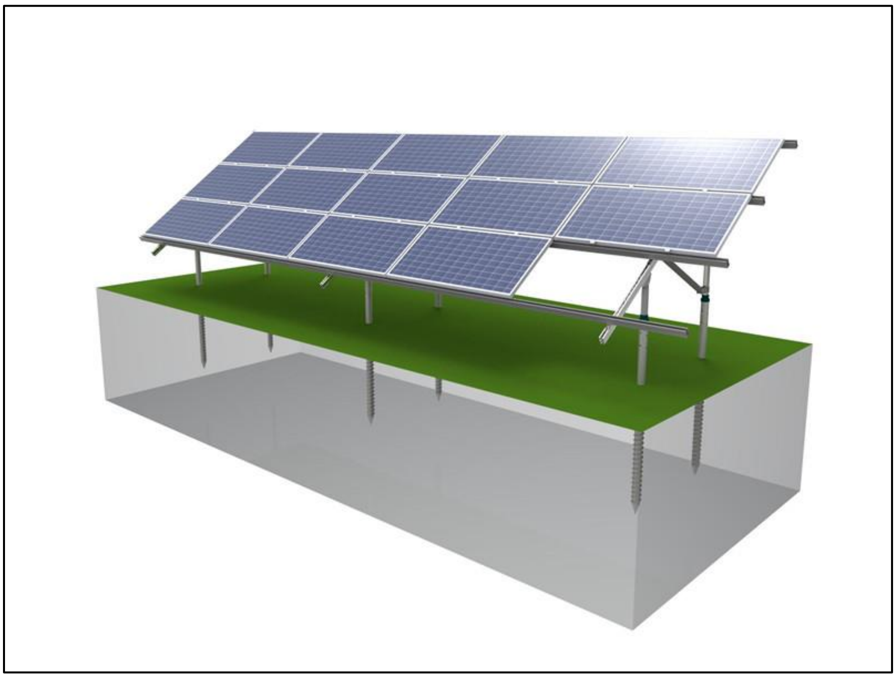


Fig 15.2.8. Steel Foundation

- **Ballasted Foundations**: Ballasted foundations are essentially large blocks of concrete or other heavy materials that are used to weigh down the solar panel system. They are typically used on sites where excavation is difficult or prohibited, such as on rooftops or in areas with high water tables.

Fig 15.2.9. Ballasted Foundation

- **Screw Pile Foundations**: Screw pile foundations are a relatively new technology that involves driving helical screw-like piles into the ground to support the solar panel system. They are particularly suitable for sites with hard soil conditions, as they require less excavation and can be installed quickly.

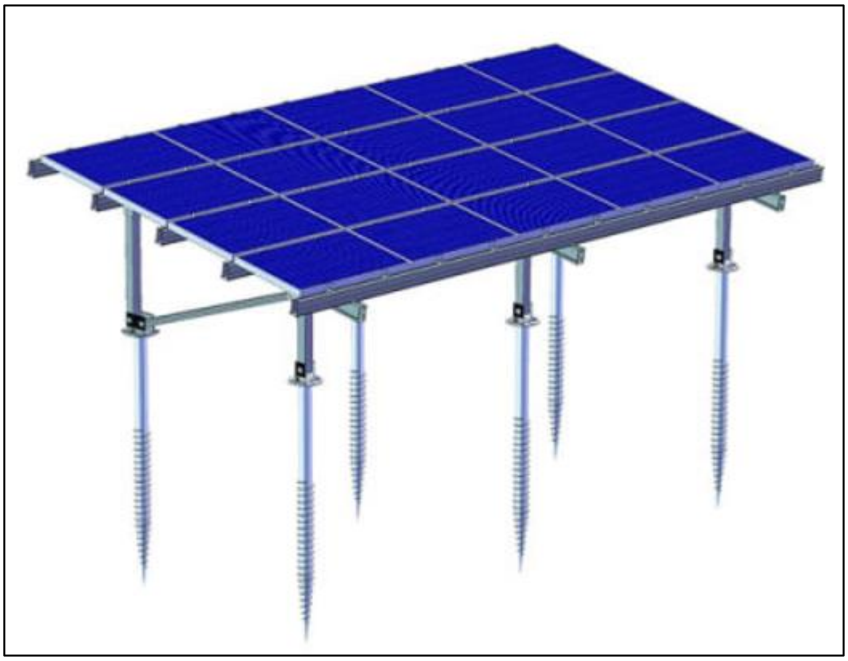


Fig 15.2.10. Screw Pile Foundation

Type of foundations for inverters and transformers:

- **Concrete pad foundation**: Transformers and inverters at solar projects frequently have concrete pad foundations. It is made out of a flat slab of concrete that is poured over a compacted subgrade. The equipment's weight and size, the soil conditions, and the required load will all affect the pad's thickness and dimensions.
- **Pier foundation:** In this type of foundation, the equipment is supported by vertical columns or piers that are buried in the earth. The size and spacing of pier foundations, which can be made of steel or reinforced concrete, will be determined by the weight and size of the equipment, the soil conditions, and the required load.
- **Steel frame foundation**: A steel frame foundation is a particular kind of foundation that supports the machinery using a steel frame. The size and spacing of the frame will depend on the weight and size of the equipment, the soil conditions, and the load requirements. The frame can be attached to the ground by driven piles or concrete footings.
- **Trench foundation**: A small trench is dug in the earth and filled with concrete to serve as the foundation for the apparatus. The size and weight of the equipment, the soil characteristics, and the required load requirements will all affect the trench's depth and width.

7.1.4 How to Prepare Structural Design Report

Every ground-mounted solar plant project needs to have a structural design study. By offering a thorough study of the structural components, foundation design, and connecting details, it plays a significant part in assuring the safety and stability of the solar plant. The study enables us to make sure that the solar plant is capable of withstanding the various environmental stresses, such as wind, snow, and seismic activity. A well-written structural design report also guarantees adherence to regional laws and regulations, which is crucial for the project's success. It offers a thorough understanding of the structural design, empowering stakeholders to make defensible choices about the project's sustainability, spending plan, and schedule.

To ensure that the final report is precise, thorough, and complies with the project criteria, it is important to follow a number of critical procedures while creating a structural design report for a ground-mounted solar plant. The following procedures can help in creating a structural design report for a solar power plant that is ground-mounted:

- **Gather project information**: The first stage in creating a structural design report for a solar power plant that is ground-mounted is to compile all the relevant project data. This contains site plans, geotechnical reports, environmental reports, architectural drawings, and any other pertinent data pertaining to the project.
- **Analyse the structural loads**: The next step is to analyse the structural loads that the solar plant will be subjected to. This includes dead loads (the weight of the solar panels, racking systems, and foundation), live loads (people and equipment), wind loads, seismic loads, and any other applicable loads.
- **Determine the design criteria**: Based on the project information and structural load analysis, the designer should determine the design criteria, including the material and size of the structural elements, connection details, and any special requirements that need to be incorporated into the design.
- **Develop the structural design**: The structural design can then be developed by the designer using manual calculations or structural analysis tools. The design must include all required information, such as the placement and dimensions of structural components, reinforcing information, and connection information.
- **Prepare the report**: The final step is to prepare the structural design report. The report should have a table of contents, a cover page, an introduction, design criteria, a structural design, the findings of the analysis, and any recommendations or unique considerations. Along with any relevant designs or drawings, the report should also include information on the foundation design for the solar panels, racking systems, inverters, and transformers.

- **Consider local codes and regulations**: It's important to consider any local codes and regulations that may impact the design of the solar plant, such as building codes, zoning regulations, and environmental requirements.

7.1.5 Preparing the Layout for Cable Trenching, Electrical Control Panel, Transformer, and VCB

Preparing the Layout for Cable Trenching in solar PV Plant

Designing the layout for cable trenching in a solar PV (photovoltaic) plant involves specific considerations to efficiently accommodate the solar panels, inverters, and associated electrical components. Here's a comprehensive guide to preparing the layout for cable trenching in a solar PV plant:

1. **Site Survey and Assessment:**
    - Understand the terrain, soil type, and drainage considerations for trenching.
    - Identify the optimal location for solar arrays, considering sunlight exposure, shading, and future expansion.
2. **Solar Panel Array Layout:**
    - Plan the arrangement of solar panels, ensuring maximum exposure to sunlight and minimal shading.
    - Design rows and configurations that allow for efficient cable routing from panels to inverters.
3. **Inverter Placement:**
    - Position inverters strategically, considering proximity to solar panels while maintaining adequate ventilation and cooling.
    - Plan for easy access for maintenance and repair of inverters.
4. **Trenching Design:**
    - Determine the routes for cable trenching, ensuring they align with the solar panel rows and inverter locations.
    - Specify trench depth and width based on the cables' specifications, considering the environmental conditions and local regulations.
5. **Cable Routing:**
    - Plan the paths for cables from the solar arrays to the inverters, ensuring organized and protected routing.
    - Use cable trays, conduits, or buried conduits within the trenches to protect the cables and enable future maintenance.
6. **Safety and Compliance:**
    - Ensure compliance with local building codes, safety standards, and regulations for trenching and cable installation.
    - Include covers, warning signs, and safety barriers for the trenches to prevent accidents.

7. **Grounding and Bonding:**
    - Design grounding systems for solar arrays and electrical components to mitigate electrical hazards and ensure system safety.
    - Plan for bonding connections in the trenching layout as required by electrical codes.
8. **Accessibility and Maintenance:**
    - Design the trench layout to allow easy access for cable installation and future maintenance.
    - Incorporate access points or junction boxes along the trench route for connection points and troubleshooting.
9. **Future Expansion Consideration:**
    - Plan the trench layout with future expansion in mind, leaving space for additional cables or equipment.
10. **Documentation and Labels:**

- Label cables, junction boxes, inverters, and other components for easy identification and maintenance purposes.
- Create detailed documentation of the trench layout, including cable types, depths, and routing information.

Layout of Cable trenching for a 1MW Solar Plant?

DC Cabling

- The power generated from the solar modules is transferred to String Combiner Box (SCB) / Junction Box and from there it is transferred to Inverter input. This total current is in the form of Direct Current only. So the cabling process is known as DC cabling.

  Majorly it is divided into two parts
  i)Cabling from SCB to Inverter input Side.
  ii)Cabling from PV Modules to Junction box or SCB

Cabling from SCB to Inverter Input Side

Study the Line Diagram clearly and identify the locations where exactly the SCB need to be installed and make a path for DC cables 240sq.mm from SCB to Inverter. Make marking to prepare a trench for Cabling, which should 3 ft X 3ft in dimensions. Now measure the distance from junction boxes to Inverter room and cut the cable and pipes accordingly. We need two cables +Ve and –Ve from each Junction box. Insert the cables in HDPE pipes and place them in the trench. Use fuming paste to close the pipe to avoid entering of rain water or mud into the pipe. Now fill the trench with sand up to to half feet and then place bricks or marbles on it and cover the total with agriculture sand.

Instead of making all this we can adopt another method for trenching make 1ft X 1ft or 1ft X 2ft trench depending upon the no of cables passing through and prepare a cannel for the cables using bricks and close the cannel by marble stones. So, it would be easy if we have problem in future. Instead of digging the trench we can open the trench whenever we required.

Now take the measurements of the SCB and prepare a structure or frame to fix the junction box. To prepare a frame take L angle or C angle steel bars and cut them according to the dimensions. Now make the markings where the nut and bolts need to be placed and drill a hole a exact marking locations. After completing the drilling operation arrange the pieces into a frame shape by using the nuts and bolts. Now weld the corner of the bars wherever two bars get connected.

Now take the arranged frame to the locations and make two feet depth holes and place the frame in it. Fill the holes with concrete up to ground level. While filling concrete make sure the structure is having the required and angles are in correct position. To confirm use sprit level and measurements. Make a pile cap or pedestal on surface of the earth at poles. Now fix the SCB/Junction box to the frame and make connections for DC cables in junction box.

Note: Paint SCB structure or frame with GI paint which will protect from rusting and make cable labeling over the trench so that we can understand where exactly the trench. Help in digging there if needed.

Designing the layout for an electrical control panel, transformer, and VCB (Vacuum Circuit Breaker) in a solar plant involves careful planning and adherence to safety standards. Here are steps and considerations to create the layout:

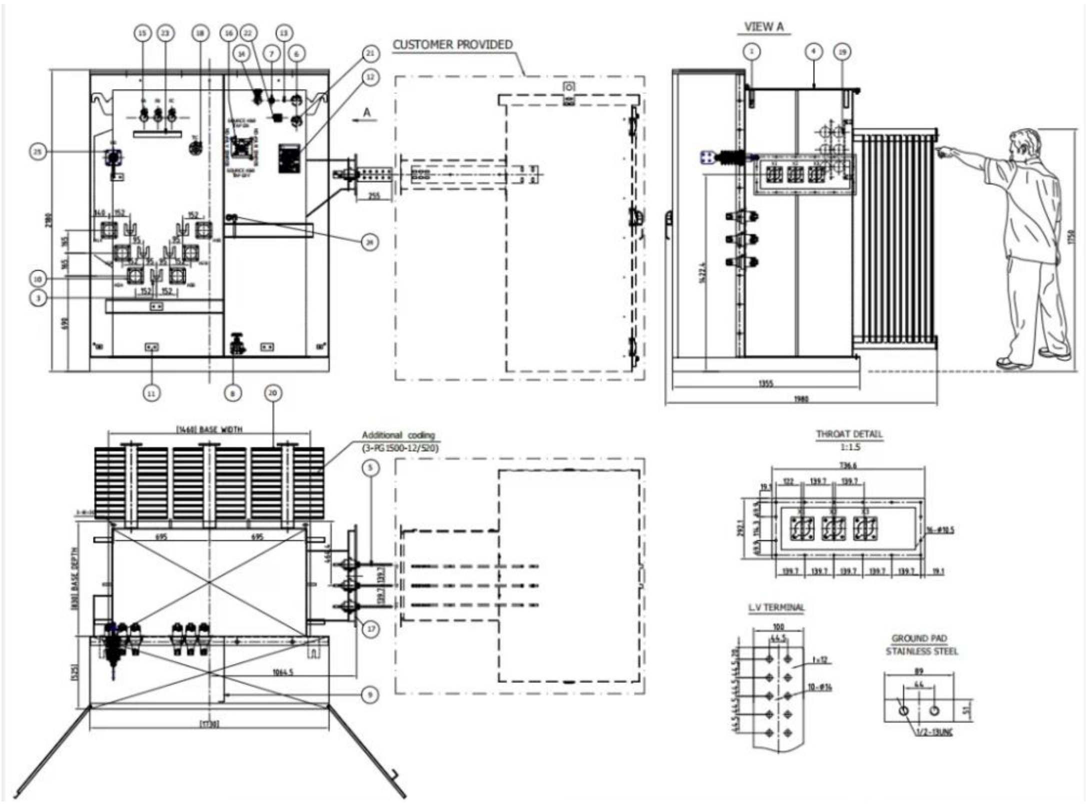
1. **Site Analysis and Requirements:**
    - Understand the solar plant's capacity, voltage, and system requirements.
    - Determine the distance between the solar panels, inverters, and the electrical control panel.
    - Assess the space available for installation and compliance with local regulations.
2. **Electrical Control Panel Placement:**
    - Choose a location close to the solar array to minimize wiring distances.
    - Ensure the control panel is easily accessible for maintenance but protected from environmental factors.
    - Provide adequate ventilation and cooling to prevent overheating of electrical components.
    - Plan for adequate space around the panel for future expansions or modifications.
3. **Transformer Placement:**
    - Place the transformer close to the solar array and control panel to minimize losses in transmission.
    - Follow safety codes regarding clearance space, especially for high-voltage equipment.
    - Ensure proper grounding and protection against moisture and extreme weather conditions.
4. **VCB Placement:**
    - Install the VCB in a location that allows easy access for maintenance and operation.
    - Keep it close to the transformer or the point of connection to the grid.
    - Ensure appropriate clearances and safety measures according to the VCB's specifications.
5. **Layout Design:**
    - Create a detailed layout diagram using CAD software or similar tools.
    - Include the positioning of the control panel, transformer, and VCB in the layout.
    - Plan cable routing paths, ensuring clear segregation of low-voltage and high-voltage lines.
    - Use cable trays or conduits to organize and protect wiring.
    - Label components and provide clear identification for easier maintenance and troubleshooting.
6. **Safety Considerations:**
    - Ensure compliance with local electrical codes and safety standards.
    - Implement safety features such as emergency shutdown procedures and warning signs.
    - Install barriers or enclosures where necessary to prevent unauthorized access.

7. **Consultation and Approval:**
    - Review the layout with electrical engineers and relevant stakeholders for input and approval.
    - Obtain necessary permits and approvals from local authorities before installation.
8. **Installation and Testing:**
    - Follow the approved layout plan during installation, adhering to safety protocols.
    - Conduct thorough testing to ensure proper functioning and compliance with specifications.
9. **Documentation:**
    - Maintain detailed documentation of the layout, installation, and testing procedures.
    - Keep records of equipment specifications, manuals, and any modifications made.
10. **Regular Maintenance:**

- Schedule regular inspections and maintenance to ensure the efficient operation of the electrical components.
- Update the layout documentation as changes or upgrades are implemented.

The Parameters for Transformer Selection

1. Rated output kVA
* Based on the inverter output rating, load diagrams and the harmonics, patterns of weather, and unusual service conditions
2. Nominal Voltage:
* Designed specifically for the Inverter Transformer to work with the Pulse voltage form of Inverters
3. Diagram of the connection between winding and vector group:
* Inverter operation isn't dependent on the group of vectors like Dy1, Dy5 and Dy11
* There is no Neutral requirement on the primary Side of LV
* Neutral point is isolated on the secondary HV side transformer
4. Electronic Shield (ES):
* Recommendation to install ES between the primary and secondary windings in order to limit the possibility of transmission in high frequency disturbances (harmonics or surges, pulses, etc.)) from the primary winding to the secondary
* Windings normally connected to circuits inverters are not rounded.



7.1.6 Preparing Pile Foundation, Column Post and Other Civil Foundation

Preparing Pile Foundation

A skilled engineer should be consulted to make sure the design complies with all relevant regulations and standards. A stable and long-lasting platform for the solar panels can be provided by a properly installed pile foundation, assuring their best performance and lowering maintenance needs.
Preparing a pile foundation for a ground-mounted solar plant installation involves several steps:

- **Site Evaluation**: Conduct a site survey to assess the terrain and soil characteristics. This survey should include soil testing to determine the soil's bearing capacity, depth to bedrock, and other geotechnical parameters.

- **Pile Selection**: Select the type of pile to be used based on the site's soil conditions, foundation loads, and other factors. Common pile types used for solar plant installations include steel H-piles, concrete piles, and screw piles.

- **Pile Design**: Design the pile foundation based on the site's soil characteristics, size of the solar panels, and other factors. The foundation should be designed to support the loads imposed by the solar panels and distribute them evenly over the soil.

- **Pile Installation**: Install the piles using the appropriate equipment, such as a pile driver or drilling rig. The piles should be driven or drilled to the required depth to ensure adequate support.

- **Pile Cap**: Install a pile cap to distribute the loads from the solar panel mounting structure evenly across the pile foundation.

- **Mounting Structure**: Install the solar panel mounting structure on the pile foundation using anchor bolts or other appropriate fasteners.

- **Electrical Grounding**: Ground the mounting structure and solar panels to ensure electrical safety.

Assessing Pile Foundation as per the soil Assessment Report

Assessing the pile foundation for a ground-mounted solar power plant is crucial to ensure the safety, durability, and optimal performance of the solar panel system. The foundation must be designed and installed to withstand the loads imposed by the solar panels, including wind and seismic loads. A well-designed foundation can also minimize settling and maintain the proper orientation and tilt angle of the solar panels, maximizing their energy output. Assessing the pile foundation can help identify potential risks and issues early on, allowing for corrective actions to be taken to mitigate them. This can help avoid costly repairs and retrofitting and ensure compliance with applicable codes and standards. By investing in the assessment and design of the pile foundation, the solar power plant can operate safely and efficiently, providing clean and sustainable energy for years to come.

To assess a pile foundation based on a soil assessment report, you will need to evaluate the report to determine the soil's bearing capacity and other geotechnical parameters. Here are the steps to follow:

- **Review the soil assessment report**: Review the soil assessment report provided by a geotechnical engineer. The report should include details on the soil conditions at the site, such as soil type, depth to bedrock, and groundwater level. Additionally, it should include soil test results to determine the soil's strength, stiffness, and other relevant parameters.
- **Determine the bearing capacity:** Based on the soil test results, determine the soil's bearing capacity. This is the maximum load that the soil can support per unit area without experiencing excessive settlement or shear failure.
- **Determine the pile type and size**: Based on the soil assessment report and the loads to be supported, determine the appropriate pile type and size. The type of pile selected should be capable of transferring the loads from the solar panel mounting structure to the soil, and the pile size should be sufficient to support the expected loads.
- **Check the pile design**: Review the pile design provided by the geotechnical engineer to ensure that it meets the applicable codes and standards. This design should specify the pile length, diameter, and spacing, as well as the required pile cap and anchor bolts.
- **Evaluate the installation**: Review the installation procedures and verify that they are consistent with the design specifications. Ensure that the piles are installed to the required depth and that the pile cap and anchor bolts are properly installed.

These steps can be used to evaluate a pile foundation using a soil assessment report to ensure it offers a sturdy and long-lasting basis for the solar panel mounting system. A trained engineer should be consulted to help with this process.

Preparing Column Post

Preparing column posts for a ground-mounted solar plant involves several steps:

- **Column Selection**: Choose the appropriate column size, material, and thickness based on the expected loads and environmental conditions at the site. The column should be strong enough to support the weight of the solar panels and resist wind and seismic loads.

- **Column Placement**: Determine the location and spacing of the columns based on the size of the solar panels and the row spacing. The columns should be spaced evenly and placed in a straight line to ensure the solar panels are properly supported.

- **Foundation Preparation**: Prepare the foundation for the column posts. The foundation should be designed to distribute the loads evenly and prevent settling. Depending on the soil conditions at the site, the foundation may require excavation, soil stabilization, or other measures to ensure stability.

- **Column Mounting**: Mount the column posts to the foundation using anchor bolts or other appropriate fasteners. The columns should be installed plumb and level to ensure the solar panels are properly supported and oriented.

- **Bracing**: Install diagonal bracing between the column posts to provide additional support and stability. The bracing should be installed at the appropriate angles and tensioned to ensure it is effective.

- **Electrical Conduits**: Install electrical conduits between the column posts to route the wiring for the solar panels. The conduits should be properly sized and secured to prevent damage and ensure reliable operation.

By following these steps, you can prepare column posts for a ground-mounted solar plant installation, ensuring a stable and durable foundation for the solar panel system. It is recommended to consult with a qualified engineer to assist in the design and installation process.

Preparing Civil Foundations for Ground Mounted Solar Plant

A ground-mounted solar power plant's civil foundations are necessary components of the installation. They support the solar panels, inverters, transformers, and other machinery required to run the facility. The safety and long-term dependability of the solar plant depend on the preparation and construction of the foundations properly. The solar power plant can run safely and effectively for many years in the future by ensuring the foundations are designed and built in accordance with the necessary regulations.

Preparing a civil foundation for a ground-mounted solar plant involves several factors, including the size of the solar panels, soil characteristics, wind loads, and seismic loads. Conducting a site survey to assess the terrain and soil characteristics is a crucial first step when preparing civil foundations for a ground-mounted solar plant. This survey should include soil testing to determine the soil's bearing capacity and its ability to support the weight of the foundation and solar panels. Additionally, it is important to determine the design criteria for the foundation, including the maximum wind speed and seismic loads the foundation must withstand. Based on these factors, the foundation should be designed to distribute the weight of the solar panels evenly over the soil to minimize settlement. Careful consideration of these design factors can ensure that the foundation provides a stable and secure base for the solar panels, maximizing their efficiency and lifespan.

Following are the general task involved in the preparation of civil foundations:

- **Excavation**: Excavate the area for the foundation to the required depth and shape.

- **Rebar Placement**: Place the reinforcing steel (rebar) in the excavated area according to the design specifications.
- **Concrete Pour**: Pour the concrete into the foundation area and allow it to cure. The concrete should be of the required strength and consistency to provide the necessary support.
- **Anchor Bolts**: Install anchor bolts in the foundation to secure the solar panel mounting structure.
- **Mounting Structure**: Install the mounting structure on the foundation using the anchor bolts.
- **Electrical Grounding**: Ground the mounting structure and solar panels to ensure electrical safety.

Note:
To know more about Civil Foundations for Ground Mounted Solar Plant, you may refer to the following link:
https://www.youtube.com/watch?v=micjDafdy1w

Title of Video:	CIVIL WORKS AND FOUNDATIONS FOR SOLAR PLANTS
By:	Green JP Consultants
Duration:	4 m 31 sec

7.1.7 Assess Design of the Civil/ Structural Allied Works of the Solar PV Power Plant

Assessing the design of the civil/structural allied works of a ground-mounted solar PV power plant is crucial to ensure that the infrastructure is safe, reliable, and meets the project requirements. Here are some guidelines and assessment criteria for various components:

- **Compound Wall/Entry Gate**: The compound wall and entry gate should be designed to provide security and restrict access to unauthorized personnel. The assessment criteria for this component would include the materials used, structural design, height, and aesthetics. The wall should be high enough to prevent unauthorized access, but not so high that it causes shading of the solar panels.

  The following figure represents the assessment criteria for assessing Compound Wall/Entry Gate:

Materials	• The materials used for the compound wall and entry gate should be durable, able to withstand weather conditions, and provide adequate security.
Structural Design	• The design should ensure stability and resistance to external forces such as wind and seismic loads.
Height	• The height of the wall should be sufficient to prevent unauthorized access but not so high that it causes shading of the solar panels.
Aesthetics	• The design should be visually appealing and match the overall aesthetics of the solar plant.

Fig.15.3.1: Assessment Criteria for Assessing Compound Wall/Entry Gate:

- **Internal Roads of solar plant**: Internal roads should be designed to provide safe and efficient access for vehicles and personnel. The assessment criteria for this component would include road width, surface materials, drainage, and safety features such as speed bumps and signage. The roads should be wide enough for vehicles to pass safely, with a surface that provides good traction and allows for efficient rainwater drainage. The following figure represents the assessment criteria for assessing the internal roads of the solar plant:

Road width:	The width should be sufficient to allow vehicles to pass safely, especially if there is a two-way traffic flow.
Surface materials:	The surface should provide good traction and allow for efficient rainwater drainage to prevent flooding and slipperiness.
Drainage:	Proper drainage should be in place to prevent water accumulation on the road surface and erosion.
Safety features:	The design should include safety features such as speed bumps and signage to ensure safe vehicle movement within the plant.

Fig.15.3.2: Assessment Criteria for Internal Roads of Solar Plant

- **Walkways between Different Rows of Modules**: Walkways between different rows of modules should be designed to provide safe and easy access for maintenance personnel. The assessment criteria for this component would include walkway width, surface materials, slope, and safety features such as handrails. The walkways should be wide enough for personnel to move freely, with a surface that provides good traction and prevents slipping. The following figure represents the assessment criteria for assessing the walkways between different rows of modules:

Walkway Width	• The width should be sufficient to allow maintenance personnel to move freely and safely.
Surface Materials	• The surface should provide good traction and prevent slipping, especially in wet conditions.
Slope	• The walkway should have a slope to prevent water accumulation and ensure efficient rainwater drainage.
Safety Features	• The design should include safety features such as handrails and signage to ensure safe movement and prevent accidents.

Fig.15.3.3 Assessment Criteria for Assessing the Walkways Between Different Rows of Modules

- **Water Distribution Network**: The water distribution network should be designed to provide reliable and efficient water supply for the plant. The assessment criteria for this component would include pipe material, diameter, pressure rating, and flow rate. The pipes should be made of materials that can withstand harsh environmental conditions and pressure, and the flow rate should be sufficient to meet the plant's water needs. The following figure represents the assessment criteria for assessing the water distribution network:

Pipe material:	The pipes should be made of materials that can withstand harsh environmental conditions, prevent corrosion, and provide adequate water supply.
Diameter:	The diameter of the pipe should be sufficient to meet the water demands of the plant.
Pressure rating:	The pipe should have a pressure rating that can handle the pressure of the water supply system.
Flow rate:	The flow rate should be sufficient to meet the plant's water needs and ensure efficient water distribution.

Fig.15.3.3: Assessment Criteria for Assessing the Water Distribution Network

- **Water Drainage System**: The water drainage system should be designed to prevent water accumulation and ensure proper drainage. The assessment criteria for this component would include slope, drainage materials, and capacity. The slope should be designed to allow for efficient rainwater drainage, with drainage materials that can withstand harsh environmental conditions and prevent clogging. The following figure represents the assessment criteria for assessing the water drainage system:

Slope:	• The slope should be designed to allow for efficient rainwater drainage and prevent water accumulation.
Drainage Materials:	• The drainage materials should be able to withstand harsh environmental conditions and prevent clogging.
Capacity:	• The drainage system should have the capacity to handle the maximum amount of rainfall and prevent flooding.

Fig.15.3.4: Assessment Criteria for Assessing the Water Drainage System

When assessing the design of civil/structural allied works of a ground-mounted solar PV power plant, it's important to consider the local environmental conditions, construction materials, and applicable regulations and standards. It's recommended to engage with qualified engineers and contractors to ensure that the design and construction meet the required standards and specifications.

7.1.8 Civil/ Structural Drawings for the Ground Mount Solar Power Plant.

As an entrepreneur considering civil/structural drawings for a ground-mounted solar power plant, here's a more focused perspective on what you might need:

1. Site Evaluation and Selection:
    - Identify potential sites for your solar power plant. Consider factors like land availability, proximity to grid connections, sunlight exposure, and any regulatory restrictions.
2. Feasibility Study:
    - Conduct a feasibility study to analyse the economic viability of the project. Evaluate costs, potential energy production, return on investment (ROI), and financing options.
3. Engaging Design and Engineering Professionals:
    - Collaborate with experienced civil and structural engineers specialized in solar power plant design. These professionals will help determine the best layout, structural requirements, and foundation designs for the plant.
4. Design Optimization:
    - Focus on optimizing the design to maximize energy production while keeping construction and operational costs reasonable. This might involve innovative layout planning to achieve higher energy output.
5. Cost-Effective Solutions:
    - Work closely with engineers to develop cost-effective structural designs without compromising safety or quality. Optimize the use of materials and construction techniques to reduce expenses.
6. Permitting and Regulations:
    - Understand local regulations and permitting processes. Ensure that your designs comply with building codes, environmental regulations, and utility interconnection standards. This is crucial for successful project implementation.

7. Modular Approach:

    - Consider a modular approach in the design, making it scalable and easy to expand in the future. This can facilitate phased construction and accommodate future growth.

8. Efficient Project Management:

    - Implement efficient project management strategies to streamline the construction process, control costs, and meet project timelines.

9. Quality Assurance and Control:

    - Implement quality assurance measures throughout the construction phase to ensure that the plant meets engineering standards and specifications.

10. Environmental Impact Consideration:

    - Incorporate sustainability practices into the project design, such as minimizing land disturbance, managing water runoff, and promoting biodiversity in surrounding areas.

11. Monitoring and Maintenance:

    - Plan for monitoring systems and regular maintenance schedules to ensure optimal performance and longevity of the solar power plant.

For an entrepreneur, balancing cost-effectiveness with quality and efficiency is essential. Collaborating with professionals, understanding the regulatory landscape, and focusing on a sustainable, optimized design will contribute significantly to the success of your ground-mounted solar power plant project.

7.1.9 Discuss the Reinforcement Material for Civil Construction of Various Foundation

Reinforcement materials for civil construction are materials that are utilised to provide buildings built as part of civil construction projects more strength and durability. For the required strength and resistance to various forces, such as tension, compression, bending, and shear, these materials are combined with concrete. Steel bars, steel mesh, and fiber-reinforced polymer (FRP) bars are the most frequently utilised reinforcement materials. The type of structure, the loads it will support, the environmental conditions, and other considerations all affect the choice of reinforcement material. For civil construction projects to be safe and last a long time, reinforcement material must be chosen and placed properly.

The type of foundation and the loads it will support will determine the choice of reinforcement material. Following are some examples:

- **Pile Foundation**: Steel bars are frequently utilised as reinforcement material for piling foundations. Prior to pouring the concrete, these steel bars are inserted into the piling borehole. The steel reinforcement aids in the pile's ability to withstand tensile forces over the course of its lifetime. The size of the pile and the loads it will support will determine the thickness and quantity of steel bars to be employed.
- **Strip Foundation**: Prior to pouring the concrete in the foundation trenches for strip foundations, steel reinforcing is inserted. The reinforcement aids in resisting the tensile forces that earth movement or settlement will subject the foundation to. The foundation's dimensions and the loads it will support determine the steel reinforcement's thickness and spacing.
- **Raft Foundation**: Steel mesh or steel bars are frequently utilised as reinforcement material for raft foundations. To enhance strength and avoid cracking, the reinforcement is inserted inside the concrete raft. The raft size and the loads it will support determine the thickness and spacing of the steel reinforcement.
- **Retaining Wall**: Steel bars are inserted horizontally into retaining walls to act as reinforcement. The steel reinforcement aids in resisting the tensile pressures that soil pressure will cause to act on the

retaining wall. The wall height and the loads it will support determine the thickness and spacing of the steel reinforcement.

In order to guarantee the stability and strength of foundations for civil construction, reinforcement material is crucial. The choice of material, as well as its thickness and spacing, are influenced by the type of foundation, the loads it will support, and the ground conditions. By using the right reinforcement material, civil building may be designed and built to withstand the demands of the environment and the loads it will support over time.

Exercise

1. Explain the importance of soil test report for designing the Mounting structure.

____

____

____

____

2. Explain how to Prepare Structural Design Report.

____

____

____

____

3. Explain the Assessment Criteria for Assessing the Water Drainage System.

____

____

____

____

Module 15 - Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant	Unit 15.1	Planning and Building a Solar PV Farm	https://www.youtube.com/watch?v=C2xYV57HVX8	
Module 15 - Interpret and prepare Civil foundation, layout, mounting structure and allied civil work in ground mount solar plant	Unit 15.2	CIVIL WORKS AND FOUNDATIONS FOR SOLAR PLANTS	https://www.youtube.com/watch?v=micjDafdy1w	

UNIT 7.2: Planning and Scheduling

Unit Objectives

At the end of this unit, participant will be able to:

- Explain how to prepare a project development plan
- Discuss the planning for construction of compounds wall or fencing, access of internal road, drainage
- Explain how to develop the overall project constructability approach for cost effective implementation
- Discuss how to plan, schedule, and monitor all material deliveries to avoid project delays

7.2.1 Project Development Plan for Ground-Mounted Solar Civil Work Construction

Project Development Plan acts as a project road map by defining all the steps that must be followed to accomplish the project's goals. A well-designed project development plan guarantees accurate planning, reduces risks and delays, makes communication easier, enhances project management, and ensures adherence to legal and environmental standards. It helps to guarantee that all facets of the construction process are taken into account and coordinated by creating a clear plan for the project. It serves as a platform for communication between all project participants, encouraging efficient collaboration and coordination. It also offers efficient project control, making it possible to monitor and track progress, identify problems and hazards, and put necessary corrective measures in place. It also aids in identifying any potential environmental effects linked to the construction process as well as ensuring compliance with all pertinent legislation and criteria. In conclusion, a well-designed PDP is essential for the timely and successful completion of any construction project. It helps to keep the project on schedule, within budget, and on pace to achieve its goals.

A Project Development Plan for a ground-mounted solar civil work construction project is created based on the following reports:

- **Site assessment report**: This report should include a detailed analysis of the site, including a summary of the terrain, soil type, water supply, accessibility, and any environmental or regulatory considerations.
- **Site plan**: Based on the information gathered in the site assessment report, a detailed site plan should be developed that outlines the proposed layout for the solar panels, support structures, and electrical wiring.
- **Risk assessment report**: Any potential risks associated with the site should be identified and documented in a risk assessment report. This report should include recommendations for mitigating these risks.
- **Environmental impact report**: If there are any potential environmental impacts associated with the site, an environmental impact report should be developed that outlines these impacts and provides recommendations for mitigating them.
- **Regulatory compliance report**: Any regulatory requirements associated with the site should be identified and documented in a regulatory compliance report. This report should include recommendations for complying with these requirements.

Preparing a Project Development Plan for Ground-Mounted Solar Civil Work Construction involves several key steps such as:

- **Project Scope and Objectives**: Clearly define the scope of the project and the objectives to be achieved. This may include determining the size and capacity of the solar plant, the estimated construction timeline, and the budget available for the project.
- **Site Assessment and Evaluation**: Evaluate the site and identify any potential issues that may impact the construction process. This includes an evaluation of the terrain, soil type, water supply, accessibility, and any environmental or regulatory considerations.
- **Design and Engineering**: Develop detailed design and engineering plans for the solar panels, support structures, and electrical wiring. These plans should be developed in accordance with industry best practices and relevant regulatory requirements.
- **Procurement and Logistics**: Develop a plan for the procurement and logistics of materials and equipment required for the construction process. This includes identifying suppliers, obtaining quotes, and coordinating the delivery of materials and equipment to the construction site.
- **Construction Process**: Develop a detailed plan for the construction process, including site preparation, foundation installation, and electrical wiring. This should include a timeline for each phase of the construction process, as well as any required inspections or permits.
- **Health and Safety**: Develop a health and safety plan that addresses the risks associated with the construction process, including potential hazards related to equipment, site conditions, and environmental factors.
- **Project Management and Control**: Develop a project management plan that outlines the roles and responsibilities of each team member involved in the construction process. This should include a communication plan, a risk management plan, and a plan for project monitoring and control.
- **Environmental Impact Assessment**: Assess the potential environmental impact of the construction process and develop a plan for mitigating any potential impacts.
- **Regulatory Compliance**: Ensure compliance with all relevant regulations and requirements, including permits, inspections, and environmental considerations.
- **Monitoring and Evaluation**: Develop a plan for monitoring and evaluating the project, including tracking progress against the project timeline, identifying issues and risks, and implementing corrective actions as needed.

7.2.2 Construction of Compounds Wall/Fencing, Access of Internal Road, Drainage

Planning for the construction of compounds wall or fencing, access of internal road, and drainage is essential for the success of any construction project, including ground-mounted solar civil work construction. These features are crucial for ensuring the security, safety, and longevity of the solar plant and its equipment. Compounds wall or fencing provides a secure boundary around the solar plant, preventing unauthorized access, theft, and vandalism. Access roads provide safe and easy access for vehicles and personnel to the solar plant, and well-designed internal roads ensure smooth movement of vehicles and equipment on the site. Proper drainage is essential to prevent water accumulation and erosion, which can cause damage to the equipment and infrastructure, and reduce the risk of accidents by providing a safe and dry environment for workers. Planning for these features also ensures regulatory compliance, avoids delays and penalties, and enhances the aesthetics of the solar plant.

Following are some of the key considerations in planning for these features:

- **Compounds wall or fencing**: Consideration should be given to the purpose, design, and materials while planning a compound's wall or fencing. In addition to securing the solar plant's perimeter, the wall or fence should also improve the appearance of the facility and deter theft, vandalism, and unauthorized access. The materials should be strong, resistant to the elements, and simple to maintain.
- **Access of internal road**: The safe and effective movement of cars and equipment on the site should be taken into account while planning the internal roads. All sorts of vehicles and equipment used in the building and upkeep of the solar plant should be able to fit on the width and length of the road. The road surface needs to be sturdy, able to endure large loads and adverse weather, and simple to maintain.

Drainage: The topography and soil characteristics of the site should be taken into account while designing drainage. The drainage system should be made to both keep employees safe and dry while also preventing water accumulation and erosion, which can harm infrastructure and equipment. Guttering, culverts, and catch basins are examples of surface and subsurface drainage elements that should be included in the system.

Note:
To know more about Construction of Solar PV Projects, you may refer to the following link:
https://www.youtube.com/watch?v=sM7ge9ahsGY

Title of Video:	How a Solar Farm is Constructed from Beginning to End
By:	Spark & Foster Films
Duration:	6 m 12 sec

7.2.3 Overall Project Constructability Approach for Cost Effective Implementation

Developing an overall project constructability approach is a critical step in ensuring the cost-effective implementation of any construction project, including ground-mounted solar civil work construction. Following are some steps to develop the overall project constructability approach for cost-effective implementation:

- **Develop a detailed project scope**: Clearly state the project scope and requirements, including design specifications, performance criteria, and construction timeline.
- **Conduct constructability reviews**: To find potential problems and areas for improvement in the project design and construction plan, conduct constructability evaluations. This may entail assessing the design of the site, the order in which construction is carried out, the choice of materials, and other elements that may have an effect on the project's budget and schedule.
- **Identify potential risks and challenges**: Identify any risks or challenges that might affect the project's budget, schedule, and quality. Finding environmental and legal restrictions, site-specific difficulties, and other elements that could have an impact on the construction process are a few examples of this.
- **Develop a cost-effective construction plan**: Create a cost-effective construction plan that includes industry best practices, efficient material and equipment consumption, and an optimised building sequence based on the constructability study and risk assessment. Finding opportunities for value engineering and other cost-cutting techniques should be part of this.

Conduct regular project reviews: Review the construction plan and implementation status frequently to spot problems early, make any adjustments to the plan, and make sure the project is on pace to fulfil its cost, time, and quality goals.

7.2.4 Plan, Schedule, and Monitor All Material Deliveries to Avoid Project Delays

Planning, scheduling, and monitoring material deliveries is a critical aspect of any construction project, including ground-mounted solar civil work construction.

Effective planning, scheduling, and monitoring of material deliveries are essential for any construction project, especially for ground-mounted solar civil work construction. It helps to ensure the timely delivery of required materials to the construction site, avoiding delays in the project schedule. Delayed material deliveries can lead to additional costs such as rescheduling labor and reordering materials. By managing the material delivery process effectively, it is possible to minimize these costs and keep the project within budget. Monitoring the quality of materials is also crucial, as it can impact the long-term success of the project. By having a detailed material delivery plan, construction teams can optimize the construction process, reduce material waste, and improve overall project efficiency. Finally, effective communication and collaboration between the construction team and suppliers can enhance the material delivery process, leading to better project outcomes. In conclusion, proper planning, scheduling, and monitoring of material deliveries are essential for the success of any construction project.

Following are some steps to plan, schedule, and monitor all material deliveries to avoid project delays:

- **Develop a detailed material delivery plan**: Create a material supply plan that details the quantity, delivery dates, and all materials required for the project. The plan should also take supplier availability and lead times for materials into account.
- **Establish delivery schedules**: Establish delivery schedules that are feasible and based on the material delivery plan. The project timeframe, location accessibility, and material handling capabilities should all be taken into account.
- **Communicate with suppliers**: Ensure your suppliers are informed of the delivery timeline and any changes to the plan by communicating with them frequently. By doing this, it will be easier to prevent delays and ensure that products are delivered on schedule and to the right place.
- **Monitor material deliveries**: Keep an eye on material delivery to make sure they arrive on time and that they fulfil the standards needed. This can involve performing routine quality control checks and keeping tabs on how material deliveries are progressing in relation to the project schedule.
- **Adjust the plan as needed**: If the project timeframe changes or is delayed, make the necessary adjustments to the material supply plan. Deliveries could need to be rescheduled, new suppliers might need to be found, or the project timeline might need to be adjusted to account for delays.

- for installation of module structures and modules, installation of inverters, transformers, earthing piles, DC/ AC power protection devices, lightning arresters, substation as per the grid codes and regulatory provisions
- Discuss how to manage schedule for each of the construction activity i.e. construction of internal roads, construction of foundations for mounting module support structures, combiner boxes, inverters, transformers, and substation etc.
- Discuss how to do successful completion of the projects by identifying innovative ways to maximize Profit
- Explain how to manage project billing cycle to ensure timely submission and payment of invoices
- Explain how to analyze the bids to ensure complete and accurate quotes for scopes of work
- List various parameters associated with local power loads/ grid dispatch
- Explain how to negotiate with subcontractors
- Discuss how to coordinate with the customers on a regular basis and provide them with the updates
- Discuss how to communicate to the customer and other key team members the contractual, financial, cost accounting, scheduling, technical and other matters related to the project.

7.2.5 Supervise and Manage

It takes careful coordination and adherence to grid regulations and regulatory requirements to supervise and manage the timetable for the construction of module structures, modules, inverters, transformers, earthing piles, DC/AC power protection devices, lightning arresters, and substations. Following is an outline of the steps involved in supervising and managing:

- **Develop a detailed installation plan**: Make a detailed installation plan including a timetable, task breakdown, and resource allocation. Take into account the order of installation, interdependence between activities, and the amount of labour, tools, and supplies needed.
- **Coordinate with suppliers and contractors**: Contact the vendors and contractors in charge of supplying and installing the various parts. Ascertain that they are familiar with the installation timeline and the grid codes and regulatory requirements.
- **Conduct regular site inspections**: Conduct regular site inspections to keep an eye on the installation progress. Track the development, standard, and adherence to the regulatory requirements and grid codes. To reduce delays and assure compliance, address any problems or deviations right away.
- **Manage resources effectively**: Ensure that all required resources—including labour, tools, and materials—are on hand on the designated date and time. Work together with contractors and suppliers to avoid any shortages or delays in the delivery of supplies and machinery.
- **Monitor and adjust the schedule**: Follow the installation's progress closely and compare it to the schedule that was originally intended. Take corrective action to address any delays or diversions to return the project to its original course. In order to do this, it could be necessary to redistribute resources, alter the schedule, or deal with any technical issues that come up during installation.
- **Coordinate with grid authorities and regulatory bodies**: To ensure that the installation of the solar plant complies with all necessary norms and regulations, maintain close contact with grid authorities and regulatory agencies. Obtain the required inspections, permits, and permissions in accordance with the legal requirements.
- **Document and report progress**: Ensure that the installation procedure is accurately documented, taking note of any variations, difficulties, and solutions. Create regular status reports to inform stakeholders and maintain project transparency.

You can ensure the effective and legal installation of module structures, modules, inverters, transformers, earthing piles, DC/AC power protection devices, lightning arresters, and substations for your ground-mounted solar civil work construction project by methodically supervising and managing the schedule for installation and adhering to grid codes and regulatory provisions.

Supervising and Managing the Installation of Components

Supervising and managing the installation of module structures and modules for a ground-mounted solar project requires careful coordination, attention to detail, and adherence to industry best practices. Following are some key steps to consider:

- **Familiarize yourself with the project specifications**: Understand the project plans, specifications, and technical requirements related to the installation of module structures and modules. This includes studying the design drawings, layout plans, and installation guidelines provided by the manufacturer.
- **Develop an installation plan**: Create a detailed installation plan that outlines the sequence of tasks, resource requirements, and timelines for the installation of module structures and modules. Consider factors such as site access, weather conditions, and any dependencies with other construction activities.
- **Coordinate with the construction team**: Work closely with the construction team, including subcontractors and installers, to ensure smooth coordination and communication. Clearly communicate the installation plan, expectations, and safety protocols to all involved parties.
- **Conduct pre-installation inspections**: Before starting the installation process, inspect the site to verify that the foundation or mounting structure is properly prepared and meets the required specifications. Ensure that the site is clear of any obstructions or hazards.
- **Ensure compliance with safety standards**: Implement and enforce strict safety protocols throughout the installation process. Provide personal protective equipment (PPE) to workers, conduct safety briefings, and ensure adherence to relevant safety regulations and guidelines.
- **Oversee module structure installation**: Monitor the installation of module structures, including the placement of mounting components, alignment, and secure attachment to the foundations. Verify that all structural connections and fastenings are done correctly to ensure the stability and durability of the structures.
- **Supervise module installation**: Coordinate the installation of solar modules onto the module structures. Ensure that the modules are handled and positioned carefully to avoid any damage. Verify that electrical connections, wiring, and grounding are performed according to industry standards and safety requirements.
- **Conduct quality control inspections**: Regularly inspect the installed module structures and modules to ensure compliance with design specifications and quality standards. Address any identified issues or deviations promptly to maintain the integrity and performance of the solar installation.
- **Document and report progress**: Maintain detailed documentation of the installation process, including photos, inspection reports, and any modifications or adjustments made during installation. This documentation serves as a record of the installation progress, assists in quality control, and provides a reference for future maintenance and troubleshooting.
- **Monitor project timelines and milestones**: Continuously monitor the installation progress against the established timelines and milestones. Address any delays or issues promptly, and adjust the schedule if necessary to ensure that installation activities stay on track.

Grid Codes and Regulatory Provisions for the Installation of Components

Grid codes and regulatory provisions for the installation of ground-mounted solar civil construction work vary depending on the specific country or region. However, there are some common elements that are typically included in these codes and provisions. Here are a few key aspects to consider:

- **Technical requirements**: Grid codes define the technical specifications and standards that must be followed during the design and installation of the solar power plant. This includes guidelines for voltage levels, power factor control, frequency regulation, harmonic limits, and grid connection requirements.
- **Safety regulations**: Regulatory provisions establish safety guidelines and requirements to ensure the safe operation of the solar installation. These regulations may cover aspects such as electrical safety, grounding and earthing, fire safety, protection against electrical faults, and compliance with building and construction codes.
- **Grid interconnection**: Grid codes outline the requirements for connecting the solar power plant to the electrical grid. This includes specifications for grid synchronization, protection systems, power quality, and compliance with grid stability requirements.
- **Metering and billing**: Grid codes often specify the metering and billing requirements for solar power plants. This includes the installation and calibration of appropriate energy meters to accurately measure the electricity generated by the solar plant and facilitate proper billing and accounting.
- **Environmental and permitting requirements**: Regulatory provisions may include environmental regulations that need to be followed during the construction and operation of the solar plant. This

can include guidelines for land use, environmental impact assessments, wildlife protection, and compliance with environmental permits.

- **Compliance and testing**: Grid codes often require the testing and verification of the solar installation to ensure compliance with the specified technical and safety requirements. This can involve performance testing, insulation testing, protection system testing, and other checks to ensure the proper functioning of the system.
- **Grid connection agreements**: In many cases, solar power plant developers need to enter into grid connection agreements with the utility company or grid operator. These agreements define the terms and conditions for grid connection, including technical requirements, power purchase agreements, and any additional obligations or rights.

7.2.6 Maximize Profit

Successful completion of projects involves not only delivering the desired outcomes but also maximizing profit. Following are some key steps to identify innovative ways to maximize profit during project execution:

- **Comprehensive project planning**: Start by creating a complete project plan that includes an analysis of the project's goals, deliverables, and timetable. Take into account elements including resource needs, purchasing tactics, and risk-reduction initiatives. A project that has been carefully designed will maximise earnings.
- **Value engineering**: Implement value engineering strategies to reduce project costs while maintaining quality. Identifying cheaper materials, design tweaks, or building techniques while retaining project specifications is part of this process. Assess project elements on a regular basis to find potential cost-saving measures.
- **Efficient resource allocation**: The distribution of project resources, such as labour, tools, and materials, should be monitored and optimised. Ensure that resources are used efficiently to reduce waste, prevent needless spending, and shorten project schedules. Review resource usage frequently, and modify as necessary.
- **Collaborative supplier and subcontractor management**: Develop strong relationships with suppliers and subcontractors to bargain for fair terms and prices. Find creative solutions and engage in open dialogue to identify more affordable options. Take into account long-term alliances that can provide competitive price, dependability, and quality.
- **Risk management**: Make a thorough risk assessment to find potential hazards and create countermeasures against them. By reducing risks, costly delays and interruptions can be avoided. Set aside money for unforeseen problems that might occur while the project is being completed.
- **Continuous cost monitoring and control**: Implement reliable cost monitoring and control technologies to keep tabs on project costs in real time. Regularly assess actual expenses to the budget to find any discrepancies or overruns. Utilise this knowledge to adopt appropriate cost-saving strategies and corrective actions.
- **Innovative technology adoption**: Investigate cutting-edge building techniques and technologies that can boost productivity, cut costs, and increase project efficiency. Use automation and digital tools to speed up procedures, increase accuracy, and reduce labour-intensive jobs.
- **Value-added services**: Identify possibilities to offer value-added services that can bring in more money. Offering maintenance contracts, energy optimisation services, or implementing sustainable features that are eligible for government subsidies or incentives are a few examples of how to do this.
- **Stakeholder collaboration**: Engage stakeholders, including as customers, investors, and project team members, to find ways to maximise profits. Encourage their opinions and suggestions because they might have insightful knowledge and experience that can be used to develop creative cost-cutting or revenue-generating solutions.
- **Post-project evaluation**: Make a thorough analysis of the finished project to pinpoint any lessons learned and areas that want improvement. This feedback can be utilised to improve current projects, streamline workflows, and spot more profit-maximizing opportunities.

7.2.7 Manage Project Billing Cycle

Managing the project billing cycle is critical for entrepreneurs as it ensures timely submission and payment of invoices, which helps in maintaining cash flow and financial stability of the business. It also ensures that the project is completed within the allocated budget and timeline, thus avoiding any potential cost overruns or delays.

By managing the project billing cycle effectively, entrepreneurs can also ensure that they are accurately tracking project expenses and revenue, which is important for tax purposes and for making informed business decisions. This can help entrepreneurs to identify areas where they can cut costs and increase profitability, as well as areas where they need to invest more resources to achieve better results.

Additionally, managing the project billing cycle can also help entrepreneurs to maintain strong relationships with clients and vendors by ensuring that invoices are submitted and paid on time. This can improve the reputation of the business and lead to more repeat business and referrals.
Following are some steps you can take to manage the billing cycle effectively:

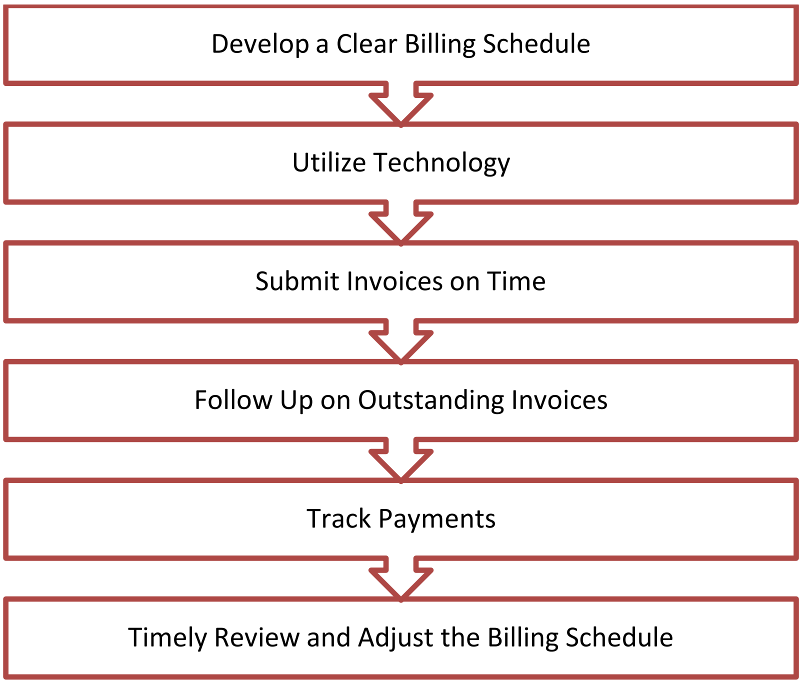


Fig.16.2.1: Some Steps to Manage the Billing Cycle Effectively

- **Develop a clear billing schedule**: Effective billing management starts with creating a detailed billing schedule. The frequency of invoicing, the billing cycle, and the terms of payment should all be listed in this schedule. Both parties will be aware of the deadlines and expectations related with the invoicing process if a clear timetable is established at the beginning of the project. This can make it easier to avoid misunderstandings and confusion and guarantee that bills are sent and paid on time.
- **Utilize technology**: An effective way for creating accurate invoices is to use project management software to keep track of the work finished on the project and the amount of time spent on each assignment. These software can track work completion in real-time, assisting in making sure that bills accurately reflect the work completed during the billing period. The software can also assist in finding potential invoicing inconsistencies or errors that can be fixed before the invoice is filed.
- **Submit invoices on time**: Effective billing administration depends on timely invoice submission. Payment delays brought on by late invoicing may negatively affect cash flow. To prevent such delays, set a deadline for submitting bills and ensure that they are delivered on time or before. The possibility of late payments can be decreased and a regular billing schedule can be maintained by consistently filing invoices on time.
- **Follow up on outstanding invoices**: In order to guarantee that unpaid invoices are paid promptly, follow-up is necessary. Sending reminders, making phone calls, or starting other forms of communication may be necessary to address any problems or discrepancies. Following up on unpaid invoices on a regular basis can also help to keep the client happy and lower the chance of a conflict or losing future business.

- **Track payments**: Monitoring funds received is a crucial part of efficient billing administration. Maintain a record of all payments made and make sure they are applied to the right invoice. This lowers the possibility of errors or conflicts and helps keep accurate financial records. Additionally, keeping track of payments made can help in spotting any discrepancies or problems with the invoicing that can be fixed before they worsen.
- **Review and adjust the billing schedule if necessary**: To ensure that invoices are sent and paid on time, it is essential to review and, if required, modify the billing schedule. The billing schedule should be modified if the project's timeline or scope changes. Work with the customer to implement any necessary revisions, and ensure that the project contract reflects those changes. Maintaining a consistent billing schedule and ensuring that bills are issued and paid on time can be achieved by periodically evaluating and modifying the billing schedule as necessary.

These steps will help you handle the project billing cycle efficiently so that bills are sent and paid on time. Both your relationship with the customers and your cash flow will benefit from this.

7.2.8 Analysing Bids

Entrepreneurs need to analyse bids to ensure that they receive complete and accurate quotes for scopes of work to make informed decisions that are aligned with their business goals and budget constraints. By analysing bids, entrepreneurs can identify the best vendors or contractors who can deliver the required work within the given timeframe and budget.

To analyse bids effectively, entrepreneurs need to follow these steps:

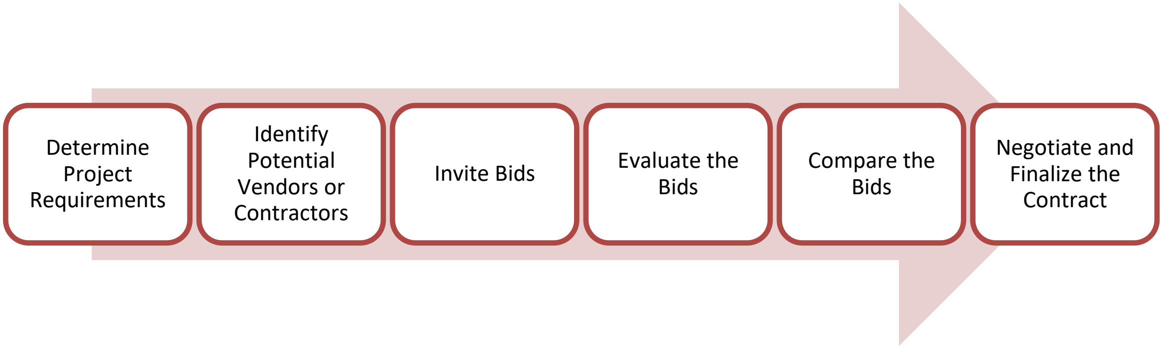


Fig.16.2.2 Steps to Analyse Bids

- **Determine project requirements**: The entrepreneur should be clear about the requirements of the project, including the scope of work, timelines, and budget.
- **Identify potential vendors or contractors**: Entrepreneurs should research and identify potential vendors or contractors who can meet the requirements of the project.
- **Invite bids**: Once potential vendors or contractors have been identified, entrepreneurs should invite them to submit bids. The bids should clearly specify the scope of work, timelines, and cost.
- **Evaluate the bids**: Entrepreneurs should review and evaluate the bids received from the vendors or contractors. The evaluation should consider factors such as the quality of work, experience, and reputation of the vendor or contractor, the proposed timeline, and the cost.
- **Compare the bids**: Entrepreneurs should compare the bids received from different vendors or contractors to identify any differences in their approach, pricing, and overall quality. This can help to determine which vendor or contractor is best suited for the project.
- **Negotiate and finalize the contract**: Once a vendor or contractor has been selected, entrepreneurs should negotiate the terms of the contract to ensure that the scope of work, time frame, and budget are clearly defined. This can help to prevent misunderstandings and disputes later on in the project.

Entrepreneurs may analyse bids efficiently by following these steps to guarantee that they get complete and accurate prices for the project scope of work. This can increase the project's chances of being finished on schedule, under budget, and with the needed degree of quality, therefore enhancing the business's chances of success.

7.2.9 Negotiation Skills

Negotiation is a method used to settle differences. The aim of negotiation is to resolve differences through a compromise or agreement while avoiding disputes. Without negotiation, conflicts are likely to lead to resentment between people. Good negotiation skills help satisfy both parties and go a long way towards developing strong relationships.

Why Negotiate?

Starting a business requires many, many negotiations. Some negotiations are small while others are critical enough to make or break a start-up. Negotiation also plays a big role inside the workplace. As an entrepreneur, you need to know not only know how to negotiate yourself, but also how to train employees in the art of negotiation.

How to Negotiate?

Take a look at some steps to help you negotiate:

- **Step 1:** Pre-Negotiation Preparation: Agree on where to meet to discuss the problem, decide who all will be present and set a time limit for the discussion.
- **Step 2:** Discuss the problem: This involves asking questions, listening to the other side, putting your views forward and clarifying doubts.
- **Step 3:** Clarify the Objective: Ensure that both parties want to solve the same problem and reach the same goal.
- **Step 4:** Aim for a Win-Win Outcome: Try your best to be open minded when negotiating. Compromise and offer substitute solutions to arrive at an outcome where both win.
- **Step 5:** Clearly Define the Agreement: When an agreement has been reached, the details of the agreement should be crystal clear to both sides, with no scope for misunderstandings.
- **Step 6:** Implement the Agreed Upon Solution: Agree on a course of action to set the solution in motion.

7.2.10 Communicating with Customers and Team Members

One should keep the following points in mind when having a conversation with a colleague or client:

- Use positive language while reporting a complaint to senior.
- Remain assertive when under pressure.
- Ask questions calmly. Do not intimidate or interrogate.
- Aim to achieve a win/win outcome for both the parties.
- Honour the commitments and promises made to resolve the issues.

Often, one needs to communicate with people at different levels, such as labourers, supervisors, team members and so on. Effective communication has an important role to play in fulfilling responsibilities. These forms of communication are:

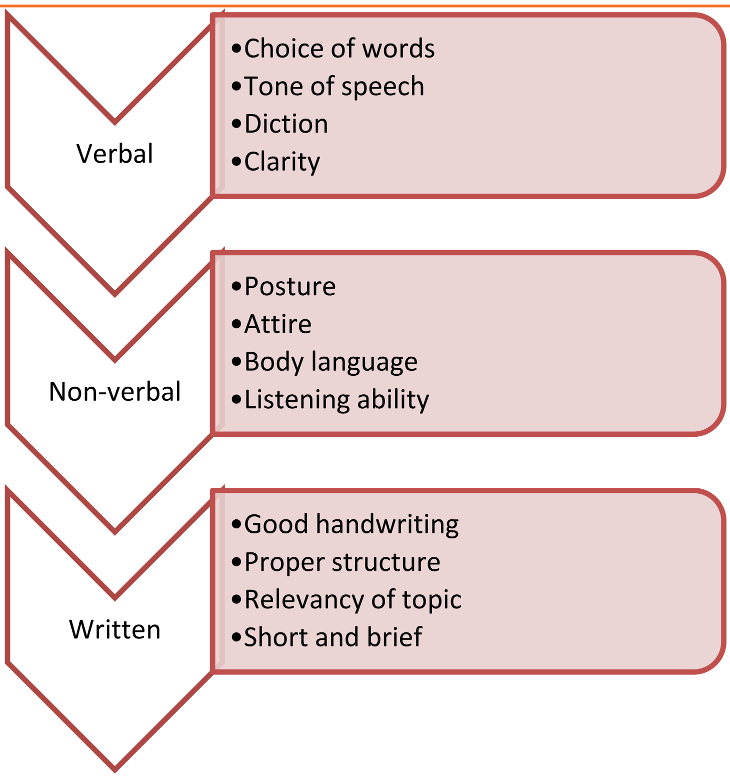


Fig. 16.2.3: Forms of communication

Some language and listening barriers are:

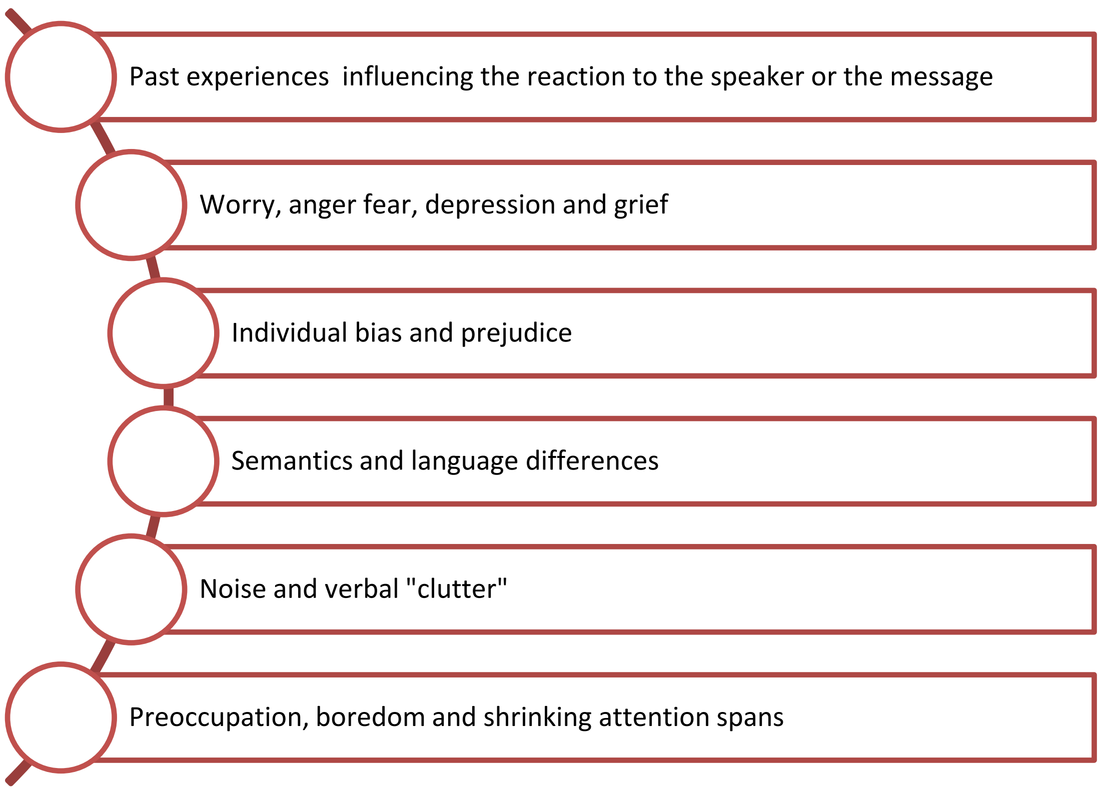


Fig16.2.5: Language and listening barriers

Effective communication depends on active and effective listening. The following figure shows some ways to develop an active listening skill:

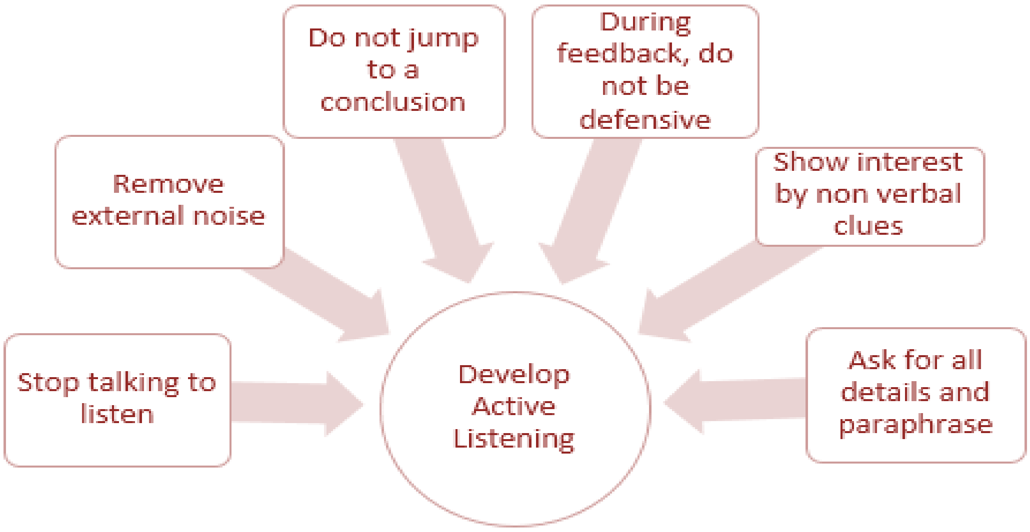


Fig. 16.2.4: Active listening skills

While interacting with clients and colleagues, one should always maintain a decorum and follow proper manners. Etiquette and manners can be easily developed by following certain practices. These practices are:

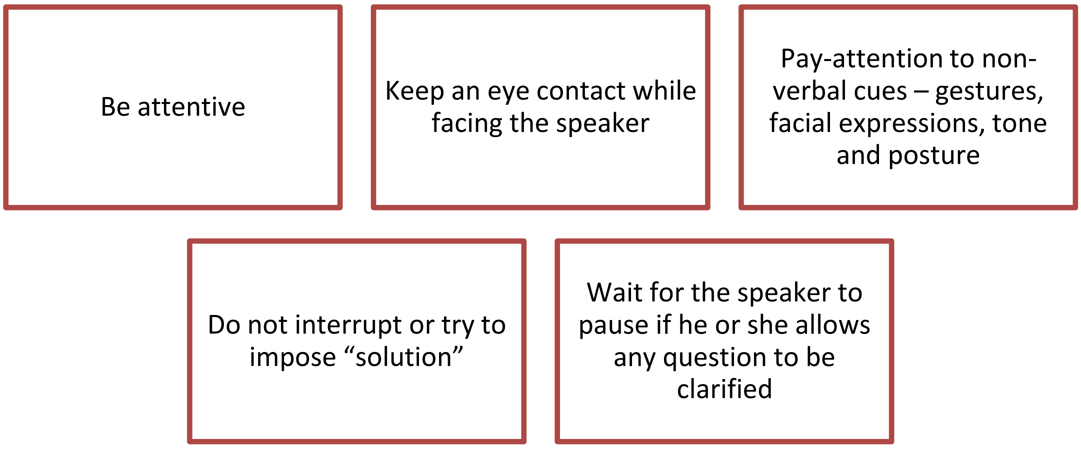


Fig. 16.2.5: Etiquette and Best practices

The conflicts that arise due to perception barriers can be solved by the following methods:

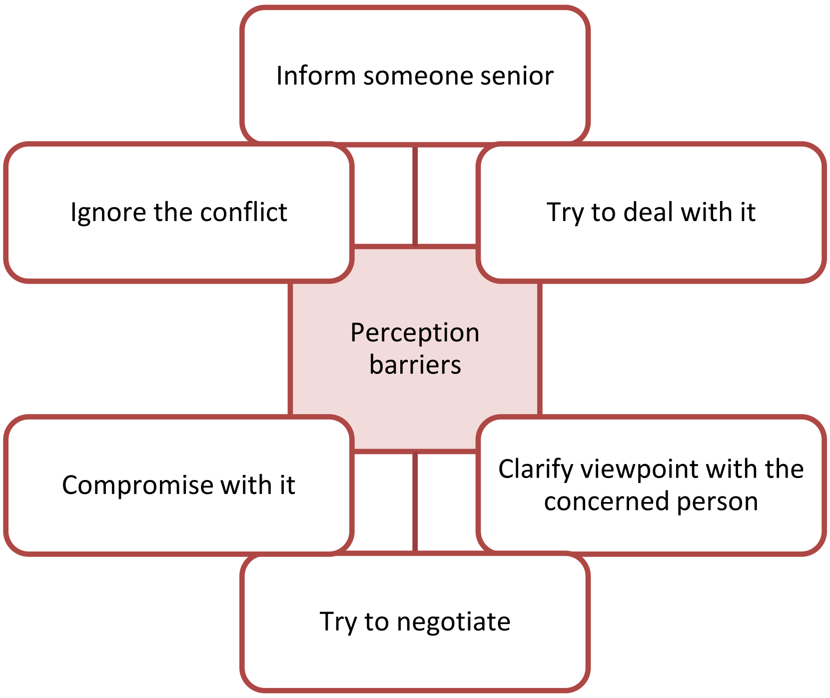


Fig. 16.2.6: Conflicts

7.2.11 Periodic Site Visits

Periodic site visits by the entrepreneur to check on the status of installation work are essential for a number of reasons, such as:

Oversight and Project Management

Quality Control

Issue Identification and Resolution

Stakeholder Engagement

Risk Mitigation

Fig.16.2.7 Reasons for Periodic Site Visits

- **Oversight and Project Management**: The entrepreneur plays a key role in overseeing the project and ensuring its successful implementation. By visiting the site regularly, they can directly monitor the progress of the installation work and ensure that it aligns with the project plan and objectives. This hands-on approach allows for effective project management and reduces the chances of miscommunication or misunderstandings between the entrepreneur and the installation team.
- **Quality Control**: The entrepreneur's presence on-site enables them to assess the quality of work being done by the installation team. They can inspect the components, construction techniques, and adherence to industry standards and specifications. This helps in maintaining high-quality standards and ensures that the solar plant is installed to the desired level of efficiency and performance.
- **Issue Identification and Resolution**: Site visits allow the entrepreneur to identify any potential issues or challenges that may arise during the installation process. By proactively addressing these issues, the entrepreneur can minimize delays, avoid costly rework, and maintain the project's timeline and budget. They can work closely with the installation team to find suitable solutions and keep the project on track.
- **Stakeholder Engagement**: The entrepreneur's presence on-site demonstrates their commitment to the project and fosters trust among stakeholders, including investors, customers, and regulatory authorities. It allows for direct communication with the installation team, fostering a collaborative environment and ensuring that everyone is aligned with project goals.
- **Risk Mitigation**: Regular site visits enable the entrepreneur to identify and mitigate potential risks associated with the installation process. This includes identifying safety hazards, addressing environmental concerns, and ensuring compliance with local regulations. By actively managing risks, the entrepreneur can safeguard the project's success and protect their investment.

In summary, periodic visits by the entrepreneur to the ground mount solar plant site(s) provide oversight, quality control, issue identification and resolution, stakeholder engagement, and risk mitigation. These visits play a crucial role in ensuring the successful implementation of the solar project and maximizing its long-term benefits.

Exercise

1. Explain the importance of soil test report for designing the Mounting structure.

____

____

____

____

2. Explain how to Prepare Structural Design Report.

____

____

____

____

3. Explain the Assessment Criteria for Assessing the Water Drainage System.

____

____

____

____

Module 7 - Planning and Scheduling	Unit 7.1	How a Solar Farm is Constructed from Beginning to End		https://www.youtube.com/watch?v=sM7ge9ahsGY	

UNIT 7.3: Interpret Drawings of DC and AC Side of Electrical Components of Ground- Mount Solar Power Plant

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to interpret the electrical single line diagram of DC side of the interconnection
2. Interpret the electrical line diagram of ACDB, LT Panel, earthing layout.
3. Discuss how to install and interconnect the Inverter, DC-Disconnect, String monitoring box (SMB)
4. Discuss how to install the DC and AC Energy Meter
5. Interpretate the electrical single diagram of AC side of the plant.
6. Explain how to prepare BOQ with the with as built drawing of electrical interconnection and internal wiring diagram of major electrical component
7. Discuss the process of Pre commissioning and commissioning activities of solar Plant in both DC and AC side.

7.3.1 Interpretation of SLD and interconnections

As an entrepreneur, interpreting the drawings of the DC (Direct Current) and AC (Alternating Current) sides of electrical components in a ground-mounted solar power plant involves understanding the key components and their functions in generating and transmitting electricity. Here's an entrepreneur-focused interpretation:

DC Side:

1. Solar Panels:
    - These are the primary components that convert sunlight into DC electricity. Panels are installed in arrays and generate electricity when exposed to sunlight.
2. String or Series Connections:
    - Solar panels are connected in strings or series to increase the system's voltage. Multiple panels in series generate higher voltage levels.
3. Combiner Boxes:
    - Combiner boxes consolidate the output from multiple strings of solar panels into a single DC output. They often include fuses or circuit breakers for safety and monitoring equipment to track performance.
4. DC Disconnects:
    - These switches or disconnects are placed between the solar array and inverters. They allow for isolating the DC power source for maintenance or emergencies.
5. Inverters:
    - Inverters are crucial components that convert DC electricity from the panels into AC electricity suitable for use in the electrical grid. They also manage the system's performance.

AC Side:

1. Inverter Output:
    - The AC output from the inverters is the converted electricity that will be integrated into the grid.

2. AC Disconnects:
    - Similar to DC disconnects, these switches or breakers are installed on the AC side to isolate the system from the grid for maintenance or safety reasons.
3. Transformer (if present):
    - If needed, a transformer on the AC side might be included to adjust voltage levels before connecting to the grid.
4. Switchgear and Protection Devices:
    - These components ensure the system's safety by protecting against electrical faults and overloads. They often include circuit breakers, relays, and surge protection devices.
5. Metering and Monitoring Equipment:
    - Meters and monitoring equipment are installed to measure the power output, monitor system performance, and ensure compliance with grid regulations.
6. Grid Connection Point:
    - This is the point where the solar power plant connects to the electrical grid. It includes the necessary switches or connections to synchronize and feed power into the grid.

For an entrepreneur, understanding these drawings is essential for grasping the overall functioning and key components of the solar power plant's electrical systems. It aids in planning, decision-making, and ensuring the successful integration of the solar power plant with the grid infrastructure.

7.3.2 Electrical line diagram of ACDB, LT Panel, earthing layout.

Understanding the electrical line diagram of the ACDB (AC Distribution Board), LT (Low Tension) Panel, and earthing layout is crucial for an entrepreneur aiming to comprehend the electrical distribution and safety systems within a ground-mounted solar power plant.

ACDB (AC Distribution Board):

1. Incoming AC Power:
    - AC power generated by the solar panels and converted by the inverters enters the ACDB.
2. Circuit Breakers or Switches:
    - Circuit breakers or switches within the ACDB control the distribution of AC power to various loads or subpanels.
3. Output Connections:
    - Outgoing connections distribute the AC power to different loads or sections within the solar power plant. This could include connections to different buildings, equipment, or other sections requiring power.
4. Protection Devices:
    - Protective devices like fuses, surge protectors, and overcurrent protection are often integrated within the ACDB to safeguard against electrical faults or surges.
5. Monitoring and Metering Equipment:
    - Monitoring devices and meters might be installed to track the flow of electricity, measure power consumption, and ensure proper functioning of the AC distribution system.

LT Panel (Low Tension Panel):

1. Incoming Power from ACDB:

    - The LT Panel receives power from the ACDB and further distributes it to various loads or sections within the solar power plant.

2. Distribution Breakers/ MCCB (Molded Case Circuit Breakers):

    - MCCBs or breakers within the LT Panel are used for circuit protection and control. They regulate the flow of electricity to different sections of the plant or various equipment.

3. Busbars or Terminals:

    - Busbars or terminals facilitate the distribution of electricity within the LT Panel. They serve as connection points for various outgoing circuits.

4. Metering and Monitoring Systems:

    - Similar to the ACDB, the LT Panel may also include meters and monitoring systems to measure power consumption, monitor system performance, and ensure safety.

Earthing Layout:

1. Earthing Electrodes:

    - Earthing electrodes are buried conductive materials (such as rods or plates) that establish a connection between the electrical system and the ground. They serve to dissipate excess electrical currents and provide safety against faults.

2. Earthing Conductors:

    - Conductors run from various electrical equipment (ACDB, LT Panel, inverters, etc.) to the earthing electrodes, creating an effective earthing system.

3. Earthing Pit and Earth Grid:

    - Earthing pits are specific locations where the earthing electrodes are installed and connected. An earth grid might be formed by interconnecting multiple earthing electrodes for better grounding effectiveness.

Understanding these diagrams from an entrepreneurial standpoint helps in comprehending the electrical distribution, safety measures, and monitoring systems within the solar power plant. It aids in ensuring proper functioning, safety compliance, and efficient operation of the electrical infrastructure.

Note:
To know more about Earthing Layout of electrical substation, you may refer to the following link:
https://www.youtube.com/watch?v=c2TXqWG2Id4

Title of Video:	Earthing Layout of electrical substation
By:	electrical mastar
Duration:	11 m 43 sec

7.3.3: Quality and workmanship of Electrical Installation

Ensuring the quality and workmanship of cable laying and other electrical installations is crucial for the safe and reliable operation of a solar PV power plant. Here are some key considerations to ensure high-quality installations:

- **Qualified and Experienced Contractors**: Engage qualified and experienced contractors who specialize in electrical installations and have a track record of successful projects. Verify their credentials, licenses, and certifications to ensure they have the necessary expertise and knowledge in electrical installations.
- **Compliance with Standards and Codes**: Ensure that all electrical installations adhere to relevant national and local electrical codes, standards, and regulations. These codes specify requirements for cable sizing, insulation, grounding, protection devices, and other aspects of electrical installations. Compliance ensures safety, performance, and longevity of the system.
- **Proper Planning and Design**: Ensure that the cable laying and electrical installation activities are based on a well-defined plan and design. This includes determining the optimal cable routes, cable lengths, cable types, and sizing based on load requirements, voltage drop calculations, and environmental conditions. Thorough planning minimizes errors and reduces the chances of rework or modifications later on.
- **Cable Selection and Quality**: Select high-quality cables that are appropriate for the application and meet the required specifications. Consider factors such as insulation material, conductor size, current-carrying capacity, environmental durability, and voltage rating. Ensure that cables are sourced from reputable manufacturers and undergo proper testing and certification.
- **Proper Cable Handling and Installation**: Ensure that cables are handled, stored, and installed properly to prevent damage or degradation. Follow manufacturer guidelines for cable bending radius, pulling tension limits, and other installation instructions. Use appropriate cable support systems, such as trays, conduits, or cable clips, to secure and protect the cables.
- **Cable Termination and Connections**: Properly terminate and connect cables to electrical equipment, such as junction boxes, switchgear, transformers, and inverters. Follow recommended techniques for cable stripping, insulation, crimping, and tightening to ensure reliable and secure connections. Perform thorough testing and inspections to verify the quality of terminations and connections.
- **Inspections and Testing**: Conduct regular inspections and testing throughout the installation process to identify any deficiencies, faults, or non-compliance. This includes visual inspections, insulation resistance tests, continuity checks, ground resistance tests, and other relevant electrical tests. Document and address any identified issues promptly to maintain high-quality installations.
- **Documentation and Record Keeping**: Maintain comprehensive documentation of the installation process, including design drawings, cable schedules, test reports, and any modifications or deviations from the original plan. This documentation provides a reference for future maintenance, troubleshooting, and system upgrades.
- **Quality Assurance and Supervision**: Implement a quality assurance program that includes on-site supervision, monitoring, and audits. Assign qualified personnel to oversee the installation activities and ensure adherence to quality standards. Conduct regular progress reviews and site visits to assess the workmanship and address any issues promptly.

7.3.4: Prepare BOQ as per built drawing of Electrical Interconnection.

Preparing a Bill of Quantities (BOQ) with as-built drawings of electrical interconnections and internal wiring diagrams of major electrical components for a ground-mounted solar power plant as an entrepreneur involves several key steps:

As-Built Drawings Review:

Begin by reviewing the finalized as-built drawings that depict the actual layout, interconnections, and internal wiring of major electrical components in the solar power plant.

Identification of Components:

Identify and list the major electrical components depicted in the drawings. This might include inverters, transformers, ACDB, LT panels, switchgear, earthing systems, cabling, and other key elements.

Quantification of Materials:

Create a detailed list of materials needed for electrical interconnections and internal wiring. This involves quantifying the cables, wires, connectors, conduits, junction boxes, busbars, fasteners, and other specific components required based on the drawings.

Specifications and Quantities: Specify the types, sizes, specifications, and quantities of each material or component required for the electrical interconnection and wiring based on the as-built drawings. Include relevant technical details and standards compliance information.

Unit Measurements and Costs: Estimate the quantities using appropriate units (lengths for cables, numbers for connectors, etc.) and assign costs for each item based on market rates, supplier quotations, or historical project data.

Labor and Installation Costs: Factor in labour and installation costs associated with the electrical interconnection and wiring. This might include labour hours required for installation, testing, and commissioning of the electrical systems.

Include Contingencies: Allow for contingencies or unforeseen costs by adding a percentage (typically 5-10%) to the total material and labour costs to accommodate any unexpected changes or additional requirements that may arise during implementation.

BOQ Documentation: Organize the BOQ document systematically, listing the materials, quantities, specifications, unit costs, total costs, and any other relevant information in a clear and comprehensive format.

Quality Assurance and Compliance: Ensure that the materials specified in the BOQ meet the required quality standards, conform to regulatory guidelines, and are approved for use in electrical installations.

Review and Verification: Review the BOQ for accuracy, completeness, and consistency with the as-built drawings. Verify quantities, costs, and specifications before finalizing the document.

By carefully preparing a detailed BOQ based on the as-built drawings of the electrical interconnections and internal wiring diagrams, an entrepreneur ensures accurate cost estimation, procurement planning, and effective execution of the electrical systems within the solar power plant. This comprehensive approach aids in budgeting, resource allocation, and successful project implementation.

How To Prepare?

Defining the Work

The first and most important step in creating a BOQ is this. The work that needs to be done is explained in detail and in simple terms so that the contractor can understand what needs to be done.

Quantity Take Off

After the work is described, the quantity is calculated and subtracted. It provides a thorough idea of the scope of the work that must be done by a contractor. The amount of work is specified in units.

Describing The Rates

The last step in creating a BOQ is to provide information about the rates of work items per unit.

Schedule of Rates (SOR), current rates, market surveys, and other considerations are used to determine each item's rate.

Finally, to determine the total amount for a specific item of work, the total quantity of work is multiplied by the unit rate of the item of work.

7.3.5: Pre commissioning and commissioning activities of solar Plant.

As an entrepreneur overseeing the pre-commissioning and commissioning activities of a solar power plant on both the DC (Direct Current) and AC (Alternating Current) sides, here's an overview:

Pre-commissioning Activities:

1. **Inspection and Verification:**
    - Conduct thorough inspections of the entire solar power plant, including the DC and AC sides, to ensure that all equipment, components, and connections are installed as per design specifications and regulatory standards.
2. **Testing of DC Side:**
    - Perform testing on the DC side, including the solar panels, strings, combiner boxes, DC cables, inverters, and other related components.
    - Verify the functionality of DC disconnects, inverters, and protection systems.
    - Check for proper grounding and continuity of the DC system.
3. **Testing of AC Side:**
    - Conduct testing on the AC side, including inverters, transformers (if present), AC distribution boards, switchgear, protection devices, and metering systems.
    - Verify the synchronization of the solar power plant with the grid (if applicable).
    - Ensure proper functioning of AC disconnects and safety measures.
4. **Functional Testing:**
    - Perform functional tests to validate the performance of the entire system under different operating conditions.
    - Check for proper communication and monitoring capabilities of control systems.
5. **Documentation and Record-keeping:**
    - Prepare detailed documentation of all pre-commissioning tests, inspections, and results. Maintain records of equipment manuals, warranties, and as-built drawings for reference.

Commissioning Activities:

1. **System Energization - DC Side:**
    - Gradually energize the DC side of the solar power plant by connecting strings and verifying the output of each string.
    - Monitor the performance of solar panels and ensure that the DC voltage and current levels are within acceptable ranges.
2. **Inverter and AC System Energization:**
    - Gradually energize the inverters and AC side of the system. Monitor the output voltage, frequency, and power quality.
    - Ensure synchronization with the grid (if connected).

3. **Load Testing:**
    - Gradually introduce loads to the system and verify its ability to handle the expected power output.
    - Monitor for any abnormalities or issues during load testing.
4. **Performance Testing and Optimization:**
    - Perform performance tests to assess the solar power plant's efficiency, output, and compliance with performance guarantees.
    - Optimize the system settings to maximize power generation and ensure optimal operation.
5. **Grid Connection Testing (if applicable):**
    - If connected to the grid, conduct grid connection tests to ensure seamless integration and compliance with grid codes and regulations.
6. **Safety and Emergency Procedures Testing:**
    - Verify the functionality of safety systems, emergency shutdown procedures, and protective devices.
7. **Handover and Documentation:**
    - Upon successful commissioning, compile a comprehensive commissioning report detailing all tests, results, and procedures performed during the commissioning process.
    - Complete necessary documentation and formalities for handover to the operations team.

Ensuring a systematic and thorough approach to pre-commissioning and commissioning activities is crucial for the successful launch and operation of the solar power plant. It involves meticulous testing, verification, and documentation to ensure compliance with standards, safety, and optimal performance of the plant.

Exercise

1. Explain how to interpret the electrical single line diagram.

2. Electrical line diagram of ACDB, LT Panel, earthing layout.

3. Explain the Procedure of Preparing a Bill of Quantities (BOQ) with as-built drawings of electrical interconnections.

Module 18 - Manage Electrical installation in Ground Mount Solar Plant	Unit 18.1	Earthing Layout of electrical substation	92	https://www.youtube.com/watch?v=c2TXqWG2Id4	

UNIT 7.4 Prepare a vendor list of potential suppliers and negotiate with them

Unit Objectives

At the end of this unit, participant will be able to:

1. Discuss how to identify manufacturers and suppliers through various sources who can meet the requirements.
2. Discuss how to manage vendor or supplier for material procurement.

7.4.1 Identify manufacturers and suppliers through various sources

Identifying manufacturers and suppliers for a Solar PV plant as an entrepreneur involves several steps to ensure you find reliable, high-quality products. Here's a comprehensive guide:

1. Define Requirements and Specifications:
Clearly outline your project's needs, including the type and capacity of solar panels, inverters, mounting structures, etc. Consider factors like efficiency, durability, warranty, and certifications.

2. Online Directories and Platforms:

- **Alibaba** and **Global Sources**: Explore these platforms for a wide range of suppliers worldwide.
- **Thomasnet**: A directory specifically for industrial product sourcing.
- **IndiaMart** (for India-based suppliers) or
- **TradeIndia**: Platforms with a focus on Indian suppliers.
- **Trade Shows and Exhibitions**: Attend relevant industry events to meet suppliers in person.

3. Manufacturer Websites:
Visit the websites of reputable solar equipment manufacturers. Look for certifications, product specifications, and customer reviews.

4. Local & International Associations:
Contact solar energy associations, such as the Solar Energy Industries Association (SEIA) in the U.S., or local equivalents in your country. They often have directories or can provide recommendations.

5. Request for Quotations (RFQ):
Send detailed RFQs to potential suppliers, including your requirements, quantities, delivery terms, etc. Evaluate their responses carefully.

6. Verify Supplier Credentials:
Check supplier credentials, certifications, product quality, and previous project experiences. Ensure they comply with industry standards.

7. Customer References and Reviews:
Ask for references from the supplier and contact their previous clients to inquire about their experience.

8. Quality Assurance and Testing:
Request product samples for testing or visit manufacturing facilities if feasible. Ensure products meet your quality standards.

9. Negotiate Terms and Conditions:
Discuss pricing, payment terms, delivery schedules, warranties, and support services. Negotiate favourable terms for your project.

10. Final Supplier Selection:
Consider factors like reliability, quality, cost, after-sales support, and proximity. Choose suppliers that best align with your project needs and budget.

7.4.2 How to manage vendor or supplier for material procurement

Managing vendors or suppliers for material procurement is crucial for the success of any entrepreneurial venture. Here's a comprehensive guide on how to effectively manage vendors or suppliers:

1. Clear Communication and Expectations:

- Establish clear communication channels and maintain open dialogue with your vendors.
- Clearly communicate your requirements, specifications, delivery schedules, and quality standards.

2. Vendor Selection and Onboarding:

- Conduct a thorough vetting process before selecting vendors. Consider factors like reliability, quality, pricing, and past performance.
- Create a formal onboarding process detailing your expectations, contractual terms, and any compliance requirements.

3. Establish Key Performance Indicators (KPIs):

- Define measurable KPIs such as delivery time, quality standards, responsiveness, and adherence to deadlines.
- Regularly evaluate vendor performance against these KPIs.

4. Contract Management:

- Create comprehensive contracts specifying terms and conditions, pricing, payment terms, penalties for non-compliance, and dispute resolution mechanisms.
- Ensure legal review and compliance with applicable laws and regulations.

5. Vendor Relationship Management:

- Foster strong relationships with your vendors based on trust and transparency.
- Regularly engage with them, address concerns, and provide constructive feedback.

6. Continuous Improvement:

- Encourage vendors to provide suggestions for improvement and innovation.
- Conduct periodic reviews and feedback sessions to identify areas for enhancement.

7. Quality Assurance and Inspections:

- Implement quality control measures to ensure materials meet your standards.
- Conduct inspections or audits periodically to ensure compliance.

8. Risk Management:

- Identify potential risks and develop contingency plans.
- Have alternative vendors or backup plans in case of disruptions.

9. Payment Management:

- Adhere strictly to agreed-upon payment terms and ensure timely payments.
- Resolve billing discrepancies promptly and maintain a transparent billing process.

10. Technology Integration:

- Implement software or tools for better vendor management, such as procurement management systems or supply chain management software.

11. Feedback and Performance Reviews:

- Gather feedback from internal stakeholders regarding vendor performance.
- Conduct regular performance reviews with vendors and provide constructive feedback.

12. Resolve Disputes Amicably:

- Address conflicts or disputes promptly and amicably through effective communication and mediation if necessary.
- Maintain professionalism and work towards mutually beneficial solutions.

Additional Tips:

- Cultivate a collaborative, win-win relationship with your vendors rather than a purely transactional one.
- Keep yourself updated on industry trends and market changes that might impact your supply chain.
- Develop a contingency plan for supply chain disruptions caused by unforeseen circumstances.

Effective vendor management is an ongoing process that requires dedication, communication, and collaboration. By implementing these strategies, you can optimize your procurement processes and build strong, reliable partnerships with your suppliers.

7.4.3 Know-how of all electrical tools utilised for system installation.

As an entrepreneur in the solar installation industry, understanding and utilizing electrical tools are essential for successful project execution. Here's an overview of some common electrical tools used in solar system installation:

1. **Multimeter**: This versatile tool is crucial for measuring voltage, current, and resistance. It helps in troubleshooting and ensuring proper electrical connections.
2. **Crimping Tool**: Used to create secure connections between wires and terminals. Proper crimping ensures good conductivity and reduces the risk of loose connections.
3. **Wire Strippers**: Essential for removing the insulation from wires without damaging the conductors. Different gauges of wires may require specific strippers.
4. **Wire Cutters**: Used for cutting and trimming wires to the required length during installation.
5. **Insulation Resistance Tester**: Helps in assessing the insulation resistance of cables and components to prevent electrical faults.
6. **Torque Wrench**: Ensures that bolts and nuts are tightened to the manufacturer's recommended torque settings, which is crucial for maintaining electrical connections.
7. **Heat Gun**: Used for heat-shrink tubing to insulate and protect wire connections from moisture and other environmental factors.
8. **Solar Panel Installation Tools**: These might include specialized equipment like solar panel lifting equipment, racking systems, and alignment tools to properly install and position solar panels.
9. **Safety Equipment**: Personal protective equipment (PPE) such as insulated gloves, safety glasses, and non-conductive footwear are essential to ensure safety during electrical work.
10. **Drills and Fastening Tools**: Necessary for securing mounting brackets, racking, and other components during installation.
11. **Cable Management Tools**: Cable ties, clips, and conduits help in organizing and securing wires to maintain a tidy and safe installation.
12. **Voltage Tester or Detector**: Used to verify the absence of voltage before working on electrical connections, ensuring safety for the installer.
13. **Soldering Iron and Solder**: Sometimes used for making connections between wires or components.
14. **Power Drill and Bits**: Necessary for drilling holes and fastening hardware during installation.
15. **Ladders and Safety Harnesses**: Essential for working at heights safely during solar panel installation.

Remember, utilizing these tools efficiently not only ensures a smooth installation process but also contributes to the safety, durability, and overall quality of the solar system setup. Regular maintenance and proper use of these tools are crucial for the success and longevity of the installed systems. Additionally, staying updated with technological advancements and industry standards is important for an entrepreneur in this field.

7.4.4 Testing and commissioning of equipment in DC and AC side of the plant.

As an entrepreneur in the solar industry, testing and commissioning of equipment on both the DC (Direct Current) and AC (Alternating Current) sides of the solar power plant are critical steps to ensure the system operates efficiently and safely. Here's an overview of the testing and commissioning process from an entrepreneurial perspective:

1. **DC Side Testing and Commissioning**:

a. **Inspection of Solar Panels**: Verify that all solar panels are correctly installed, free from damage, and oriented optimally for sunlight exposure.
b. **String Testing**: Conduct continuity tests to ensure proper wiring connections within each string of solar panels. Use a multimeter to check voltage and current levels.
c. **DC Isolation Testing**: Test the isolation of the DC side components (panels, strings, and inverters) from the ground and ensure there are no leakages using an insulation resistance tester.
d. **Inverter Testing**: Verify the correct installation and functionality of inverters. Test for output voltage, frequency, and ground fault protection.
e. **Array Grounding System Testing**: Ensure the effectiveness of the grounding system to mitigate the risk of electrical hazards.
f. **DC System Performance Check**: Measure the actual output of the DC side under standard conditions to ensure it matches expected values based on design.

2. **AC Side Testing and Commissioning**:

a. **Inverter Grid Connection Testing**: Verify the inverter's connection to the grid and check synchronization with the grid frequency and voltage levels.
b. **AC Isolation Testing**: Ensure the AC side components (inverters, transformers, switchgear) are isolated from the ground and perform insulation resistance tests.
c. **Protection System Testing**: Test protective devices like circuit breakers, relays, and fuses to ensure they operate correctly in case of overloads or faults.
d. **Metering and Monitoring Systems**: Validate the accuracy of energy meters and monitoring systems to track power generation accurately.
e. **Grid Interconnection Testing**: Coordinate with utility companies to conduct tests and obtain necessary approvals for grid interconnection.

3. **Documentation and Compliance**:

a. **Record-Keeping**: Maintain comprehensive records of all tests and inspections conducted during the commissioning process.
b. **Regulatory Compliance**: Ensure that the installation complies with local regulations, standards, and codes. Obtain necessary certifications and approvals.
c. **Handover Documentation**: Prepare detailed documentation for the client, including operation and maintenance manuals, warranties, and as-built drawings.

4. **Safety Measures**:

a. **Safety Protocols**: Ensure that all testing and commissioning activities are performed following strict safety protocols to prevent accidents or injuries.
b. **Training**: Provide training to the client's personnel on operating and maintaining the solar system safely and efficiently.

In conclusion, thorough testing and commissioning procedures are essential for delivering a high-quality, reliable solar power system to clients. It not only ensures the system's performance but also builds trust and credibility for your business in the solar industry. Regular maintenance and periodic inspections after installation further contribute to the system's longevity and customer satisfaction.

7.4.5 how to commissioned the transformer and other HT equipment.

Commissioning high voltage (HT) equipment like transformers involves a series of steps to ensure proper installation, testing, and operation. Here's a general guideline from an entrepreneur's standpoint:

1. **Preparation and Planning:**
    - Ensure you have the necessary permits and approvals for the installation.
    - Review the manufacturer's specifications, installation guidelines, and commissioning procedures for the specific equipment.
    - Plan the commissioning schedule, considering safety precautions, availability of manpower, and necessary tools.
2. **Installation:**
    - Employ qualified personnel or contractors to install the equipment following manufacturer guidelines and local regulations.
    - Verify that the installation is done correctly, including wiring, connections, and grounding.
3. **Inspection and Pre-commissioning Checks:**
    - Conduct thorough inspections to ensure everything is in place and according to specifications.
    - Check for any visible damages or discrepancies that might affect the equipment's performance.
    - Perform pre-commissioning tests such as insulation resistance tests, transformer turns ratio tests, and polarity checks.
4. **Energization and Initial Startup:**
    - Before energizing, confirm that all personnel are clear of the hazardous zones and that proper safety measures are in place.
    - Gradually energize the equipment following the prescribed procedures to avoid sudden surges or faults.
    - Monitor parameters during initial startup, such as voltage, current, and temperatures, to ensure they're within acceptable ranges.
5. **Functional Testing:**
    - Conduct functional tests to ensure the equipment operates as expected. For a transformer, this might include load testing, vector group verification, and voltage regulation tests.
    - Perform control and protection system checks to ensure they're functioning correctly.
6. **Documentation and Reporting:**
    - Maintain detailed records of all commissioning activities, including test results, observations, and any adjustments made during the process.
    - Generate a comprehensive commissioning report summarizing the entire process, highlighting any issues encountered and their resolutions.

7. **Training and Handover:**
    - Provide necessary training to the operating staff on the proper operation, maintenance, and safety procedures related to the commissioned equipment.
    - Hand over the commissioned equipment along with all relevant documentation to the client or responsible personnel.

Remember, commissioning high voltage equipment requires specialized knowledge and expertise. It's crucial to involve qualified professionals and adhere strictly to safety protocols throughout the process. Additionally, compliance with local regulations and standards is essential for successful commissioning and operation.

7.4.6 how to get the CEIG/Electrical inspection done at site.

As an entrepreneur seeking to get the CEIG (Central Electricity Inspectorate for Government) or an electrical inspector's inspection done at your site, you'll need to follow certain steps to ensure compliance with regulations and facilitate a smooth inspection process:

1. **Preparation:**
    - Review and ensure that the installation of electrical equipment complies with the relevant regulations, standards, and guidelines set by the Central Electricity Authority (CEA) or any other relevant authority in your region.
    - Gather all necessary documentation related to the installation, including drawings, specifications, test reports, and compliance certificates.
2. **Application for Inspection:**
    - Contact the local electrical inspectorate or the designated authority responsible for conducting inspections in your area.
    - Submit an application for inspection, mentioning the details of your electrical installation and requesting a visit from the CEIG or electrical inspector.
3. **Coordination and Communication:**
    - Coordinate with the inspectorate or authority to schedule the inspection visit. Ensure that you and your team are available on the scheduled date and time.
    - Communicate any specific requirements or details related to the installation to the inspectorate beforehand.
4. **Readiness for Inspection:**
    - Prepare the site for inspection by ensuring that the installation is complete, and all safety measures are in place.
    - Have all relevant documents, drawings, test reports, and certificates readily available for the inspector's review.
5. **On-Site Inspection:**
    - During the inspection, accompany the inspector and provide any necessary assistance or information they may require.
    - Address any queries or concerns raised by the inspector and take note of any recommendations or corrective actions suggested.
6. **Compliance and Rectification:**
    - If any deficiencies or non-compliance issues are identified during the inspection, take prompt action to rectify them as per the inspector's instructions.
    - Document the corrective measures taken and be prepared for a re-inspection if necessary.

7. **Follow-Up and Compliance Verification:**
    - After rectifying any identified issues, request a follow-up inspection to verify compliance with the standards and regulations.
    - Ensure that all necessary corrections have been made before requesting the follow-up inspection.
8. **Obtaining Certification:**
    - Upon successful inspection and compliance verification, the CEIG or electrical inspectorate may issue a certification or approval for the electrical installation.

Remember, adherence to regulations and standards is crucial for the safety and efficiency of electrical installations. It's essential to maintain clear communication, provide necessary documentation, and promptly address any concerns raised during the inspection to ensure compliance and obtain the necessary approvals or certifications.

Note:
To know more about Application Process for CEIG Approval, you may refer to the following link:
https://www.youtube.com/watch?v=PlOwZWMSBVY

Title of Video:	Application Process for CEIG Approval
By:	Enterclimate
Duration:	6 m 41 sec

Exercise

1. Explain how to identify manufacturers and suppliers through various sources who can meet the requirements.

____

____

____

____

2. Explain how to commissioned the transformer and other HT equipment.

____

____

____

____

3. Explain how to get the CEIG/Electrical inspector inspection done at site.

____

____

____

____

Module 19 - Manage Electrical installation in Ground Mount Solar Plant	Unit 19.2	Application Process for CEIG Approval	113	https://www.youtube.com/watch?v=PIOwZWMSBVY	

Skill India
कौशल भारत - कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

8. Operation and maintenance of Solar Rooftop PV Plant

SCGJ | SKILL COUNCIL FOR GREEN JOBS

UNIT 8.1: Tools Required for PV System Maintenance

UNIT 8.2: Preventive Maintenance of PV System

UNIT 8.3: Troubleshooting and Maintenance

SGJ/N4126

Key Learning Outcomes

At the end of this module, you will be able to:

1. Clean the solar panels periodically
2. Inspect the solar PV system periodically
3. Troubleshoot to identify faults in the system
4. Report and document completion of work
5. Follow quality and safety procedures

UNIT 8.1: Tools Required for PV System Maintenance

Unit Objectives

At the end of this unit, you will be able t o learn about the maintenance

8.1.1 List of Tools for Maintenance Activities

Solar systems generally require special tools. Therefore, it is important that all essential tools, spares and consumables are kept in the site ready for use. A list of such tools and materials is listed below. Person responsible for O&M of solar systems must be familiar and equipped with these tools and equipment and they must be kept in a secured location and maintained properly. Measuring instrument must be checked regularly for its functionality and accuracy.

Table 8.1: List of tools and materials required for O&M of solar systems

Tools required for PV system maintenance		
First aid kit	Needle nose pliers	Compass
System service logbook	Linesman pliers	Flashlight
Datasheet & O&M manual	Diagonal cutters	Sun Pathfinder
This manual	DC soldering iron	Safety goggles
Paper/Pencil	Hacksaw	Rubber gloves
Multimeter	Battery terminal cleaner	Combination square
Clampon ammeter	Battery terminal puller	Small container
Hydrometer	Clamp spreader	Caulking gun
Screwdrivers	Utility knife	Needle nose pliers
Nut drivers 1/4in and 5/16in	Hammer	Wire strippers
Measuring tape (25m)	Cell water filler	Crimping tool
Angle measuring device	Cleaning brush	

8.1.2 Maintenance Schedule and Logbook

- An equipment logbook incorporating the maintenance schedules will be provided to the technicians
- A loose-leaf folder can be used as the system log book with individual sheets added for each item.
- Log books are useful because the historical information they contain can show changes over time, as well as abnormal variations from the usual, indicating a problem in the making.

Table 8.2: Maintenance schedule and logbook

		
First aid kit	Flash light	Cauliking gun
		
Multimeter	Clampon ammeter	Hydrometer
		
Screwdrivers	Nut drivers	Crimping tool set
		
Compass	Sun pathfinder	Angle gauge

		
Battery safety accessories	**Battery water filler**	**Battery maintenance kit**

Figure: Important tools required for testing of PV system

8.1.3 Important to Note

Use relevant worksheet for inspection and testing given in this chapter

- Conduct the tests on a clear sunny day
- Commissioning should begin at the PV array so that if there are any issues with the wiring it can be rectified before the inverter is switched on and prevent damage to the inverter.
- Before commissioning a system, the installer should ensure:
    - All strings are in segments to prevent accidental arcs (i.e. leave one of the module interconnections open).
    - All fuses are removed.
    - All circuit breakers and isolators are in the 'off' position (including the AC and DC isolators at the inverter).
    - That the inverter is turned off.

UNIT 8.2: Preventive Maintenance of PV System

Unit Objectives

At the end of this unit, you will be able to:

1. Perform the weekly, monthly and annual maintenance of the PV modules, PV Array, batteries, inverter, wiring system, earthing and lightning protection components
2. Monitor interaction of the solar PV system with the grid

8.2.1 What is a Maintenance Schedule?

A sample maintenance schedule is presented below to indicate typical frequencies of maintenance actions.

Table 10.3: Sample Maintenance Schedule

Sl. No.	Maintenance Work	Frequency
1	Ensure security of the power plant	Day-to-day
2	Inspect and clean the PV modules from dust and other dirt like bird's dropping etc. as and when required	Weekly
3	Monitor power generation and export	Daily (Remotely)
4	Keep the inverters clean to minimize the possibility of dust ingress	Quarterly
5	Ensuring all electrical connections are kept clean and tight.	Half-yearly
6	Check mechanical integrity of the array structure	Annually
7	Check all cabling for mechanical damage	Annually
8	Check output voltage and current of each string of the array and compare to the expected output under the existing conditions	Annually
9	Check the operation of the PV array DC isolator	Annually

8.2.2 PV Module Maintenance

Performance of a PV system is highly affected when solar modules are not kept clean. Studies show that energy generation of a PV system can go down up to 20% or more if modules are not cleaned regularly. It is recommended to the modules are inspected on weekly basis and as and when required should be cleaned from dust and bird dropping etc. Areas that are generally dusty and polluted would require more frequent inspections and cleaning.

How to clean PV modules?

- Inspect the site prior to cleaning work is performed. Note any hazardous conditions and danger.
- Cover all electrical equipment, i.e. inverters and combiner boxes, beneath area to be cleaned with covers prior to cleaning.
- Inspect surface of solar modules prior to cleaning. All accumulated dirt's to be removed from module surface.

- Use ladder wherever required to reach surface of solar modules to be cleaned.
- Use a wiper with a sponge or brush to clean modules. Use clean water at minimal pressure. No chemicals or abrasive cleaning shall be applied to clean solar modules.

Fig. 10.2.1 Cleaning of modules using a wiper

- Use of hard water for long time will damage the modules by forming a coating on the modules. Avoid using water with hardness more than 200ppm.
- It is advisable to condition the module cleaning water through water softener prior to application. This will minimize mineral deposit on the module.
- If dirt like bird droppings cannot be removed easily use brush with soft bristle to remove such dirt.
- Perform cleaning in early morning or wait until evening, to avoid thermal shock to glass.
- Perform washing only when modules are not in direct sunlight, when the sun is positioned below the horizon.

Fig. 8.2.2 Cleaning modules using a mop

Check PV modules for physical damage:

- Inspect array for broken modules at least once in a year.
- If there is any broken module report to the supervisor or engineer and replace it with appropriate module.
- To replace the broken solar module, isolate the string by switching OFF DC isolator at the string and remove thebroken module. Replace with a correct module and connect. In case if there is no DC isolator on individual string,

switch OFF the main AC isolator between mains and inverter, then switch of the array DC isolator and replace the broken module.

Fig. 8.2.3 Broken module in a PV array

Check PV modules for performance:

- Check output voltage and current of each string of the array and compare to the expected output under the existing conditions.
- Verify output from the array (Isc and Voc and if possible, Imp and Vmp)

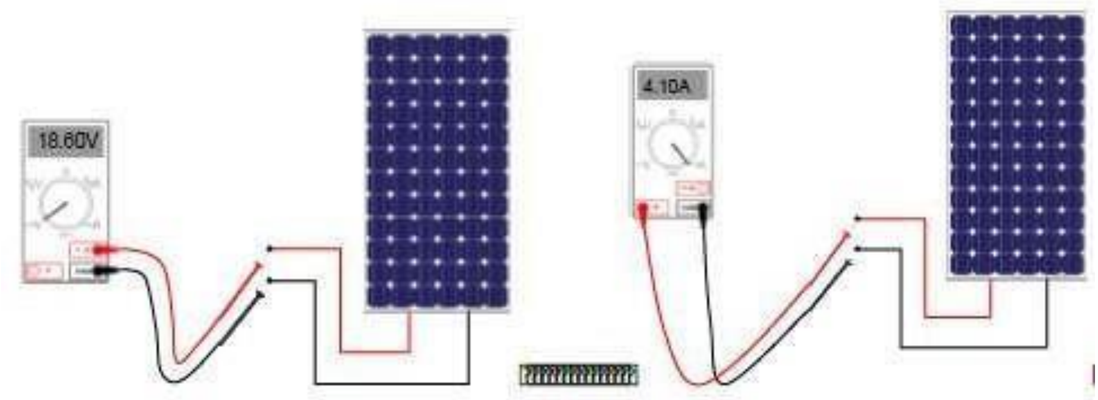


Fig. 8.2.4 Checking current and voltage of PV modules

Picture credit: GSES

Inspection and maintenance of Solar Arrays

- Use a DC clamp-on ammeter to determine the array output current during a sunny weather.
- Conduit and connections must all be tight and undamaged. Look for loose, broken, corroded, vandalized, and otherwise, damaged components. Check close to the ground for animal damage.

Measuring open circuit voltage

- Measure the open circuit voltage of the array as shown in the figure below. Compare the measured amount of open circuit voltage from the array against the manufacturer's specifications.

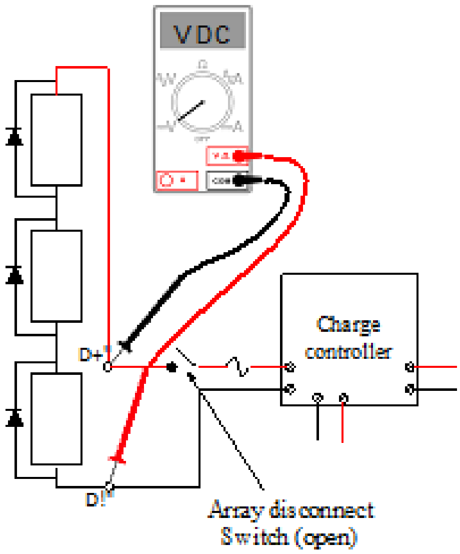


Fig. 8.2.5 Measuring the open circuit voltage of array

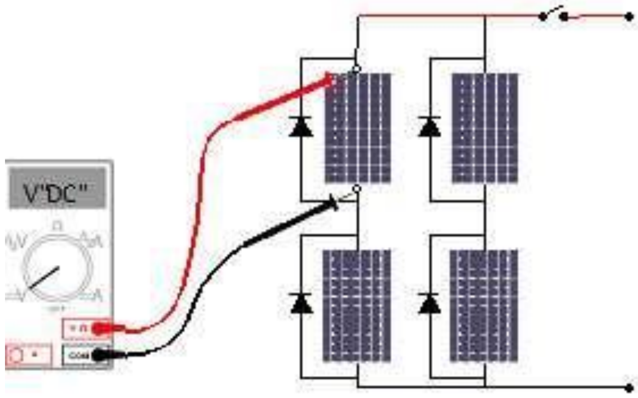


Fig. 8.2.6 Measuring the open circuit voltage of module

Picture credit: GSES

Short circuit current:

- If your DC meter has leads, connect them to the positive and negative terminals of each module and set the meter to the 10A range.

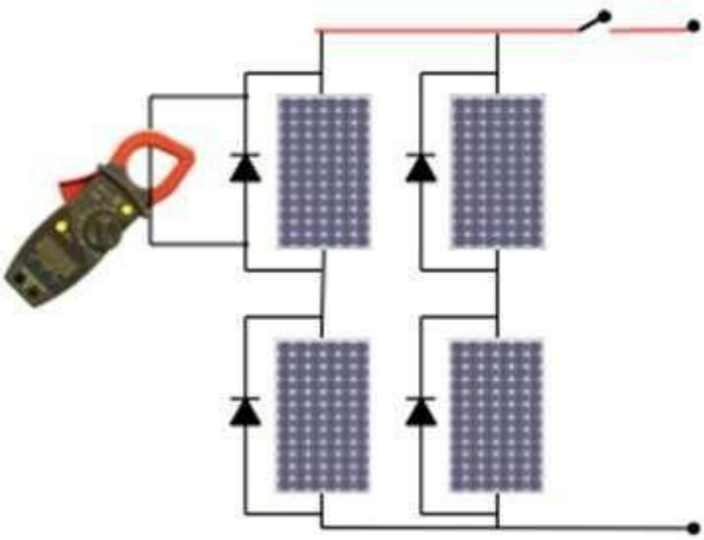


Fig. 8.2.7 Measuring module short circuit current

Picture credit: GSES

8.2.2 Monitor Power Generation

Monitor power generation and export to grid in case of a grid-connected system. This can be done by observing and recording daily energy generation report sent automatically by the inverter communication system. If the inverter communication system does not work, report it to the supervisor or engineer.

Fig. 8.2.8 Daily report of energy generation from a grid connected inverter

8.2.3 Maintenance of Batteries

Weekly maintenance of batteries:

- Clean the battery terminals, surface and surrounding clean
- Observe battery state of charge (SOC) using hydrometer
- In case of VRLA battery use voltmeter to measure voltage to check corresponding SOC.

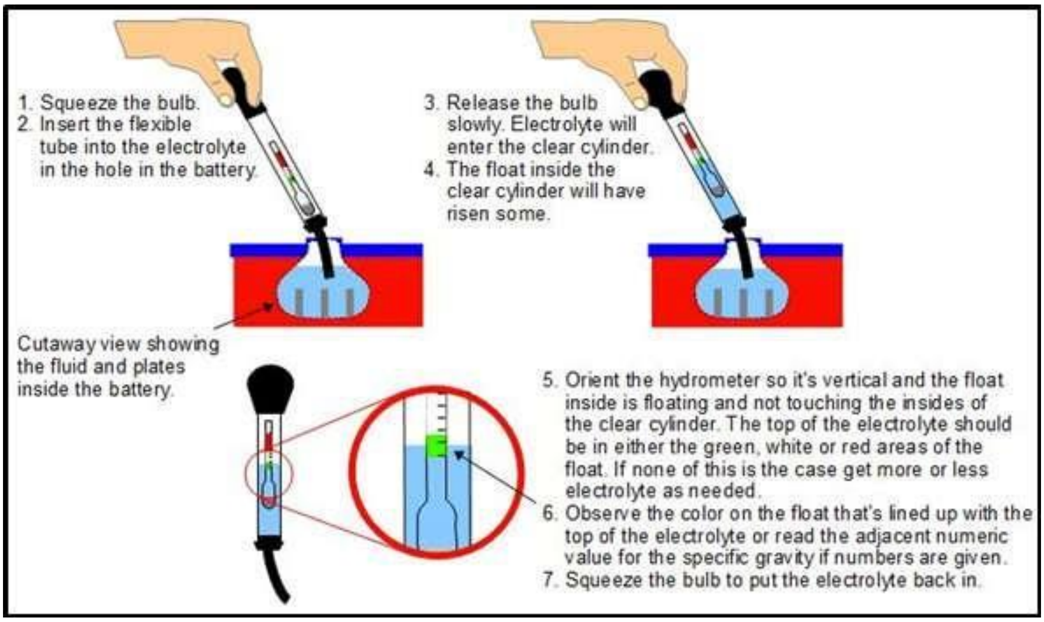


Fig. 8.2.9 Maintenance of battery by testing the electrolyte

Table 8.4: Typical battery voltages as function of state of charge

SOC	Specific Gravity	Battery Voltage	
		12 volt	24 volt
100%	1.265	12.68	25.35
90%	1.250	12.60	25.20
80%	1.235	1Z52	25.05
70%	1.225	12.44	24.88
60%	1.210	12.36	24.72
50%	1.190	12.28	24.56
40%	1.175	12.20	24.40
30%	1.160	12.10	24.20
20%	1.145	12.00	24.00
10%	1.130	11.85	23.70
0 %	1.120	11.70	23.40

Monthly Maintenance

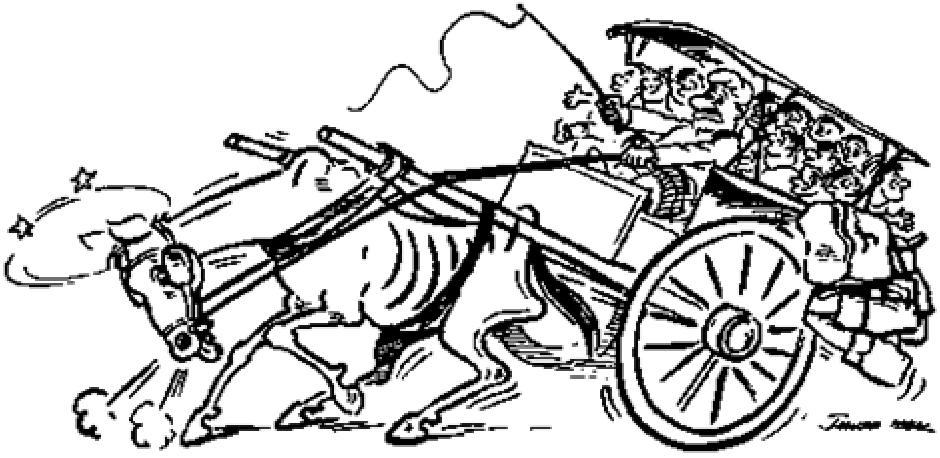
- If flooded lead acid batteries are used check electrolyte level and top up if required. Wipe electrolyte residue from the top of the battery

Fig. 10.2.10 Top up of batteries

- Inspect all terminals for corrosion and loosened cable connections. Clean and tighten as necessary. After cleaning, add anti-oxidant to exposed wire and terminals.

Fig. 8.2.11 Cleaning of batteries terminals to avoid corrosion and loose connections

- Check if new loads have been added and system is overloaded



Checking state of charge

A hydrometer describes the state of charge by determining the specific gravity of the electrolyte. Usually, the specific gravity of electrolyte is between 1.120 and 1.265. At 1.120, the battery is fully discharged. At 1.265, it is fully charged. This procedure is discussed in details in unit 6.6.

Determine state of charge through actual load test:

Use a DC voltmeter (or a multi-meter)

1. Operate the system loads from the batteries for five minutes. Turn off the loads and disconnect the batteries from the rest of the system.
2. Measure the voltage across the terminals of every battery

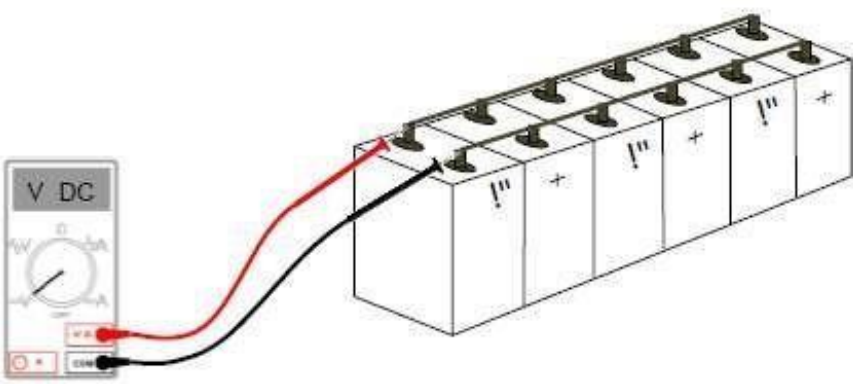


Fig. 8.2.12 Battery load tests using a multimeter (Voltmeter)

Photo credit: GSES

Table 8.5: Open circuit voltages and corresponding states of charge for deep cycle lead acid batteries during a load test

Open Circuit Voltages		State of Charge	
2.12 or more	6.36 or more	12.72 or more	100%
2.10 to 2.12	6.30 to 6.36	12.60 to 12.72	75-100%
2.08 to 2.10	6.24 to 6.30	12.48 to 12.60	50-75%
2.03 to 2.08	6.90 to 6.24	12.12 to 12.48	25-50%
1.95 to 2.03	5.85 to 6.90	11.70 to 12.12	0-25%
1.95 or less	5.85 or less	11.70 or less	0%

8.2.4 Maintenance of Inverter

Inspection and maintenance of Inverters:

- Remove dust or dirt, inspect system wiring for poor connections.
- Check the operation of the inverter at the time of the inspection.
- Measure and record the current draw of the inverter in both idling and operating states.
- Check all inverter wiring for loose, broken, corroded, or burnt connections or wires. Look for potential accidental short circuits or ground faults.
- Look if any object blocks inverter room ventilation and restricts free airflow for natural cooling of inverter. Remove such obstruction or object.

Fig. 10.2.13 Inverter maintenance

8.2.5 Maintenance of Cables, Connectors and Switches

- Visually check all conduit and wire insulation for damage.
- Check for loose, broken, corroded, or burnt wiring connections.
- Check if all equipment’s are connected with proper wire and conduit.
- Make sure all wiring is secured, by gently but firmly pulling on all connections.
- Check all terminals and wires for loose, broken, corroded, or burnt connections or components.

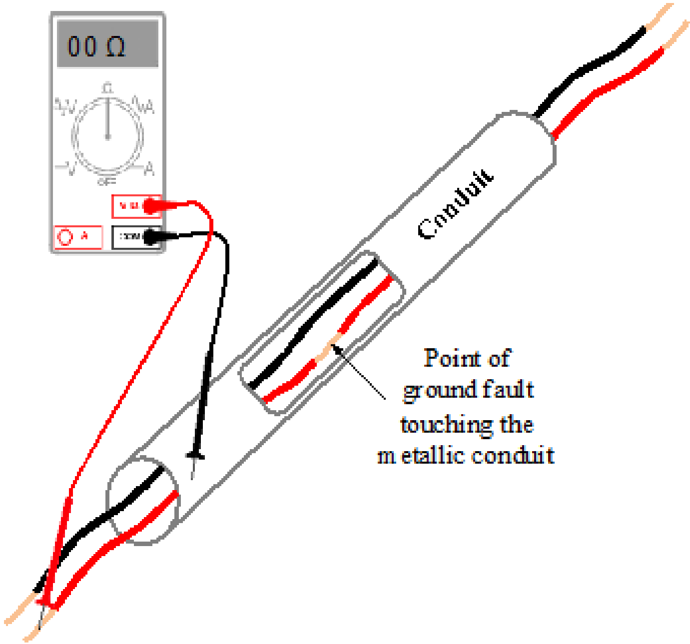


Fig. 8.2.14 Finding a ground fault

Picture credit: GSES

- Check the operation of the PV array DC isolators
- Check array wiring for physical damage and wind chafing

Fig. 8.2.15 Physically damaged wires

- Check array mounting hardware for tightness and corrosion

Fig. 8.2.16 Array mounting hardware to be checked

Inspection and maintenance of Earthing and Lightning Protection:

- Use an ohmmeter of multimeter to check the continuity of the entire grounding system.
- Make sure that all module frames, metal conduit and connectors, junction boxes, and electrical components chassis are earth grounded.

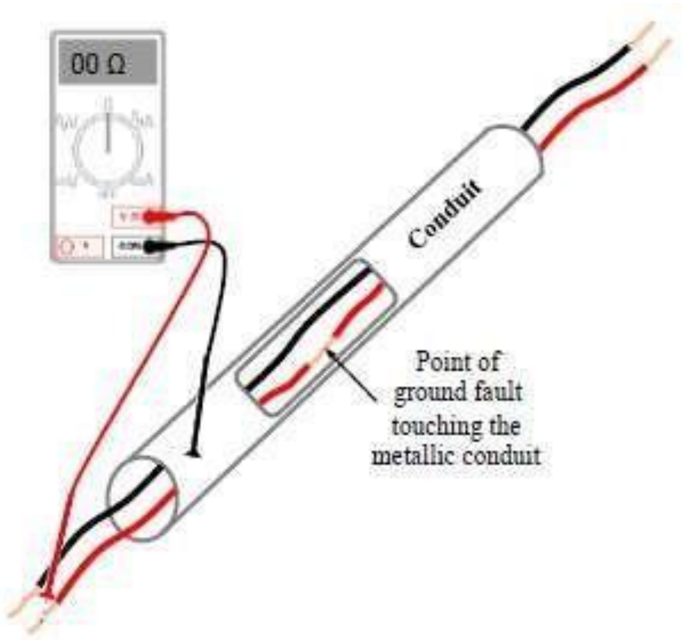


Fig. 8.2.17 Finding a ground fault

Picture credit: GSES

UNIT 8.3: Troubleshooting and Maintenance

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the faults in the system when there is an interruption in power generation
2. Perform regular checks like looking for dust, shade, etc., which might interrupt power output
3. Check current output for each string and identify the string which gives a low / undesired power output
4. Identify the faulty module in the string by shading the modules and checking the output using ammeter reading
5. Perform sequentially the standard troubleshooting activity to identify faults when there is power supply interrupting the grid
6. Check for working conditions of fuses and circuit breakers
7. Check the service panel connections
8. Check the cables and ensure that there is no damage
9. Check the wire connection to inverter and identify for any damage in wire connection
10. Inform the inverter service technician if there is a circuit board level fault for further repair
11. Escalate the issue to superiors if faults cannot be identified

Load does not operate at all	Check if switches are turned off or are in the wrong position	Put all switches in correct position
	Check if system circuit breakers or fuses are blown	Reset circuit breaker or replace fuse
	Check if load is too large for the system	Reduce load size
	Check if there is shadow on solar array	Remove shadow
	Check if weather is cloudy	Wait till weather is sunny and battery gets charged
	Check if load itself is defective	Repair or replace load
Load operates poorly or not at all	Check voltage drop in the system due to small and long wire	Use bigger wire
	Check if system is overloaded	Reduce load size or operating time
	Check if there is a ground fault or a defective diode	Correct ground faults or defective didoes

	Check if wiring or connections are loose, broken burned, or corroded	Repair or replace damaged wiring or Connections.
	Check if there is any short-circuit	Repair short circuits
	Check if small "phantom" load keeps inverter idling, draining battery	Turn off phantom load or replace it with one not requiring PV power
	Check if wiring polarity is reversed	Correct wiring polarity

8.3.2 Troubleshooting with Batteries

Table 10.7: Batteries - Troubleshooting guidelines, check-list and recommended action

Battery voltage remain low	Check if load is too large	Reduce load size
	Check if load operates too long	Reduce operating time
	Batteries are too cold	Insulate battery enclosure
	Check if there is shadow	Remove shadow
	Check if weather is cloudy for several days	Wait till weather is sunny and restrict use of load
	Check if load it is defective	Repair or replace load
Battery do not accept charge	Check if load is too large	Reduce load size
	Check if load operates too long	Reduce operating time
	Check if there is shadow	Remove shadow
	Check if weather is cloudy for several days	Wait till weather is sunny and restrict use of load
Low electrolyte level	Check if battery is overcharged	Add distilled water
Voltage loss overnight even when no loads are on	Check if blocking diode is faulty	Replace diode
High water loss due to overcharging	Check if batteries are overcharged	Repair or replace charge controller
Electrolyte leakage	Check if battery container is broken or leaking	Replace battery/Report dealer/manufacturer
When Battery voltage constantly remain high	Check if charge controller is removed of faulty	Replace the charge controller with lower charge termination setting
	Check if battery capacity is too small for array	Increase battery capacity
	Check if charge controller is misadjusted	Adjust charge controller

8.3.3 Troubleshooting with Charge Controllers

Table 10.8: Charge controllers - Troubleshooting guidelines, check-list and recommended action

Erratic controller operation and/or loads being disconnected improperly	Timer not synchronized with actual time of day	Either wait until automatic reset nest day or disconnect array wait 10 seconds and reconnect array

	Electrical "noise" from the inverter	Connect inverter directly to batteries, put filters on load
	Low battery voltage	Repair, replace battery
Erratic controller operation and/or improper load disconnection	High surge from load	Use large wire to load or add batteries in Parallel
	Otherwise, faulty charge controller, possibly from lightning damage	Repair or replace charge controller and check system grounding
	Adjustable low voltage disconnect set incorrectly	Reset low voltage setting
	Load switch in wrong position on Controller	Reset switch to correct position
	Charge controller has low voltage disconnect feature	If necessary, replace charge controller with one with a low voltage disconnect feature
Fuse to array blows	Array short circuited with batteries still Connected	Disconnect batteries when testing array's short circuit current
	Current output of array too high for charge controller	Replace charge controller with one with a higher rating
Fuse to load blows	Current draw of load too high for charge controller	Reduce load size or Increase charge controller size
	Surge current draw of load too high for charge controller	Reduce load size or Increase charge controller size

8.3.4 Troubleshooting with Inverters

Table 8.9: Inverters - Troubleshooting guidelines, check-list and recommended action

No output from the inverter	Check switch, fuse or circuit breaker open, blown or tripped or wiring broken or corroded	Close switch, replace or reset fuse or circuit breaker or repair wiring or connections
	Check low voltage disconnect on inverter or charge controller open	Allow batteries to recharge
	Check if there is time delay on inverter start-up from idle	Wait a few seconds after starting loads
Loads operating improperly	Excessive current draw by load	Reduce size or loads
	Defective inverter	Replace inverter/report to dealer/ manufacturer
Motors operating at wrong speeds	Inverter not equipped with frequency control	Replace inverter with one equipped with frequency control/report to dealer/ manufacturer
Inverter circuit breaker trips	Load operating or surge current too high	Reduce size of loads or replace inverter with one of larger capacity/report to dealer/manufacturer
Inverter DC circuit breaker trips	Inverter capacitors not charged up on initial start up	Report to dealer/manufacturer

8.3.5 Troubleshooting with Solar PV Arrays

Table 8.10: Solar PV Array - Troubleshooting guidelines, check-list and recommended action

No current from array	Check switches, fuses or circuit breaker open, blown or tipped or wiring broken or corroded	Close switches, replace fuses, reset circuit breakers, repair or replace damaged wiring
	Check if module is damaged or broken	Report
No voltage from array	Check switches, fuses or circuit breaker open, blown or tipped or wiring broken or corroded	Close switches, replace fuses, reset circuit breakers, repair or replace damaged wiring
	Check if module is damaged or broken	Report
Array voltage low	Some modules shaded	Remove source of shading or relocate the Array
	Some array interconnections broken or corroded	Repair interconnections
	Defective bypass or blocking diode	Repair defective diodes
	Some modules damage or defective	Replace affected modules
	Full sun not available	Wait for sunny weather
	Modules are dirty	Clean modules
	Array tilt or orientation incorrect	Correct tilt/ orientation
	Check if module is damaged or broken	Report
	Some modules in series disconnected	Reconnect
	Bypass diode defective	Replace diodes
	Wiring from too long or small	Use large wire
	Check if module is damaged or broken	Report
Array current is low	Check if full sun is not available	Measure current at noon
	Check if modules are dirty	Clean modules
	Check if array tilt or orientation incorrect	Correct tilt/orientation
	Check if battery is fully charged	Wait till battery is charged
	Check if module is damaged or broken	Report
	Check if array is not giving expected load current at full sun	Report

Exercise

1. Name few tools and equipments required for maintenance of PV systems.
2. How often do you need to clean the solar modules and what time of the day you should clean them?
3. How often batteries require topping up and what type of water will you use?
4. During inspection you observe that state of charge of the batteries are very low for long period. What might be the reason and how do you diagnose it?
5. How do you check if modules are working satisfactorily?

6. What might be the consequence of shorting the batteries and how will you avoid that?
7. What are the safety resources required for battery maintenance?
8. What is catalytic recombination cap/ ceramic vent plug and how is it fixed?
9. Why you should not use detergent to clean PV modules?
10. What happen if you use hard water to clean the modules for long time?
11. How will you clean sticky dirt from the modules?
12. What equipment you will use to check if there is shadow possibility in any point of time in a year?
13. How will you check for the conditions of mounting and its stability to hold solar panels?
14. How do you identify faults in a PV system when there is an interruption in power generation?
15. How do you monitor performance of a PV system?

Notes

Skill India
कौशल भारत-कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

SCGJ | SKILL COUNCIL FOR GREEN JOBS

9.Operation and maintenance of Solar Ground Mounted PV Plant

SGJ/N4126

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Understand the purpose and scope of operation and maintenance activities,
2. Identify different maintenance approaches,
3. Oversee inspection and maintenance of plant components,
4. Troubleshoot common faults, and ensure optimum plant performance through proper maintenance practices.
5. Explain how to check the integrity and working condition of all connections, fuses and circuit breakers.

UNIT 9.1: Operation and Maintenance of Solar Ground Mounted PV Plants

Unit Objectives

At the end of this unit, participant will be able to:

1. Explain how to handle and utilize tools and equipment for O&M available at site.

9.0.0 List of Tools for Maintenance Activities

Solar systems generally require special tools. Therefore, it is important that all essential tools, spares and consumables are kept in the site ready for use. A list of such tools and materials are is listed below. Persons responsible for O&M of solar systems must be familiar and equipped with these tools and equipment and they must be kept in a secured location and maintained properly. Measuring instrument must be checked regularly for its functionality and accuracy.

Table 7.1: List of tools and materials required for O&M of solar systems

Operation and maintenance play a critical role in maintaining the efficiency and reliability of solar power plants throughout their operational life. The primary objective of O&M is to ensure that the plant continuously generates electricity at the expected performance level while maintaining safety and equipment integrity. Effective maintenance reduces unexpected breakdowns, improves energy yield, lowers repair costs, and enhances the return on investment. Ground mounted solar plants are exposed to various environmental conditions such as dust, temperature fluctuations, humidity, rain, and wind, making regular maintenance essential for sustained performance.

Maintenance activities in solar power plants are generally categorized as preventive, predictive, corrective, and breakdown maintenance. Preventive maintenance involves scheduled inspections and servicing activities carried out at regular intervals to avoid equipment failures. Predictive maintenance uses performance data, monitoring systems, and diagnostic tools to identify potential issues before they become critical. Corrective maintenance involves repairing identified faults that affect plant performance, while breakdown maintenance is performed after equipment failure has occurred. The selection of an appropriate maintenance strategy depends on plant size, operating conditions, equipment criticality, and maintenance resources available.

Operation and maintenance personnel must be familiar with the various tools and equipment used during inspection and maintenance activities. These include multimeters, clamp meters, insulation resistance testers, thermal imaging cameras, earth resistance testers, torque wrenches, cleaning equipment, and personal protective equipment. Proper use of these tools helps technicians identify abnormalities, verify equipment health, and ensure compliance with safety requirements. Regular calibration and maintenance of testing instruments are important to ensure accurate measurements and reliable results.

During routine maintenance, the condition of junction boxes, combiner boxes, cables, connectors, fuses, and circuit breakers should be carefully inspected. The integrity of these components is critical for maintaining uninterrupted power flow from the solar modules to the inverter. Loose connections, damaged insulation, corrosion, moisture ingress, and overheating can significantly affect plant

Type of Maintenance	Description	Typical Frequency	Main Benefit
Preventive Maintenance	Scheduled inspection and servicing	Monthly/Quarterly	Reduces equipment failures
Predictive Maintenance	Data-based monitoring and diagnostics	Continuous	Early fault detection
Corrective Maintenance	Repair of identified faults	As Required	Restores system performance
Breakdown Maintenance	Repair after equipment failure	Unscheduled	Restores plant operation
Condition-Based Maintenance	Maintenance based on equipment condition	Periodic Monitoring	Optimizes maintenance cost

Table 9.1 Types of Maintenance in Solar Ground Mounted PV Plants

UNIT 9.2: Inspection, Troubleshooting and Performance Optimization

Unit Objectives

At the end of this unit, participant will be able to:

- Explain about inspection, Troubleshooting and Performance Optimization

Inspection, Troubleshooting and Performance Optimization

The operation and maintenance team must regularly verify that all cables, connectors, junction boxes, combiner boxes, and inverter connections comply with approved design drawings and as-built documentation. Proper cable continuity and insulation resistance are essential for efficient power transmission and electrical safety. Visual inspections should identify signs of mechanical damage, rodent attacks, loose terminations, overheating, corrosion, and insulation deterioration. Any discrepancies observed during inspections should be recorded and addressed promptly to prevent further degradation of system performance.

Troubleshooting is an important aspect of operation and maintenance activities. Faults may occur due to damaged cables, defective modules, blown fuses, inverter alarms, communication failures, or environmental factors. The troubleshooting process begins with identifying symptoms, collecting operational data, performing inspections, and isolating the source of the problem. Once the root cause is identified, corrective actions should be initiated or escalated to the appropriate technical personnel. Timely fault resolution minimizes downtime and prevents energy generation losses.

Earthing and lightning protection systems are critical safety components that require periodic inspection and testing. Maintenance personnel should verify that earthing conductors, earth pits, bonding connections, and lightning arrestors are installed according to approved drawings and remain in good condition. Earth resistance measurements should be performed periodically to ensure compliance with applicable standards. Any deterioration in the earthing system can compromise equipment protection and personnel safety.

Module cleaning is another essential maintenance activity, particularly in areas with high dust levels. Dust accumulation, bird droppings, leaves, and other contaminants can reduce solar radiation reaching the modules and decrease energy generation. Cleaning schedules should be established based on local environmental conditions, and cleaning should be carried out using approved methods that do not damage the module surface. In addition to cleaning, vegetation growth and nearby objects should be monitored and controlled to prevent shading losses.

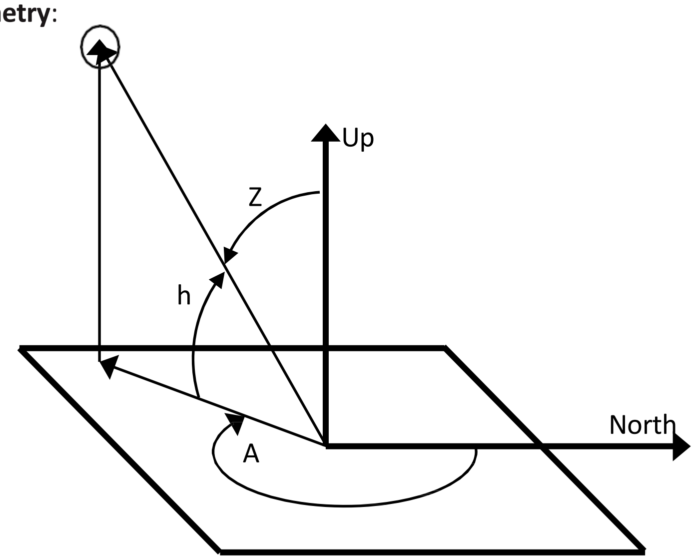
A well-managed operation and maintenance program combines scheduled preventive maintenance with corrective maintenance activities whenever faults occur. Modern monitoring systems allow operators to track plant performance, identify deviations from expected generation, and take corrective action quickly. Best industry practices include maintaining detailed maintenance records, conducting periodic performance analysis, following safety procedures, and implementing continuous improvement measures. Successful solar power plants achieve high availability and optimum performance through disciplined maintenance planning and execution.

Fault Identified	Possible Cause	Corrective Action
Low Energy Generation	Module soiling or shading	Clean modules and remove shading source
String Failure	Loose connection or damaged cable	Inspect and repair cable connections
Inverter Trip	Grid fault or protection issue	Check inverter alarms and protection settings
High Temperature at Connection Point	Loose termination	Tighten connections and inspect components
Blown Fuse	Overcurrent or short circuit	Identify fault source and replace fuse
Communication Failure	Network or monitoring issue	Verify communication cables and settings
High Earthing Resistance	Poor earth pit condition	Improve earthing system and retest
Water Ingress in Junction Box	Damaged gasket or sealing	Replace seals and ensure proper enclosure protection

Table 9.2 Inspection, Troubleshooting and Performance Optimization

Terminologies Used in Solar Industry

- **Radiation**: The electromagnetic energy which is produced by the sun and transmitted through photons is termed as solar radiation.
- **Solar Irradiation (Energy)/Insolation**: It is the sun's radiant energy incident on a surface of unit area. Irradiance is measured in watt-hour/m^2 or kW-h/ m^2
- **Equinox**: Literally means "equal night", a day when the number of hours of daylight equals the number of hours of night. The vernal equinox, usually March 21, signals the onset of spring, while the autumnal equinox, usually September 21, signals the onset of autumn.
- **Solstice**: A Day when the sun is at the highest point in the sky (summer solstice, 21 June) or at the lowest point in the sky (winter solstice, 21 December).
- **Zenith Angle**: The angle between the sun's rays and the vertical direction. It is the complement to the solar altitude or solar elevation, which is the altitude angle or elevation angle between the sun's rays and a horizontal plane.
- **Azimuth Angle**: The angle between the horizontal direction (of the sun, for example) and a reference direction (usually north, although some solar scientists measure the solar azimuth angle from due south).
- **Elevation**: The elevation angle is the angular height of the sun in the sky measured from the horizontal. The elevation is 0^0 at sunrise and 90^0 when the sun is directly overhead.
- **Angle of Incidence**: The angle made by a ray (solar energy) with a line perpendicular to the surface. For example, a surface that directly faces the sun has a solar angle of incidence of zero, but if the surface is parallel to the sun (for example, sunrise striking a horizontal rooftop), the angle of incidence is 90^0.
- **Solar Geometry**:



H = elevation angle, measured up from horizon

Z = zenith angle, measured from vertical

A = azimuth angle, measured clockwise from north

- **Air Mass**: The air mass is the path which light takes through the atmosphere normalized to the shortest possible path length (that is, when the sun is directly overhead). The air mass measures reduction in the power of light as it passes through the atmosphere and is absorbed by air and dust. The air mass is defined as
    - Air Mass = 1/ cos θ
    - Where θ is the angle from vertical (Zenith Angle)
- **Solar Noon**: The time at which the position of the sun is at its highest elevation in the sky. At this time, the sun is either due south (typically in the northern hemisphere) or due north (typically in the southern hemisphere).
- **Solar Thermal Technology**: It refers to the direct conversion of solar energy into heat energy.
- **Solar Photovoltaic (PV) Technology**: It refers to the direct conversion of solar energy into electrical energy.
- **Solar PV Cell**: It is defined as the semiconductor device that directly converts Light energy into DC (direct current) electricity,
- **Solar PV Module**: It is defined as the series connected assembly of solar PV cells to generate DC electricity.
- **Solar PV Array**: It is defined as the connected (series/parallel or both) assembly of solar PV modules to generate DC electricity.
- **Electrical Grid/Grid**: It is defined as the interconnected network which delivers electricity from suppliers to consumers.

Material Receipt

Material Receipt at the Site

Date ____________ | Name of the driver : ____________________

Site Location ____________________ | Vehicle No : ____________________

Customer Name : __________________ | Time of Delivery : ____________ __________

Delivery Challan/Gate Pass No : ______________________ | Invoice No : ____________ | Date of Invoice ____________

Sl No.	Bill of Materials as per Design document	Quantity as per BOM	Material received at the site	Quantity Received	Condition of the material received based on Visual inspection	Remarks
1	Solar PV Modules 540Wp	10 nos				
2	Mounting Structure	nos				
3	MMS Hardware	nos				
4	Bolts, Washers, & Nuts	nos				
5	Cable 6Sqmm	mts				
6	Cable 4Sqmm	mts				
7	Cable 2.5Sqmm	mts				
8	MC4 Connectors					
9	Inverter's					
10	Batteries					

Receiver's Signature
Solar PV Engineer / Site Supervisor

Verifier's Signature
Preferably Customer / Representative

Driver's signature
Mobile No :

Site Assessment/FIR Sample Format

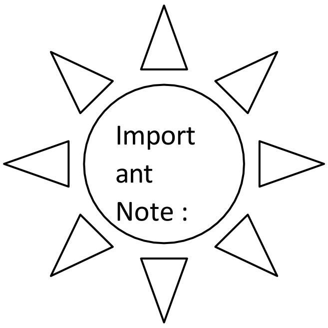
Name of the Customer			
Contact Address			
Contact No & Email address			
Site Address for Installation			
Site Latitude	(GPS Coordinates)		
Site Longitude			

Rooftop Photographs to be pasted below (Use GPS Camera only)	
1(from East)	
2(from west)	
3(from North)	
4(from South)	

Observations / Remarks on the site which will the Designer/Technical person to understand the site better. (For Example There is a building on the south whose shadow is covering the roof/ there is a tree / anything that has to be considered to be either positive/negative impacting the performance of the SPVRT). It should cover availability of space for movement of men and material to the rooftop.	

ENERGY AUDIT

Data to be collected for an Off Grid SPVRT System.				
Sl No.	Load Description	No Of Hours of Operation - 1	Wattage - 2	Energy Demand
1	Fan			
2	Tube Lights (LED / T5 / Electronic Choke/ Copper Choke / Nylon Choke)			
3	CFL			
4	Water Pump			
5	Television			
6	Desk Top / Lap Top			
			Total Energy Demand	



SPV BDE should always think of how to benefit the customer,
He / She can only do it when they keep updating the knowledge on latest electronic & electrical Appliances launched in the market and the BEE Star Ratings. If possible should try to undergo energy auditors course if eligible / atleast read and digest the study material of the course.

Measurements (Offgrid):

Sl No.	Description	Distance from the SPVRT in meters
1	SPVRT to Inverter	
2	Inverter to Battery	
3	Inverter to Termination Point	
4		

MONTHLY AVERAGE ENERGY CONSUMPTION

Data to be collected for an Grid Interactive SPVRT System.			
Sl No	Month & Year	Consumption in Kwhrs	Photocopy of the Utility bill Collected (Yes / No)
1			
2			
3			
4			
5			
6			
7			
8			
Total Annual consumption			
Annual Average consumption. (Total/12)			

*This data will help you arrive at the energy generation demand, which in turn will result in determining SPVRT System capacity.

Measurements (Grid Interactive SPVRT):

Sl No.	Description	Distance from the SPVRT in meters
1	SPVRT to Inverter	
2	Inverter to Termination Point (Normally Bi-Directional Meter point / Gross Meter point where the safety switch gear is also available).	
3	Cable lengths from Array to the Inverter placement point	
4	Height of the Building (Approx)	

Additional Information sheet

Notes

Skill India
कौशल भारत-कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

N·S·D·C
National Skill Development Corporation
Transforming the skill landscape

SCGJ | SKILL COUNCIL FOR GREEN JOBS

10. Maintain Personal Health& Safety at Project Site

SGJ/N0106

Key Learning Outcomes

At the end of this module, the participant will be able to:

1. Explain the requirements for safe work area.
2. Explain the importance of administering first aid.
3. Identify the personal protective equipment used for the specific purpose.
4. Identify the hazards associated with photovoltaic installations;
5. Identify work safety procedures and instructions for working at height.
6. Explain the importance of Occupational health & Safety standards and regulations for installation of Solar PV system.
7. Incorporate good housekeeping practices and infection control guidelines.

UNIT 10.1: Workplace Safety

Unit Objectives

At the end of this unit, participant will be able to:

1. Recognize the potential sources of accidents
2. Classify the safety policies
3. Identify the responsibilities of managers and employees regarding safety
4. List the general safety guidelines
5. Recognize the safety measures and precautions

10.1.1 Potential Sources of Accidents

People are becoming aware of the fact that a healthy workforce is profitable for the employers as well. A safe, healthy and positive work environment can boost the morale and work-life balance of the employees and, as a result, be beneficial for the business.

Health refers to the general well-being of a person. A healthy person, who in this case is a PV engineer, is free from all diseases and has a sense of mental, physical and social well-being. A healthy PV engineer can:

- Help to prevent absence from work
- Improve work performance
- Demonstrate work responsibility
- Reduce accidents and work-related ill health

Just as an installation of a safe PV system takes place according to a building's code and standards, similarly, an engineer's safety is also important. Few of the safety considerations are:

- Work area
- Equipment involved and usage of tools
- Forms of personal protection
- Safety hazards information

PV system installation involves various safety hazards mainly as a result of an electric shock or a fall from a rooftop. A significant/theoretical understanding of electrical systems and respective safety hazards associated with it is essential in order to work safely with PV systems.

An engineer must be aware of the potential sources of accidents so that he/she can avoid them or ensure minimizing their impact at the workplace.

The engineer needs to:

- Spot and report potential hazard timely.
- Follow rules related to hazardous materials as per the company's policy.
- Handle all equipment and tools with care.
- Avoid mishaps that may be a result of dangerous gases, chemicals or sharp tools; and potential hazards that may be an outcome of probable injuries like minor burns or cuts.

The following figure lists the types of hazards:

Safety hazards	Biological hazards	Chemical hazards	Physical hazards	Ergonomic hazards

Fig. 10.1.1: Types of Hazards

Safety Hazards

Death or any type of illness or injury caused due to unsafe conditions are categorized under safety hazards. The following figure lists some instances of safety hazards:

- Spillage on floor or a person tripping due to cords lying loose across the floor
- Working at varied heights or on a roof with the help of ladders or scaffolds
- A person accidently touching any unguarded machinery or moving parts of a machinery
- Electrical hazards as a result of frayed cords, improper wiring or missing ground pins
- Restricted spaces
- Hazards related to machinery (boiler safety, lockout/tagout or forklifts)

Fig. 10.1.2: Instances of safety hazards

Biological Hazards

Any biological substance that may threaten health of humans as a result of toxins or viruses are known as biological hazards. A person may be exposed to biological hazards in the following cases:

- Schools, colleges and universities
- Day care facilities,
- Hospitals, laboratories and nursing homes
- Outdoor occupations

The following figure lists the types of ill- health effects caused by biological hazards:

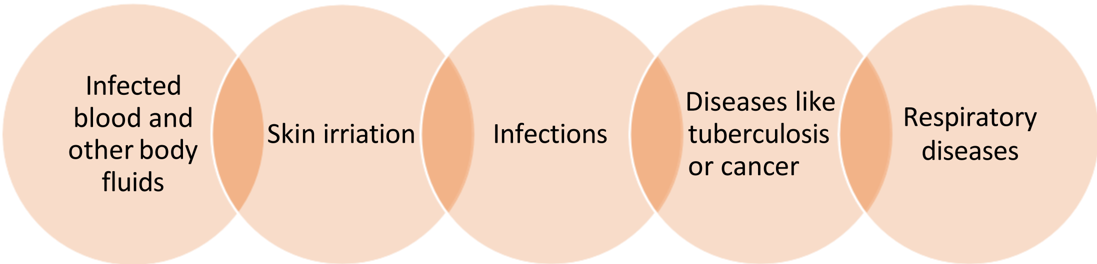


Fig. 10.1.3: Types of ill-health effects caused by biological hazards

Physical Hazards

An occupational hazard caused by environmental factors is termed as a physical hazard. It includes hazards such as:

- Radiation caused by radio waves, microwaves or EMFs
- Sunlight/ultraviolet rays' exposure
- Extreme temperatures, be it hot or cold
- Noise pollution

Ergonomic Hazards

Ergonomic hazards occur due to single/multiple factors within the working environment that pose a threat to the musculoskeletal system of an individual. An uncomfortable workstation leading to wrong sitting postures, repetitive movement of a body part causing sprain or strain, muscle sores, etc., are categorized under ergonomic hazards. The following figure lists some instances that may cause ergonomic hazards:

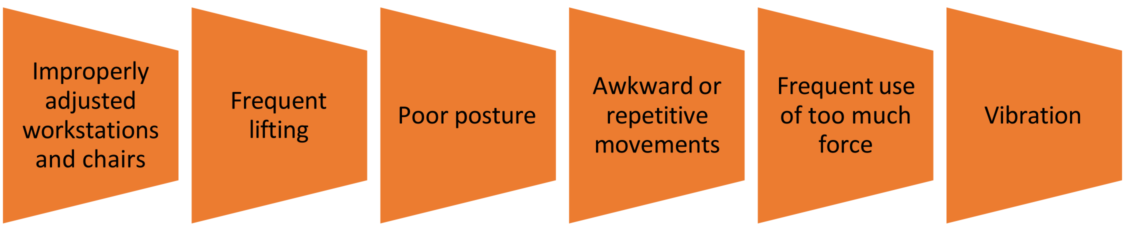


Fig. 10.1.4: Instances that may cause ergonomic hazards

Chemical Hazards

Exposure to chemicals at a workplace is the main cause of chemical hazards. Exposure to chemicals can be due to working around items that involve chemical preparations in any state solid, liquid or gas. Not all chemicals pose a threat, but there may be workers who are sensitive to even the mildest or non-toxic forms of chemical that is termed healthy. A person can be exposed to chemicals by inhalation of fumes, ingestion or poisoning.

The following figure lists some chemicals that one should be aware of:

- Exposure to cleaning agents, acids, paints or their solvents (dangerous if stored in unlabelLed containerS)
- Exposure to vapors and fumes as a result of welding
- Exposure to toxic gases like propane, helium, acetylene and carbon monoxide
- Exposure to flammable materials like explosive chemicals and gasoline
- Pesticides

Fig. 10.1.5: Some chemicals that can cause hazards

10.1.2 Safety Policies

A health and safety policy is a written declaration made by an employer. It states the company's commitment for safeguarding the health and safety of the workers and also is an assurance to the public. It is a signed document made by the management related to the health and safety of the employees. A workplace requires a health and safety policy for the following reasons:

- To show complete commitment towards their health and safety of employees
- To prove to the employees that safety performance and work performance are in harmony with each other
- To give a clear statement of the company's objectives, principles, plans, ideas and procedures
- To increase buy-in through all divisions of the company

- To provide a definite outline of the accountability of the organization for the health and safety of the workers and also give a clear idea of the responsibility of the employer and the employees.
- To abide by the Occupational Health and Safety Act
- To define practices and processes to be adhered at the workplace to avoid injuries and diseases

EHS Policy

The following figure highlights some important points from EHS policy guidelines:

Provide a conducive work environment for all employees, ensuring their health and safety while taking all relevant steps to protect the environment.

Protect the environment by conserving natural resources through elimination of waste generation and prevention of environmental pollution.

Protect the health and safety of all employees, visitors, contractors or suppliers at the plant.

Comply with all applicable regulatory requirements legally that relate to occupational and environmental health and safety.

Make continual improvements in protecting environment and occupational health and safety performance by improving processes, introducing and investing in new technologies and upgrading competence and awareness of the employees.

Fig. 3.1.6: Some important points from EHS policy

EMS Policy

To minimize the risk of 'environmental impacts', some objectives and targets are set. The following are the primary objectives and goals set at ISO14001- certified plants:

- Minimization of wastes and enhancement of recycling rate
- Advancement of energy and resource conservation
- Adequate supervision of chemical substances
- Promotion of environmentally friendly products

The following table lists some other objectives and targets of EMS policy:

Aspect	Objectives	Targets
Use of paper	Reduction in use of paper	Reduction in use of paper by 25% within a year
Consumption of electricity	Reduction in the consumption of electricity	Reduction in the consumption of electricity by 20% within 1 year (based on current year consumption)
Use of cleaning material	Reduction in the use of hazardous cleaning material	Reduction in the usage of cleaning material by 25% within a year
Use of hazardous chemicals such as solder paste, epoxy potting and tin-lead solder	Increased use of environmentally friendly alternative chemicals	Reduction in the usage of these chemicals by 5% within a year

Table 3.1.1: EMS policy

Health and Safety Policy

The following table lists some of the objectives and key measures of health and safety policy:

Priority Actions	Measures/Key Performance Indicators
Make workplace free from injuries and illnesses	
• Improve workstation ergonomics • Reduce manual handling slips, trips and falls • Provide EHS training, safety audits and participation	• Number of incidents/accidents and near misses • First aid incidents • Number of days lost
Improve Occupational Health and Safety Assessment Series (OHSAS) knowledge throughout the organization	
• Provide EHS training, as per defined needs, of at least 6 hours to each employee • Improve OHSAS knowledge by employee participation	• Number of hours of training given to each employee in a year • Number of suggestions received from workers to improve workplace safety besides internal audit findings and safety audit findings

Table 3.1.2: Health and Safety Policy

Responsibilities towards Safety Policies

All the members in an organization have their own responsibilities to ensure workplace safety. There is a safety committee in an organization. All the workers and managers as well as the safety committee members need to carry out their responsibilities related to safety. The following figure highlights the responsibilities towards safety policies:

Safety Committee Responsibilities

- Detect unsafe conditions and work practices by inspecting the facility regularly
- Participate actively in health and safety training programmes and evaluate the programme's effectiveness
- Plan improvements to existing procedures, health and safety rules, and plan safety promotions/incentives
- Assessing personal protective equipment (PPE) needs
- Overseeing emergency response preparedness and tests
- Performing job safety analysis and serving as a resource for questions on safety
- Immediately investigate workplace accidents

Responsibilities of Managers

- Familiarize employees with the OHS program, ensuring its effective implementation in the plant and enforce all company safety rules consistently and fairly
- Beware of all safety considerations on introduction of a new process/procedure, machine/material to the workplace and providing safety training to employees before assigning duties
- Give maximum support to programmes and committees that promote health and safety
- Review and complete all reports related to an accident/incident, when required, to ensure fulfilment of all requirements so that appropriate action is taken to prevent recurrence
- See that all injuries, even minor ones, are treated immediately and referred to a medical facility, if necessary
- Inspect the work area regularly to look out for unsafe conditions and work practices and ensuring that unsafe or harmful conditions found in the course of inspections or incident investigations are remedied without delay
- Conduct risk assessments for potential hazards from ergonomic hazards, violence at the workplace or emergencies to minimizing risks.
- Conduct a safety orientation for all new employees, as well as for existing employees who are learning a new role and including the risks associated with the role
- Ensure that all employees are healthy at work

Responsibilities of Workers

- Work with care and attention at all times and wear protective equipment as needed or required
- Operate equipment/machinery after receiving the proper instructions
- Report minor/major injuries/symptoms to the manager to enable action towards prevention and take possible remedy action to ensure that no one gets hurt
- Encourge co-workers to work safely and assist them to have a safe working environment by suggesting actions to improve the effectiveness of the programme on health and safety
- Maintain equipment in good, safe working order and check the usage of guards and safety devices on equipment as per the standards
- Adhere to all safety rules and regulations
- Take necessary steps to improve health and attendance as every employee has an obligation to meet the attendance requirements of the job; When an employee shows extraordinarily long absences, the rest of the team can be placed in a compromising position and conditions could become unsafe

Fig. 3.1.6: Responsibilities towards safety policies

10.1.3 General Safety Guidelines

An employee should obey some general guidelines as listed in the following figure to ensure a safe and healthy workplace:

- Never bypass or disable a safety device or system.
- Operate company machinery and vehicles only if authorized and trained.
- Maintain good housekeeping in work area.
- Keep stairs, aisles and hallways clear of unnecessary litter or material.
- Use the appropriate tool for a job, as trained by the supervisor or job safety analysis (JSA) report or as per the manufacturers' instructions.
- Inspect the area and equipment as required by the supervisor.
- Ensure that no illegal drugs/ alcohol are in use as per the terms of the company's drug and alcohol policy.
- Report to the immediate supervisor prior to the start of work if taking prescribed medication that might have some work restriction.
- Ensure no firearms are present on the property, including the parking lot or on the jobsite.
- Wear seatbelt at all times when a company vehicles is to be operated.
- Report all equipment maintenance issues to the supervisor.
- Dress appropriately for duties.
- Avoid horseplay to keep oneself/co-workers away from any danger.
- Never engage in hostile actions against other employees.
- Report all hostile actions of co-workers to the supervisor.
- Smoke in designated areas.
- Maintain good personal hygiene.

Fig. 10.1.7: General safety guidelines

10.1.4 Safety Measures and Precautions

Safety procedures and measures are dependent on the type of work. There may be a need for electrical safety, fire safety or mechanical safety for the engineer. The following figure lists the general measures an employee should be aware of, to ensure safety:

Daily safety instructions

- Take safety measures to prevent accidents.
- Ensure zero accidents while at work.
- Avoid damaging components caused by negligence in electrostatic discharge (ESD) procedure.
- Ensure no loss for company due to safety negligence.
- Ensure proper maintainance of machine and work process for achieving quality output as per the company standards.

Before starting work

- Plan and discuss requirement of work to be done.
- Consider potential hazards and measures to be taken.
- Confirm permission to isolate (use a permit system if relevant).
- Isolate the electrical equipment or circuit.
- Place a "DANGER, DO NOT OPERATE" tag.
- Put up safety barriers when required.
- Use the correct earthing equipment.
- Cover and insulate a nearby live apparatus.
- Check test instruments and get authorization to do the work.

When working

- Use safety observers when required.
- Always wear PPE.
- Never rely on memory.
- Connect the earth and neutral conductors first.
- Check the isolation points before resuming work after a break.
- Check and clean the tools that are used regularly.
- Use non-conducting tape measures.

After completion of work

- Check if tools are left after work completion.
- Remove own earthing equipment.
- Notify all personnel involved that the equipment will be energized.
- Hand in the work permit (if relevant).
- Remove "DANGER, DO NOT OPERATE" tags.
- Switch off all machineries.
- Remove and store all PPE properly.

Fig. 10.1.8: General safety measures for an employee

Electrical Safety

One of the most potential causes of an accident may be an electric shock for working professionals on site. The precautions are of utmost importance when it comes to the job role of a Solar PV Engineer.

To ensure electrical safety, the engineer needs to check the following points:

- Adequate wiring
- Appropriate electrical equipment used along with its correct label and capacity
- Working and good condition of equipment
- Listing of current breaks for the circuit breaker
- Unexposed electrical parts
- Overhead power-lines to be out of contact range from work area
- Proper insulation of wires
- Double insulated or grounded electrical systems and tools
- No overloaded circuits
- Removal of damaged power tools/equipment
- Usage of appropriate PPE by employees
- Usage of appropriate tools by employees
- Labelled and correct usage of chemicals
- Ladders do not conduct electricity
- Dry area without any standing water
- Securely installed equipment
- Equipment is not exposed to possible overheating due to poor air circulation or covering of the ventilation device.

Fig. 10.1.9: Ensuring electric safety

Some electrical safety points while installation and maintenance of a PV system are mentioned in the following figure:

Handling PV modules exposed to sunlight

- Read the warnings mentioned at the back side of the PV modules for handling them.
- As PV modules produce electricity when they are exposed to sunlight, precautions should be taken so that the person carrying or handling the PV modules does not get a shock.
- The positive and negative terminals of the PV modules should be kept away from each other to avoid the shorting of terminals when the PV module is under sunlight.

Handling batteries in PV system

- Always remember to disconnect the batteries before moving or handling them.
- Avoid shorting the terminals of batteries to avoid severe damage.
- While checking the battery voltage with multimeter, always check if the multimeter is in voltage mode. If the multimeter in current mode is connected to the battery terminals, then the short circuit to the battery terminals will take place. This can result in severe damages and also cause fire.

Safe distance from exposed electrical conductors

- Knowledge of the location of an exposed electrical conductor helps in keeping a safe distance from it, but at the same time, knowing the potential in that conductor is also very important.

Undertake proper work practice in wet or damp locations containing electricity

- Working area selection is a must for electrical safety which involves checking the location for sufficient space, conducting material around and also, if the place is wet or damp.

The proper lockout/tag-out procedures for electrical equipment and systems

- Lockout/tag-out procedure is always required for electrical equipment and systems to ensure that only one person is responsible for the operation of the equipment or system.

Cleaning, adjusting, or repairing equipments without unplugging them from mains

- Unplugging from live mains is essential before doing maintenance work or handling live conductors or equipment to ensure electrical safety else it may lead to severe accidents.

Fig. 10.1.10: Some electrical safety points while installation and maintenance of a PV system

Fire Safety

It is essential to ensure safety from fire whether a professional is working onsite or offsite. To ensure fire safety, the engineer should ensure the following:

- In case of fire, the glass of the nearest manual call point must be broken and people must be alerted. Alternatively, emergency phone numbers must be dialed to give information about the location of the fire or a fire bell must be used.
- An attempt must be made to extinguish fire with the help of available firefighting equipment.

- If familiar with the plant/machinery/equipment affected by fire, try to avoid spread of fire or isolate it.
- Avoid using water to extinguish fire in case the source of fire is from electrical power. Instead switch the mains off.
- Hose reel and hydrants must not be used at any other time except in case of a fire.
- "No Smoking" signboard must be put up and adhered to at all times.
- A fire drill is normally carried out every six months. It is important to educate and train people to be ready for firefighting during a fire.

After completion of the drill, one should:

- Record the total evacuation time.
- Silence the alarms.
- Bring the fire alarm system back to its normal operating condition.
- Re-evaluate and discuss concerns arising during the fire drill.
- Keep records and notes of the fire drill and update the evacuation checklist report.

Mechanical Safety

While installation of a PV system, mechanical safety for the onsite working professional is the key point to be considered. The working professional should carry mechanical safety equipment while installing the PV modules on the rooftop. There may be a few minor hazards which include burns, or cuts from sharp tools.

Site Safety

The engineer should review the site safety plan and abide by the following guidelines to ensure jobsite safety:

Ensure that the workplace is not messy to avoid the possibility of tripping over.

Take necessary precautionary measure if the workplace is a sloped roof with clutter, as the possibility of falling off the roof significantly increases.

Ensure that the tools are kept in their proper places to avoid any injuries.

Use sunscreen, stay well-hydrated and wear l ight-colored clothes, on bright sunny days while working on a roof top. The chances of getting sunburnt and heat exhaustion increases in such a case.

Use gloves while handling anything that may be rough, too hot, sharp or that might splinter. Special insulating gloves must be worn if working with live voltages.

Ensure proper carrying of tools or materials as dropping tools or materials on oneself, someone else, or on sensitive equipment or materials is hazardous. A hazard is dropping conductive tools across battery terminals.

Ensure proper procedure during installation to reduce or eliminate hazards. When a PV system is being assembled, it presents the possibility of shock to personnel. Improperly installed systems may result in shock or fire hazards after a period of time due to wiring or arcing faults.

Fig. 10.1.11: Ensuring site safety

Safety Regarding Battery Installation

An engineer needs to ensure the safe installation the battery in a PV system and review the maintenance plan. The following figure lists the possible hazards and the precautions to be taken during battery installation:

Sulphuric Acid	Gas	Electrical Shock
• The battery contains sulphuric acid that could cause burns and other injury. Sulphuric acid flows out if a battery is damaged. • For this purpose, appropriate precaution by the means of wearing apron, wearing gloves and protective glasses should be taken. • In case of contact with sulphuric acid, wash thoroughly with water and consult a doctor for treatment.	• There is a possibility of an explosive gas being generated in a battery. • Combustibles flame or spark must be avoided in the battery room. • A battery room must be equipped with a fire extinguisher. • All the tools required during an installation must be wrapped using an insulated electrical tape to reduce the scope of short circuit. • Tools, sundries and other conductive things should not be left on the battery.	• A system including multiple cells may create high voltage. • During the installation of a battery, danger of electrical impact must be avoided.

Fig. 10.1.12: Ensuring safety during battery installation

Use of Safety Gears and PPE

PPE are specially made to protect workers from:

- Injuries caused by impacts of electricity
- Electrical hazards
- Heat and chemicals
- Other occupational safety hazards

The following figure lists the PPE used at workplace:

Fig. 10.1.13: PPE

PPE is the last measure of control when worker exposure to the safety hazards cannot be eliminated by feasible work practices or engineering control.

The following figure lists some responsibilities of employer and the employee regarding use of PPE:

Responsibilities of employer	Responsibilities of employee
• Assess hazards at the workplace • Provide PPE • Determine the use the PPE • Ensure protective helmet for employees at all times to avoid head injuries	• Use PPE as per the instructions received in the training • Inspect condition of PPE regularly • Maintain PPE and keep it in a clean/reliable condition

Fig. 10.1.14: Responsibilities regarding PPE

Protective clothes refer to clothing designed specially to protect workers from potential hazards. Lab coats and ballistic vests worn by electricians, scientists and law enforcement officials respectively fall under this category. The different items of the PPE can either be worn individually or in complete sets.

The following figure highlights different types of protections:

Skin protection

- The most common type of occupational diseases are related to skin and their treatment is expensive.
- Skin damage may be due to severe cuts, punctures, burns, abraisions or extreme temperatures.
- Any form of PPE acting as a barrier between the skin and agent of exposure is considered a skin protector.
- Gloves are an essential PPE for skin protection. For example: heat resistant gloves, cut-resistant gloves and chainsaw gloves.

Eye protection

- Example of eye injury is when solid particles such as wood chips, cement, metal slivers, or sand get into the eye.
- Smaller and larger particles in smoke, such as broken glass, also account for particulate matter causing eye injuries.
- Goggles provide better protection than safety glasses and are effective in preventing eye injury from a dusty environment, chemical splashes, and welding.

Hearing protection

- Industrial noise is an occupational hazard, though it is not visible to the eye.
- About 82% of occupational hearing loss cases occurred to workers in the manufacturing sector.
- Hearing protection must be used whenever an employee's noise exposure exceeds an 8-hour time-weighted average (TWA) sound level of 90 dBA. Noise levels above 115 dBA require control measures for any duration.
- PPE for hearing protection consists of earplugs and earmuffs.

Foot protection

- It may be due to exposure of foot to heavy objects such as barrels or tools that might roll onto or fall on employees' feet, sharp objects such as nails or spikes that might pierce ordinary shoes and molten metal that might splash on feet; it may be due to hot, wet or slippery surfaces.
- Safety shoes have impact-resistant toes and heat-resistant soles and may be electrically conductive for use in explosive atmosphere or nonconductive for protection from electrical hazards.

Fi10.1.15: Types of protection

3.1.7 Reporting Incidents

Reporting any accident or hazard to the supervisor is very important. An engineer should know the reporting procedure.

The following figure lists the types of incidents:

Dangerous occurrence Any occurrence that poses a threat to the safety of people at a workplace and there is imminent risk of death or serious injury to someone	**First aid** The provision of initial care for an illness or injury and is usually performed by non-expert, but trained personnel to a sick or injured person until definitive medical treatment can be accessed
Hazard A source/situation, potentially a harm in terms of human injury or ill-health, damage to property, damage to the environment or a combination of these	**Hazard identification** The process of recognizing that a hazard exists and defining its characteristics
Illness Any physical or mental ailment, disorder, defect or morbid condition, which can be of sudden or gradual development and also includes the aggravation, acceleration, exacerbation or reoccurrence of any pre-existing disease	**Incident** Any unplanned event resulting in or having a potential for injury, ill health, damage or other loss
Injury Any physical or mental damage to the body caused by exposure to a hazard	**Lost time injury** A work related injury which results in a person being absent from work for at least one full shift
Medical treatment injury A work related injury which results in treatment provided by a qualified health professional	**Near miss** An incident that does not produce an injury or disease but has the potential to do so
Risk The likelihood and consequence of an injury or harm occurring	**System failure** Systematic processes that fail to manage the task, activity, process or problem

Fig. 10.1.16: Types of incidents to be reported to a supervisor

The following figure lists the incident reporting procedure:

An employee who encounters a near miss or an accident should report the incident to his superior immediately.

→ If the employee is injured, the concerned superior should ensure the injured is rendered first aid and sent to a clinic/ hospital for treatment, if necessary.

→ If the employee is amputated, his/her bleeding should be controlled at once. Then call for an ambulance at Tel: 102.

→ The superior should immediately inform human resources department, then conduct an investigation of the accident by interviewing injured and/or witness (es).

→ The department concerned has to implement the preventive measures accordingly. Operation or machine involved may need to be suspended/stopped if the situation warrants.

→ The accident statistics chart should be updated till the last accident occurrence.

Fig. 10.1.18: Incident reporting procedure

TIP

Incident reporting is a procedure for reporting incidents. It is essential, so that all near misses/accidents/dangerous occurrences/occupational diseases do not go unnoticed and that preventive measure can be taken to prevent similar reoccurrences.

10.1.8 Precautions for Fire Hazards

While working upon solar systems, an engineer must know how to ensure safety from fire accidents. To ensure fire safety, an engineer must participate in safety workshops or fire drills organised by the organisation. There must be a "No Smoking" signboard in sensitive areas in the organisation. A solar PV Engineer should perform the following steps in case of a fire:

- The glass of the nearest manual call point should be broken and people must be alerted. Alternatively, emergency phone numbers should be dialed to give information about the location of the fire or a fire bell must be used.
- An attempt should be made to extinguish the fire with the help of available firefighting equipment.

- If familiar with the plant/machinery/equipment affected by fire, efforts should be made to avoid spread of fire or isolate it.
- It should be kept in mind that water should not be used to extinguish fire in case the source of fire is from electrical power. Instead, the mains should be switched off.
- Hose reel and hydrants should not be used at any other time except in case of a fire.

10.1.9 Fire Drill

A fire drill is a practice of vacating a building in case of an emergency. A fire drill exercise is shown in the following image:

Fig. 3.1.18: Fire drill exercise

The points shown in the following figure should be kept in focus while conducting a fire drill:

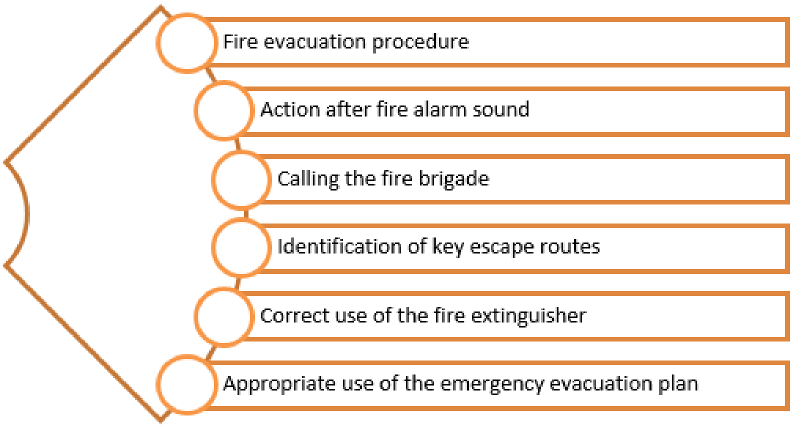


Fig. 3.1.19: Precautions of fire drill

After completion of the drill, the in-charge person should do the following actions:

- Note the practice completing time
- Silence the alarms
- Set the fire alarm system back to its normal working condition
- Discuss and solve concerns arise during the fire drill
- Take all the notes and records of the drill and update the earlier report

10.1.10 Using Fire Extinguisher

The most important equipment to prevent fire accident is the fire extinguisher because fire is a very serious hazard that one faces in the industry.

It can be used immediately to put out a small fire till the specialist fire engine arrives, and consequently save lives and property. The working of fire extinguisher is shown in the following image:

Fig. 3.1.20: Use of fire extinguisher

The following image shows the type of fire extinguisher to be used for different classes of fire:

Fig.10.1.21: Different types of fire extinguisher

UNIT 10.2: Workplace Hazards

Unit Objectives

At the end of this unit, participant will be able to:

1. List different hazards present at the workplace.
2. Identify the safety requirements at workplace.
3. List the precautionary requirements for health at workplace.

10.2.1 Workplace Hazards

A hazard is a source that may impact a person's health in an adverse way. Risk is the chance of being harmed when a person comes in touch with a hazard. The following image shows an example of a hazard and a risk:

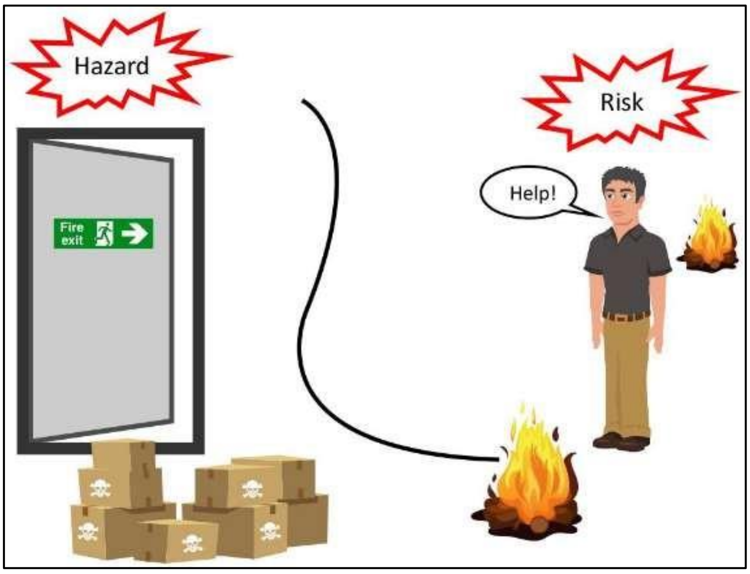


Fig. 10.2.1: Example of hazard and risk

An engineer must know about all the potential sources of accidents so that he or she can take steps to overcome these as quickly as possible. The precautions that need to be taken are as follows:

- Identify and report any potential hazard on time
- Always follow the precautionary rules related to harmful materials as per the company's policy
- Timely report any abnormality of an equipment to the concerned person
- Handle the tools with utmost care

The possible cause of accidents may be unsafe activities and conditions as listed in the following figure:

Unsafe Acts	Unsafe Conditions
• Using equipment carelessly	• Improper safety at workplace
• Using a defective equipment	• Damaged tools or equipment
• Not using protective wears	• Crowded work place
• Not paying attention	• Poor maintenance of workplace
• Not following the instructions	• Poor working conditions
• Avoiding safety precautions	• Poor health conditions

Fig. 10.2.2: List of unsafe acts and conditions at workplace

Chemical Hazards: While working with solar panels, an engineer deals with a very diverse kind of components and e-waste. Some of this may be extremely reactive and this increases the chances of any chemical disease or infection. An engineer is expected to deal with such harmful e-waste with utmost care and only after wearing protective wears to remain safe. Also, this type of e-waste must be dealt in a separate environment to avoid any risk of spread of harmful diseases to other people.

10.2.2 Health and Sanitation Requirements at Workplace

Safety of health and proper sanitation is the topmost priority of every employee while working at a place. Every employer makes a written declaration of its health and safety policy. The policy states the company's commitment to provide proper health and sanitation facilities to its workers.

A workplace requires a health and safety policy for the following reasons:

- To show complete commitment towards the health and safety of the employees
- To prove to the employees that they are safe at their workplace
- To give an overview of the company's objectives, principles, plans, ideas and procedures
- To set the accountability of the organisation for the health and safety of the workers
- To abide by the Occupational Health and Safety Act
- To follow all health policies related to any injury to an employee

In an organisation, there are several policies that are listed and should be followed by every employee. Similarly, an engineer must follow these policies.

The following figure lists some of them:

Environment, Health and Safety (EHS)	Environmental Management System (EMS)	Occupational Health and Safety (H and S)	National Building Code (NBC)

Fig. 10.2.3: Different safety requirements

Environment, Health and Safety (EHS)

EHS is a discipline which provides proper study and implementation of environmental protection and safety at work in practical ways. This is issued by the organisation as per the regulations set by higher authorities to ensure proper conditions of work for all employees.

Environmental Management System (EMS)

EMS refers to the management of an organisation's environmental program in a holistic and brief manner. It involves the structure, planning and resource development of an organisation for developing and implementing policies aimed at protection of the environment.

Occupational Health and Safety (H and S)

This is a policy specially designed to maintain safety and to handle emergency situations at workplace. This policy sets the standards that an employer must follow while making employees work at a particular workplace. This includes the instructions for timely mock drills and setting up of necessary and appropriate signboards at the workplace.

National Building Code (Fire and Safety)

Indian National Building Code 2005 is a document which is created to make sure that the buildings are safe from fire. This document provides comprehensive information about the standards which should be followed for safety purposes while constructing a building.

The following points can be seen in the NBC for fire and safety:

- **Categories of Buildings:** Buildings are classified in categories to put an accurate safety system. Residential buildings are in category A, industrial buildings are in category G, business buildings are in category E and storage areas are in group H.
- **Residential Buildings:** These are divided further in categories as per the accommodation provided.
- **Dangers to Avoid:** According to this code every building should be constructed, equipped and maintained to avoid fire.
- **Rules for Exits:** The building should have a proper exit to facilitate escape in case of fire and it should be as per the capacity of the building.
- **Mandatory Fire Safety Drills:** Fire safety drills should be performed timely to create awareness in people about the actions to be performed in real scenarios.

UNIT 10.3: Standard Safety Practices

Unit Objectives

At the end of this unit, participant will be able to:

1. List the personal protection practices.
2. Identify the standard safety practices at workplace.
3. Explain the improvement opportunities at workplace.

10.3.1 Personal Protection Practices

Personal protective equipment protects an employee from a hazard which he or she is exposed to while working. The equipment is required for protection from skin burn and eyesight and his hearing capacity.

The following figure lists the components of these equipment:

Rubber Gloves:	• To protect hands from hazardous and chemical substances
Flame Proof Aprons:	• To protect body from any burn in case of fire
Helmets/Hard Hats:	• To protect head from any falling components and in case of a fall from a height
Ear Defenders/Plugs:	• To protect ears from loud noise created while working
Safety Boots:	• To protect feet from sharp edged material
Knee Pads:	• To protect knees from damage in case of a fall
Particle Masks:	• To prevent any gaseous substance from entering into body through inhalation
Glasses/Goggles:	• To protect eyes from outside particles and dust

Fig. 10.3.1: Different protective

It is the responsibility of the employer to provide all these equipment to every concerned person and tell them the importance of each and every protective gear. Similarly, the employees have the responsibility of following all instructions related to these equipment and use/ maintain them properly.

3.3.2 Standard Safety Operations at Worksite

An engineer must know about warning signs and their meanings to ensure proper safety at workplace.

The following figure depicts some common warning symbols along with their meanings:

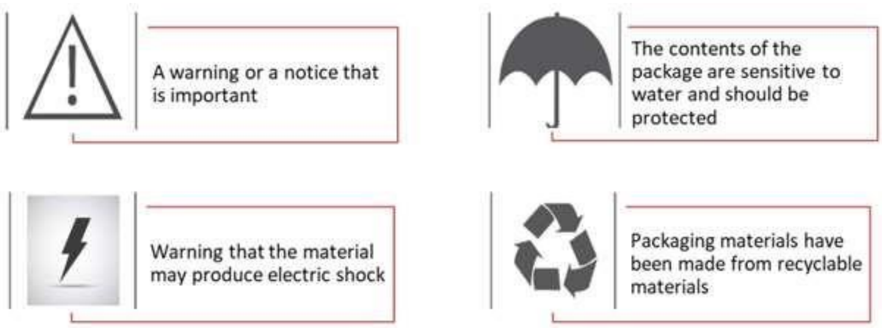


Fig. 3.3.3: Standard safety signs

The following image depicts the standard safety signboards:

No Smoking   Fire Exit   No Entry

Fig. 3.3.4: Standard safety signboards

Apart from all these standard signs, there are some other signs which are generally visible in every kind of industry as shown in the following image:

Fig. 3.3.5: Danger signs

10.3.3 Improvements in Health and Safety at Workplace

Implementing safety practices and improvement in them is not a one go process. Instead, one should keep on bringing positive changes in all these practices. Such policies should change from time to time so that they include all the ongoing and frequent activities taking place in the surroundings. All these improvements can be done in the following ways:

1. **Inspection of Workplace and Formulation of a Plan –** To ensure safety at workplace, the manager or the owner must conduct inspection of the workplace from time to time. The safety can best be ensured by consulting the staff as they have proper knowledge about the surroundings and can easily identify the potential hazards.

2. **Proper Training –** Safety of a workplace not only depends upon the safety equipment installed there but also on the people working there. All employees must be aware about the potential accidents and the way to control or avoid such incidents. To keep them aware, it is the employers' responsibility to provide them timely training.

3. **Good Communication –** Proper communication with employees is a key to avoid all kinds of confusions and to give them confidence to present their views. This will ensure that they will be able to easily approach their manager in case of any unexpected mishap and ensure proper safety at the workplace.

UNIT 10.4: Safety Actions

Unit Objectives

At the end of this unit, participant will be able to:

1. List steps of first aid.
2. Prepare a report of periodic checks.
3. Explain how to provide help to heart attack victims.

10.4.1 Administer First Aid

It is important to be aware of the first aid techniques to provide help in an emergency to the co-workers. It may be required to move an injured person, to provide first aid or to perform rescue operation to minimise loss.

Basic Techniques of Bandaging

The following figure shows the steps for bandaging:

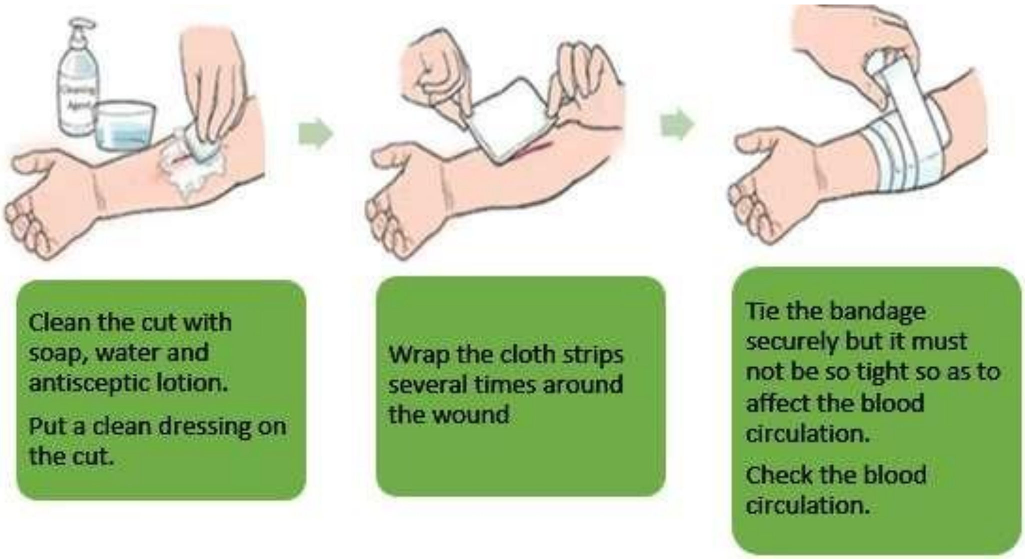


Fig. 3.4.1: First aid procedure

First Aid to Victims of Heart Attack

Heart attack or a cardiac arrest may be caused due to an electric shock. The engineer may need to provide first aid to the victim before emergency services arrive. He/she may need to perform the process of Cardio Pulmonary Resuscitation (CPR) which involves compressing the heart manually to pump blood and assisting in breathing to provide oxygen to the person.

The steps for performing the CPR are as follows:

Step 1: Check for pulse on the wrist and the neck.

Fig. 3.4.2: Checking the pulse

Step 2: Check for breathing and see if the person is conscious.

Step 3: Ask someone to call for medical help.

Step 4: Position the victim, making sure that the person's face is upwards. If the person has an injury on the head or the neck, aids must be provided without moving him/her.

Step 5: Apply pressure on the centre of the chest.

Fig. 10.4.3: Applying pressure

10.4.2 Report of Periodic Checks

The potential hazards, safe methods of working and the precautions to be taken should be documented properly to have a safe system. The document report should include the details as shown in the following figure:

Name	Date/Time of Incident	Date/Time of Report	Location
Environment Conditions	Persons Involved	Sequence of Events	Injuries Sustained
Damage Sustained	Actions Taken	Witnesses	Signature of Supervisor/ Manager

Fig. 3.4.4: Important points of a report

10.4.3 Documented Safety Protocols at Workplace

Documents

The engineer and every employee should be aware of different documents to know the safety instructions and procedures while working.

The following figure lists some of the documents:

- Fire notices such as fire signs and symbols as well as fire evacuation strategies
- Accident reports that mention the accidents that may occur, number of previous accidents and actions taken for them
- Safety instructions for equipment and their proper handling procedures
- Company policies and documents regarding health and safety
- Government notices regarding health and safety guidelines

Fig. 10.4.5: Important documents regarding

10.4.4 Housekeeping

Good housekeeping is proof of good management and shows that engineers are concerned about the health and safety employees on a daily basis. A workplace well in order adds to the safe working environment and minimizes obstacles that are a threat to health and safety.

The following figure lists the benefits of good housekeeping:

- Prevents fire and accidents
- Gives control to employees
- Increases production
- Gives employees the freedom to move
- Gives employees pride
- Protects products and equipment
- Reduces waste

Fig. 10.4.6: Benefits of good housekeeping

Note:
To know more about Benefits of Good House keeping, you may refer to the following link:
https://www.youtube.com/watch?v=EU0ZcSQPwE4

Title of Video: Housekeeping Training Video

By: SheelTalks

Duration: 4 m 27 sec

Exercise

1. Prepare a checklist for workplace health and safety procedure.

2. List the ways to prevent and control infection.

3. Identify the hazards associated with photovoltaic installations.

4. List the purpose of housekeeping.

SCGJ SKILL COUNCIL FOR GREEN JOBS

Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

[Employability skills workbook]

Skill India
कौशल भारत-कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

Annexure

SCGJ | SKILL COUNCIL FOR GREEN JOBS

Module No.	Unit No.	Topic Name	Page no.	URL	QR Code(s)
Module 1 - Introduction to Solar PV Sector in India	Unit 1.1	Types of Solar PV Systems	13	https://www.youtube.com/watch?v=FurOeegcA3Q	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.1	Types of Solar Panel	13	https://www.youtube.com/watch?v=qIpRaQkzqAw	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Business Matters | Where does India stand on solar energy targets	31	https://www.youtube.com/watch?v=wAxxyl-3_uc	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Introduction to the Renewable Energy Financial Model	31	https://www.youtube.com/watch?v=n0MF0iZEOes	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.2	Solar Energy vs. Fossil Fuels: A Comparison	32	https://www.youtube.com/watch?v=u-cq7TwK3Xk	
Module 1 - Introduction to Solar PV Sector in India	Unit 1.3	Understanding Grid Interactive Systems	34	https://www.youtube.com/watch?v=zr9AlfA85ME	

Module 2 -	Unit 2.6	Keys to Solar Marketing and Business Success	81	https://www.youtube.com/watch?v=W3QuS47UrpQ	
Module 10 - Maintain Personal Health & Safety at Project Site	Unit 3.4	Housekeeping Training Video	123	https://www.youtube.com/watch?v=EU0ZcSQPwE4	
Module 4 - Market Research and Cost Estimation of Rooftop Solar Project	Unit 4.3	Cost benefit analysis of Rooftop PV system	141	https://www.youtube.com/watch?v=cTJNPOa1hrI	
Module 5 -	Unit 5.1	Mounting Solar Panels	153	https://www.youtube.com/watch?v=QujOA2TJm3k	
Module 5 -	Unit 5.1	How to read and understand Electricity Bill	167	https://www.youtube.com/watch?v=qiBvhNKd3JY	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.2	Detailed Solar Inverter Installation: A Step-by-Step Guide	197	https://www.youtube.com/watch?v=2D8bMRZ280I	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.2	Permanent Roof Anchor Installation	204	https://www.youtube.com/watch?v=nw-ZM7hLfxU	
Module 6 - Installation of Solar PV rooftop plant	Unit 6.3	Solar Plant Earthing	206	https://www.youtube.com/watch?v=GQMgMZn3WBU	

Module 6 - Installation of Solar PV rooftop plant	Unit 6.4	How to test a solar panel.	213	https://www.youtube.com/watch?v=eh6hzMb69gY	
Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.2	Testing of cable/ wire	223	https://www.youtube.com/watch?v=cgqzezYaYkc	
Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.3	Equipment Grounding for PV Systems	233	https://www.youtube.com/watch?v=-kjM9VWKHnk	
Module 7 - Operation and Maintenance of Solar PV Rooftop Plant	Unit 7.3	Solar Operations and Maintenance	239	https://www.youtube.com/watch?v=pmWlvUWkeQo	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.1	Solar module BOM	257	https://www.youtube.com/watch?v=TraDKXKvPa8	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.3	Rooftop Solar PV Systems	264	https://www.youtube.com/watch?v=gX4MmPkaWDk	
Module 8 - Manage Solar PV Project Lifecycle	Unit 8.3	How to Troubleshoot Solar PV System Assets	273	https://www.youtube.com/watch?v=MLNVA_iusAs	

Skill India
कौशल भारत - कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

N·S·D·C
National Skill Development Corporation
Transforming the skill landscape

Please Scan/ Click QR Code to access e book

SCGJ SKILL COUNCIL FOR GREEN JOBS

Address: 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri New Delhi - 110021
Email: info@sscgj.in
Web: www.sscgj.in
Phone: 011-41792866
CIN No: 00000000